Dissertation

Spatial modeling of charging infrastructure for the traffic flow-based charging demand of battery-electric vehicle fleets

in fulfilment of the requirements for the degree of
Doktorin der technischen Wissenschaften

submitted at the
Energy Economics Group
Faculty of Electrical Engineering and Information Technology
Technische Universität Wien

by

Dipl.-Ing.$^{\text{in}}$ Antonia Golab

Supervisor:

Prof. Dr. Johann Auer
Technische Universität Wien

Reviewer and Examiner:

Prof.$^{\text{in}}$ Dr.$^{\text{in}}$ Sonja Wogrin
Technische Universität Graz

Prof. Dr. Patrick Plötz
Karlsruher Institut für Technologie

Vienna, March 2026

Abstract

Planning charging infrastructure for battery-electric vehicle fleets at large geographic scales requires understanding how charging demand is geographically allocated—and how fleet adoption dynamics, consumer-specific behavior, and spatially varying charging costs shape this allocation. This thesis advances this understanding through four research contributions across passenger car and commercial vehicle fleets in both regional and long-distance transport. Across these segments, the analyses endogenously determine the geographic allocation of charging demand and capture how this is influenced by fleet turnover and adoption patterns, consumer-specific behavior, and spatially varying charging costs.

The three distinct methodologies developed in this thesis share a common basis in spatially resolved traffic flow and origin-destination data, capturing the mobility of vehicle fleets and thus their spatially flexible allocation of charging demands. The method design reflects trade-offs between spatial resolution, temporal scale, and model scope, as required by the respective research questions—ranging from single-year station siting at individual highway segments to multi-year fleet turnover optimization at the regional level and hourly charging load profiling for electricity system integration.

The quantitative analyses are applied to different European case studies at regional, national, and international geographic scopes. Considering the contributions jointly rather than in isolation, three overarching insights emerge: First, across both passenger car and commercial fleet analyses, driving range has a limited impact on required charging capacity and infrastructure costs, with expansion converging towards densification of the charger network rather than initial siting. The spatial density of charging sites emerges as a key factor for enabling equitable battery-electric adoption. Second, the spatial flexibility in charging demand allocation proves substantial—spatially differing charging costs redistribute demands across national borders, while transit flows generate regional demand heterogeneity that domestic fleet projections alone would miss. Third, income-specific behavioral differences that are captured using the value of time produce divergent charging preferences, indicating that a homogeneous consumer representation oversimplifies infrastructure planning.

Overall, the findings indicate that charging infrastructure planning at large geographic scales needs to shift from coverage-based siting toward operations-aware capacity planning, accounting endogenously for the interplay between capacity sizing, station utilization, and vehicle operation. Expansion targets for passenger cars must be grounded in spatial accessibility rather than installed capacity, as spatially dense infrastructure is essential for enabling adoption across all income groups. For commercial fleets, the spatially flexible allocation of charging demands must be leveraged as a demand-side resource for the electricity system, particularly along international corridors where charging demand is shaped by infrastructure and cost configurations across national boundaries. Achieving this requires coordinated action from transport policy makers, infrastructure planners, and electricity system operators beyond the national level.

Contents

Abbreviations

BET battery-electric truck

BEV battery-electric vehicle

HDT heavy-duty truck

LDT light-duty truck

OD origin-destination

1. Introduction

1.1. Background: Electrification of road transport sector

Decarbonizing the transport sector is an urgent priority for achieving the European Green Deal goals [1]. In 2023, the transport sector accounted for 29% of total European greenhouse gas emissions, with road transport alone responsible for around 73% of these transport-related emissions [2]. In response, European countries have committed to reducing greenhouse gas emissions from the transport sector by 90% by 2050 [3]. To achieve this significant reduction over the next two and a half decades, zero-emission drivetrain technologies need to be rapidly adopted on a large scale. At the forefront of these technologies stands the battery-electric drivetrain.

Direct electrification of road transport is widely regarded as the cost-optimal pathway for reducing fossil fuel demand in the sector [4]. This is primarily driven by the superior energy efficiency of electric drivetrains compared to internal combustion engines and the declining costs of electricity from renewable energy sources [5]. Moreover, the battery-electric drivetrain has achieved significant technological and market maturity over the last decade, which is marked by substantial advances in charging speed, driving range, and continuously decreasing battery costs [6]. The shift towards electrification is already evident through market shares across all road transport segments: In 2023, battery electric vehicles accounted for nearly 15% of all new car sales, while 40% of newly bought buses were electric and 10% of light and medium-haul commercial vehicles were battery-electric [7]. Yet these growing market shares have only begun to transform the existing fleet: as of 2023, just 1.8% of the EU passenger car fleet, 0.1% of trucks, and 2.5% of buses were electric [8]. This gap between electrified fleet numbers and total fleet composition underscores both the progress made and the scale of transformation still required.

The trajectory towards electrified road transport fleets is guided by two key policy frameworks at the EU level. A foundational document is the EU Sustainable and Smart Mobility Strategy, published in 2020 [9], which presents a comprehensive vision for decarbonizing all transport modes and sets the milestone of a 90% reduction in transport-related greenhouse gas emissions by 2050. For road transport, the strategy identifies two primary pathways: the uptake of zero-emission drivetrain technologies, primarily battery-electric vehicles, and the shift of transport demand towards lower-carbon modes such as rail.

The legislative translation of this strategy is the Fit for 55 Package, adopted in 2021 [10], which targets a reduction of net greenhouse gas emissions by at least 55% by 2030 relative to 1990 levels. Among its most significant provisions for road transport is the requirement of 100% CO_2 emission reduction for new passenger cars from 2035. Central to the infrastructure dimension of this package is the Alternative Fuels Infrastructure Regulation (AFIR; Regulation EU 2023/1804) [11], which sets binding targets for the expansion of charging infrastructure for battery-electric vehicles. AFIR operates along two dimensions: distance-based targets, which mandate minimum charging capacity at regular intervals along the Trans-European Transport Network (TEN-T) [12], and national targets, which require minimum publicly accessible charging capacity proportional to the number of registered battery-electric vehicles in each Member State.

Charging infrastructure is a fundamental prerequisite for road transport electrification as its availability directly influences vehicle adoption by alleviating range anxiety among passenger car drivers [13]. A key variable in the planning is the geographic placement which shapes the convenience and cost-effectiveness of charging for both private and commercial users [14]. Further, siting charging stations at locations of high utilization supports cost-effective infrastructure investments [15]. However, determining *where* charging capacity is needed and in what quantities remains a non-trivial problem. Capacity sizing must be balanced against infrastructure utilization; the integration with the electricity system adds another layer of complexity; and limited empirical experience—due to currently low adoption numbers—constrains the understanding of how these factors interact [16].

1.2. Problem statement: The geographic allocation of charging infrastructure

A central factor underlying the complexity of geographic charging infrastructure planning is the *mobility* of vehicles. Unlike stationary energy consumers—buildings, industrial sites—whose electricity demand is bound to a fixed grid connection point, vehicles consume energy at varying locations along their travel routes. A vehicle traveling from Vienna to Munich, for example, may recharge at any of several stations along the corridor, and this choice depends on remaining battery range, infrastructure availability, cost, and personal preferences. Battery-electric vehicles are therefore *spatially flexible* in their charging demand allocation: they can charge at different locations along their travel route rather than being bound to a single point. This property is what makes planning for charging infrastructure fundamentally different from that of other energy-consuming sectors, as the resulting charging demand is not geographically predetermined. The challenge is further amplified at large geographic scales, where vehicles routinely traverse regional and national boundaries, as is the case along major European transport corridors.

This leads to the central problem addressed in this work: the geographic allocation of charging demand—and, consequently, the sizing and siting of infrastructure capacities—for aggregate battery-electric vehicle fleets at large geographic scales, spanning regional, national, and international transport networks.

This problem is relevant to three decision-making perspectives: First, transport policy makers who design regulatory frameworks, set targets for charging infrastructure deployment such as the AFIR, and initiate tendering processes for charging infrastructure. Second, transport and charging infrastructure planners, for example road network operators, who must translate these targets into concrete investment and siting decisions—determining where and how much charging capacity to deploy, including tendering for charging point operators and investments in local grid connection capacity. Third, electricity grid and system operators who require long-term estimates of future charging loads to plan grid reinforcement and to integrate demand-side flexibilities for system balancing and congestion management.

The central problem of geographic charging capacity allocation can be decomposed into three interconnected dimensions:

(i) The identification of locations and required capacities for charging infrastructure,

(ii) the influence of spatial configuration on charging demand and driver behavior, and

(iii) the integration of charging operations with the electricity system.

This thesis addresses these dimensions with four research contributions (Contributions 1–4), focusing in particular on **(i)** and **(ii)** and touching upon **(iii)**. Contributions 1 and 3 focus on dimension **(i)**, addressing the identification of charging infrastructure locations and capacities for passenger cars on highway networks and for long-haul trucks along international corridors, respectively. Contribution 2 addresses dimension **(ii)**, examining how the spatial configuration of public charging infrastructure influences battery-electric vehicle (BEV) adoption across income groups and neighboring regions. Contribution 4 spans dimensions **(i)** and **(iii)**, estimating the spatio-temporal distribution of commercial fleet charging demand and its potential for integration with the electricity system.

The methodological foundation for addressing these dimensions is a traffic-flow-based representation of charging demand, derived from origin-destination data coupled with network topology. Origin-destination data indicate the volume of passengers or freight transported between two regions, from which traffic flow by location can be deduced while also capturing the geographic extent of fleet movements. This representation is suited to address the central problem of geographic charging capacity allocation as it captures the spatially flexible charging demand allocation of large traveling fleets. Crucially, it inherently accounts for interregional and transit traffic — vehicles that pass through or across regions and national borders — which fleet-projection-based approaches typically do not account for. Fleet-projection-based methods estimate future charging demand by projecting the number of registered vehicles per region, typically based on market diffusion curves, vehicle stock turnover models, or national fleet forecasts, and then deriving charging needs from the locally registered fleet size. Because these projections are anchored to vehicle registration data at fixed geographic units, they do not capture the movement of vehicles across regional or national boundaries and therefore underestimate charging demand in transit regions while potentially overestimating it at vehicle registration locations. The following section presents how this approach is applied across four research contributions.

	Motivation | Problem Statement		Research Question	Scientific publication
passenger car fleets	• What are required fast-charging capacities along Austria's highway network for long-distance travel with BEV passenger cars and where are these located under different decarbonization scenarios? • Future charging infrastructure demand affected by different decarbonization scenarios for the transport sector (electrification share, technological improvments, shift to other travel modes,. ...)	**RQ1**	What are required fast-charging capacities along Austria's highway network for battery-electric passenger cars under different decarbonization scenarios and where are these located ?	Golab, A., Zwickl-Bernhard, S., & Auer, H. (2022). Minimum-Cost Fast-Charging Infrastructure Planning for Electric Vehicles along the Austrian High-Level Road Network. *Energies, 15(6)*, 2147. https://doi.org/10.3390/en15062147
	• Battery-electric vehicles are currently luxory item that are used in higher-income level classes • Unexplored impact of different public charging infrastructure roll-out strategies in two dimensions: on BEV adoption in different income levels and on neighbouring regions (spillover effect)	**RQ2**	To what extent do the speed and spatial distribution of public infrastructure expansion influence BEV adoption rates among various income groups and across neighboring regions?	Golab, A., Zwickl-Bernhard, S., & Auer, H. (2025). Public charger expansion: Impacts across income classes and beyond regional borders [Manuscript submitted for publication]. *Transportation Research Part D: Transport and Environment.*
commercial fleets	• Cost differences in the installation of charging infrastructure and charging by country and electricity market zone indicate potential for "charging tourism" • Unknown magnitudes of potential charging infrastructure capacity allocations being affected by these differences	**RQ3**	What magnitude of charging infrastructure capacities is affected by geographically varying network fees and electricity prices in the cost-optimal planning of charging infrastructure allocation for long-haul battery-electric trucks?	Golab, A., Bakker, S., Zwickl-Bernhard, S., & Auer, H. (n.d.). Spatial patterns of charging demand distribution in inter-zonal long-haul truck electrification [Manuscript submitted for publication]. *Energy*.
	• Future impact of charging demand of battery-electric commercial fleet and potential for demand-side flexibility • Quantification of large-scale charging demand of battery-electric fleet at hourly temporal and NUTS-2 level under the consideration of transport flows within the countries and transit traffic in Austria	**RQ4**	What is the expected charging demand of the commercial BEV fleet spatio-temporally distributed in Austria in 2040?	Golab, A., Loschan, C., Zwickl-Bernhard, S., & Auer, H. (2025). The value of flexibility of commercial electric vehicle fleets in the redispatch of congested transmission grids. *Energy*, *316*, Article 134385. https://doi.org/10.1016/j.energy.2025.134385

Figure 1.1.: Overview of the four research questions (RQ1–4): each row summarizes the motivation, research question, and corresponding scientific publication. Contributions are grouped by vehicle fleet type — passenger car fleets (RQ1, RQ2) and commercial fleets (RQ3, RQ4).

1.3. The thesis' contributions and addressed research questions

The research questions address existing gaps in the literature on charging infrastructure planning, as detailed later in Chapter 2. The thesis formulates four research questions (RQ1–4), each addressed by a dedicated first-authored scientific journal paper (Contributions 1–4). Figure 1.1 provides an overview.

Contribution 1

Golab, A., Zwickl-Bernhard, S., & Auer, H. (2022). Minimum-cost fast-charging infrastructure planning for electric vehicles along the Austrian high-level road network. *Energies*, *15*(6), 2147. https://doi.org/10.3390/en15062147

Contribution 2

Golab, A., Zwickl-Bernhard, S., & Auer, H. (2026). Public charger expansion: Impacts across income classes and beyond regional borders. *Transportation Research Part D: Transport and Environment.* In press.

Contribution 3

Golab, A., Bakker, S., Zwickl-Bernhard, S., & Auer, H. (n.d.). Spatial patterns of charging demand distribution in inter-zonal long-haul truck electrification. *Energy.* Submitted, March 2026.

Contribution 4

Golab, A., Loschan, C., Zwickl-Bernhard, S., & Auer, H. (2025). The value of flexibility of commercial electric vehicle fleets in the redispatch of congested transmission grids. *Energy*, *316*, 134385. https://doi.org/10.1016/j.energy.2025.134385

Figure 1.2 illustrates the contextualization of the contributions in the existing literature. The research questions address gaps across different BEV application segments, each characterized by distinct requirements for charging infrastructure allocation. Contributions 1 and 2 address the passenger car segment, while Contributions 3 and 4 focus on commercial vehicle fleets. Contributions 1 and 3 focus on charging infrastructure for long-distance transport along highway and international corridor networks. Contribution 2 centers on regional charging infrastructure and its role in enabling broader BEV adoption. Contribution 4 spans both long-distance and regional transport, addressing multiple commercial vehicle segments and their distinct charging patterns.

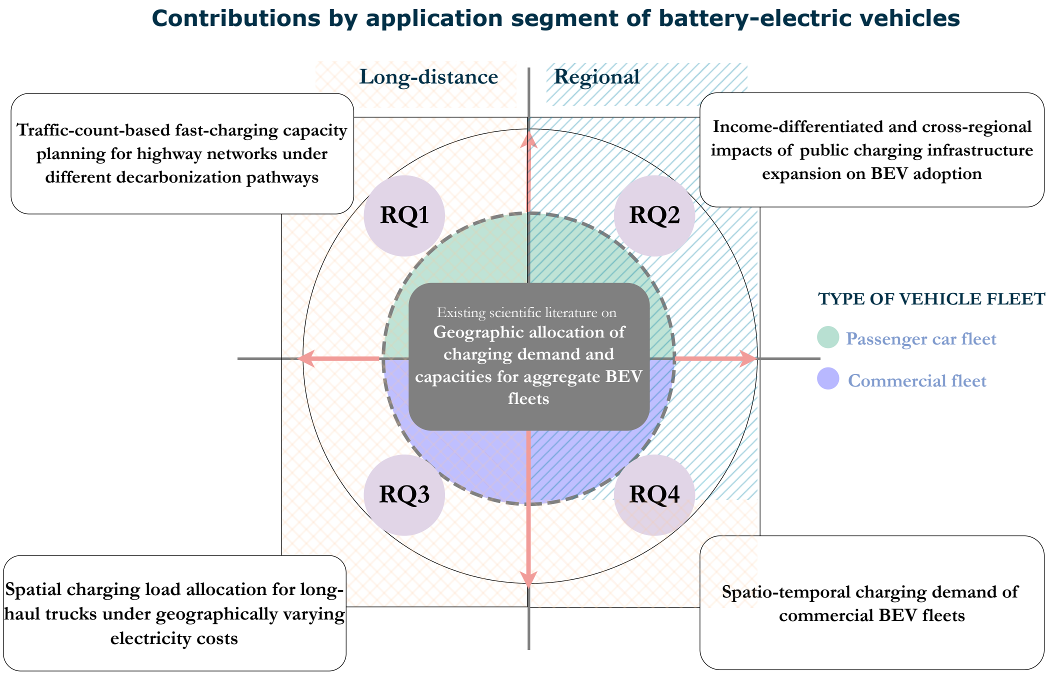

Figure 1.2.: Contributions 1–4 by application segment of battery-electric vehicles. The four research questions (RQ1–4) are positioned along two axes: long-distance vs. regional transport (horizontal) and passenger car vs. commercial fleet (vertical). The central area represents the existing scientific literature on the geographic allocation of charging demand and capacities for aggregate BEV fleets, which the contributions extend.

1.3.1. Contribution 1 (RQ1)

Contribution 1 addresses the spatial allocation of fast-charging infrastructure for battery-electric passenger cars along highway networks. Long-distance travel by car depends on the availability of en-route charging, yet planning such infrastructure involves considerable uncertainty regarding future demand [16]. This demand is shaped by the technical characteristics of the electrified fleet — notably battery capacity, driving range, and charging power — and by the broader transport sector decarbonization pathway, including the rate of BEV adoption, modal shift towards rail and public transport, and changes in overall transport demand [17]. The challenge lies in ensuring sufficient geographic coverage to enable long-distance electric mobility while avoiding overcapacity that results from overly optimistic or misaligned assumptions. The research question addressed is:

Research Question 1 (RQ1): What are the required fast-charging capacities along Austria's highway network for battery-electric passenger cars in 2030 under different decarbonization scenarios, and where are these allocated?

To address this, a charging station allocation problem for highway networks is formulated that combines a node-based optimization structure with flow-based demand representation. Charging demand is defined at discrete nodes corresponding to specific service areas along the network, while the flow of vehicles between nodes captures the spatial flexibility in demand allocation based on traffic direction and driving range. This formulation leverages granular data on service area positions and traffic flow counts along the Austrian highway network. Fast-charging capacities are planned for 2030 under different scenarios of BEV fleet shares, road traffic load, driving range, and charging power levels. Extensive sensitivity analyses are conducted to identify which parameters most significantly affect charging capacity requirements, thereby addressing the uncertainties arising from different transport sector decarbonization pathways.

1.3.2. Contribution 2 (RQ2)

Contribution 2 focuses on public charging infrastructure, which plays a key role in enabling the adoption of passenger BEVs at a large scale. Early adopters of BEVs are predominantly higher-income households with access to home charging [18]. Achieving the long-term goal of broad BEV adoption across all socio-economic backgrounds [14], however, increases the reliance on publicly accessible charging infrastructure. Lower-income households, who are more likely to lack private charging access and to travel longer distances to reach affordable charging facilities [19], are particularly dependent on the availability of public charging infrastructure.

Along with this large-scale adoption, the degrees of freedom in the geographic allocation of charging activity increase with growing driving range [20]. Empirical studies have shown that local public charging infrastructure generates cross-border impacts on BEV adoption in neighboring regions, effects that are likely to be amplified as growing driving ranges allow vehicles to charge in more distant locations [21, 22].

Bringing these aspects together in the context of public charging infrastructure expansion at the regional level, the second contribution addresses the following overarching research question:

Research Question 2 (RQ2): How does public charging infrastructure expansion influence the adoption of passenger BEVs across income groups and neighboring regions?

This is decomposed into two sub-questions:

RQ2.a: To what extent does the spatial density of public charging infrastructure influence battery-electric vehicle adoption across consumers with different income levels?

RQ2.b: How does a local rapid charging infrastructure expansion affect the adoption of battery-electric vehicles in neighboring regions with slower charging infrastructure expansion?

To address these questions, a system-cost optimization of passenger car fleet decarbonization is formulated. Intangible costs that vary by income class are introduced, capturing differential tolerances for both detouring time to recharge and charging duration. The key methodological contribution is modeling detour time as an endogenous decision variable, explicitly linking public charging infrastructure expansion to increased spatial charger density and, consequently, to reduced consumer travel time. To capture cross-regional interactions, the analysis is conducted at the regional level (NUTS-3[1]) using traffic flow and socio-economic data from the Basque Country, Spain. This approach accounts for commuter traffic patterns and the potential to

[1] The Nomenclature of Territorial Units for Statistics (NUTS) is a hierarchical classification of geographic regions used by the European Union for statistical purposes [23]. NUTS-0 corresponds to countries, NUTS-1 to major socio-economic regions, NUTS-2 to basic regions for the application of regional policies, and NUTS-3 to small regions for specific diagnoses.

flexibly allocate charging activity between trip origins and destinations, thereby quantifying income-class-dependent preferences for charger utilization and cross-border impacts.

1.3.3. Contribution 3 (RQ3)

Contribution 3 examines the spatial patterns of charging demand distribution for long-haul battery-electric trucks along the Scandinavian-Mediterranean corridor. For fossil-fueled trucks, country-level differences in diesel taxation have significantly influenced the geographic allocation of fueling activity across borders [24]. As charging costs are location-dependent — varying with electricity-zone-dependent electricity prices and local grid charges [25], [26] — this phenomenon will, with the adoption of battery-electric trucks, shift to the allocation of charging demands. This can lead to high concentrations of charging demand at unfavorable locations in the electricity system, for example where insufficient grid capacity causes congestion. This leads to the following research question:

Research Question 3 (RQ3): To what extent do geographically varying charging costs — arising from electricity price differentials and grid charges across market zones — affect the spatial patterns of charging demand distribution for international long-haul battery-electric trucks?

To address this question, a cost-optimization model jointly optimizes fleet turnover, modal shift between road and rail, and the spatial allocation of charging demands along the Scandinavian-Mediterranean corridor — the longest TEN-T Core Network Corridor, spanning seven countries with distinct electricity market zones — under four fuel and charging cost distribution scenarios. The cost-optimized charging allocation is compared against a heuristic benchmark based on mandatory break locations to quantify the degree of spatial flexibility in truck charging demands. The analysis further examines how modal shift to rail reshapes the geographic concentration of charging demand, and how localized cost instruments — electricity price and grid charge premiums — can redistribute charging loads across border regions.

1.3.4. Contribution 4 (RQ4)

With the electrification of the commercial fleet, significant peak charging power levels are expected. Assessing the impact of these demands on electricity grid and system operation — including dispatch, congestion management, and redispatch — requires charging load profiles at at least hourly temporal resolution, spatially disaggregated to the grid-relevant level. The potential for shifting charging demands temporally to minimize required investments into grid capacity and balancing energy has been demonstrated by many studies [27]. However, quantifying this flexibility potential requires spatially-differentiated estimates of future charging loads at hourly resolution that account not only for regional vehicle fleets but also for transit freight traffic — a critical consideration for European electricity systems where market zones align with national borders while substantial cross-border freight flows generate charging demand. Contribution 4 addresses this challenge by estimating hourly charging load profiles across multiple commercial vehicle segments (light-duty trucks, heavy-duty trucks, and buses) for Austria in 2040:

Research Question 4 (RQ4): What is the expected spatio-temporal distribution of charging demand from the commercial battery electric vehicle fleet in Austria in 2040?

For heavy-duty trucks, the methodology adopts a traffic-flow-based approach, which—unlike fleet-projection-based methods—is able to capture charging demand from interregional and transit transport: vehicles operating across provincial boundaries and international freight passing to, from, and through the country. This approach distinguishes between en-route charging along transport corridors and depot-based charging at trip origins and destinations—a differentiation particularly important given that a substantial share of truck traffic on Austrian highways originates outside the country and would be missed by domestic fleet projections. In contrast, light-duty trucks and buses, which operate primarily within regional boundaries, are modeled based on provincial fleet characteristics and typical driving patterns. This differentiated modeling approach enables capturing the geographic heterogeneity in charging demand magnitudes and temporal charging load patterns across Austria's regions.

1.4. Outline of the thesis

The remainder of this thesis is structured as follows. Chapter 2 reviews the relevant literature and identifies the gaps addressed by each contribution. Chapter 3 presents the methodological approaches developed for the four contributions. Chapter 4 presents the results of the respective case studies. Chapter 5 provides a discussion and synthesis of the findings across all contributions, including limitations. Chapter 6 draws conclusions and formulates implications for the practical implementation of charging infrastructure siting. Chapter 7 outlines directions for future research.

2. Review of State-of-the-Art & Progress Beyond

The following literature review is composed from the literature reviews of the four underlying publications [28, 29, 30, 31], and extended where necessary to provide a coherent narrative across the research questions of this thesis.

The body of scientific literature on the electrification of road vehicles and charging infrastructure planning is extensive. In the following, selected papers are presented that help in understanding the state-of-the-art methodologies and most relevant findings, and that provide the context for the contributions of this thesis.

The literature review is split into two parts. The first part, Sections 2.1 and 2.2, focuses on the electrification of the road sector and the role of charging infrastructure in the adoption of private (Section 2.1) and commercial (Section 2.2) battery-electric vehicles. The second part, Section 2.3, provides insights into the methodological approaches used to answer the research questions related to the demand-driven geographic charging capacity planning, which is the methodological foundation this thesis builds on. In Section 2.4, the core contributions of this work to the existing scientific literature are stated by journal publication.

2.1. Electrification of the private passenger cars sector

2.1.1. Towards large-scale electrification of passenger car fleets

The pathways towards a zero-emissions passenger transport sector encompass three complementary strategies: reducing travel demand, shifting demand towards low-carbon modes, and transitioning to zero-emission technologies [5, 32, 33]. While the first two levers require substantial behavioral and structural changes, technological transformation in the private motorized passenger sector allows for preserving individual mobility patterns and flexibility of current car ownership [34].

For technological substitution, the transformation towards a passenger car fleet of battery-electric vehicles is established as the most cost-efficient way to decarbonize [5]. This is extensively demonstrated: Studies on total cost of ownership show the cost benefits of battery-electric passenger cars, stemming mainly from lower maintenance and fuel costs due to improved drivetrain efficiency [35]. Life-cycle analyses show significant reductions in carbon emissions from the use of battery-electric vehicles, best achieved with renewable electricity sources [36]. This sector-overarching cost-efficiency is also evident in studies that apply energy system models to quantify decarbonization pathways under various future policy scenarios [37]. The performance parameters of the battery-electric drivetrain, particularly the driving range and charging speed, have significantly improved during the last decades, and this, together with future trajectories, solidifies the system-wide benefits of the large-scale adoption of battery-electric vehicles [38].

By the end of 2024, approximately 58 million battery-electric passenger cars were in use world-wide, compared to 1.47 billion fossil-fueled passenger cars, placing the battery-electric vehicle still in its early adoption phase [6]. This phase is predominantly driven by higher-income households with access to home charging [18], for whom the purchase cost barrier — historically significant [39] — is offset through subsidization schemes across many European countries [40] and continuously decreasing vehicle costs, with cost parity now achieved for several car segments [6]. Yet for lower-income households, adoption remains out of reach due to several persisting barriers: (1) limited availability in affordable car segments, with most BEVs concentrated in larger, premium categories [41], (2) declining purchase subsidies in many policy frameworks [6], and (3) reliance on the second-hand market, which remains underdeveloped for BEVs [42, 43].

Schiaroli and Fraccascia [44] empirically analyze socio-behavioral barriers and motivators to BEV adoption across three European countries, identifying range anxiety as one of the most prominent obstacles. Range anxiety encompasses the fear of insufficient charging infrastructure, the perception of limited driving range compared to internal combustion engine vehicles [45], and the inconvenience of charging times that substantially exceed conventional refueling [13], [46]. While these concerns have been partially mitigated over the past decade — through significant improvements in battery technology that have increased range and accelerated charging [47], and the fact that most travel routes fall well within a modern BEV's range [48] — charging infrastructure remains a critical factor. Baden [49] conduct a comprehensive review on range anxiety mitigation and emphasize that the targeted allocation of charging infrastructure is key, with capacities aligned to drivers' needs at convenient locations. Crucially, the authors argue that this allocation must be based on geographic and behavioral segmentation, as different EV usage patterns shape fundamentally different requirements for charging infrastructure.

2.1.2. Role of charging infrastructure roll-out for passenger cars

Charging infrastructure is categorized by the power level, the use case, and access type [50]. *Home* charging is typically at low voltage levels, providing charging directly to the car while parked at home. Home charging is currently the most popular method, as existing BEV owners are predominantly homeowners for whom it provides the most convenient and cost-efficient charging option [51]. Often, BEVs at home chargers are integrated directly into the home energy management system, and co-optimized with home PV electricity production, battery storage, and/or other home appliances [52, 53]. Similar to home charging, *workplace* charging provides charging at a regular daily parking place. However, workplace charging availability remains limited, as provision falls short due to unclear benefits to companies [54, 55].

Public charging provides the opportunity to recharge en route, thereby alleviating range anxiety. Most importantly for the current adoption stage, it enables BEV owners without access to home charging to recharge their vehicles [50, 56]. Functionally, public charging infrastructure resembles conventional fueling stations but is typically located at parking facilities due to longer charging times. The geographic placement of public charging stations aims to achieve both high spatial coverage and convenience in integrating charging into regular mobility patterns [57]. Convenience charging stations are often co-located with destinations for other activities, such as shopping [58], leading to their placement at so-called *points of interest*, e.g., shopping malls [59]. *Slow* charging offers an opportunity to recharge during longer stays and daily downtime, day or night, delivering comfort comparable to home or workplace charging. *Fast* charging, i.e., charging at higher power levels using DC technology, enables rapid recharging, e.g., during errands or en route [60]. From a business perspective, the challenge lies in balancing investment costs and competitive pricing against uncertainty in expected charging demand and its growth at individual sites [61]. Therefore, consumers currently face higher charging prices at public charging stations [25, 52].

Simultaneously, there is an access gap in home charging that correlates with income level [14, 62]. This makes lower-income groups especially dependent on public charging infrastructure and, consequently, exposed to higher charging prices than car owners with home chargers. However, charging point operators are more likely to allocate charging stations in high-income communities due to greater certainty about BEV adoption and charging demand [63, 64]. This creates a paradox: those most dependent on public charging have the least access to it. Therefore, equity in charging-station allocation is an important factor, alongside subsidization and other monetary incentives, to support rapid adoption across all income classes [14].

Understanding these equity implications requires examining how charging infrastructure influences BEV adoption more broadly. In scientific literature, the relationship between local charging infrastructure development and the uptake of BEVs remains unclear. There exists a wide range of choice models that relate the decision to purchase a BEV to socio-economic variables and reveal the influence of factors such as income level, home ownership, household size, a person's attitude towards renewable technologies, and the availability of charging infrastructure [65, 66]. Range anxiety is directly related to the absence of charging infrastructure, and poses a psychological barrier to the adoption of BEVs [13]. However, empirical studies show that the relationship between public charging infrastructure deployment and BEV uptake is more nuanced. Illmann and Kluge [67] find an evident but weak relationship between the deployment of public charging infrastructure and the uptake of BEVs in German communities. Another finding of their study is that the uptake is more affected by the charging speed provided at the charging stations, rather than the number of chargers themselves. Similar is found in the work of White et al. [68] who conduct an analysis for three U.S. metropolitan areas. In their work, no direct impact of range anxiety on the BEV adoption is found. The authors observe that higher spatial densities of public charging infrastructure increase the perceived social pressure to adopt BEVs. Globisch et al. [69] confirm through questionnaires that lower charging duration is valued over higher spatial density and lower fees, while Gebauer et al. [70] demonstrate that widespread DC fast-charging significantly improves potential users' perception of electromobility.

Public charging infrastructure deployment creates spatial spillover effects that extend beyond regional boundaries—a pattern consistent with the network nature of transport infrastructure more broadly [71, 72, 73]. Empirical evidence demonstrates these cross-regional impacts for battery-electric vehicle adoption: Qian et al. [21] find that fast-charging infrastructure in centrally located regions positively influences BEV adoption in neighboring areas, while Sheng et al. [22] show that in New Zealand public charging infrastructure in neighboring regions has a more significant impact on local BEV adoption than locally installed infrastructure itself. Both studies conclude that effective charging infrastructure planning requires spatially holistic approaches that account for cross-border effects rather than treating regions as isolated systems. However, these analyses are based on observed historical patterns; prospective assessments of how strategic infrastructure deployment decisions affect BEV adoption across regional boundaries have not been conducted.

2.2. Battery-electric vehicle adoption in commercial fleets

2.2.1. Electrification across commercial fleet segments

Currently, the electrification of the commercial fleet is in significantly earlier stages than the private passenger vehicle sector, though substantial progress has been made: 40% of buses and 10% of light and medium commercial vehicles sold in 2023 were battery-electric [74]. The core difference to the passenger sector is cost-efficiency as the primary driving force, rather than social aspects and personal freedom considerations in private passenger car ownership [75]. While the three fundamental decarbonization levers—reducing demand, shifting toward low-carbon modes, and replacing conventional drivetrains—apply to commercial transport as well [76], practical constraints limit their application. Modal shift from road to rail freight faces substantial barriers: cost-competitiveness only at longer ranges, bottlenecks in rail infrastructure capacity requiring high investment, and limited flexibility due to track-based operations and last-mile delivery challenges [77, 78]. Against this background, direct electrification of drivetrains has emerged as the primary pathway to decarbonize commercial road transport [79, 80].

The technology pathway toward electrification, however, differs markedly from the passenger car sector due to the greater diversity in vehicle segments, particularly regarding daily ranges and payload requirements [81, 82]. For light-duty commercial applications, battery-electric vehicles have become the clear solution for reducing fossil fuel consumption, mirroring developments in the passenger car sector [83]. For heavy-duty applications, particularly long-haul trucking, the technology landscape initially appeared less certain. Zero-emission options narrow to battery-electric vehicles and hydrogen fuel-cell electric vehicles, with early analyses suggesting that battery costs increase substantially for long-haul applications due to payload losses from heavy batteries needed to overcome long ranges while transporting heavy goods [84]. Noll et al. [82] analyze total cost of ownership across different truck segments in European countries, finding that while cost benefits for light- and medium-duty battery-electric drivetrains are clear, cost-competitiveness for long-haul heavy-duty applications varies substantially by country, driven primarily by operational expenditures, country-dependent tolls, and relative differences between diesel and electricity prices. Recent technological developments, however, have shifted this picture considerably. Plötz [80] argue that the core challenges of implementing battery-electric trucks for long-haul applications—particularly long charging times and limited range—have been largely overcome. Their analysis concludes that battery-electric technology will play the central role throughout all relevant road transport segments, including heavy-duty trucks, while hydrogen fuel-cell electric vehicles will remain in niche applications due to weak cost-effectiveness.

Despite this technological convergence toward battery-electric drivetrains, significant barriers to adoption remain. Scherrer et al. [85] identify two fundamental challenges for fleet operators based on empirical evidence from the German logistics sector: establishing adequate high-power charging infrastructure represents a major deployment hurdle, while operational planning must accommodate substantial flexibility in charging timing and location. This flexibility requirement stems from persistent uncertainties regarding the availability and reliability of public charging infrastructure—a constraint that forces fleet operators to maintain adaptive operational strategies rather than implementing fixed routing and scheduling patterns. These operational challenges create planning uncertainties that extend beyond vehicle acquisition decisions to encompass fundamental questions about charging infrastructure requirements and their integration with existing logistics operations [86, 87].

2.2.2. Charging infrastructure for battery-electric trucks

The transition to battery-electric vehicles in the commercial sector requires integrating time-consuming charging activities into operational schedules and logistics processes [88]. This integration challenge is compounded by the need to balance operational flexibility with cost-effectiveness: while the most economical approach involves low-power overnight charging at fleet depots [89], this strategy requires fleet operators to invest in grid connection reinforcement and on-site charging infrastructure [90], with costs potentially offset through on-site renewable generation and smart charging strategies [91, 92].

The diversity of commercial vehicle operations necessitates different charging infrastructure strategies. Al-Hanahi et al. [93] categorize charging infrastructure into three functional types based on operational requirements: (1) Depot charging—at fleet facilities, yards, and industrial sites—serves vehicles that return daily to a fixed location, enabling cost-effective overnight charging but requiring substantial upfront infrastructure investment. (2) Public charging infrastructure addresses vehicles with variable schedules requiring intermediate charging away from depots, supporting multi-shift operations and providing backup capacity for fleets with power-limited depot infrastructure. (3) Highway charging infrastructure supports long-haul operations through both fast charging during mandatory 45-minute driver breaks (currently at 350 kW, advancing toward 1 MW "megawatt charging") and slow charging during extended rest periods. The adequate driving ranges of modern battery-electric trucks enable strategic integration of these charging opportunities into logistics operations [94], though doing so requires careful spatial planning of infrastructure deployment along freight corridors. These different infrastructure types create distinct spatial planning challenges: while depot and public charging respond primarily to local demand patterns, highway charging infrastructure must align with long-distance freight flows along major corridors—including substantial international transit traffic that may not be captured in domestic fleet projections.

2.3. Modeling of charging demand, charging infrastructure capacities, and roll-out

2.3.1. Graph-based charging infrastructure allocation for highway networks

Charging infrastructure allocation problems are commonly formulated based on a graph representation with nodes representing locations of potential charging sites, and edges representing connecting road infrastructure. This approach enables high geographic resolution in modeling the spatial relationships between charging sites and demand locations [95, 96]. Metais et al. [95]

differentiate these graph-based optimization tools by how the charging demand is derived from vehicle activity. The authors differentiate between node-based, flow-based[1], and tour-based approaches: Node-based approaches use a discrete expression of charging demand at nodes, which can be, for example, derived from local population density. In the flow-based application, the demand is expressed via vehicle flow numbers, often derived from origin-destination (OD) data. A concrete example is the flow-capturing approach by Kuby and Lim [97]. Charging stations are allocated along edges, and under cost minimization, the goal is to "capture" traffic flows, ensuring that in the optimized configuration, all vehicles have the opportunity to charge along their respective routes. The third category comprises tour-based approaches, which consider individual agents in the system and, therefore, model individual vehicle charging demand at high resolution. These concepts are suited to different charging applications and entail different types of input data and computational complexities. In general, the literature suggests that node-based approaches are well suited for urban areas, while flow-based approaches are better suited for highway networks [96, 98].

Building on the flow-based framework, several studies have developed optimization models for highway charging infrastructure. Wang et al. [99] present a multistage equilibrium model utilizing origin-destination data to determine optimal highway charging infrastructure placement, targeting reductions in charging duration, waiting times, driver expenses, and infrastructure installation costs. In a flow-based framework, Jochem et al. [98] seek to minimize the total number of charging points while determining their capacity requirements according to the traffic flow each station would serve. Similarly adopting a flow-based methodology, Yan [100] develops a two-stage genetic algorithm with multi-objective optimization to reduce both infrastructure construction expenses and charging costs incurred by drivers. Napoli et al. [101] introduce an iterative algorithm for station placement that establishes the spacing between consecutive charging facilities based on vehicle driving range capabilities. A sequential site selection approach is developed by Csiszár et al. [102], determining charging station locations according to traffic flow patterns and the population density of adjacent settlements.

More recent methodological developments have integrated on-site capacity planning into the optimization framework. Rose et al. [103] develop a node-capacitated flow-capturing approach based on origin-destination data that plans charging stations at discrete nodes while inherently planning on-site capacity based on past driven distances within the optimization, applying this to Germany. Building on this approach, Nugroho et al. [104] adopt the node-capacitated flow refueling location model to design fast-charging infrastructure for inter-city travel in the Greater Jakarta Area, analyzing different expansion scenarios for various electric vehicle ranges. Rehman et al. [105] extend this framework by proposing a capacitated flow refueling location model that jointly optimizes fast-charging station placement and on-site battery storage sizing.

Across these studies, vehicle driving range consistently serves as the primary constraint for determining the maximum permissible spacing between network charging stations. Csiszár et al. [102] examine how driving range influences infrastructure requirements, demonstrating that longer ranges enable a fixed charging network to serve a greater total number of vehicles. Kavianipour et al. [106] investigate the combined effects of enhanced driving range and charging power on infrastructure needs, revealing that technological advancements reduce infrastructure

[1]In the paper, it is referred to as *path*-based, but in the broader literature it is more commonly termed *flow*-based.

investment requirements as improved vehicle capabilities diminish the necessary number and capacity of charging points. Using mixed-integer linear programming, Wang et al. [99] evaluate optimal infrastructure deployment under budget constraints, identifying a saturation threshold in flow coverage—the proportion of BEVs successfully charged along their routes—beyond which further increases in driving range do not alter infrastructure requirements. Rehman et al. [105] find the strongest impact of range variations between 100-300 km, with moderate effects beyond this threshold. Similarly, Nugroho et al. [104] observe in a sensitivity analysis across different BEV sizes and battery capacities that larger BEVs require only 6% fewer total stations in projections between 2025-2035, suggesting diminishing returns from range increases.

2.3.2. Modeling the impact of charging infrastructure expansion on passenger BEV adoption

Research examining optimal infrastructure expansion and its influence on BEV adoption can be classified into three literature streams: first, system dynamics modeling approaches that capture complex interactions among various system components [107, 108]; second, agent-based modeling frameworks featuring fine-grained spatial resolution in both charging station placement and driver mobility patterns [109, 110]; third, large-scale optimization methodologies mapping cost-efficient transport sector decarbonization trajectories, typically from a system-wide perspective [19]. Gao et al. [108] employ a system dynamics framework to project urban BEV adoption trajectories. The relationship between vehicle adoption and public charging deployment emerges through utility function optimization, incorporating variables such as average fueling station access distance, refueling duration, vehicle range, and total ownership expenses. An agent-based methodology presented by Wolbertus et al. [110] explore comparable research questions while also concentrating on urban BEV uptake. Unlike Gao et al. [108], this work models the adoption-infrastructure interaction at finer geographic granularity, specifically at the district scale. The BEV purchase decision is formulated as dependent on walking distance to the nearest charging point. Consumer segmentation by neighborhood reflects an assumption of consistent EV purchase propensity within geographic areas. Wolbertus et al. [110] compare single versus hub deployment strategies. The *single* approach distributes new charging stations broadly, whereas the *hub* strategy concentrates multiple charging points at centralized locations. Their findings indicate that distributed deployment positively influences BEV uptake, while concentrated hub placement demonstrates substantially weaker effects on adoption rates.

Regarding the third literature category, numerous studies address limitations inherent in total-cost-minimization frameworks [19, 111]. These limitations particularly concern consumer behavioral dimensions, including technology selection, modal preferences, and travel patterns. Common solutions involve coupling multiple models or augmenting individual model capabilities [112]. In their review paper, Venturini et al. [113] identify various methodologies for incorporating behavioral elements. For technology choice modeling, they highlight two particularly effective approaches: consumer segmentation to reflect realistic adoption dynamics, and integration of behavioral patterns alongside intangible (non-monetary) cost factors. Tattini et al. [19] implement a comparable methodology that considers income-differentiated consumer segments, assigns monetary valuations to mode- and technology-dependent time consumption, and constrains purchase decisions by household expenditure limitations. Luh et al. [112] segment consumers by residential type and incorporate diverse charging infrastructure categories. Their analysis examines how various charging access options affect BEV deployment and passenger transport emissions reduction. Results demonstrate that home charging availability critically influences adoption patterns. Public slow and fast charging become essential for widespread uptake when private charging access remains constrained. A subsequent publication by these authors [114] extends this analysis by incorporating travel time valuation, encompassing both recharging periods and detour time required to access charging facilities. This follow-up work emphasizes improvements in public transport utilization and the associated travel duration implications across different trip categories.

2.3.3. Consideration of different income levels in charging infrastructure planning

Methodological frameworks for charging infrastructure planning rarely differentiate consumers by income level, though several recent studies have begun addressing equity considerations. Loni and Asadi [115] propose a multi-objective model that minimizes charging site deployment and operation costs while maximizing both charging demand fulfillment and accessibility as a measure of social equity access. Batic et al. [116] develop a graph-neural network approach that deploys public charging infrastructure with explicit focus on establishing equitable network coverage, directly addressing accessibility gaps in low-income areas. Voinea et al. [117] propose an optimization-based framework that begins with spatio-temporal forecasting of adoption patterns across consumer segments defined by BEV model classes—specifically, more affordable short-range BEVs versus long-range premium models—which correlate with adopters' income levels. The proposed framework is applied to the spatial resolution of postal code level. Building on such segmentation approaches, Wang et al. [118] quantify charger access disparities derived from indicators including median household income, employment rate, and ethnicity, introducing an equity measure directly into their objective function to minimize access disparities between consumer groups within a multi-objective optimization. The charging infrastructure is planned explicitly for workplaces, while considering the spatial resolution of neighborhoods. Across these studies, income-level differentiation relies on spatial proxies that correlate neighborhood characteristics with income levels, rather than on behavioral parameters that capture income-dependent differences in charging preferences.

2.3.4. Overarching charging station deployment and battery-electric truck adoption at large-scale

Recent research advances dynamic infrastructure planning approaches that integrate fleet electrification trajectories with charging network development, without sacrificing spatial precision in site allocation. Karam et al. [77] examine this integration from a fleet operator's economic perspective within Denmark's borders. Their methodology employs network-based road representation combined with joint optimization of fleet operations costs and infrastructure investment decisions. The analysis tracks fleet conversion through 2040, determining when and where charging capacity additions become economically justified alongside vehicle replacements. Bakker et al. [119] broaden this scope considerably in their analysis of Norwegian freight transport evolution. Their framework addresses long-term infrastructure investment to support technological diversification and modal transitions in goods movement at the national scale. Operating at NUTS-2 geographic resolution, they model coordinated pathways for freight decarbonization encompassing both zero-emission vehicle adoption and modal shifts toward lower-carbon alternatives such as rail, alongside the corresponding expansion of transport networks and energy supply infrastructure. Their approach determines refueling and recharging capacity requirements directly from freight volumes served by each powertrain technology at each location, treating energy infrastructure sizing analogously to transport mode capacity planning. Building on this foundation, Pagnier et al. [120] integrate the Norwegian freight model from Bakker et al. [119] with a comprehensive energy system representation. Their coupled analysis reveals substantial interdependencies between heavy-duty electrification levels and charging infrastructure demands, demonstrating that spatially differentiated electricity pricing significantly influences optimal decarbonization strategies for the freight sector. However, these studies operate within national boundaries. The spatial allocation of charging demand along international corridors—where electricity costs and grid charges vary across market zones—has not been addressed in an optimization framework.

2.3.5. Traffic-flow based modeling of charging profiles of commercial fleets for large-scale integration to the electricity system

To explore the potential benefits of electric vehicle flexibility provision—both for private passenger transport and commercial purposes—at large scales for the electricity system, various studies have estimated charging demand from fleet projections, driving patterns, and charging behavior assumptions, though differing substantially in their spatial resolution, fleet composition, and methodological frameworks. At the national scale, Li et al. [121] determine future charging loads for China by simulating daily charging behaviors based on regional fleet projections and driving patterns for private battery-electric vehicles, buses, taxis, and commercial light-duty vehicles, though notably excluding freight trucks. Similarly, Suski et al. [122] model charging demand for electric buses, cars, and two-wheelers in Malé, Maldives, incorporating charging flexibility into a capacity expansion framework, but also omitting heavy-duty commercial transport. While alternative methodologies exist—such as spatially explicit approaches that allocate charging demand based on observed vehicle movements [123] or traffic flow data [124]—these are primarily applied to passenger vehicle fleets at local or metropolitan scales. The spatial allocation of heavy-duty truck charging demand, which differs fundamentally from passenger vehicle patterns due to long-haul operations and corridor-specific infrastructure requirements, receives limited attention in the context of large-scale electricity system integration. In particular, fleet-projection-based approaches—which estimate charging demand from locally registered vehicle stocks—inherently miss the interregional and transit freight traffic that constitutes a substantial share of highway charging demand, especially in European transit countries. A traffic-flow-based approach, grounded in origin-destination data and observed traffic volumes, is needed to capture this demand component.

2.4. Contribution to the progress beyond state-of-the-art

Against this background, the novelties of this work are stated in the following, organized by research question.

2.4.1. Contributions to Research Question 1

- A novel charging station allocation problem is formulated for highway networks that combines the node-based optimization structure with flow-based demand representation. This allows for direct leverage of openly available granular data on service area positions and densely sampled traffic flow counts along the highway network.

- Fast-charging infrastructure capacities are planned for the Austrian highway network for 2030 under different scenarios of BEV shares in the passenger car fleet, road traffic load, BEV driving range, and charging power levels. Extensive sensitivity analyses identify the parameters with the greatest impact on charging capacity requirements. The scenarios address uncertainties in demand resulting from the different transport sector decarbonization pathways.

2.4.2. Contributions to Research Question 2

- A fundamentally different approach to representing income-group-specific requirements in passenger car charging capacity modeling is introduced. Rather than relying on spatial proxies, income groups are differentiated through values of time, reflecting behavioral heterogeneity regardless of residential location. This allows distributional consequences of charging infrastructure coverage decisions to be examined beyond neighborhood-level access disparities.

- The detour charging time, i.e., the extra walking or driving time to reach the nearest charging station, is used to link the impact of infrastructure roll-out to BEV adoption. In this way, exhaustive analysis based on high-resolution geographic data is avoided, while the densification of public charging infrastructure is still represented and connected to BEV adoption.

- The spillover effects between regional charging infrastructure expansion and BEV adoption are analyzed across neighboring regions within a techno-economic optimization framework. Previous studies of such cross-regional spillovers have relied exclusively on empirical data to explain observed patterns but have not assessed the prospective impact of strategic infrastructure deployment on the electrification of the passenger car fleet.

2.4.3. Contributions to Research Question 3

- The *spatial* flexibility in the allocation of charging demand for the aggregate long-haul battery-electric truck fleet is examined at an international scale, enabling quantification of the effect of geographically varying charging costs — including electricity prices and grid charges — on the cost-optimal spatial distribution of charging loads.

- The case study covers the Scandinavian-Mediterranean corridor, one of the most freight-intensive in the Trans-European Transport Network and, as the longest of its Core Network

Corridors, characterized by extensive long-haul transport routes spanning from Northern to Southern Europe. Traversing multiple countries with distinct electricity prices and grid charges, this corridor provides a well-suited case for examining cost-optimal transport decarbonization under both the transition to battery-electric trucks and potential modal shift to rail.

2.4.4. Contributions to Research Question 4

- Origin-destination data are used to determine spatio-temporal patterns of charging demand for the future battery-electric truck fleet, extending beyond exclusively fleet-projection-based methods to account for charging demand from interregional and transit trucks.
- The study focuses on Austria and, thereby, addresses the lack of empirical spatio-temporal charging patterns for diverse commercial fleets at sub-national resolution for Central European countries. Austria's central geographic location and substantial commercial transit traffic make it representative of conditions across comparable countries in the region.

3. Methods

3.1. Overview

The methodologies developed in this thesis address different aspects of the geographic allocation of charging capacities for electric vehicles. Three distinct optimization and modeling approaches are formulated, as addressing each research question necessitates different model features and fleet representations (see Table 3.1 and Figure 3.1). Model 1 determines the optimal siting and sizing of fast-charging stations along highway networks in a static setting, i.e., for a single planning period. Model 2 performs a joint optimization of the long-term transition of vehicle fleets and the deployment of charging infrastructure over a multi-year horizon. It is built around a core linear program that is extended with consumer income differentiation and detour time modeling for passenger cars (RQ2) and with state-of-charge tracking, mandatory break regulations, and modal shift for long-haul trucks to rail (RQ3). Model 3 generates hourly charging load profiles from traffic flow data for commercial vehicle fleets with transport routes of both local and interregional extent.

The three models reflect a trade-off between spatial resolution, temporal scope, and the set of endogenous variables. Model 1 operates at the highest spatial resolution (individual highway segments and service areas) but considers only a single planning period and takes charging demand as exogenous. Model 2 extends the scope to multi-year fleet turnover and joint optimization of infrastructure deployment at a coarser regional level. Model 3 shifts the focus to hourly temporal resolution to capture the intra-week patterns of charging load relevant for electricity system integration.

Table 3.1.: Overview of methodological designs in this thesis. Model 2 serves as the core formulation and is extended differently for RQ2 and RQ3.

	Model 1 (RQ1)	**Model 2 – RQ2**	**Model 2 – RQ3**	**Model 3 (RQ4)**
Method type	MILP	LP / MILP	LP	Deterministic simulation
Vehicle segment	Passenger cars	Passenger cars	Heavy-duty trucks	Commercial fleet (heavy-duty truck (HDT), light-duty truck (LDT), buses)
Charger type	Fast charging only	Home, work, public (slow & fast)	Public en-route (fast)	Depot, en-route, opportunity
Demand representation	Charging demand defined by location	Endogenous BEV adoption by OD pair	Endogenous BEV adoption by OD pair	Exogenous BEV fleet sizes by type, location & OD pair
Infrastructure decisions	Siting & sizing of stations	Capacity expansion by type & location	Capacity expansion by type & location	None
Temporal scope	Single year	Multi-year (2020–2050)	Multi-year (2020–2050)	Representative week
Geographic scope	National highway network	Sub-national (NUTS-2/3)	International corridor	National
Spatial resolution	Service areas / grid cells	NUTS-2/3 regions	NUTS-2 regions	NUTS-2 regions
Key outputs	Station locations, number of charging points per station	BEV adoption by income class, charger capacity by type & region	Charging demand allocation along corridor, modal split road/rail	Hourly charging profiles by location & fleet type

Figure 3.1 illustrates how the three models relate to the central challenge of charging capacity planning. The three models are applied as stand-alone analyses. The following sections describe each model in detail. As the three models are independent, each section uses its own nomen-

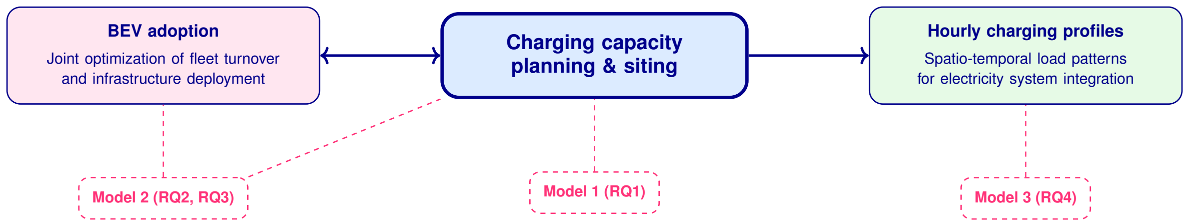

Figure 3.1.: Overview of the three methodological approaches and their relation to the central challenge of charging capacity planning.

clature; symbols may be reused across sections with different meanings. The methodological descriptions are based on the underlying journal publications, Golab et al. (2022), Golab et al. (2026), Golab et al. (n.d.), and Golab et al. (2025).

3.2. Static charging infrastructure planning

3.2.1. Problem description

The model is designed to cost-optimally plan fast-charging infrastructure along highway networks. The planning encompasses the geographic allocation, i.e., the siting, and the sizing of charging capacities. The model reflects cost-optimal decisions from the perspective of an infrastructure planner seeking to determine a fast-charging infrastructure set-up for ensuring charging demand coverage of battery-electric passenger cars traveling long distances. This decision-making could, for example, apply to a highway network operator preparing tenders for charging service providers, or to a public authority defining minimum infrastructure requirements. The considered costs encompass two components: the onsite preparations to enable the support of large capacities at a location and the hardware and construction costs associated with the installation of individual charging points. The planning refers to a defined network topology described by intersections, distances along highway segments, and positions of service areas, where potential sites for charging stations are exclusively service areas.

3.2.2. Model features and assumptions

The *HighCharge* model is formulated as a mixed-integer linear program (MILP). The underlying methodology is based on a graph representation of the highway network (Figure 3.2). Road connections between two junctions or between a junction and a network endpoint are referred to as *segments*, and the vertices as *nodes*. Two types of nodes are distinguished: nodes representing possible sites for charging station installation along segments (*node type 1*), and nodes representing highway junctions or endpoints (*node type 2*). At each node, a charging demand exists which can be covered at the node itself or at adjacent nodes within a defined range. The model allocates sites for charging station installation and determines the optimal number of charging points at each site.

The charging infrastructure is planned in consideration of the traffic flow and technological parameters of the vehicle's battery. The following key assumptions related to the representation of charging demand are made:

- The BEV fleet traveling along the highway network is treated as a homogeneous quantity, allowing to consider accumulated charging demand and translating this into the optimal sizing of a charging station. Based on this assumption, the technical parameters of an *average* BEV are assumed, such as average driving range ($\overline{dist}_{range,BEV}$), energy consumption ($\overline{d}_{spec,BEV}$) and charging capacity ($\overline{P}_{charge,BEV}$).
- All charging demands, $\sum_{n,k} \hat{D}_{n_k}$, result from the energy consumption of BEVs driving along the highway and need to be compensated for in total by charging stations built along the highway network.
- Highway charging infrastructure is primarily used by long-distance drivers as BEV owners mostly charge at home or at work.
- A fast-charging infrastructure along a highway network is designed based on peak demands, including seasonal peak demand ($\hat{f}_{l_k}$) and hourly peak demand, during a day (γ_h).

3.2.3. Input-Output relations

Figure 3.2 displays the input-output relations of the modeling framework. In order to capture the spatially granular charging demand along the highway network, the inputs are grouped into four categories: (i) the composition of the car fleet is represented through the share of BEVs, the share of long-distance drivers, and traffic load data; (ii) the technological characteristics of BEVs are captured through energy consumption, driving range, and charging power; (iii) constraints and costs of the charging infrastructure are included through the peak capacity of charging stations, grid connection constraints, and infrastructure costs; and (iv) the physical layout of the road network is represented through the graph topology described above. These inputs feed into a demand calculation step that derives the charging demand at each node of the network. The *HighCharge* optimization model then determines at which nodes charging stations are installed and how many charging points each station is equipped with.

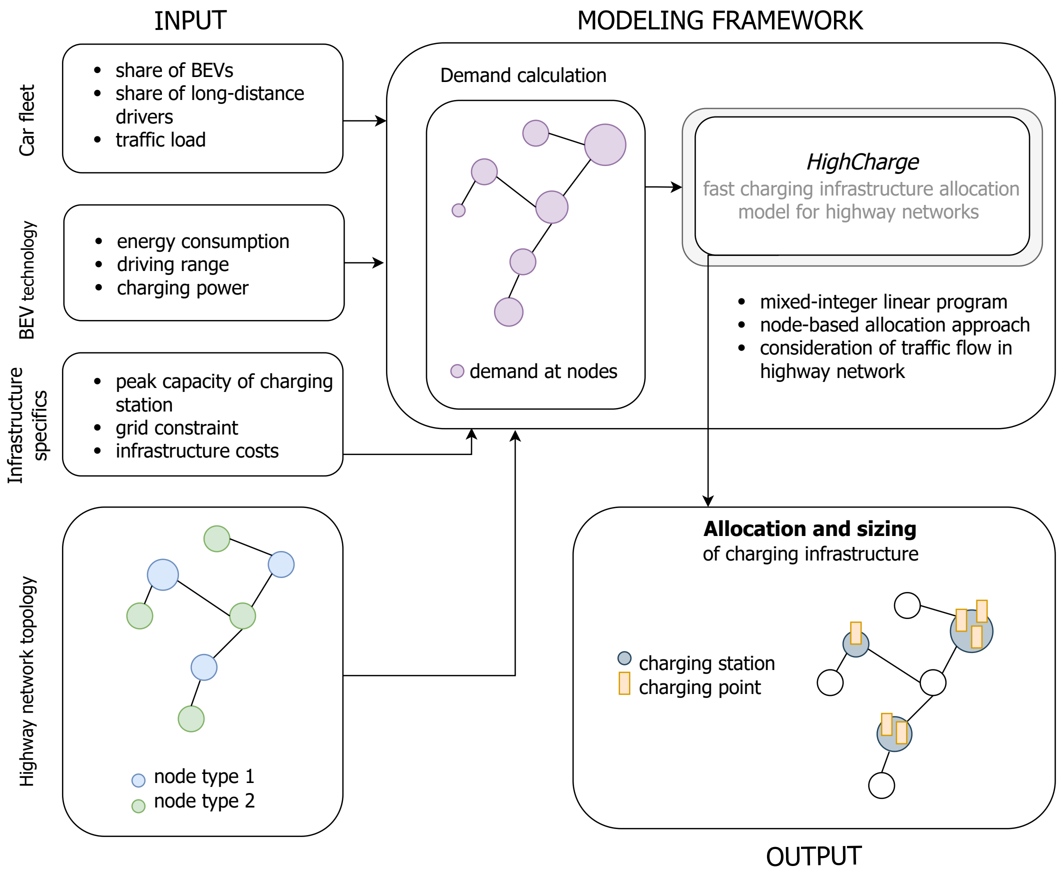

Figure 3.2.: Overview of the modeling framework for RQ1. Input parameters feed into a demand calculation and the *HighCharge* optimization model, which determines the allocation and sizing of charging infrastructure along the highway network.

3.2.4. Mathematical formulation

Table 3.2 lists the nomenclature used in the following equations.

Table 3.2.: Nomenclature for the *HighCharge* model (RQ1).

Symbol	Description	Unit / Domain
Sets and indices		
l	Node	–
k	Highway segment	–
l_k	Node l on segment k	–
n_k	Node with charging demand on segment k	–
A_{n_k}	Set of nodes within distance $dist_{max}$ of node n_k	–
s_{l_k}	Segment to which node l_k belongs	–
Decision variables		
X_l	Whether a charging station is installed at node l	$\{0, 1\}$
Y_{l_k}	Number of charging points at node l_k	$\mathbb{Z}_{\geq 0}$
$\hat{E}_{l_k}^{charged,n_k}$	Energy charged at node l_k to cover demand from n_k	kWh
$\hat{D}_{l_k}^{in,n_k}$	Demand transferred into node l_k from adjacent node	kWh
$\hat{D}_{l_k}^{out,n_k}$	Demand transferred out of node l_k to adjacent node	kWh
Parameters		
c_X	Cost of installing a charging station (site preparation)	€
c_Y	Cost per charging point (hardware and construction)	€

Symbol	Description	Unit / Domain
$\hat{D}_{n_k}$	Charging demand at node n_k	kWh
$\overline{dist}_{range,BEV}$	Average driving range of BEV	km
$\overline{d}_{spec,BEV}$	Average specific energy consumption of BEV	kWh/km
$\overline{P}_{charge,BEV}$	Average charging power of BEV	kW
$dist_{max}$	Maximum distance for demand coverage	km
r_{l_k,l'_k}	Rationing parameter for traffic flow splitting at junctions	$[0, 1]$
P_{max}	Maximum power capacity at a station	kW
$\hat{P}_{CP}$	Power per charging point	kW
$\hat{f}_{l_k}$	Seasonal peak demand factor	–
γ_h	Hourly peak demand factor	–

The objective function of the optimization model is to minimize the total charging infrastructure investment costs:

$$\min_{X_l, Y_{l_k}, \hat{E}_{l_k}^{charged,n_k}, \hat{D}_{l_k}^{in,n_k}, \hat{D}_{l_k}^{out,n_k}} c_X * \sum_l X_l + c_Y * \sum_{l_k} Y_{l_k} \tag{3.1}$$

The charging infrastructure is designed in such a way that all demand $\sum_{n,k} \hat{D}_{n_k}$ in the network is covered. For each node the following energy balance is defined:

$$\hat{E}_{l_k}^{charged,n_k} - \hat{D}_{l_k}^{demand,n_k} = \hat{D}_{l_k}^{out,n_k} - \hat{D}_{l_k}^{in,n_k} \quad : \forall n_k, l_k \in A_{n_k} \tag{3.2}$$

Set A_{n_k} encompasses all nodes l_k which are within the distance of $dist_{max}$ to node n_k. Within this set radius around a node n_k, the energy demand stemming from node n_k, $\hat{D}_{n_k}$ has to be covered. Between the nodes l_k, which are part of A_{n_k}, the demand is shifted and for this, the following holds for adjacent nodes l_k and l'_k:

$$\hat{D}_{l_k}^{out,n_k} * r_{l_k,l'_k} = \hat{D}_{l'_k}^{in,n_k} \quad : \forall n_k, \{l_k, l'_k\} \in A_{n_k} \tag{3.3}$$

For parameter r_{l_k,l'_k}, the following applies:

$$r_{l_k,l'_k} = \begin{cases} 1, & \text{if } s_{l_k} = s_{l'_k} \\ < 1, & \text{otherwise} \end{cases} \tag{3.4}$$

$$\sum_{l_k} r_{l_k,l'_k} = 1 \tag{3.5}$$

This rationing parameter is, therefore, only lower than 1, if node l is a junction point and in this case, the $\hat{D}_{l_k}^{out,n_k}$ needs to be split to simulate traffic flow partitioning at a junction point. For example, if three segments meet and it is assumed that traffic flow is split equally between these segments at the junction, then $r = 1/2$.

The demand coverage by a charging point and charging station is defined through:

$$\sum_{n_k} \hat{E}_{l_k}^{charged,n_k} \leq \sum_{k} Y_{l_k} * \overline{P}_{charge,BEV} \quad : \forall l_k \tag{3.6}$$

$$\frac{P_{max}}{\hat{P}_{CP}} * X_l \geq \sum_{k} Y_{l_k} \quad : \forall l \tag{3.7}$$

Equation 3.7 establishes the relation between the sizing of a charging station, i.e., the number of charging points, and the binary variable, whether a charging station is installed at node n_k.

3.3. Joint optimization of fleet transition and charging infrastructure deployment

3.3.1. Problem description

Taking the perspective of a social planner, the model minimizes total transport system costs over a multi-year horizon. These costs encompass vehicle purchase and maintenance, charging costs, infrastructure investment and operation, and travel-time-related costs. The model jointly optimizes three interdependent decisions: (1) the composition and turnover of the vehicle fleet across drivetrain technologies over time, (2) the geographic allocation and sizing of charging infrastructure of different types and at different locations, and (3) the spatial allocation of charging activity along travel routes. This joint optimization captures the interdependence between infrastructure availability and vehicle adoption. The geographic scope is represented as a graph, where nodes correspond to regions and edges to transport connections between them. Transport demand is given exogenously as a set of origin-destination trips. The vehicle stock is tracked by generation to capture technological improvements over time and the resulting heterogeneity of technical parameters within the fleet, e.g., in specific energy consumption.

This core formulation serves as the foundation for both RQ2 and RQ3. For RQ2, it is extended with consumer-group differentiation by income class, monetary budget constraints, and a mixed-integer formulation linking charging infrastructure expansion to fueling detour times. For RQ3, it is extended with state-of-charge tracking along routes, travel time accounting under mandatory driver break regulations, and modal shift to rail.

3.3.2. Model features and assumptions

The basis of the methodology follows the classic formulation of the Network Design Problem [125]. Table 3.3 provides an overview of the state-of-the-art methods that are integrated here, together with corresponding extensions for the present work. The core model is formulated as a linear program with the objective of minimizing the total costs related to the vehicle stock, charging, and charging infrastructure, and other transportation costs, encompassing investments, operation and maintenance costs. It is important to note here that while the model formulation is general enough to accommodate different fueling technologies, the applications in RQ2 and RQ3 focus exclusively on battery electric vehicles and charging infrastructure.

The core features and functionalities of this model are:

- **Network design and transport demand coverage:** The geographic relation between multiple neighboring regions is conceptualized based on a graph representation. Nodes represent regions and edges represent the connections between neighboring regions. The modeling framework is designed based on origin-destination data. The travel demand is given exogenously by a set of trips that are defined by their origin-destination pair, purpose, distance, and consumer group on an annual level. These trips can be covered using different vehicle types, drivetrain technologies, and fuels. For trips that cross multiple geographic locations, i.e., for inter-regional trips that cross at least one other region, there can be a definition of multiple possible travel paths between the origin-destination pair.

- **Vehicle stock modeling:** The vehicle stock is modeled based on the numbers of different vehicle types and technologies. The *generation* of a vehicle is a particularly important dimension here. It defines the year in which a vehicle was purchased on the market. Each generation is associated with different technological attributes which allows to capture the technological improvements in the battery size and charging speed of BEVs while also modeling the presence of vehicles with variable technological attributes within the fleet.

- **Sizing of different types of charging infrastructure:** Charging infrastructure is placed at nodes of the network. It is sized based on the geographic allocation and differentiated by type, which may vary by the provided fueling power, maximum utilization rate, and availability. The expansion of fueling infrastructure capacity follows a defined set of investment periods.

- **Spatially flexible allocation of charging activity:** For a given trip, there exists a set of different opportunities to charge, e.g., at work, at a public charging station, or at home. The decision on where the charging demand is covered is endogenous and can change annually.

Decarbonization of the transport sector is represented through a carbon price that is included in the cost of fossil fuels. This carbon price increases the relative cost of conventional drivetrain technologies over time, thereby driving the cost-optimal transition towards electric vehicles within the system cost minimization.

This core model is extended for RQ2 (Section 3.3.5) and RQ3 (Section 3.3.6) with application-specific features described in the respective sections.

Table 3.3.: Overview of integrated model functionalities drawn from state-of-the-art methodology. '-' implies that the included modeling feature is integrated without significant adaptations from the initial concepts. The listed sources are references to research papers that describe or apply these modeling approaches.

Modeling feature	State-of-the art references	Adoption/extension in this work
Network design and transport demand coverage	[125]	-
Consumer groups of different income-class levels and level of service measure	[19], [114]	-
Vehicle stock modeling	[126], [127]	-
Sizing of different types of charging infrastructure	[119], [112]	-
Spatially flexible allocation of charging activity	[124]	-

3.3.3. Input-Output relations

Figure 3.3 provides a conceptual overview of the input-output relations of the core optimization model. The inputs are grouped into five categories: (i) the *travel demand* is specified through regional and inter-regional origin-destination trip data that define the transport activity to be served; (ii) the *vehicle types, drivetrain technologies, and fuels* characterize the available mobility options and their technological attributes; (iii) the *graph topology*, consisting of nodes (regions), edges (transport connections), and route definitions, represents the spatial structure of the transport network; (iv) the *parameters on charging infrastructure expansion*, differentiated by charging power, define the technical constraints and maximum possible speed of build-up; and (v) the *cost parameters*, including fuel prices, technology costs, fueling infrastructure costs, and route-related or location-specific costs, define the economic framework of the optimization.

The model produces four main categories of outputs: the allocation of *trips per vehicle type and drivetrain technology*; the resulting *vehicle stock shift* over time; the spatially resolved *charging demand by location*; and the required *charging infrastructure expansion* across the network.

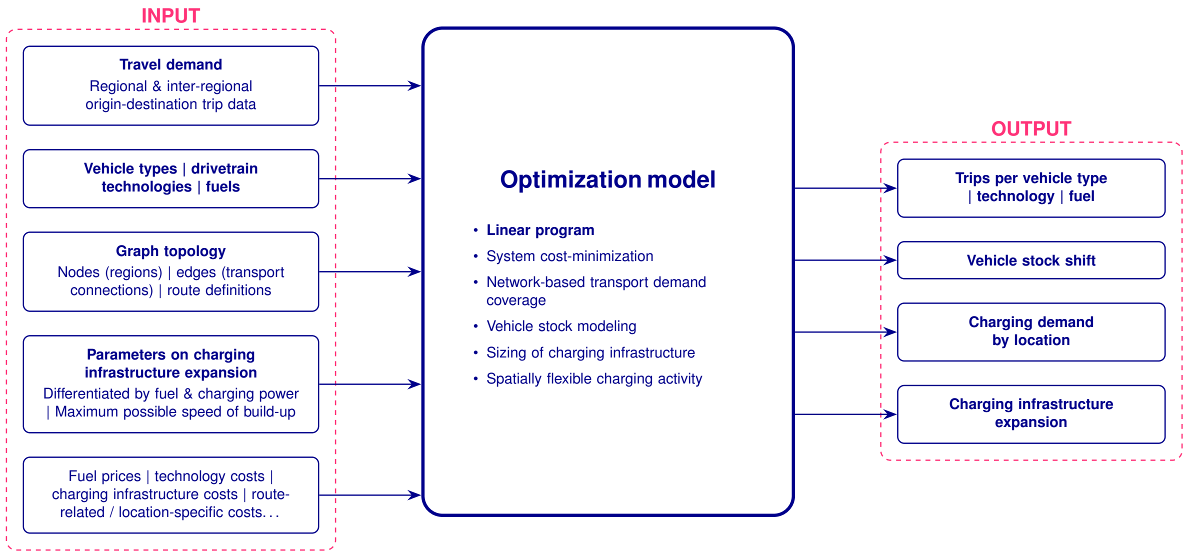

Figure 3.3.: Input-output relations of the core optimization model.

Table 3.4 contains the relevant nomenclature used in the following equations.

Table 3.4.: Nomenclature for the core model formulation.

Sets and indices	Description		
$y \in \mathcal{Y}$	year		
$r \in \mathcal{R}$	trip between origin-destination (OD) pair		
$e \in \mathcal{E}$	geographic location		
$v \in \mathcal{V}$	type of vehicle		
$t \in \mathcal{T}_v$	drivetrain technology fueled by a specific fuel		
$g \in \mathcal{G}$	year that vehicle was introduced to market		
$l \in \mathcal{L}$	types of fueling infrastructure (by capacity)		
$\mathcal{Y}^{exp}$	set of years of investment decisions		
$\mathcal{R}_e$	set of OD pairs whose trip goes through location e		
$\mathcal{L}_t$	types of infrastructure by drivetrain technology		
$\mathcal{L}^{vt}$	types of charging infrastructure available for vehicle type v and technology t		
$\mathcal{E}_r$	set of geographic locations along OD pair r		
$\mathcal{E}_{rl}$	geographic locations along OD pair r with available charging type l		
Decision variables		**Unit**	
f_{yrvtg}	number of person-trips	-	$[0, \infty[$
h_{yrvtg}	number of vehicles	-	$[0, \infty[$
h^{+}_{yrvtg}	newly purchased vehicles	-	$[0, \infty[$
h^{-}_{yrvtg}	number of vehicles leaving the vehicle stock	-	$[0, \infty[$
$s_{yrvtgle}$	energy fueled	kWh	$[0, \infty[$
$q^{+,\text{charg\_infr}}_{ytle}$	added charging infrastructure	kW	$[0, \infty[$
Parameters			
γ_l	factor between annual peak hour and total annual demand	–	$[0, 1]$
$Q^{\text{charg\_infr}}_{tle}$	initial charging infrastructure	kW	$[0, \infty[$
$\hat{Q}^{\text{charg\_infr}}_{tle}$	upper limit for charging infrastructure capacity	kW	$[0, \infty[$
D^{spec}_{vtg}	specific energy consumption	kWh/km	$[0, \infty[$
W_{vtg}	load factor	persons/vehicle	$[0, \infty[$
L^{annual}_{vtg}	annual range	km	$[0, \infty[$
L_r	length of trip	km	$[0, \infty[$
C^{CAPEX}_{yvtg}	expenditure costs for vehicles	€/vehicle	$[0, \infty[$
F_{yr}	exogenous travel demand	trips	$[0, \infty[$
H^{init}_{rvtg}	initial vehicle stock	vehicles	$[0, \infty[$
Λ_{vtg}	maximum vehicle lifetime	a	$[0, \infty[$
α^{h}_{t}	Bass-diffusion innovation rate parameter	–	$[0, 1]$
β^{h}_{t}	Bass-diffusion imitation rate parameter	–	$[0, 1]$

3.3.4. Mathematical formulation

In the following, we[1] elaborate on the mathematical formulation of the core model and the extensions from state-of-the-art methodologies.

Objective function

The objective function comprises costs for the vehicle stock ($C^{\text{vehicle\_stock}}$), required fueling (C^{fueling}), and costs for fueling infrastructure ($C^{\text{infrastructure}}$). All cost components are discounted to net present value using the discount rate δ. The objective function is defined as follows:

$$\underset{f,h,h^+,h^-,s,q^+}{minimize} \; z = \sum_{y} \frac{1}{(1+\delta)^{y-y_0}} \left(C_y^{\text{vehicle\_stock}} + C_y^{\text{fueling}} + C_y^{\text{infrastructure}} \right) \tag{3.8}$$

The vehicle stock cost $C_y^{\text{vehicle\_stock}}$ sums the capital expenditures for newly purchased vehicles across all regions, vehicle types, technologies, and generations. The fueling cost C_y^{fueling} sums the charging and fueling costs for all energy fueled across locations. The infrastructure cost $C_y^{\text{infrastructure}}$ sums the investment and operation costs for the expansion of charging infrastructure across all types and locations.

Demand coverage

The vehicle stock is sized to cover all car trips and the required fueling activity is allocated among the visited geographic locations of a car trip and different fueling infrastructure types. Decision variable f expresses the number of trips covered by a specific vehicle type and technology-fuel pair of a specific generation. The sum over all vehicle types, technology-fuel pairs, and vehicle generations equals the exogenously given travel demand:

$$\sum_{v \in \mathcal{V}} \sum_{t \in \mathcal{T}_v} \sum_{g \in \mathcal{G}} f_{yrvtg} = F_{yr} \qquad : \forall y, r \tag{3.9}$$

Sizing of fueling infrastructure

Fueling infrastructure is expanded for each investment period, $y' \in \mathcal{Y}^{exp}$, and sized based on an assumed factor which represents the ratio between the annual demand peak and the annual total demand, γ_l:

$$Q_{tle}^{\text{charg\_infr}} + \sum_{y' \in \mathcal{Y}_y^{\text{exp}}} q_{y'tle}^{+,\text{charg\_infr}} \geq \gamma_l \sum_{r \in \mathcal{R}_e} \sum_{g \in \mathcal{G}} s_{yrvtgle} \qquad : \forall y, t, e, l \tag{3.10}$$

[1]Throughout this chapter, the first-person plural “we” is used where text is directly based on the underlying journal publications.

s refers to the annual fueling demand. The expansion of fueling capacities is also limited by maximum values:

$$Q_{tle}^{\text{charg\_infr}} + \sum_{y' \in \mathcal{Y}_y^{\text{exp}}} q_{ytle}^{+,\text{charg\_infr}} \leq \hat{Q}_{tle}^{\text{charg\_infr}} \qquad : \forall y, t, l, e \tag{3.11}$$

Vehicle stock sizing and turnover

Based on the trips covered by a vehicle type of a technology, the vehicle stock is sized as follows:

$$h_{yrvtg} \geq \frac{L_r}{W_{vtg} L_{vtg}^{\text{annual}}} f_{yrvtg} \qquad : \forall y, r, v, t, g \tag{3.12}$$

$$h_{yrvtg} = h_{yrvtg}^{\text{exist}} + h_{yrvtg}^{+} - h_{yrvtg}^{-} \qquad : \forall y \in \mathcal{Y} \setminus \{0\}, r, v, t, g \tag{3.13}$$

$$h_{yrvt(g=y-\Lambda_{vtg})}^{-} = h_{yrvt(g=y-\Lambda_{vtg})} \qquad : \forall y, r, v, t \tag{3.14}$$

Equation 3.12 expresses the sizing of the vehicle stock via a vehicle's annual range L^{annual} and load factor W. With equations 3.13 and 3.14, the vehicle stock is aging. Vehicles exit the vehicle stock when a maximum lifetime is reached.

Fleet initialization

The vehicle stock in the initial year y_0 is initialized from exogenous data. The carried-over stock depends on the relationship between the model year y, the vehicle generation g, the initial year y_0, and the vehicle lifetime Λ_{vtg}:

$$h_{yrvtg}^{\text{exist}} = \begin{cases} H_{rvtg}^{\text{init}} & \text{if } y = y_0,\ g < y_0,\ y_0 - g \leq \Lambda_{vtg} \\ h_{(y-1)rvtg} & \text{if } y > y_0,\ g < y,\ y - g \leq \Lambda_{vtg} \\ 0 & \text{if } g = y \text{ (new generation)} \\ 0 & \text{if } y - g > \Lambda_{vtg} \text{ (lifetime exceeded)} \end{cases} \tag{3.15}$$

where H_{rvtg}^{init} is the exogenously given initial vehicle stock for pre-existing generations. For the initial year, pre-existing generations are initialized from data; for subsequent years, the stock is carried over from the previous year. Vehicles of generation $g = y$ (newly produced) have no carry-over. New vehicle purchases are restricted to the current generation ($h^{+}_{yrvtg} = 0 \; \forall g \neq y$).

Spatially flexible allocation of charging demand

Fueling demand is derived from the number of trips f and via the specific demand D^{spec} of the vehicle:

$$\sum_{l \in \mathcal{L}_t} \sum_{e \in \mathcal{E}_{rl}} s_{yrvtgle} = \frac{D^{\text{spec}}_{vtg} \cdot L_r}{W_{vtg}} f_{yrvtg} \qquad : \forall y, r, v, t, g \tag{3.16}$$

Technology transition rate

The year-over-year change in the vehicle stock of a drivetrain technology t is bounded by a Bass-diffusion-type constraint to prevent unrealistically rapid technology transitions:

$$\begin{aligned} \left| \sum_{v} \sum_{g} h_{yrvtg} - \sum_{v} \sum_{g} h_{(y-1)rvtg} \right| \leq \alpha^h_t \sum_{v,t',g} h_{yrvt'g} + \beta^h_t \sum_{v} \sum_{g} h_{(y-1)rvtg} \\ : \forall y \in \mathcal{Y} \setminus \{y_0\}, r, t \end{aligned} \tag{3.17}$$

where α^h_t and β^h_t are the maximum rate-of-change parameters relative to the total vehicle stock and the previous stock of the respective technology.

3.3.5. MILP extension for regional application (RQ2)

3.3.5.1. Extended problem description and model features

The RQ2 extension addresses the role of public charging infrastructure roll-out in the adoption of BEVs across consumers of different income classes. Current BEV adoption is concentrated among high-income households with access to home charging, while the long-term decarbonization of the passenger car fleet requires adoption across all socio-economic backgrounds. As BEV adoption broadens, the central role of home charging shifts towards public charging infrastructure, whose spatial density and speed of deployment become determining factors for adoption.

To capture these dynamics, the core model is extended in two ways. First, consumer groups differentiated by income class are introduced. Each consumer group is characterized by a specific value of time (VoT), a monetary budget for vehicle purchases, and a share of home ownership. The VoT monetizes the travel time perceived by the consumer and rises with income level. The monetary budget constrains the expenditures for vehicles and home charging infrastructure over a defined purchase cycle. This differentiation allows the model to endogenously determine

BEV adoption pathways that vary across income classes, reflecting their heterogeneous economic constraints and time valuations.

Second, the detouring time needed to reach the closest charging station is introduced as an endogenous decision variable. This links the expansion of public charging infrastructure to the spatial density of the charger network and, consequently, to a reduction in consumers' total travel time. Because the detour time depends on the discrete number of charging stations deployed at a given location, this extension transforms the core LP into a mixed-integer linear program (MILP). The combination of income-specific parameters and endogenous detour time enables the analysis of how charging infrastructure deployment strategies affect BEV adoption differently across income classes and across neighboring regions.

In summary, the RQ2 extension adds the following features to the core model:

- **Consumer groups by different income-class levels and level of service measure:** Consumer-specific parameters allow differentiation between income class levels in two dimensions: the value of time (VoT) and the monetary budget. The VoT relates to the monetary value of travel time as perceived by the consumer. The travel time is the level of service (LoS) that is provided by a drivetrain technology, measured in hours. The value of time rises with the income level of a consumer. The monetary budget refers to the budget of a consumer that is spent on the purchase of a new vehicle and is defined together with a time horizon, reflecting the frequency of the purchase of a new vehicle.

- **Fueling detour time[2]:** The detouring time needed to reach the closest charging station is included as a decision variable that is tied to the expansion of the charging infrastructure. This extends the core LP to a mixed-integer linear program and allows linking the investments in charging infrastructure to an improved level of service.

Figure 3.4 provides an overview of the extended input-output relations. Compared to the core model (Figure 3.3), the additional inputs include income-class-specific parameters (VoT, monetary budgets, home ownership shares) and the initial detour time assumptions. The additional outputs include BEV adoption shares by income class and the evolution of detour times by location.

Table 3.5 lists the additional symbols introduced by the RQ2 extension.

Table 3.5.: Additional nomenclature for the RQ2 extension.

Sets and indices	Description
$u \in \mathcal{U}$	consumer group
$i \in \mathcal{I}_l$	index for detour time reduction potential for type of charging infrastructure l
$\mathcal{L}^{nothome}$	charging infrastructure that is not allocated at home of vehicle owner
$\mathcal{L}^{home}$	charging infrastructure that is allocated at home of vehicle owner
S_j^u	$= \{y_m \in Y \mid j \in [m, m+\tau_u-1]\}, m = 1, ..., \|Y\| - \tau_u + 1$ subsets of time periods for which a monetary budget is defined by consumer group

Decision variables		Unit
$n_{yurvtgle}$	number of charging vehicles	$[0, \infty[$

[2]The term *fueling* is used here because the detour time applies to all drivetrain technologies considered in the model, including conventional vehicles detouring to fossil fuel infrastructure. For battery-electric vehicles, this corresponds to the additional travel time required to reach the nearest public charging station.

b_{yiel}	VoT-weighted detour cost at location e for charging type l in year y	€	$[0, \infty[$
w_{yiel}	binary decision variable indicating whether detour reduction i is active	-	$\{0, 1\}$
$q^{+,\text{charg\_infr,home}}_{yurtle}$	added charging infrastructure at home	kW	$[0, \infty[$
$q^{\Delta+,\text{charg\_infr}}_{ytle}$	continuous residual capacity beyond the selected discrete capacity level cap^{lb}	kW	$[0, \infty[$
budget^{+}_{yur}	budget overrun	€	$[0, \infty[$
budget^{-}_{yur}	budget underrun	€	$[0, \infty[$
Parameters			
LoS^{f}_{yrvt}	level of service for trip along route r using vehicle type v and technology t	h	$[0, \infty[$
VoT_{u}	value of time of consumer group u	€/h	$[0, \infty[$
$Q^{\text{charg\_infr,home}}_{urtle}$	initial charging infrastructure at home	kW	$[0, \infty[$
$\hat{Q}^{\text{charg\_infr,home}}_{utle}$	upper limit for charging infrastructure capacity at home (by consumer group)	kW	$[0, \infty[$
$\text{Cap}^{\text{tank}}_{vtg}$	tank capacity	kWh	$[0, \infty[$
speed_{vt}	average driving speed by vehicle type and technology	km/h	$[0, \infty[$
$\text{fueling\_time}_{yrvt}$	fueling time for a trip along route r	h	$[0, \infty[$
α_{iel}	reduction $i = 1, \ldots, I$ of detour time	–	$[0, 1]$
cap^{lb}_{iel}, cap^{ub}_{iel}	lower and upper bounds of capacity level i which tie to a reduction of detour time of α_{iel}	kW	$[0, \infty[$
τ_{u}	number of years between consecutive vehicle purchase decisions, representing the vehicle replacement period	a	$[0, \infty[$
β	relative threshold for budget under- or overrun	–	$[0, 1]$
B^{init}_{el}	initial detour time	h	$[0, \infty[$
Budget_{u}	monetary budget of consumer group	€/trip	$[0, \infty[$
F_{yur}	exogenous travel demand (by consumer group)	trips	$[0, \infty[$

3.3.5.2. Extended mathematical formulation

The RQ2 extension introduces consumer groups $u \in \mathcal{U}$ as an additional index dimension to all core decision variables. All core variables and constraints are extended accordingly; for example, f_{yrvtg} becomes f_{yurvtg}, h_{yrvtg} becomes h_{yurvtg}, and so forth. Correspondingly, the exogenous travel demand F_{yr} becomes F_{yur}. The extension proceeds in two steps: first, the core LP is extended with consumer group differentiation; second, the LP is transformed into a MILP by introducing endogenous detour time.

LP extension: consumer groups and intangible costs

The objective function (Equation 3.8) is extended with intangible costs ($C^{\text{intangiblecosts}}$) and penalty costs ($C^{\text{penaltycosts}}$) for budget overrun or underrun:

$$\underset{f,h,h^{+},h^{-},s,q^{+},\text{budget}^{+},\text{budget}^{-}}{minimize} \quad z = \sum_{y} \frac{1}{(1+\delta)^{y-y_0}} \Big(C^{\text{vehicle\_stock}}_{y} + C^{\text{fueling}}_{y} + C^{\text{infrastructure}}_{y} + C^{\text{intangiblecosts}}_{y} + C^{\text{penaltycosts}}_{y} \Big) \tag{3.18}$$

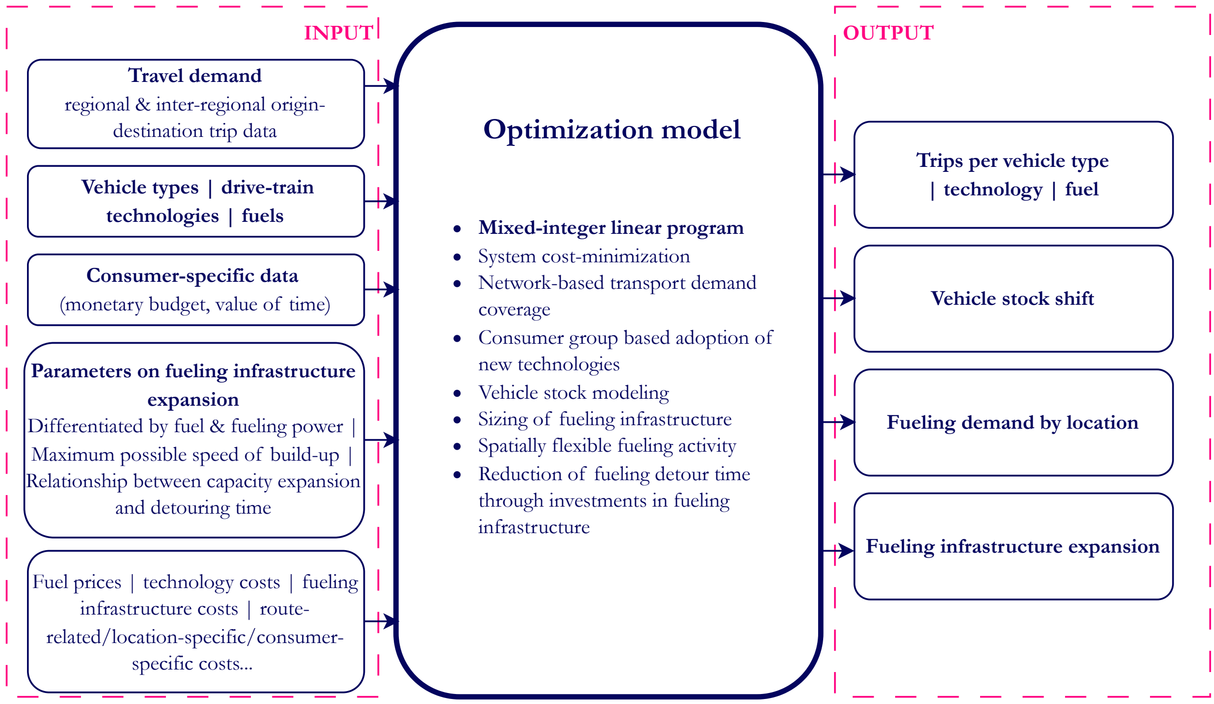

Figure 3.4.: Extended input-output relations of the optimization model for the regional application (RQ2), including consumer group differentiation and fueling detour time.

The intangible costs ($C^{\text{intangiblecosts}}$) represent non-monetary costs that are monetized in the objective function by weighting them with the consumer-group-dependent VoT. The non-monetary costs refer to in particular the total travel time associated with different vehicle types and drivetrain technologies, i.e., the level of service:

$$C^{\text{intangiblecosts}} = \sum_{u,r,v,t,g} \text{VoT}_u \cdot \text{LoS}^f_{yrvt} \cdot f_{yurvtg} \tag{3.19}$$

The parameter LoS is the level of service, i.e., total travel time, associated with a vehicle type and drivetrain technology for a trip:

$$\text{LoS}^f_{yrvt} = \frac{L_r}{\text{speed}_{vt}} + \text{fueling\_time}_{yrvt} \quad : \forall y, r, v, t \tag{3.20}$$

The infrastructure sizing constraint (Equation 3.10) is split into separate constraints for public ($\mathcal{L}^{\text{nothome}}$) and home ($\mathcal{L}^{\text{home}}$) charging infrastructure:

$$\begin{aligned} Q^{\text{charg\_infr}}_{tle} + \sum_{y' \in \mathcal{Y}^{\text{exp}}_y} q^{+,\text{charg\_infr}}_{y'tle} &\geq \gamma_l \sum_{r \in \mathcal{R}_e} \sum_{u \in \mathcal{U}} \sum_{g \in \mathcal{G}} s_{yurvtgle} \quad : \forall y, t, e, l \in \mathcal{L}^{\text{not home}} \\ Q^{\text{charg\_infr,home}}_{urtle} + \sum_{y' \in \mathcal{Y}^{\text{exp}}_y} q^{+,\text{charg\_infr,home}}_{y'urtle} &\geq \gamma_l \sum_{g \in \mathcal{G}} s_{yurvtgle} \quad : \forall y, u, t, r, e, l \in \mathcal{L}^{\text{home}} \end{aligned} \tag{3.21}$$

The expansion of home charging infrastructure is limited by upper bounds that vary by consumer group:

$$\sum_{r\in\mathcal{R}_u}\left(Q_{urtle}^{\text{charg\_infr,home}}+\sum_{y'\in\mathcal{Y}_y^{\text{exp}}} q_{yurtle}^{+,\text{charg\_infr,home}}\right)\leq \hat{Q}_{utle}^{\text{charg\_infr,home}} \quad :\forall y,u,t,l,e \tag{3.22}$$

Monetary budgets vary by consumer group. These are defined by the temporal frequency of car purchases, i.e., every τ years. The budget constrains expenditures for vehicles and fueling infrastructure installed at home, which is the left side of these constraints:

$$\begin{aligned}&\sum_{y\in\mathcal{S}_j^u}\left(\sum_{vtg} C_{yvtg}^{\text{CAPEX}} * h_{yurvtg}^{+} + \sum_{tl}\sum_{y'\in\mathcal{Y}_y^{\text{exp}}} q_{yurtle}^{+,\text{charg\_infr,home}}\right)\\ &\leq \beta * \tau_u * \text{Budget}_u * \sum_{y\in\mathcal{S}_j^u}\sum_{vtg} f_{yurvtg} + \sum_{y\in\mathcal{S}_j^u} \text{budget}_{yur}^{+} \quad :\forall u,r,j\end{aligned} \tag{3.23}$$

$$\begin{aligned}&\sum_{y\in\mathcal{S}_j^u}\left(\sum_{vtg} C_{yvtg}^{\text{CAPEX}} * h_{yurvtg}^{+} + \sum_{tl}\sum_{y'\in\mathcal{Y}_y^{\text{exp}}} q_{yurtle}^{+,\text{charg\_infr,home}}\right)\\ &\geq \beta * \tau_u * \text{Budget}_u * \sum_{y\in\mathcal{S}_j^u}\sum_{vtg} f_{yurvtg} - \sum_{y\in\mathcal{S}_j^u} \text{budget}_{yur}^{-} \quad :\forall u,r,j\end{aligned} \tag{3.24}$$

Equation 3.23 expresses the lower limit for the expenditures, which relate to the investments in new vehicles (left-hand side of the equation). Equation 3.24 imposes an upper limit on these expenditures. A threshold for the budget is included with the factor β. Budget underrun and overrun, budget_{yur}^{-} and budget_{yur}^{+}, are decision variables that are penalized in the objective function and leveraged with a factor in the term $C^{\text{penaltycosts}}$ (Equation 3.18).

MILP extension: endogenous detour time

The second extension step introduces the detouring time needed to reach the closest charging station as an endogenous decision variable, transforming the LP into a MILP. The objective function (Equation 3.18) is further extended with the VoT-weighted detour cost b_{yiel}:

$$\begin{aligned} z = \sum_{y} \frac{1}{(1+\delta)^{y-y_0}} \Big(& C_y^{\text{vehicle\_stock}} + C_y^{\text{fueling}} + C_y^{\text{infrastructure}} \\ & + C_y^{\text{intangiblecosts}} + C_y^{\text{penaltycosts}} + \sum_{i,e,l} b_{yiel} \Big) \end{aligned} \tag{3.25}$$

The number of charging vehicles n is derived from the fueled energy:

$$n_{yurlge} = \sum_{v,t} \frac{1}{\text{Cap}_{vtg}^{\text{tank}}} s_{yurvtgle} \quad :\forall y,u,r,g,e,l\in\mathcal{L}^{vt} \tag{3.26}$$

We introduce lumpiness in the sizing of the infrastructure, which ties the total installed charging capacity to discrete detour-reduction levels. The total installed capacity at a location — defined as the initial capacity $Q^{\text{charg\_infr}}_{tle}$ plus the cumulative expansion $\sum_{y'} q^{+,\text{charg\_infr}}_{y'tle}$ — is decomposed into a discrete component selected by the binary variable w_{yiel} and a continuous residual:

$$
\begin{aligned}
Q^{\text{charg\_infr}}_{tle} + \sum_{y' \in \mathcal{Y}^{\text{exp}}_{y}} q^{+,\text{charg\_infr}}_{y'tle} = \sum_{i} w_{yiel} \cdot cap^{\text{lb}}_{iel} + \sum_{y' \in \mathcal{Y}^{\text{exp}}_{y}} q^{\Delta+,\text{charg\_infr}}_{y'tle} \quad &: \forall y, t, e, l \in \mathcal{L}^{\text{not home}} \\
w_{yiel} = 1 \implies cap^{\text{lb}}_{iel} \le Q^{\text{charg\_infr}}_{tle} + \sum_{y' \in \mathcal{Y}^{\text{exp}}_{y}} q^{+,\text{charg\_infr}}_{y'tle} \le cap^{\text{ub}}_{iel} \quad &: \forall y, i, e, l \in \mathcal{L}^{\text{not home}} \\
\sum_{i} w_{yiel} = 1 \quad &: \forall y, e, l \in \mathcal{L}^{\text{not home}}
\end{aligned}
\tag{3.27}
$$

The left-hand side of the first equation is the total installed capacity. The right-hand side decomposes it into the discrete lower bound cap^{lb}_{iel} of the active detour-reduction level and a continuous residual $q^{\Delta+,\text{charg\_infr}}$ that captures capacity above this threshold. The indicator constraint (second line) enforces that when $w_{yiel} = 1$, the total installed capacity lies within the bounds $[cap^{\text{lb}}_{iel},\ cap^{\text{ub}}_{iel}]$ associated with detour-reduction level i. This ties charging infrastructure expansion to a discrete reduction of detour time.

The VoT-weighted detour cost is defined only for charging types not allocated at home, for $l \in \mathcal{L}^{\text{not home}}$. Based on an assumed charging detour time B^{init}_{el} that is initially needed to reach the nearest charging station, we model the relative reductions of the detour time aggregated over all charging vehicles that directly result from the expansion of the charging infrastructure capacities. The decision variable b_{yiel} represents the total monetized detour cost at location e for charging type l in year y, aggregated across all consumer groups and charging events. It is defined via an indicator constraint:

$$
w_{yiel} = 1 \implies b_{yiel} \ge B^{\text{init}}_{el} \cdot (1 - \alpha_{iel}) \cdot \sum_{u,r,v,t,g} \text{VoT}_{u} \cdot n_{yurvtgle} \quad : \forall y, i, e, l \in \mathcal{L}^{\text{not home}} \tag{3.28}
$$

α_{iel} represents the ith reduction of the detour time at capacity level $[cap^{\text{lb}}_{iel},\ cap^{\text{ub}}_{iel}]$. The indicator constraint enforces that when $w_{yiel} = 1$, b_{yiel} captures the VoT-weighted detour time for all charging events at location e. The use of indicator constraints, which are natively supported by the JuMP modeling language in Julia, avoids the numerical difficulties associated with choosing appropriate big-M values.

For better understanding of the computation of this equation, we provide the following example: At location e, there are 1000 charging events for consumers with $\text{VoT}_u = €15/\text{h}$ ($n_{yurvtgle} = 1000$). Due to the total installed public charging capacity at location e, the reduction $\alpha_{(i=7)el}$ is active which reduces the initial detouring time B^{init}_{el} by 80%. B^{init}_{el} equals 60 minutes. Based on this, the monetized detour cost is $b = 60\,\text{min} \cdot (1 - 0.8) \cdot 1000 \cdot €15/\text{h} = €3{,}000$.

3.3.6. Extension for long-distance transport (RQ3)

3.3.6.1. Extended problem description and model features

The RQ3 extension addresses the spatial patterns of charging demand allocation in inter-zonal long-haul truck electrification. As battery-electric trucks (BETs) travel along international corridors passing through multiple electricity market zones, their charging demand can be allocated flexibly across geographic locations. This spatial flexibility arises because charging costs vary along the corridor due to differences in electricity prices and grid connection fees between countries. The cost-optimal geographic allocation of charging demand is therefore not fixed a priori but emerges endogenously from the optimization.

To capture these dynamics, the core model is extended with the following features:

- **State of charge tracking:** The battery state of charge is tracked along the sequence of geographic locations visited on each route, constraining where trucks can charge based on battery capacity and energy consumption between stops.

- **Travel time accounting under mandatory break regulations:** Travel time is accumulated along routes, accounting for driving time, charging time, and mandatory driver rest periods following EU driving time regulations. Charging during mandatory breaks can partially or fully overlap with the required rest period, reducing additional time cost, while charging at other locations or beyond the break duration extends trip duration and thereby increases costs.

- **Modal shift to rail:** Freight demand can be served by road or shifted to rail, subject to a corridor-wide maximum rail share and a rate-of-change constraint reflecting infrastructure deployment limits.

- **Geographically varying charging costs:** Charging costs differ by location based on market-zone-dependent electricity prices. Additionally, geographically varying annual grid charges are considered in the capacity expansion of charging infrastructure.

Figure 3.5 provides an overview of the extended input-output relations. Compared to the core model (Figure 3.3), the additional inputs include battery and charging parameters (capacity, charging power, SOC constraints), mandatory break regulations following EU driving time rules, modal shift parameters for road-to-rail substitution, and geographically varying electricity prices and grid charges. The additional outputs include the charging demand allocation by market zone, state of charge and travel time profiles along routes, and the modal split between road and rail.

3.3.6.2. Extended mathematical formulation

The RQ3 extension applies the core model to long-haul freight transport. This entails several changes in the index structure compared to the RQ2 formulation. Consumer groups (u) and

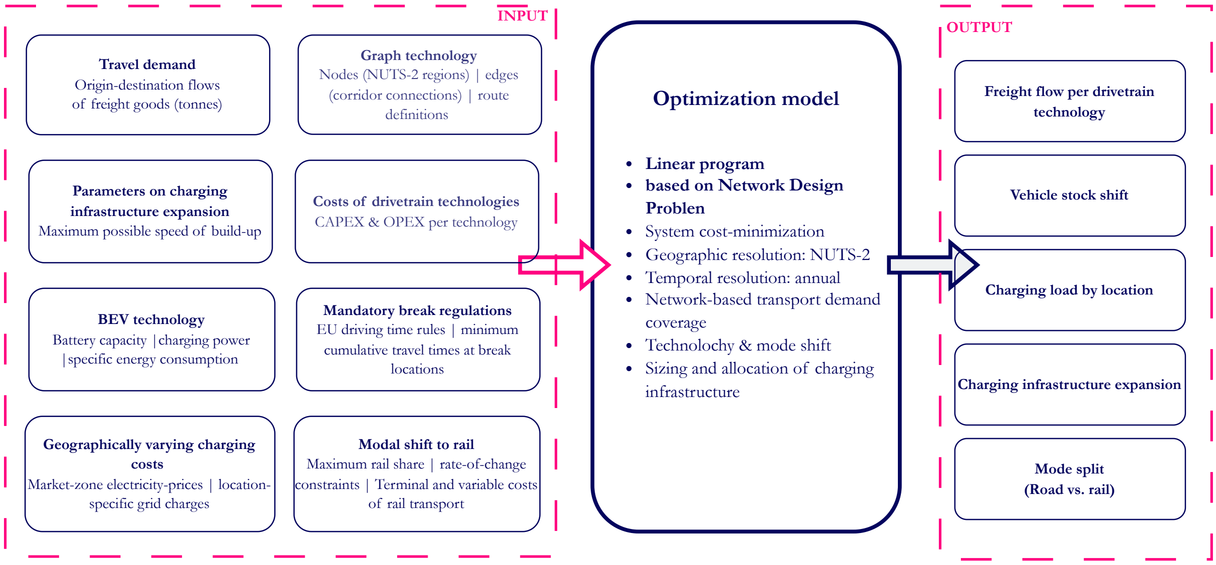

Figure 3.5.: Extended input-output relations of the optimization model for the long-distance freight transport application (RQ3), including state of charge tracking, mandatory break regulations, modal shift, and geographically varying charging costs.

vehicle types (v) are dropped, as the model considers a single vehicle type (trucks). A mode index $m \in \mathcal{M}$ is introduced to distinguish road and rail transport. The flow variable f_{yrtg} represents annual freight tonnage rather than person-trips, and the load factor W_{tg} represents payload in tonnes rather than persons per vehicle. New decision variables track the battery state of charge (soc), accumulated travel time ($travel\_time$), and break time not used for charging ($break\_time^{+}$) along each route. Table 3.6 lists the additional nomenclature.

Table 3.6.: Additional nomenclature for the RQ3 extension.

Sets and indices	Description		
$m \in \mathcal{M}$	transport mode (road, rail)		
$\mathbf{E}_r$	ordered set of geographic locations along route r		
$e_{r,i}$	i-th geographic location along route r		
Decision variables		Unit	
soc_{yrtge}	state of charge of truck fleet at location e	kWh	$[0, \infty[$
$travel\_time_{yrtge}$	accumulated travel time of fleet at location e	h	$[0, \infty[$
$break\_time^{+}_{yrtge}$	time on mandatory break not used for charging	h	$[0, \infty[$
f_{yrmtg}	freight flow by mode	t/a	$[0, \infty[$
Parameters			
SOC^{init}	initial state of charge (fraction of battery capacity)	–	$[0, 1]$
SOC^{final}	minimum final state of charge (fraction of battery capacity)	–	$[0, 1]$
Q^{batt}_{tg}	battery capacity	kWh	$[0, \infty[$
$\hat{P}_{tg}$	charging power	kW	$[0, \infty[$
$speed$	average driving speed	km/h	$[0, \infty[$
W_{tg}	payload capacity (tonnes per vehicle)	t	$[0, \infty[$
$TravelTime^{min}_{e_{r,j}}$	minimum cumulative travel time at node j	h	$[0, \infty[$
$\bar{\alpha}^{\text{mode}}$	maximum corridor-wide rail share	–	$[0, 1]$
α_f, β_f	rate-of-change parameters for modal shift	–	$[0, \infty[$
L_r	length of route r	km	$[0, \infty[$
F^{rail}_y	total rail freight flow in year y	t·km	$[0, \infty[$
F^{total}_y	total freight flow in year y	t·km	$[0, \infty[$
Δy	time step between consecutive modeled periods	a	$[0, \infty[$

State of charge

Equations 3.29 track the battery state of charge along the spatial sequence of locations visited on each route.

$$
\begin{aligned}
&soc_{yrtge_{r,1}} = SOC^{init} \cdot Q^{batt}_{tg} \cdot \frac{f_{yrtg}}{W_{tg}} \qquad \forall\, y, r, t \in \mathcal{T}^{\text{BET}}, g \\
&soc_{yrtge_{r,i}} = soc_{yrtge_{r,i-1}} + s_{yrtge_{r,i}} - D^{\text{spec}}_{tg} \cdot L_{e_{(i-1,i)}} \cdot \frac{f_{yrtg}}{W_{tg}} \qquad \forall\, y, r, t \in \mathcal{T}^{\text{BET}}, g,\ e_{r,i} \in \mathbf{E}_r \setminus \{e_{r,1}\} \\
&soc_{yrtge_{r,|\mathbf{E}_r|}} \geq SOC^{final} \cdot Q^{batt}_{tg} \cdot \frac{f_{yrtg}}{W_{tg}} \qquad \forall\, y, r, t \in \mathcal{T}^{\text{BET}}, g \\
&soc_{yrtge_{r,i-1}} - D^{\text{spec}}_{tg} \cdot L_{e_{(i-1,i)}} \cdot \frac{f_{yrtg}}{W_{tg}} \geq 0 \qquad \forall\, y, r, t \in \mathcal{T}^{\text{BET}}, g,\ e_{r,i} \in \mathbf{E}_r \setminus \{e_{r,1}\} \\
&soc_{yrtge_{r,i}} \in \left[0,\ Q^{batt}_{tg} \cdot \frac{f_{yrtg}}{W_{tg}}\right] \qquad \forall\, y, r, t \in \mathcal{T}^{\text{BET}}, g,\ e_{r,i} \in \mathbf{E}_r
\end{aligned}
\tag{3.29}
$$

All terms are fleet-aggregate quantities: the factor f_{yrtg}/W_{tg} converts freight flow into the number of vehicle trips, so soc and s represent the total energy stored and charged across all trips in that year. At each location, the state of charge is increased by charged energy s and reduced by the electricity consumed while driving from the previous location ($e_{r,i-1}$) to the current one ($e_{r,i}$). The fourth line ensures that the state of charge before charging is non-negative at every intermediate node, preventing infeasible en-route depletion.

Travel time

Travel time results from driving time between nodes, charging time at stops, and break taking. Similar to the state of charge, the travel time of the fleet along a route is tracked by:

$$
\begin{aligned}
&travel\_time_{yrtge_{r,1}} = 0 \qquad \forall\, y, r, t \in \mathcal{T}^{\text{road}}, g \\
&travel\_time_{yrtge_{r,i}} = travel\_time_{yrtge_{r,i-1}} + \frac{L_{e_{(i-1,i)}}}{speed} \cdot \frac{f_{yrtg}}{W_{tg}} + \frac{\sum_{l \in \mathcal{L}_t} s_{yrtgle_{r,i}}}{\hat{P}_{tg}} + break\_time^{+}_{yrtge_{r,i}} \\
&\qquad \forall\, y, r, t \in \mathcal{T}^{\text{road}}, g,\ e_{r,i} \in \mathbf{E}_r \setminus \{e_{r,1}\}
\end{aligned}
\tag{3.30}
$$

As with the SOC equation, all terms are fleet-aggregate quantities scaled by f_{yrtg}/W_{tg}. At each location, the accumulated travel time increases by three components: the driving time between consecutive nodes ($L_{e_{(i-1,i)}}/speed$), the charging time ($s/\hat{P}_{tg}$), and any remaining mandatory break time not used for charging ($break\_time^{+}$). The variable $break\_time^{+}_{yrtge}$ ensures that the total stop duration at a break location equals the regulated break duration, regardless of how much of that time is spent charging.

Mandatory break taking

At locations where EU driving time regulations require a mandatory break, the accumulated travel time must meet a minimum threshold:

$$
travel\_time_{yrtge_{r,j}} \geq TravelTime^{min}_{e_{r,j}} \cdot \frac{f_{yrtg}}{W_{tg}} \quad \forall\, y \in \mathcal{Y},\ r \in \mathcal{R},\ t \in \mathcal{T}^{\text{road}},\ g \in \mathcal{G},\ e_{r,j} \in \mathbf{E}_r
\tag{3.31}
$$

where $TravelTime_{e_{r,j}}^{min}$ is the minimum cumulative travel time at node j, comprising the driving time and the mandatory break time that must have been taken by that point along the route.

Modal shift constraints

The maximum share of rail freight is bounded corridor-wide at each modeled year:

$$\sum_{r\in\mathcal{R}}\sum_{t\in\mathcal{T}_{m=\mathrm{rail}}}\sum_{g\in\mathcal{G}} f_{yrmtg}\cdot L_r \;\leq\; \bar{\alpha}^{\mathrm{mode}}\sum_{r\in\mathcal{R}}\sum_{m\in\mathcal{M}}\sum_{t\in\mathcal{T}_m}\sum_{g\in\mathcal{G}} f_{yrmtg}\cdot L_r \quad \forall\, y\in\mathcal{Y} \tag{3.32}$$

where $\bar{\alpha}^{\mathrm{mode}}$ represents the maximum corridor-wide rail share in tonne-kilometres. The rate of modal shift between consecutive periods is bounded by:

$$F_y^{\mathrm{rail}} - F_{y-\Delta y}^{\mathrm{rail}} \;\leq\; \alpha_f\cdot F_y^{\mathrm{total}} + \beta_f\cdot F_{y-\Delta y}^{\mathrm{rail}} \quad \forall\, y\in\mathcal{Y}\setminus\{y_0\} \tag{3.33}$$

where F_y^{rail} and F_y^{total} denote total rail and total freight flows in year y, respectively. Parameters α_f and β_f limit the period-over-period growth of rail, reflecting the pace at which intermodal terminal capacity and rail network upgrades can realistically be deployed.

3.4. Spatio-temporal charging load modeling

3.4.1. Problem description

This methodology is part of a broader framework for investigating how electrified commercial fleets affect dispatch operations and redispatch costs in the electricity system (Figure 3.6). Figure 3.6 gives an overview of the three building blocks of the framework. The charging load profiles produced by the first block serve as input to the subsequent dispatch and redispatch models (see Appendix E for details on the electricity market integration and results). The methodological contribution of this thesis is the first building block: the bottom-up modeling of spatio-temporal charging load profiles (highlighted in Figures 3.6 and 3.7). The intended redispatch application determines the required resolution of the charging load profiles—spatially at the federal-state level (NUTS-2) and temporally at hourly granularity for a representative week. The subsequent electricity market modeling is part of the broader study presented in the underlying publication but is not a contribution of this thesis.

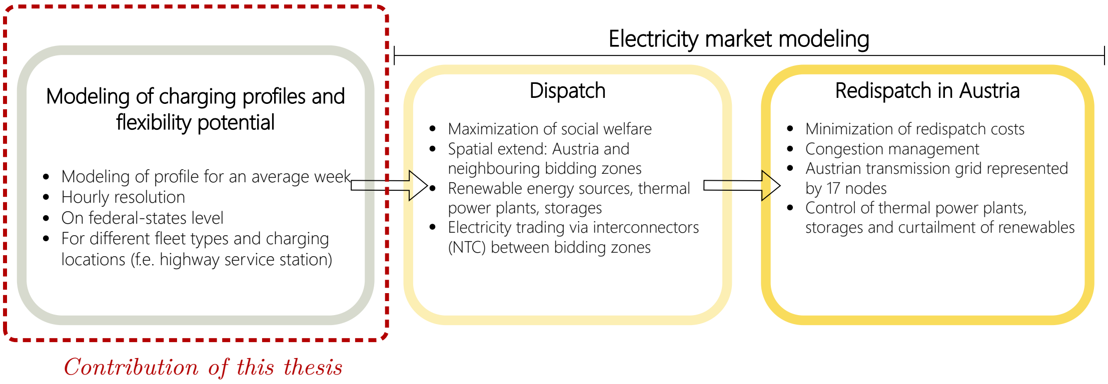

Figure 3.6.: Overview of the building blocks of the framework. The dashed box highlights the building block that constitutes the methodological contribution of this thesis.

The framework operates on three geographic layers ($\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3$; see Figure 3.7), mirroring the three building blocks of the method in Figure 3.6. The charging load modeling presented here operates on $\mathcal{G}_1 = (\mathcal{N}_1, \mathcal{E}_1)$, a network of regions and route connections of the commercial vehicle fleet, where charging opportunities exist at multiple nodes.

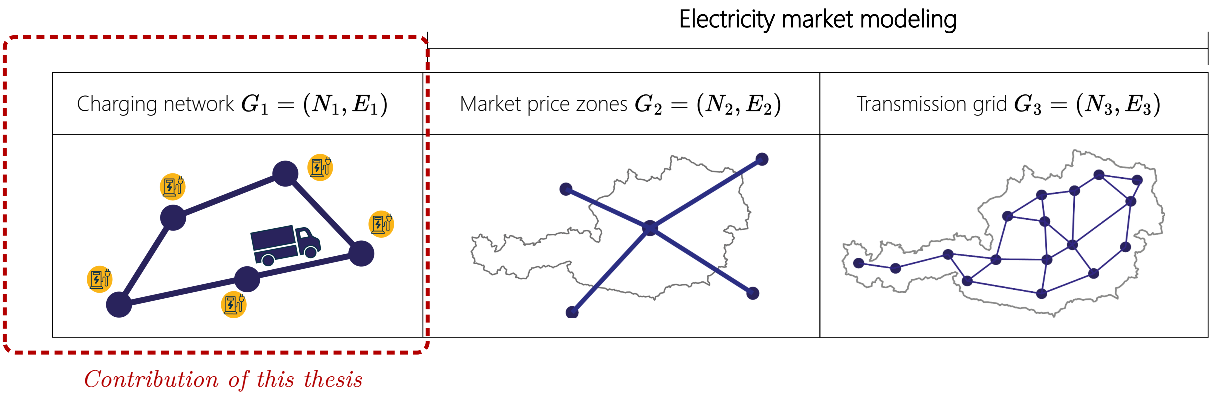

Figure 3.7.: The three layers of geographic information used throughout the framework. The dashed box highlights the layer central to the contribution of this thesis.

The charging load estimation distinguishes two cases based on the geographic scope of vehicle operation: *local* fleets, whose charging load is assigned to a single node in $\mathcal{G}_1$ (i.e., the region in which the vehicle operates), and *interregional* fleets, whose vehicles travel between multiple nodes and may charge at any visited node along their route.

3.4.2. Charging load estimation for local transport fleets

For local transport, the charging load of a vehicle is assigned to exclusively one node in $\mathcal{G}_1$. We aggregate all vehicles of a specific type to a fleet $f \in \mathcal{F}^{\text{local}}$ and the energy load for node n is estimated as follows:

$$D_{f,n}^{\text{BEV}} = \overline{D_f^{\text{spec,BEV}}} \cdot dist_f \cdot k_{f,n} \quad \forall f \in \mathcal{F}^{\text{local}}, n \in \mathcal{N}_1 \tag{3.34}$$

$$\sum_{l\in\mathcal{L}^f} D^{\mathrm{BEV}}_{f,n,l} = D^{\mathrm{BEV}}_{f,n} \quad \forall f \in \mathcal{F}^{\mathrm{local}}, n \in \mathcal{N}_1 \tag{3.35}$$

The charging load during the observed time period is derived from the distance $dist_f$ that is driven by a vehicle of fleet f, its energy consumption $\overline{D_f^{spec,\mathrm{BEV}}}$ and the amount of the vehicles of fleet type f operating at node n $k_{f,n}$ is. The charging load is assigned to different charging locations $\mathcal{L}^f$ that are available to the fleet. These are, for example, depots at node n. Each charging location type is characterized by an installed maximum charging power $\mathrm{CAP}^{\mathrm{BEVmax}}_{f,n,l}$. The particular distribution of the charging load to the charging location is conducted based on the assumed charging strategy for this fleet.

3.4.3. Charging load estimation for interregional transport fleets

For interregional transport, vehicles are assumed to travel between regions and charge at more than one node. In this case, a set of routes is given for the fleet $R^f = \{R_1, R_2, \ldots R_I\}, i = 1, 2, \ldots I$. The routes are all allocated in graph $\mathcal{G}_1$. The charging load is determined and assigned to nodes $n \in \mathcal{N}_1$ for each route separately and later aggregated at each node for all considered types of charging locations. Each route R_i is traveled by a part of the fleet, $f_i \in \mathcal{F}^{\mathrm{inter}}$. Charging locations are either allocated along the route $\mathcal{L}_f^{\mathrm{enroute}}$ or at origin and destination nodes $\mathcal{L}^{f,\mathrm{OD}}$. In accordance with this, the nodes contained in a route are categorized into two sets: $\mathcal{N}_i^{\mathrm{enroute}}$ and $\mathcal{N}_i^{\mathrm{OD}}$.

$$\begin{aligned} &D^{\mathrm{BEV}}_{f_i} = dist_i \cdot \overline{D^{\mathrm{spec,BEV}}}_f \cdot k_i \\ &= \sum_{n\in\mathcal{N}_i^{\mathrm{enroute}}} \sum_{l\in\mathcal{L}_f^{\mathrm{enroute}}} D^{\mathrm{BEV}}_{f_i,n,l} + \sum_{n\in\mathcal{N}_i^{\mathrm{OD}}} \sum_{l\in\mathcal{L}_f^{\mathrm{OD}}} D^{\mathrm{BEV}}_{f_i,n,l} \\ &\forall i = 1, 2, \ldots I, f \in \mathcal{F}^{\mathrm{inter}} \end{aligned} \tag{3.36}$$

k_i defines the number of vehicles traveling on the route R_i in the observed period. The charging load for this route is distributed between the locations positioned along the route and at the origin or destination nodes. The particular assignment of the magnitude of charging load to the nodes, $D^{BEV}_{f_i,n,l}$ depends further on the assumed charging strategy.

Nodal aggregation and temporal distribution We aggregate the charging load for each fleet type and the respective locations on node-level:

$$D^{\mathrm{BEV}}_{f,n,l} = \begin{cases} D^{\mathrm{BEV}}_{f,n,l} & \text{if } f \in \mathcal{F}^{\mathrm{local}} \\ \sum_i D^{\mathrm{BEV}}_{f_i,n,l} & \text{if } f \in \mathcal{F}^{\mathrm{inter}} \end{cases} \tag{3.37}$$

The plug-in periods are during periods of operational downtime of the vehicles, during operation times only when needed. Independently of obtaining the absolute magnitude for each fleet, the charging load is assigned to timestep $t \in \mathcal{T}$:

$$D^{\text{BEV}}_{f,n,l,t} = g(D^{\text{BEV}}_{f,n,l}, t) \tag{3.38}$$

$g(\cdot)$ indicates here the function assigning the amount of charging load to each time, resulting in a charging load defined for each time step, $D^{BEV}_{f,n,l,t}$.

4. Results of case studies

This chapter presents the case studies and results of the four contributions underlying this thesis. Each section corresponds to one contribution and draws on the respective publication: Golab et al. (2022) for Contribution 1 (Section 4.1), Golab et al. (2026) for Contribution 2 (Section 4.2), Golab et al. (n.d.) for Contribution 3 (Section 4.3), and Golab et al. (2025) for Contribution 4 (Section 4.4).

4.1. Fast-charging infrastructure network capacities for passenger cars along Austria's highway network in 2030 (RQ1)

The following case study and results are taken from the journal publication by Golab et al. (2022).

4.1.1. Case study: Austrian highway network 2030

The proposed modeling framework is applied to Austria's highway network by modeling a cost-optimal charging infrastructure expansion for 2030 under different future scenarios. In this study, year 2030 has been chosen mainly due to the two following reasons: first is that the aim of the analysis of this work is to illustrate short-term infrastructure requirements (short term in terms of infrastructure planning), and second is that it is a significant year in decarbonization plans for the transport sector as the Paris Declaration on Electro-Mobility explicitly states a 20% electrification of global road transport by 2030 [128].

To extrapolate future scenarios, model input parameters were set, which describe the current status of Austria in 2021. Table 4.1 presents the selected values. Until the end of December 2021, $76,539$ electric vehicles have been registered in Austria [129], which make up for 1.5 % of all the registered passenger cars in Austria. In order to determine traffic load on highways during peak hour and the share of long-distance drivers, the most recently acquired data (Austrian-wide) on mobility patterns was used [130]. Similar to the work of Jochem et al. [131], it was assumed that all car trips taking longer than 45 min and are longer than $25km$ or car trips longer than $50km$ are most probable to include driving on highways or motorways. Furthermore, trips were classified as long-distance if the drive was at least $100km$ long, following the common definition of *long-distance* travel [132]. Based on the starting times of these trips, it was assumed that during the hour of peak demand on Austrian highways 12% of the daily traffic takes place and 24% of the traffic are long-distance travelers.

Traffic counts were obtained from ASFINAG [133], which provide averaged weekly traffic counts for up to 276 positions along Austria's highways and motorways. Their data encompasses counts for vehicles of two categories, namely: vehicles weighing up to 3.5 tons and those weighing greater than 3.5 tons. As no differentiation is made between light-duty vehicles and passenger cars in the category of < 3.5 tons, it was assumed that all vehicles of this category are passenger cars, as light-duty transport will also be transitioned to electromobility [134]. Data for the year 2019 was chosen to be representative for 2021 as the complete data set for 2021 has not been published as of the time of the conduction of this analysis, and the 2020 dataset cannot be used due to being affected by the month-long lockdowns during the COVID pandemic.

To derive charging demand at each node of the network, the sparse traffic count positions are mapped onto each highway segment and spatially interpolated using a generalized regression neural network (GRNN) [135]. The GRNN approximates a continuous function of peak daily traffic load as a function of distance along each segment and driving direction. This function is then evaluated at each node position to obtain the local traffic load, and integrated between consecutive nodes to compute the accumulated energy demand that must be covered by charging infrastructure at each node.

The technical parameters $\overline{d}_{spec,BEV}$, $\overline{dist}_{range,BEV}$, and $\overline{P}_{charge,BEV}$ were set in such a way that they would represent an average Austrian BEV. For this, the technological attributes of the respective top 10 sold cars during the years 2019, 2020 and 2021 were used to evaluate the weighted average values (see Appendix B.3 for the complete list of car models and their technical attributes). The maximum distance for demand coverage is derived as $dist_{max} = 4/5 \cdot 0.6 \cdot \overline{dist}_{range,BEV}$, where the factor 4/5 accounts for reduced driving range at highway speeds [51, 101] and 0.6 reflects charging processes starting at a state of charge of 20% up to 80% [136].

Infrastructure cost component c_X is particularly difficult to set here as these onsite preparation costs may significantly vary based on the location of the site. Indications for the approximation of this value were drawn from [131], [61] and [137]. The installation costs of one charging point, c_Y, was assumed to be €60,000 for the hardware of a charging pole with $\hat{P}_{CP} = 150kW$ — which currently enables the fastest charging of passenger cars; an added construction costs to be €7,000 [137, 138]. Furthermore, it was assumed that the maximum installed capacity at a charging station is $12MW$.

The shape of the Austrian highway and motorway network was mapped based on geographic data by OpenStreetMap contributors [139]. The positions of the service areas were gathered based on the information drawn from ASFINAG [140] and complemented with geographic information from OpenStreetMap contributors [139]. The retrieved highway network used throughout the analysis has an overall length of approximately 2,200 km and consists of 55 segments. Moreover, there are 249 nodes in total, 139 of which represent service areas and 110 junctions. A detailed description of the data preparation steps is provided in Appendix B.5.

4.1.1.1. Future scenarios

Four quantitative scenarios for Austrian highway charging infrastructure expansion are outlined. These were developed in the course of the Horizon 2020 research project openEntrance [141]. The scenarios outline pathways to climate change mitigation by reaching the $1.5°C$ or $2.0°C$ targets and are referred to as the *Societal Commitment* (SC), *Techno-Friendly* (TF), *Directed Transition* (DT) and *Gradual Development* (GD) scenarios. The former three pathways aim to mitigate climate change by keeping the temperature rise to a maximum of $1.5°C$ as specified in the Paris

Table 4.1.: Model input parameters reflecting the status quo in Austria in 2021.

Base case	
Input parameter	Value
BEV share ϵ	1.5%
Share of traffic load during peak hour γ_h	12%
Share of long-distance drivers μ	24%
Share of overall traffic load compared with the survey year of traffic counts α	100%
traffic count data $\hat{f}_{n_k}$	maximum recorded daily traffic counts 2019
energy consumption of an average BEV in the car fleet $\overline{d}_{spec,BEV}$	$0.24kWh/km$
driving range of an average BEV in the car fleet $\overline{dist}_{range,BEV}$	$340km$
charging capacity of an average BEV $\overline{P}_{charge,BEV}$	$81kW$
peak capacity of a charging point $\hat{P}_{CP}$	$150kW$
maximum installed charging capacity at a charging station P_{max}	$12MW$
investment costs for installation a charging station c_X	€$40,000$
investment costs for installation of one charging point $c_{Y,150kW}$	€$67,000$

Agreement, the latter to $2.0°C$. Within these scenarios, the extent of societal engagement, implementations of technological novelties and the strong presence of political interventions in climate change mitigation vary [see also 142, 143]. These scenarios were developed to outline mitigation paths for different economic sectors in Europe and are used here to align the analysis described in this paper into the large-scale context of decarbonization. Based on this, different developments of the transport sector relevant to the present work are projected:

- **Societal Commitment (*SC*):** Within this scenario, politics are strongly intervening which is met by wide-spread societal acceptance, triggering behavioral changes in the face of awareness of the necessity of climate change mitigation. While this scenario is characterized by a reduction in energy demand due to behavioral changes, societal engagement supporting circular economy, and new market solutions, it is assumed that no significant technological breakthroughs appear. This translates in the transport sector to an increased modal shift to sharing concepts and public transport, which causes a significant decrease in individual passenger road transport.

- **Techno-Friendly (*TF*):** This setting combines the appearance of major technological breakthroughs and strong societal engagement, which results in an increased top down push effect in the application of new technologies that improve energy efficiency. Simultaneously, similarly as in the SC scenario, there is a strong social commitment driving an increased modal shift away from individual passenger car transport.

- **Directed Transition (*DT*):** Similarly as in the TF scenario, there is a strong active policy push supporting new technology options. While there are major technological developments, the social commitment to adopting such developments is missing. This results in the moderate growth of BEV share throughout the years and a decreased modal shift, but registered BEVs of the Austrian car fleet still show similar technological improvements as in the TF scenario.

- **Gradual Development (*GD*):** This scenario represents the projection of less ambitious climate change mitigation goals. It embodies the exertion of all three dimensions, namely, social engagement, technological breakthroughs, and significant political interventions, only a weaker extent of each. Therefore, while BEV penetration will grow to some degree, and technological improvements will appear, no changes in mobility patterns are expected for this pathway.

Based on these scenario descriptions, we[1] embed projections on changes in the modal split and developments in the BEV technology in European climate mitigation pathways for 2030. Parameters ϵ and α are specified based on the climate change mitigation goals for Austria's passenger transport sector described in the governmental document "Austria's Mobility Master Plan 2030" (*AMMP*) [134]. One of these key goals is to reduce motorized private transport by 31% between the years 2018 and 2040 to meet the $1.5°C$ target set by the Paris Agreement. In the SC scenario where strong societal and political engagement act together, this goal will be met by 2030. The AMMP further specifies that by 2030, 100% of newly registered passenger vehicles must be exclusively powered by an electric engine. In the SC and TF scenarios, it is assumed that the share of new registrations of electric vehicles will gradually increase until 100% in 2030 due to the strong awareness of the necessity of electromobility by the society. For the *DT* and *GD* scenarios, the steady growth rate as experienced in 2020-2021 is projected until 2030, which will result in an EV share of 27%[2]. Further details on the BEV share projections are provided in Appendix B.4.

Table 4.2 presents the projected values. The presence of major technological breakthroughs is expressed by a significant development in the driving range of BEVs and increasing charging power, which is in line with the current major focus in BEV technology research on battery technology with the goal of allowing faster charging and longer trips [144, 145]. According to the study by Thielmann et al. [145], there is a great potential in battery storage research suggesting that technological breakthroughs in the next years could lead to the sale of BEVs with a driving range of up to $1000km$. Based on this, the sale of BEVs with an average driving range of up to $800km$ is projected for 2030 in the TF and DT scenarios, whereas in the SC scenario, this range is assumed to be $450km$, being slightly larger than the ranges of BEVs currently on the market . To reflect a weak extent of technical developments in the GD scenario, the increase in the average driving range of up to $600km$ is projected for 2030.

To compare the results of the scenarios, a similar peak power for charging points is assumed for all four scenarios. The study by Andersen et al. [144] projected $\hat{P}_{CP} = 350kW$ to be the predominant peak charging power by 2030. The costs for the hardware of a charging pole are estimated to be €120,000, as well as an additional construction cost of €7,000 [146, 147]. To

[1] Throughout this chapter, the first-person plural "we" is used where text is directly based on the underlying journal publications.

[2] This was calculated under the assumption that the size of the Austrian car fleet will not grow further from 2021.

reflect different improvements in charging efficiency, technological breakthroughs are assumed to lead to charging power levels of up to $\overline{P}_{charge,BEV} = 315kW$, given that the efficiency will increase up to 90%, as for charging poles with peak capacity of $50kW$ nowadays [148]. Today, one of the passenger car models with the highest charging power is Porsche Taycan with $\overline{P}_{charge,BEV} = 197kW$. For the SC and GD scenarios, this is assumed to be adopted by all BEV vehicles in 2030.

Other model input parameters are assumed to be similar as in 2021, using the values from Table 4.1. Table 4.3 summarizes the overall projected values describing the average Austrian car fleet in 2030 which were extrapolated based on the assumption that these parameters would gradually change in a linear way until 2030 on the basis of the values for 2021 .

Table 4.2.: Car fleet parameters and average technical parameters of an average BEV on the market in 2030 for different scenarios (BEV share ϵ, share in road traffic α, average driving range $\overline{dist}_{range,BEV}$, average charging power $\overline{P}_{charge,BEV}$).

Model parameters	**Projections for 2030 under different scenarios**			
	Social Commitment	*Techno-Friendly*	*Directed Transition*	*Gradual Development*
$\epsilon(\%)$	33	33	27	27
$\alpha(\%)$	69	83	83	100
$\overline{dist}_{range,BEV}(km)$	450	800	600	800
$\overline{P}_{charge,BEV}(kW)$	200	315	315	200

Table 4.3.: Model parameter values projected for the scenarios of climate change mitigation, set for year 2030 (BEV share ϵ, share in road traffic α, average driving range $\overline{dist}_{range,BEV}$, average charging power $\overline{P}_{charge,BEV}$, peak charging power of a charging point $\hat{P}_{CP}$, costs of installment of one charging point c_Y).

Model parameters	**Input parameters for scenarios 2030**			
	Societal Commitment	*Techno-Friendly*	*Directed Transition*	*Gradual Development*
ϵ (%)	33	33	27	27
α (%)	69	83	83	100
$\overline{dist}_{range,BEV}$ (km)	420	670	660	520
$\overline{P}_{charge,BEV}(kW)$	166	248	243	164
$\hat{P}_{CP}(kW)$	350	350	350	350
$c_{Y,350kW}$ (€)	127,000	127,000	127,000	127,000

4.1.1.2. Charging demand calculation

The charging demand at each candidate location along the highway network is derived from empirical traffic count data rather than from synthetic demand assumptions. The following describes the procedure used to assign spatially resolved charging demand to each node in the network.

To each node n in the network, a charging demand is assigned for each direction k from which it is accessible. For this, daily traffic counts are utilized to estimate the number of cars passing at all node positions n and driving in direction k. To take into account the demand of maximum traffic load to the charging infrastructure in the analysis, the maximum daily traffic count values are obtained from the measurements performed for a span of one year. Then, the following is performed for each segment s in the direction k of the highway network: All traffic counter positions lying on s are mapped onto the segment, and a function expressing traffic load in dependence of the distance in the traveling direction is approximated using the general regression neural network (GRNN) model, which is able to estimate various forms of underlying functions using Gaussian functions given few input data points [135]. Using the function, $GRNN_{sk}(dist)$, the peak daily traffic loads are approximated for all nodes n, which lie on segment s and are accessible from the driving direction k:

$$\hat{f}_{n_k} = GRNN_{s_k}(dist_{ns_k}) \tag{4.1}$$

To obtain the charging demand at node n_k resulting from the energy consumed along the distance driven since passing the last node $n-1_k$, the integral between these distances is calculated and multiplied by the specific energy demand $\overline{d}_{spec,BEV}$ by one BEV:

$$\hat{D}_{ns_k} = \overline{d}_{spec,BEV} \cdot \epsilon \cdot \mu \cdot \gamma_h \cdot \alpha \int_{dist_{n-1}}^{dist_n} GRNN_{s_k}(dist)\,\mathrm{d}\,dist \tag{4.2}$$

ϵ describes the share of BEVs in the car fleet, whereas μ expresses the share of long-distance drivers. Factor γ_h expresses the share of daily travels appearing during the peak hour of a day. Parameter α is introduced as a factor that allows to express relative changes in the overall traffic load along highways between the year during which the traffic counts were obtained and the year for which the demand is modeled; this allows to take into account the decrease in road traffic due to activity reduction or modal shift.

4.1.2. Expansion of fast-charging infrastructure along Austrian highway network under different scenarios for 2030

In this section, the most relevant results are presented. It is divided into two parts: First, we elaborate on the expansion requirements for the existing fast-charging infrastructure along Austria's highway network under different future scenarios for 2030. Details on the results obtained for the DT scenario, for which the costs of fast-charging infrastructure expansion are the lowest, are presented. Subsequently, the results of the modeled expansion given four different scenarios, SC, TF, DT and GD, are compared based on parameters describing the costs and the required charging infrastructure expansion. To gain better insight into how the observed differences between the scenarios come to place, selected input parameters of the scenario result with the highest infrastructure expansion costs for 2030, which is the GD scenario, are altered. In the second part of this section, the focus shifts from the required infrastructure expansion for 2030 to the analysis of changing infrastructure requirements in the face of technological development and increasing demand. First, the effect of change in the driving range of an average BEV is observed and, second, that of increasing demand by the rising share of BEVs in the Austrian car fleet.

Under the consideration of existing charging points with the peak capacity of $\hat{P}_{CP} = 350kW$, which is assumed to be the predominant charging capacity of fast-charging infrastructure for 2030, and the assumption that the investment of on-site preparation (c_X) has been made for all existing charging stations with charging points allowing fast-charging, the expansion of the current charging infrastructure along the Austrian highway network is modeled under different climate mitigation scenarios.

4.1.2.1. Austria's fast-charging infrastructure for 2030 under the Directed Transition scenario

Table 4.4 and Figure 4.1 display the modeling results for the expansion of the current $350kW$ charging infrastructure along highways in Austria for 2030 under the DT scenario. In Figure 4.1, the charging infrastructure expansion is expressed through additionally needed capacities. The expansions at existing charging points are indicated in dark green, and required capacities at newly installed charging points are indicated in orange . The charging stations in the lighter shades of green and orange indicate the required expansion of $350-4900kW$, i.e., by 1-14 charging points, and charging stations in the darker shades installations of additional $5200 - 10500kW$, i.e., 15 - 28 additional charging points. The total infrastructure expansion costs are estimated to be €54 Mio., which are translated to €/kW 369 and €39 per registered BEV in the Austrian car fleet in 2030. While currently, 40 charging points with $\hat{P}_{CP} = 350kW$, i.e., $14MW$, are installed along Austrian highways, an additional $+146MW$ of charging capacity are required until 2030. Further, additional 24 charging stations are needed, of which 10 are positioned within the dense part of the highway network in the East of Austria, in the vicinity of Vienna. The results also indicate the need for the installation of new charging stations near the Austrian border in order to cover all demand within the borders, which mostly results from the model formulation enforcing all coverage of demand within the network. Aside from this, there exist three charging stations that do not need any infrastructure expansion.

Table 4.4.: Cost parameters and charging infrastructure attributes resulting from the expansion of the existing $\hat{P}_{CP} = 350kW$ charging infrastructure under the *Directed Transition* (DT) scenario.

	Nb. of charging stations	Total capacity	Specific capacity costs	Specific costs per BEV	Total expansion costs
DT scenario 2030	54	$160MW$	€/kW 369	€/BEV 39	€54 Mio.

4.1.2.2. Comparison of the results from different future scenarios

Table 4.5 presents a comparison of key parameters describing the required infrastructure expansion under different future scenarios for Austria in 2030. The following observations are made:

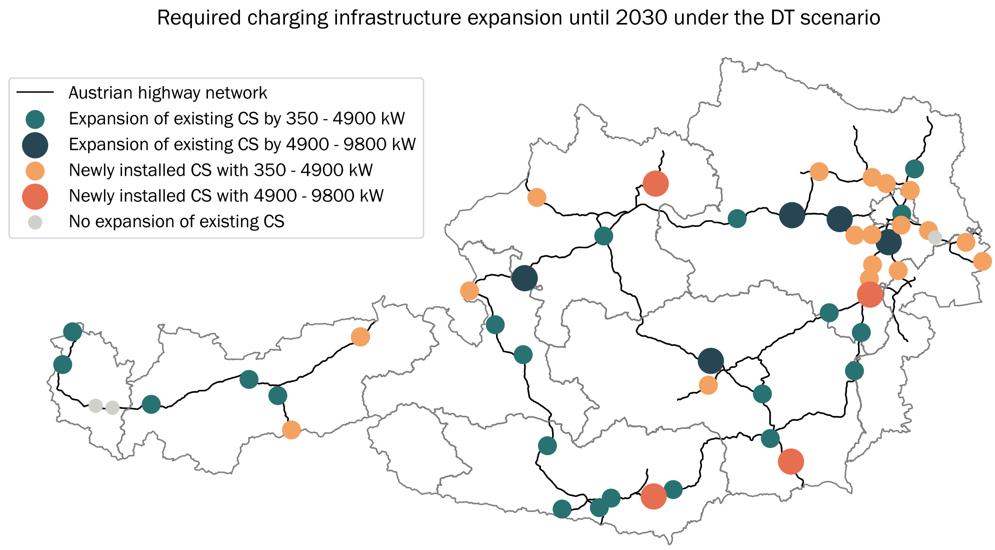

Figure 4.1.: Required expansion of fast-charging infrastructure along Austrian highways in 2030. Sizing of capacities that need to be added at currently existing and nonexisting charging stations (CS) given the *Directed Transition* (DT) scenario.

- The expansion under the DT scenario, for which the input parameters are set based on the assumption of a strong presence of political incentives pushing technological developments, results in the lowest costs of charging infrastructure expansion (€54 Mio.).

- There is a relative difference of up to +85% between the scenario causing the minimum expansion costs (DT) and the highest costs arising in the GD scenario, within which weaker climate change mitigation measures are assumed.

- The number of charging stations remains in a similar range for all scenarios, varying between 53 and 56. The specific infrastructure expansion costs per kW remain also very stable at around €/kW 368.

- The specific costs per BEV range between 39 and 72 and are the lowest in the TF and DT scenarios. The common trait of these two scenarios is the presence of technological breakthroughs leading to higher driving ranges and charging capacity of BEVs.

Table 4.5.: Comparison of the results for expansion of fast-charging infrastructure expansion along Austrian highways under different scenarios for 2030. The lowest specific costs per BEV and lowest total infrastructure expansion costs are highlighted.

	Scenarios 2030			
Model output	*Societal Commitment*	*Techno-Friendly*	*Directed Transition*	*Gradual Development*
Nb. charging stations	54	53	54	56
Total capacity (MW)	238	192	160	285
Specific capacity costs (€/kW)	368	368	369	367
Specific costs per BEV (€/BEV)	49	39	39	72
Total infrastructure expansion costs (€)	85 Mio.	68 Mio.	54 Mio.	100 Mio.
Rel. change of costs to DT scenario	+57%	+26%	-	+85%

4.1.2.3. Cost-reduction potentials in the Gradual Development scenario

To shed light into why the infrastructure expansion costs in the GD scenario are up to +85% higher than in the other scenarios and how the high specific costs per BEV of €/BEV 72 come to place, selected input parameters are altered and the effects on the costs are observed. These alterations reflect developments which are absent in the GD scenario but present in the others and which essentially lead to a cost reduction in charging infrastructure expansion investments. The following projections are drawn from the other scenarios and based on these, input parameters are altered in the GD scenario: a medium decrease in road traffic by -17% until 2030 as in the TF and DT scenarios, a major decrease by -31% as in the SC scenario, technological breakthroughs as in the TF and DT scenarios altering the driving range of an average BEV sold in 2030 to be $660km$, and having an average charging capacity of $243kW$ at a charging point with $\overline{P}_{CP} = 350kW$. Further details on these alterations are displayed in Table 4.6.

Figure 4.2 and Table 4.7 display the resulting cost reductions and changes in the required fast-charging infrastructure in response to the respective parameter changes. The lowest infrastructure expansion costs are achieved through the increase of BEV charging power which causes a large decrease in the required capacity as the efficiency of charging increases. Similarly, high cost reduction is accomplished by decreasing the overall traffic load which essentially reduces the demand for charging infrastructure. No cost reduction results from the increase in BEV driving range. Figure 4.2 illustrates these changes in costs in response to the specific parameter alterations visually.

Table 4.6.: Description of input parameter alterations reducing costs in the *Gradual Development* (*GD*) scenario. The parameter alterations are based on the other three scenarios: *Societal Commitment* (*SC*), *Techno-Friendly* (*TF*), *Directed Transition* (*DT*).

Parameter change	**Description (reference scenario)**	**Altered input parameter**	**Value in GD scenario**	**Updated value**
Medium decrease in road traffic	The overall road traffic load is subject to a reduction of -17% (DT, TF).	α	100%	83 %
Major decrease in road traffic	The overall road traffic load is subject to a reduction of -31% (SC).	α	100%	69%
Increase in driving range	The driving range of BEVs being sold in 2030 is increased to $1000km$ (DT, TF).	$\overline{dist}_{range,BEV}$	$520km$	$660km$
Increase in charging power	The average charging capacity of BEVs sold in 2030 is projected to be $315kW$ (DT, TF).	$\overline{P}_{charge,BEV}$	$164kW$	$243kW$

Table 4.7.: Changes in fast-charging infrastructure and associated investment costs in the face of different cost-reduction measures as described in Table 4.6. The highest cost reductions are highlighted.

		Cost-reduction measures			
	GD scenario 2030	Medium decrease in road traffic	Major decrease in road traffic	Increase in driving range	Increase in charging power
Nb. of charging stations	54	54	54	55	54
Total capacity (MW)	285	238	197	286	139
Total expansion costs (€)	100 Mio.	82 Mio.	68 Mio.	100 Mio.	66 Mio.
Rel. change	-	-18%	-32%	-0%	-34%

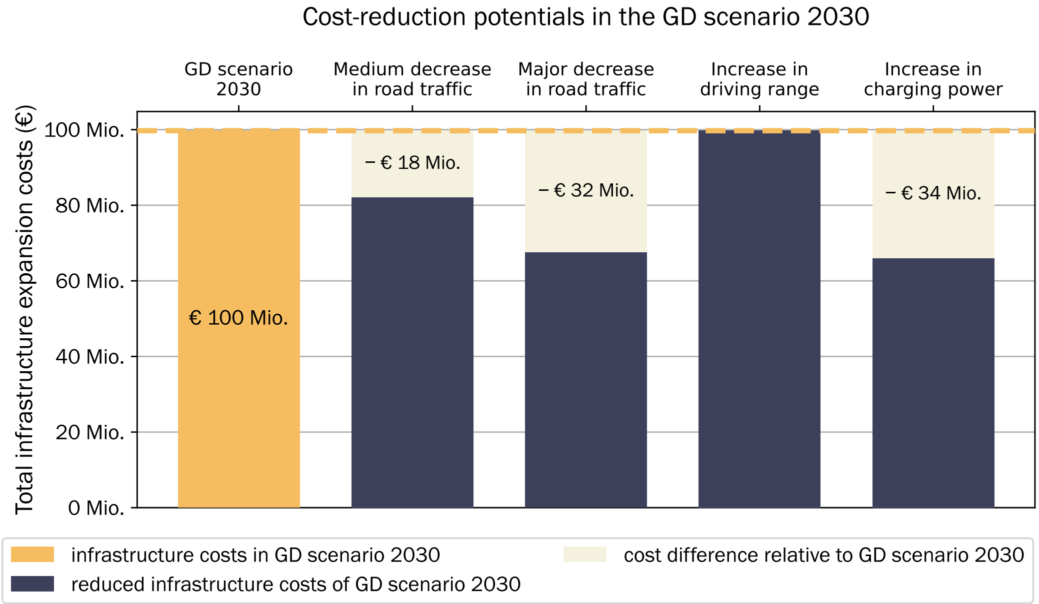

Figure 4.2.: Cost reduction potentials in the *Gradual Development* (GD) scenario. Detailed descriptions of the measures are in Table 4.6.

4.1.3. Sensitivity analyses on the requirements for fast-charging infrastructure

In the following, the mere infrastructure requirements in 2030 without regards to the charging infrastructure currently in place are analyzed. This is done to better understand how requirements for the fast-charging infrastructure change as a function of technological BEV parameters and in response to growing demand. First, the effect of the average driving range of an average BEV in the Austrian car fleet is observed in the context of the TF scenario during which major developments in battery technology are expected. Second, the change in modeled charging infrastructure depending on the BEV share in the car fleet within the SC scenario, which reflects high societal commitment to BEV uptake, is demonstrated.

4.1.3.1. Increasing driving range in the Techno-Friendly scenario

The driving range is altered between 200 and 1400km. Figure 4.3 presents the changing distribution of the numbers of the charging points at the charging stations and total number of charging stations in response to gradually increasing driving range in the TF scenario. In this figure, the blue boxplots display the distribution of charging points (CP) along the charging stations (CS). The gray dashed line indicates the maximum possible number of charging points at a charging station which is 34. This is given by the maximum installed capacity of 12MW at a charging station. The dark red connected points indicate the total number of charging stations in the modeled charging infrastructure. Table 4.8 presents the change in the key parameters for the driving ranges of 200, 800 and 1400km and gives an impression on how the key parameters of the modeled fast-charging infrastructure change in the course of the conducted sensitivity analysis. Given the assumed development in the TF scenario, the average driving range of BEVs in the Austrian car fleet is projected to be 670km by 2030 and reach approximately 1000km by around 2040.

With the increasing range, energy demand can be shifted further away from the node where it originates, which results in wider solution space during the optimization. Overall, the infrastructure costs stay stable between €71 Mio. and €72 Mio. throughout this parameter alteration. The total number of charging stations decreases from 55 at 200km to a minimum of 38 at 1100km. The installed capacity stays constant. Starting at the range of 300km, fully occupied charging stations occur throughout the sensitivity analysis. Overall the highest costs of investments are at 200km and drop to €71 Mio at the range of 500km.

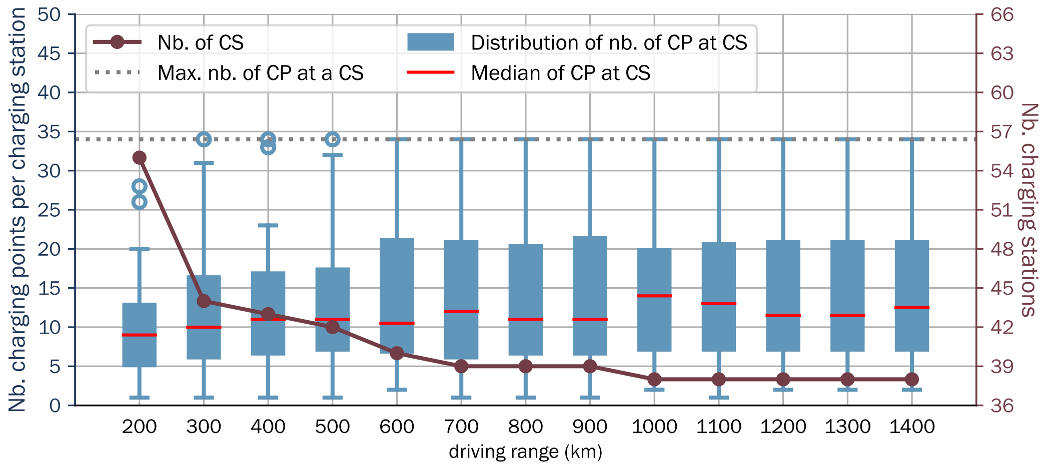

Figure 4.3.: Sensitivity analysis on driving range: impact of the increase of BEV driving range on the distribution of the number of charging points (CP) at charging stations and the total number of charging stations (CS).

During this analysis, the model was run for each of the ranges between 200 and 1400km every 100km and, therefore, in total, for 13 times. During all of these model runs, at 34 of all service areas, charging station installations occur in each of the model runs. The top sub-figure in Figure 4.4 illustrates these charging stations in black. The bottom illustration shows which of these are currently existing charging stations (in bright green) and which are not (in red). Overall, 11 of these 34 permanent charging stations are existing charging stations.

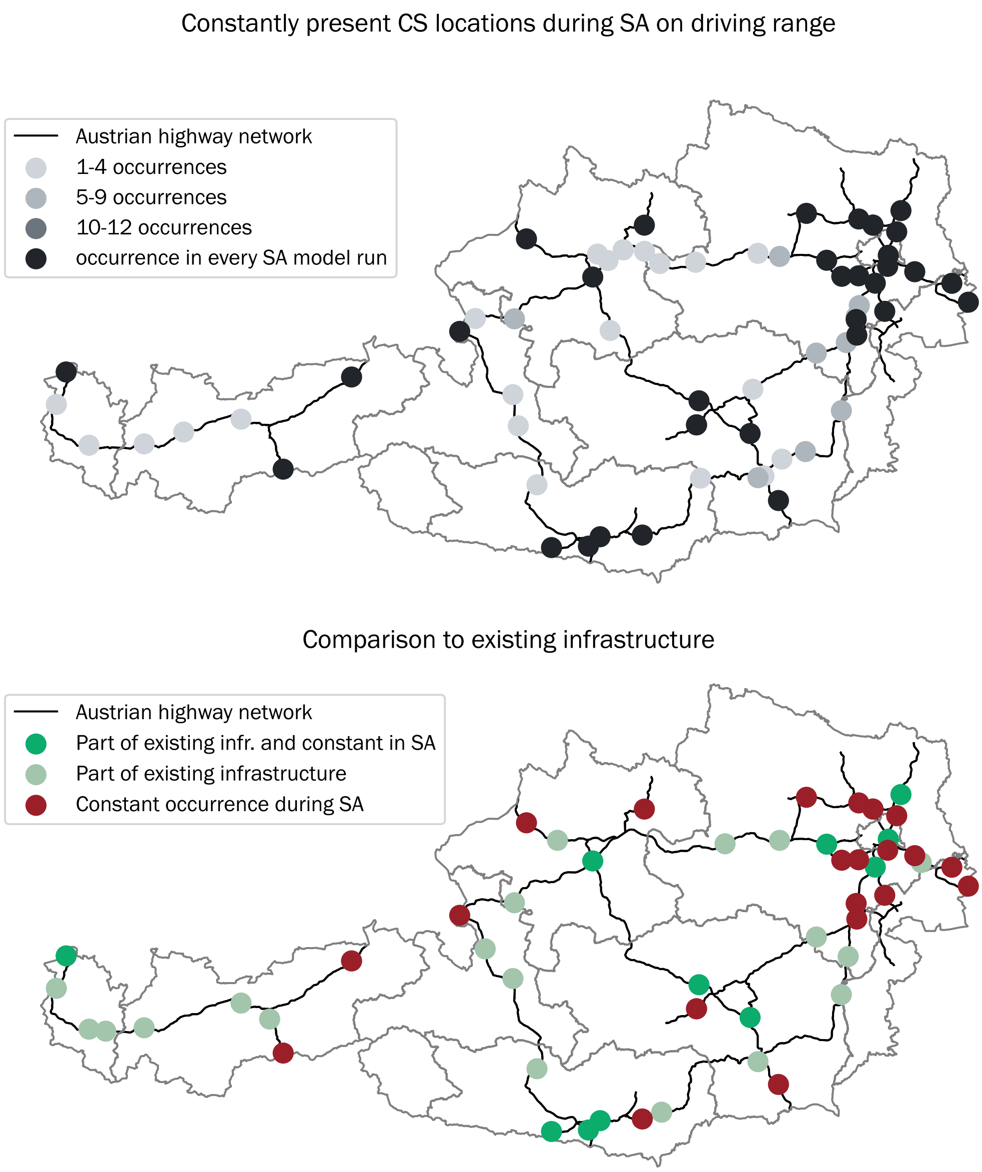

Figure 4.4.: Frequent charging point allocations across driving range sensitivity runs. Top: charging stations (CS) that appear in at least one model run. Bottom: CS that appear in all model runs and are part of the existing infrastructure.

4.1.3.2. Increasing share of BEVs in the Societal Commitment scenario

Figure 4.5 presents the results of the sensitivity analysis on the share of BEVs in the Austrian car fleet under the SC scenario for 2030. The top-left sub-figure presents the total installed capacity of the fast-charging infrastructure as a function of rising share of BEVs. The top-right sub-figure displays the change in the number of charging stations. Boxplots illustrating the distribution of the number of charging points at charging stations are presented in the bottom-left sub-figure. The timeline in this figure is projected based on the assumption that as of 2030, all registered cars will be BEVs and a BEV has a lifetime of 10 years which would lead to an Austrian car fleet that is 100 % electric by 2040. Table 4.8 presents the values of parameters describing the key features of modeled infrastructure for the different shares of BEVs, 10%, 50% and 100%.

The required capacity for fast-charging is rising linearly with the share of BEVs. The number of charging stations is significantly increasing between 40% to 100%, indicating the requirement for a densification of the fast-charging infrastructure. This effect of densification can be attributed to the increasing amount of fully occupied charging stations which is visible in the bottom left sub-figure of Figure 4.5. The overall costs increase by a factor of about 10 between 10% and 100% share of BEVs, so does the required capacity.

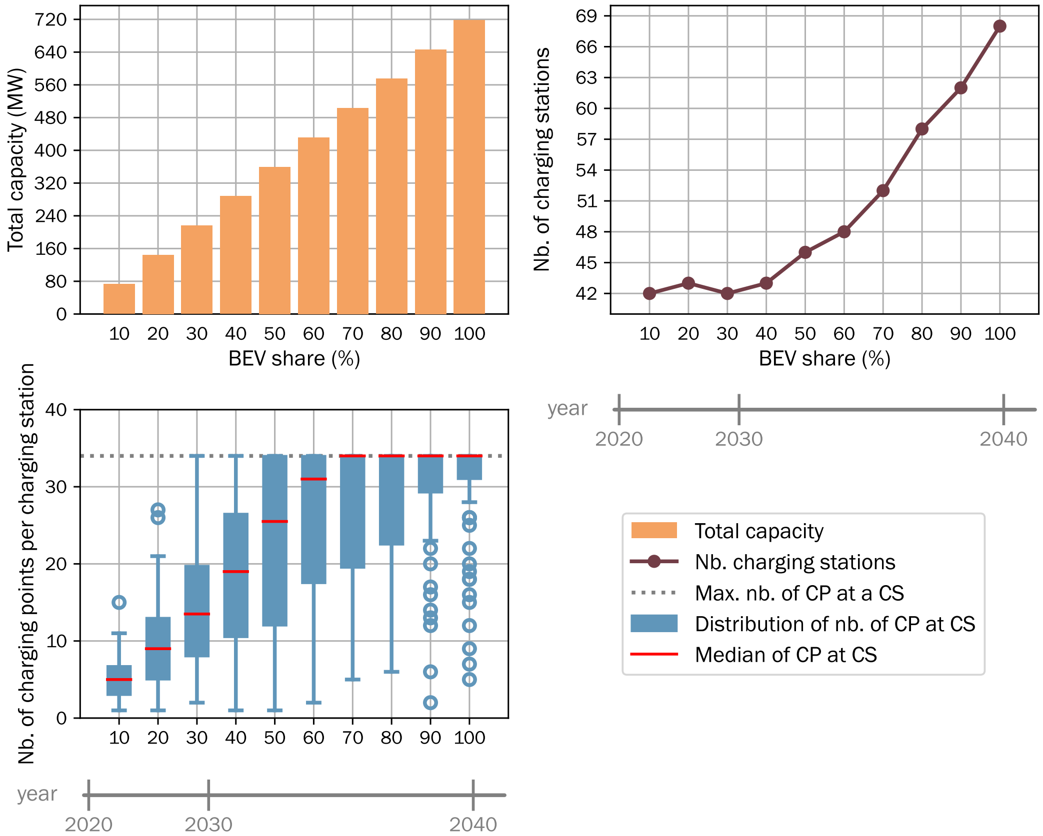

Figure 4.5.: Sensitivity analysis of increasing BEV share. Top-left: total required fast-charging capacity. Top-right: number of charging stations (CS). Bottom-left: distribution of charging points (CP) per station.

Table 4.8.: Key parameters describing the results of the sensitivity analyses on the driving range in the *Techno-Friendly* (*TF*) scenario and share of BEVs in the *Societal Commitment* (*SC*) scenario.

	driving range $\overline{dist}_{range,BEV}$ **(TF)**			**share of BEV** ϵ **(SC)**		
	$200km$	$800km$	$1400km$	10%	50%	100%
Nb. of charging stations	55	39	38	42	46	68
Total capacity (MW)	192	192	192	73	360	718
Total investment costs (€)	72 Mio.	71 Mio.	71 Mio.	28 Mio.	132 Mio.	263 Mio.

4.2. Impact of charging infrastructure roll-out on passenger car adoption across income groups and regions (RQ2)

The following case study and results are taken from the journal publication by Golab et al. (2026). The extended nomenclature and mathematical formulation are provided in Appendices C.1 and C.2, respectively. Computational details are provided in Appendix C.6.

4.2.1. Case study: Passenger car fleet of the Basque Country

4.2.1.1. Basque Country

The geographic extent of this analysis of this paper is the autonomous community Basque Country (in Spanish: *País Vasco*, NUTS code: ES21). This region is located in the North of Spain. The current passenger car vehicle stock is in the magnitude of 1.08 Mio. (2022) [149]. In 2023, the electrification rate of the passenger car fleet was at 0.05% [150]. The Basque Country is classified as a NUTS-2 region in the EV NUTS classification scheme [23] and consists of three NUTS-3 regions: Araba/Álava[3] (ES211), Guipuzcoa (ES212), Bizkaia (ES213). Figure 4.7 displays the relative geographic allocation of the regions and of the capital cities. For RQ2.a, we represent the NUTS-2 region as one node. For RQ2.b, we derive a graph-based conceptualization with three nodes representing each of the NUTS-3 regions and three edges representing the transport connections between neighboring regions.

Origin-destination data We use the modeled passenger car trips by ETISplus database [151] which holds information about: origin, destination zones (NUTS-3 level [23]), trip purposes, as well as route lengths and shortest-path route information for trips between all regions. A subset including NUTS-3 regions of the Basque Country is extracted. We further exclude all trips that go beyond the border of the case study area. This corresponds to the selection of 99% of all trips originating from there. Based on this, we set the system boundaries of the analysis equal to the boundaries of the Basque Country.

The ETISplus data were calibrated for transportation demand in 2010. To consider the growth in passenger car trips since then, we use the relative growth of the passenger car fleet since then [152]. The subsequent growth of the travel demand until 2050 is extrapolated based on historic GDP growth since 2020 [153].

Trips are categorized by the following purposes: *Business*, *Private*, *Vacation*, and *Commuting*. Trip purposes of the category *Business* are taken into account in the sizing of fueling infrastructure, but are not assigned to explicit consumer groups of different income levels. *Private* trips are defined as non-business trips with a duration of up to four days, while *Vacation* refers to trips that take more than four days. *Commuting* includes daily trips for working and studying purposes. Figure 4.6 displays the initial data trip count by purpose for local trips, i.e. within a NUTS-3 region, and inter-regional trips between NUTS-3 regions within the Basque Country.

[3] For the readability of the article, we refer to the region by its Basque language, Araba.

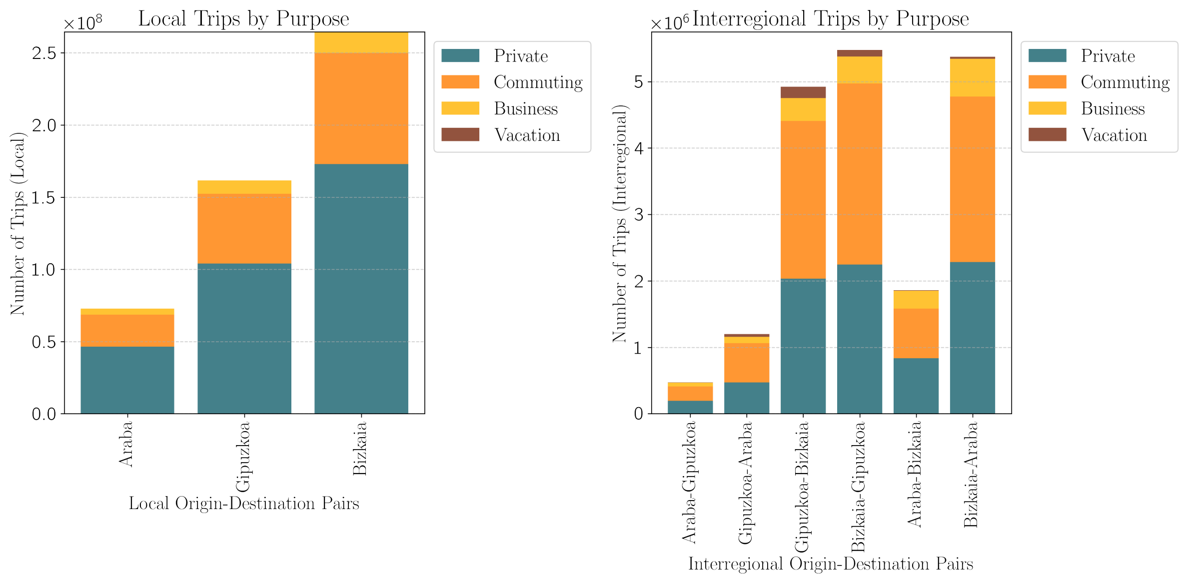

Figure 4.6.: Passenger car trips by purpose: Within a NUTS-3 region (*left*) and between NUTS-3 regions (*right*)

In the underlying data set by ETISplus, route lengths are given for the connections between NUTS-3 regions but not for trips within a region. To compensate for this missing information, we perform the assignment of route lengths (see Table 4.9).

Table 4.9.: Path lengths assigned to trips within a region (same origin and destination on NUTS-3 level)

		Average trip length (km)		
Region type	Share of total trips within a region	Private	Commuting	Business
Urban	21%	7.4	10.4	13.0
Rural	79%	9.7	17.1	23.2

Introduction of income-class specific parameters The key assumptions in the introduction of the income-class specific parameters is that the central difference in travel patterns by income class is the motorization rate, by monetary budget and home ownership [154, 155]. The income classes are characterized by having different monetary budgets for the purchase of vehicles and different VoT values. We consider five different income classes: *Very high income*, *High income*, *Medium low income*, *Low income* and *Very low income* class, mirroring the income class classification of EUROSTAT via five quintiles [156]. The most important preprocessing steps related to the income classes include the distribution of trips among them, the determination of monetary budgets and limitations for home-based charging infrastructure. We, therefore, characterize the five consumer groups by the motorization rate, VoT, average monetary budgets for new vehicles, the frequency of vehicle purchase, and the share of home ownership. Assumptions for these are summarized in Table 4.10.

Mattioli et al. [157] define a relation between motorization rate and purchasing power on NUTS-2 region level, derived from EUROSTAT data. We use this as a proxy to determine a relationship between the purchasing power standard (PPS) of the income class and the motorization rate, together with the average motorization rate in the Basque Country. This relationship is: Motorization rate $= 0.02 \times$ PPS $+ 79$. The VoT parameter is reported to vary significantly based on the monetary budgets. We use the value used in the case study by Tattini et al. [19] who apply these values in the context of a Danish case study. Due to the lack of published data for VoT values calibrated for different income classes in Spain or the Basque Country, we assume similar values for the present case study as found for Denmark and adapt these based on relative purchase power numbers, applying the factor $\times(92/139)$ [158]. A detailed derivation and sensitivity analysis of the VoT parameter is provided in Appendix C.9.

Table 4.10.: Income-class-specific assumptions for the Basque Country case study.

Income class	*Purchase Power Standard*	*Motorization rate (nb. of veh/1000)*	*Value of Time (€/h)*	*Average monetary budget over 10 years (€)*	*Purchase frequency (a)*	*Assumed home ownership*
Very low	10081	338	6.60	4160	10	30%
Low	19685	425	12.77	9282	8	55%
Medium low	27932	500	18.93	13910	6	85%
High	37910	599	25.10	18824	4	100%
Very high	61086	800	31.27	26545	2	100%

A statistic for the UK details expenditures for purchases [159] [160], displaying an approximately linear relationship between level of income and expenditure for purchases, along with how much is spent on the primary market. Based on this, we determine the monetary budgets for a new vehicle from the primary market for the *Very high*, *High* and *Medium High* income class, while we use the vehicle purchase price from the secondary market for the budget quantification of income class levels *Low* and *Very low*. Home ownership is extrapolated from the average Spanish home ownership rate reported in [161].

4.2.1.2. Drivetrain technologies and fuels

When it comes to the selection of the drivetrain technologies, we simplify it to two options: fossil-fueled vehicles and plug-in battery-electric vehicles. In contrast to this choice, studies in the literature have included larger technology portfolios by considering different electric car technologies, such as hydrogen fuel-cell or hybrid plug-in electric. The reason for the limitation to two technologies is twofold: Plug-in battery-electric vehicles have been largely proven to be cost-wise and environmentally the most viable, with a promising outlook on further cost decreases [162]. Therefore, most policies focus on supporting this technology. We reduce the fuels used to *fossil fuel*, including diesel and petrol, and electricity. Cost parameters for electricity are drawn from marginal costs for the energy system optimization for Spain performed with GENeSYS-MOD [163]. The average fuel price is based on [164]. Costs of the drivetrain technologies are drawn from a detailed modeling of the manufacturing of all components by [162]. We reduce the complexity of different car sizes to one representative vehicle type as related parameters to the choice of the car size - for example, household size, housing type [165] - are not part of the modeling. Therefore, different sizes are excluded to avoid introducing a modeling bias due to insufficient complexity of the model. Table 4.11 summarizes all assumptions on cost parameters and sources. The cost assumptions are based on the work by Grube et al. [162] who have evaluated a wide range of different drivetrain technologies for passenger cars and different passenger car sizes, including small, medium, and SUV. Here, we derive the costs for an *average*-sized vehicle by averaging the costs using weights that correspond to the magnitudes of the sizes

Table 4.11.: Overview of a selection of assumed parametrization for passenger cars of BEVs and fossil-fueled internal-combustion engine vehicles (ICEV).

	2025	2035	2045	Reference
BEV (first-hand market)				
CAPEX (€)	18,000	16,100	15,200	[162]
fuel costs (€/kWh)	0.05	0.03	0.03	[163]
Specific consumption (kWh/100km)	16.9	13.4	12.7	[162]
Battery capacity (kWh)	56	69	70	[162]
ICEV (first-hand market)				
CAPEX (€)	12,000	12,800	13,200	[162]
fuel costs (€/kWh)*	0.08	0.12	0.18	[162]
Specific consumption (kWh/100km)	58.1	44.9	36.9	[162]
carbon price (€/tCO2)	52	113	250	[167]
Charging infrastructure				
Level II: CAPEX (€/kW)		108		[25]
DCFC: CAPEX (€/kW)		350		[25]
OPEX		30% of CAPEX		

*without carbon pricing

in the current vehicle stock, assuming that this distribution would be constant throughout the optimization horizon. The data on different vehicle sizes represented in the statistic for different engine sizes in the Spanish vehicle stock are retrieved from [166]. The costs for the maintenance are increased at a rate of 3% per year for the first three years, and then by 10% per year.

To also allow for purchases from the second-hand vehicle market, purchasing options for vehicles of older generation are allowed for. For each technology, we introduce three groups of older vehicles with decreased investment costs but higher costs for maintenance and operation. The technological performance of vehicles available on the second-hand market is a function of their counterpart on the first-hand market and their average age. These three age-groups are: *1-5*, *6-10* and *11+ years*. The maximum lifetime of a vehicle is set to 25 years. An initial vehicle stock with vehicles of different ages is assumed based on data from [166]. The cost for vehicles on the second-hand market are calculated assuming a 65% drop in resale price after the first year and after this, an annual 10% cost decrease. The maintenance costs follow the annual increase in number as described above.

4.2.1.3. Types of charging infrastructure

For fossil-fueled vehicles, we assume that the fueling infrastructure with sufficient capacities has been established. Therefore, for combustion-engine passenger cars, no expansion of the fueling infrastructure is required.

For charging infrastructure, we introduce four different types: Home (Level II), Workplace (Level II), Public slow (Level II), and Public fast (DCFC) (based on [50]). Tables 4.12 and 4.13 summarize relevant assumptions. The reduction in fueling detour time is relevant for public infrastructure. In the case of fast charging (DCFC), the fueling time is considered in addition to the level of service (similar as in the work by Luh et al. [114]), while for the other types of infrastructure, we assume that there the charging is during typical down times of the vehicle and therefore does not impact the level of service. Initial home charging capacity (40 MW in 2020) reflects the early adopter profile of BEV owners, who were predominantly homeowners with dedicated parking and charging infrastructure. This assumption is consistent with empirical findings that early BEV adopters had home charging access rates exceeding 90% [51]. A sensitivity analysis on different home charging expansion scenarios is provided in Appendix C.10.

Table 4.12.: Overview on the introduced types of charging infrastructure and most important assumptions.

Type of charging infrastructure	Peak power level	Location	Max. utilization rate	Fueling detour time	Extra fueling time
Home (Level II)	11 kW	home	2%		
Workplace (Level II)	11 kW	workplace	20%		
Public slow (Level II)	22 kW	public	25%	×	
Public fast (DCFC)	50 kW +	public	25%	×	×

Table 4.13.: Assumptions for the sizing of initial capacities and upper limits for the expansion of charging infrastructure.

Type of charging infrastructure	Sizing of initial charging infrastructure	Limitation for capacity installation
Home (Level II)	based on current observations in the share of charging processes at home chargers (88%, [51])	Home charging infrastructure can be installed in 30% of owned homes.
Workplace (Level II)	based on current observations in the share of charging processes at workplace chargers (8.8%, [51])	based on today's charging demand coverage at work and scaled by current total number of passenger cars in the given region.
Public slow (Level II)	current BEV-to-charging point ratio for public slow chargers [168, 169]	No upper limit
Public fast (DCFC)	current BEV-to-charging point ratio for public fast chargers [168, 169]	No upper limit

. The maximum expansion speed of charging infrastructure is assumed to be 30% which is in line with reported expansion speed by [6].

To ensure that next to public slow charging, capacities for fast charging are installed to reduce range anxiety and enable recharge in cases of emergency (f.e. battery not sufficiently charged), we introduce a constraint imposing a minimum requirement for fast charging depending on the capacity installed for slow charging, which is set at the proportion of 2. This number is derived from the relative difference of the peak power level (the current value in Spain is 4.6 [168]).

4.2.2. Charging infrastructure roll-out scenarios

Figure 4.7 illustrates the setup for analyzing different scenarios of the roll-out of public charging infrastructure. Illustrations 1 and 2 address the setup to answer RQ2.a: two expansion strategies are compared, the *Concentrated* and the *Distributed expansion*. For this, the case study area is considered as one NUTS-2 node, while for addressing RQ2.b, the area is modeled on NUTS-3 level with three nodes.

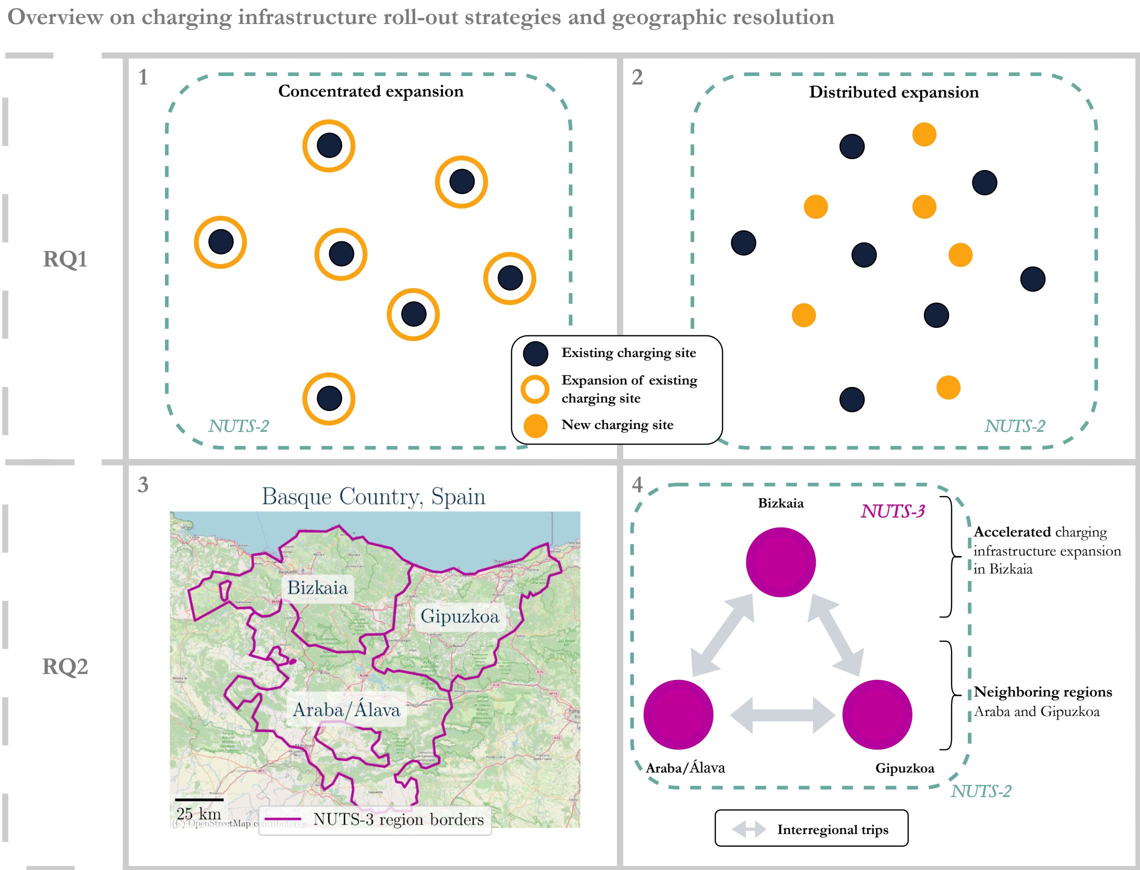

Figure 4.7.: Overview of the scenario setups for the analysis. The top row (illustrations 1 and 2) displays the contrasting approaches to the densification of public charging infrastructure, addressing research question RQ2.a. In the bottom row, illustration 3 displays a map of the geographic area of Basque Country by its three NUTS-3 regions, together with the corresponding capitals. In illustration 4, it is implied which pace of charging infrastructure roll-out is assumed for the NUTS-3 regions — addressing research question RQ2.b.

The spatial densification of charging infrastructure is addressed using the mixed-integer extension of the optimization model. For this, the trip data is accumulated to the NUTS-2 level.

To quantify the interaction between charging infrastructure investments and the detouring time required to reach the closest public charging station, a function between the average detouring time and installed charging capacities needs to be defined:

$$\alpha = g\left(q^{fuel\_infr,+}\right) \tag{4.3}$$

α is the factor for the relative decrease of the fueling detour time and is a function of additionally built capacities. By formulating this function, we directly assume that the chargers are spatially evenly distributed throughout the regions' road network. By doing this, we neglect the different degrees of urbanization and the option of allocating charging sites at points of interest, which would decrease detour time in reality. We do this with the premise that for a share of 100% BEV penetration of the passenger car fleet, a spatially even distribution of charger availability is a necessity, to be equally accessible for all. Further, this assumption indicates that there are no malfunctions of charging points which could lead to extended detouring times. The function $g()$

represents this relationship and varies for different charging infrastructure expansion strategies, which are explained in the following:

- *Distributed expansion:* Charging sites with singular charging stations are installed with the aim of achieving a dense charging network. Only when a maximum number of charging sites is reached, the number of charging points at the charging stations is expanded.
- *Concentrated expansion:* The focus lies on expanding the number of stations at charging sites. A maximum is set for the average number of charging stations at charging sites. New stations are only installed when existing sites are expanded to their maximum.

Figure 4.7 illustrates these strategies (Illustrations 1 & 2): Filled circles of dark blue color indicate initial charging infrastructure sites; orange color indicates the allocation of the installation of new charging points. The two strategies are contrasting as the *Concentrated expansion* leads to significantly lower increases in the density of charging stations versus *Distributed expansion* rapidly leads to a dense charging infrastructure. This boils down to the installed charging points ($n^{\text{points per site}}$) per charging site:

$$n^{\text{sites}} = \left\lfloor \frac{\text{installed capacity}}{n^{\text{points per site}}} \right\rfloor \tag{4.4}$$

We simplify the shape of an area using the rectangular form and translate the detour distance to a closest available charging station using the following definition:

$$\begin{aligned} d^{\text{aerial}} &= \frac{1}{2}\sqrt{\frac{\text{area}}{n^{\text{sites}}}} \\ \text{detour distance} &= \mu \times d^{\text{aerial}} \end{aligned} \tag{4.5}$$

Factors μ is a value ≥ 1 and translates the aerial distance to the driving distance in the road network (a sensitivity analysis on the circuity factor μ is provided in Appendix C.7). Finally, this is translated to detouring time, B:

$$B = \text{detour distance} \times \frac{1}{\text{driving speed}} \tag{4.6}$$

Three scenarios are defined for the analysis of spatial densification via different values for the parameter $n^{\text{points per site}}$, reflecting the *Distributed* roll-out scenario ($n^{\text{points per site}} = 2$), the *Concentrated* roll-out scenario ($n^{\text{points per site}} = 40$), and one between these two extreme cases with a balanced ten charging points per site ($n^{\text{points per site}} = 10$).

For the definition of suitable ranges in charging infrastructure expansion, we define a minimum detour time that is possible. This is set to five minutes[4]. No greater reductions of the initial

[4]The five minute threshold is based on an educated guess by the authors. This is based on the assumption that lower times for detouring (time required to reach the charging station **and** back from the charging station) are unrealistic. Further, a lower limit is set to avoid very small values in the optimization which could lead to

detour time are possible. Figure 4.8 displays reduction curves for different cases of $n^{\text{points per site}}$ for both, fast and slow public charging. Details on the discretization of these continuous functions for the MILP formulation are provided in Appendix C.5; the validation of the assumed functional shape against real charger data is documented in Appendix C.8.

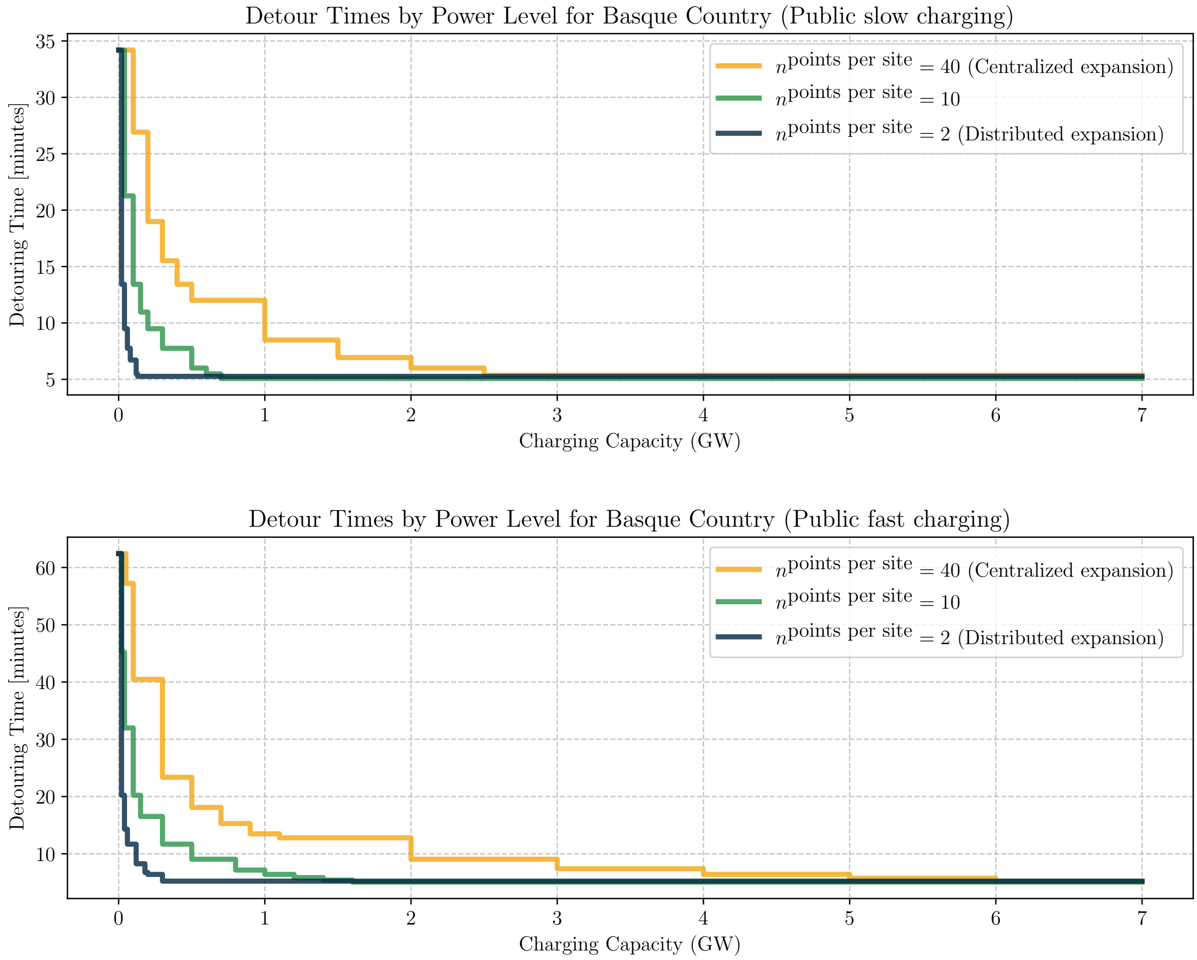

Figure 4.8.: Detour time reduction as a function of installed charging capacity for slow charging (top) and fast charging (bottom), under three spatial densification strategies ($n^{\text{points per site}} = 2, 10, 40$).

The setup for the analysis of RQ2 is displayed in the bottom row of Figure 4.7: We consider BEV adoptions in the three NUTS-3 level regions of the Basque Country and focus on the accelerated charging infrastructure expansion in Bizkaia, while observing the neighboring regions, Araba and Gipuzkoa. For this analysis, we assume the charging infrastructure capacities exogenously by including hard constraints for the infrastructure capacities. Table 4.14 displays an overview of the designed scenarios for the analysis. *Baseline Roll-out* defines a reference scenario based on the disaggregated expansion pathway to the region. The *Central Acc. Roll-out* is designed to analyse the relative changes in electrification compared to this reference scenario caused by an increase of charging infrastructure by 30%. In the scenarios *Neighbor Slowed Roll-out* and *Central Acc. + Neighbor Slowed Roll-out*, the neighboring regions have slower expansion than the reference, and the charging capacity roll-out pathway is reduced by 30%. These two scenarios differ again by first assuming the reference expansion pathway and in the second, the accelerated

increased computational complexity in the solution of the MILP.

expansion with a charging capacity increased by 30% to the reference.

Table 4.14.: Scenarios for analysing RQ2. The cross indicates the assumed expansion pathway for the regions in each scenario.

Scenario name (RQ2)	**Region name**	*Expansion pathway* -30%	Reference	+30%
Baseline Roll-out	Bizkaia		×	
	Gipuzkoa		×	
	Araba		×	
Central Acc. Roll-out	Bizkaia			×
	Gipuzkoa		×	
	Araba		×	
Neighbor Slowed Roll-out	Bizkaia		×	
	Gipuzkoa	×		
	Araba	×		
Central Acc. + Neighbor Slowed Roll-out	Bizkaia			×
	Gipuzkoa	×		
	Araba	×		

The reference expansion pathway is determined based on a model run, during which the constraint of 100% decarbonized passenger car fleet is imposed. The resulting total charging capacity values for the investment periods between 2020 and 2050 are included as hard constraints for the charging capacity expansion. Figure 4.9 displays the reference values by region.

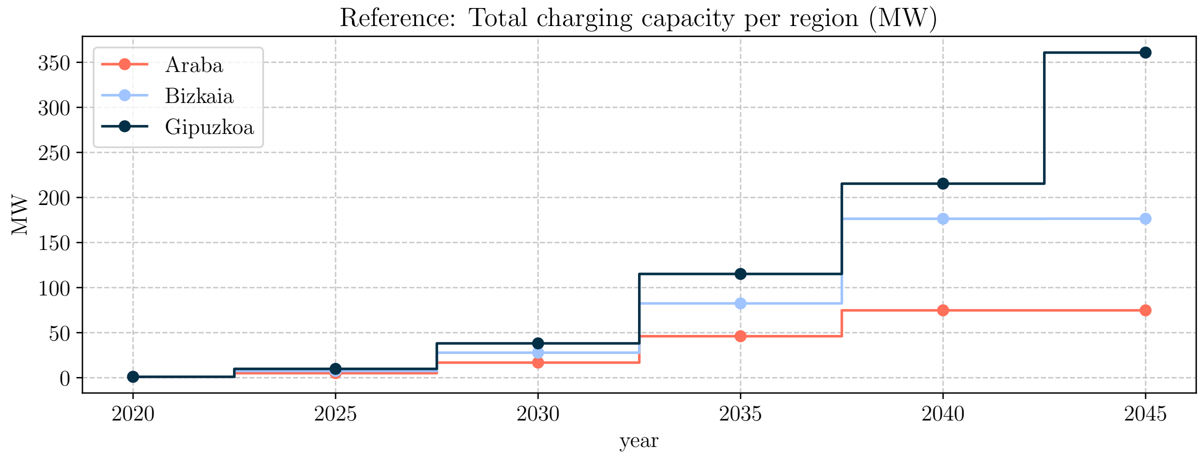

Figure 4.9.: Reference expansion pathway for scenario design of different charging capacity build-up scenarios.

4.2.3. Income-class dependent impact of spatial densification of public charging infrastructure

The results in this section are obtained using the MILP extension of Model 2 (Section 3.3.5.2), which endogenizes the fueling detour time as a function of charging infrastructure expansion.

Figure 4.10 displays two snapshots of the development of the share of electrification in the passenger car fleet by consumer group: years 2030 and 2040. $n^{\text{points per site}}$ indicates how many charging points are installed in newly built charging stations as of 2020, implying the degree of spatial densification of the public charging infrastructure network. The development of technology turnover is different among the consumer groups. In 2030 and 2040, the electrification in the vehicle fleet belonging to the *Very high income* and *High income* consumer groups is significantly higher, at the level of around 20.7-38.4% in 2030 and 29.5-61.9% in 2040, than that of the vehicle fleet belonging to the other consumer groups. The share of electrification for consumer groups *Medium low income*, *Low income* and *Very low income* rises to the levels of 1.7-5.8% in 2030, then mostly drops to 0.0-2.2%.

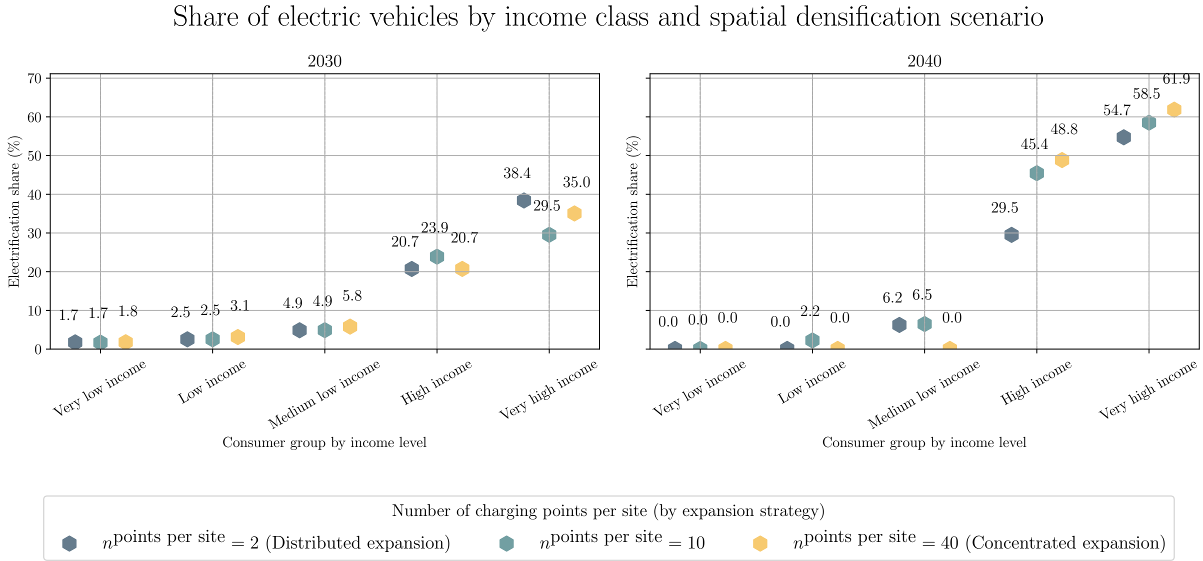

Figure 4.10.: Share of battery-electric vehicles in the passenger car fleet in 2030 and 2040 in consumer groups of different income levels for roll-out strategies of public charging infrastructure defined via the number of charging points per site ($n^{\text{points per site}}$).

The relative order of the number of BEVs in the fleet by consumer groups is not changed by the different roll-out strategies. However, significant differences in the adoption are visible between the different densification strategies: In the *Concentrated expansion*, the share drops to 0.0% for the *Medium low income* consumer group by 2040, while, in the other densification scenarios, the share rises to 6.2 and 6.5%. A significant gap is visible between the *Distributed* and *Concentrated expansion* for the *High income* consumer group in 2040; at lower spatial density of the public charger network, the *High income* consumer group has an increase of +19.3%. A similar trend is visible for the *Very high income* group, for which the gap is +7.2%. Overall, there is no clear correlation to be deduced between the number of points per site and the development of the electrification share in the fleets owned by the consumer groups of different income levels. Figure 4.11 displays differences in the development of the gap in the share of electrification between the *Distributed* and *Concentrated* scenarios. It shows that the highest overall impacts are on *High income* and *Medium low income*. Here, the *Concentrated expansion* again shows accelerated BEV adoption for higher income classes, while the *Distributed expansion* favors adoption in the *Medium low income* consumer group. In later years (around 2043), the *Concentrated expansion* shows accelerated adoption in the *Very low income*, *Low income*, and *Medium low income* consumer groups.

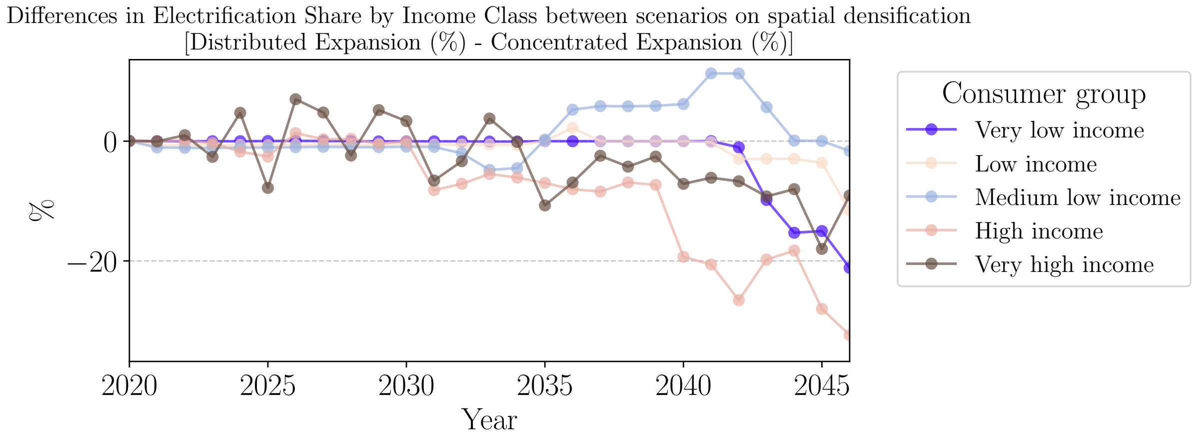

Figure 4.11.: Maximum differences in the electrification share between scenario $n^{\text{points per site}} = 2$ *(Distributed expansion)* and $n^{\text{points per site}} = 40$ *(Concentrated expansion)* by consumer groups over time.

To gain a better understanding of the variation in the electrification across consumer groups, we take a look at the three key output parameters that describe the charging infrastructure expansion decisions and their utilization by scenario. Figure 4.14 displays expanded charging infrastructure capacities together with the development of the utilization rates of public infrastructure for all scenarios. Figure 4.12 shows how the average detour time of public charging infrastructure is reduced by these capacity expansions. Figure 4.13 illustrates the charger utilization by income class, showing how much energy is charged by charger type and income class in years 2030, 2040, and 2050.

The following observations are the most relevant here:

- With an increase in the parameter $n^{\text{points per site}}$, higher investments in public charging infrastructure are made earlier. Simultaneously, home charging capacity expands more rapidly with lower values of $n^{\text{points per site}}$. The *Concentrated expansion* shows significantly greater fast charging capacity expansion — 56% more than the $n^{\text{points per site}} = 10$ scenario and 184% more than $n^{\text{points per site}} = 2$. Home charging capacities are subsequently increased to reach levels similar to those in the $n^{\text{points per site}} = 2$ and $n^{\text{points per site}} = 10$ scenarios by 2040.

- The sharp expansion in the $n^{\text{points per site}} = 40$ scenario coincides with the sharp increase in adoption in the consumer groups of lower income classes (as indicated in the energy consumption patterns by income class displayed in Figure 4.13), indicating that the increased charging capacities particularly accelerate the electrification of the passenger car fleet owned by consumers of low level income.

- Work charging capacities are consistent across all scenarios. The capacities are early on expanded to the maximum possible limits.

- The utilization rates of public charging infrastructure diverge substantially between the slow and fast charging types. For both charger types, maximum utilization is not reached. Across all scenarios, the utilization rates of fast chargers are mostly lower than 5% until 2035-2040. Later in the optimization horizon, these values increase, while the utilization of slow chargers peaks around 2035-2040 and, simultaneously with the increase of fast charger utilization, decreases to lower values, i.e., 9-17%. Moreover, in the *Concentrated expansion*, fast charger utilization remains at 0% until 2040 and increases significantly from 2041 onward. This coincides with the late rapid uptake of BEVs in low-to-medium income consumer groups.

- The diverging impacts on detour times via the detour fueling function by scenario are displayed in Figure 4.12. For $n^{\text{points per site}} = 2$, early on, the average detour times of around 15 minutes for both slow and fast charging are reached. In the case of $n^{\text{points per site}} = 10$, the detour time for public slow charging is constant and not reduced throughout the optimization horizon. The detour time for public fast charging is reduced to around 32 minutes. In scenario $n^{\text{points per site}} = 40$, the detour time for public fast charging is not reduced until 2050.

- In all scenarios, the utilization of the installed charger capacities differs by income group and charger type. Looking at the right column in Figure 4.13, the coherence between BEV adoption and the absolute volumes of charged energy is evident. For high spatial density scenarios ($n^{\text{points per site}} = 2$ and $n^{\text{points per site}} = 10$), fast charging covers charging demand of lower income groups, while in scenario $n^{\text{points per site}} = 40$, fast chargers cover demand starting in 2041. Home charging covers demand in the consumer groups *Very High* and *High income*. For $n^{\text{points per site}} = 10$ specifically, fast charging has lower detour times than slow charging (see Figure 4.12), leading the *High income* consumer group to use fast charging in 2040 and 2050.

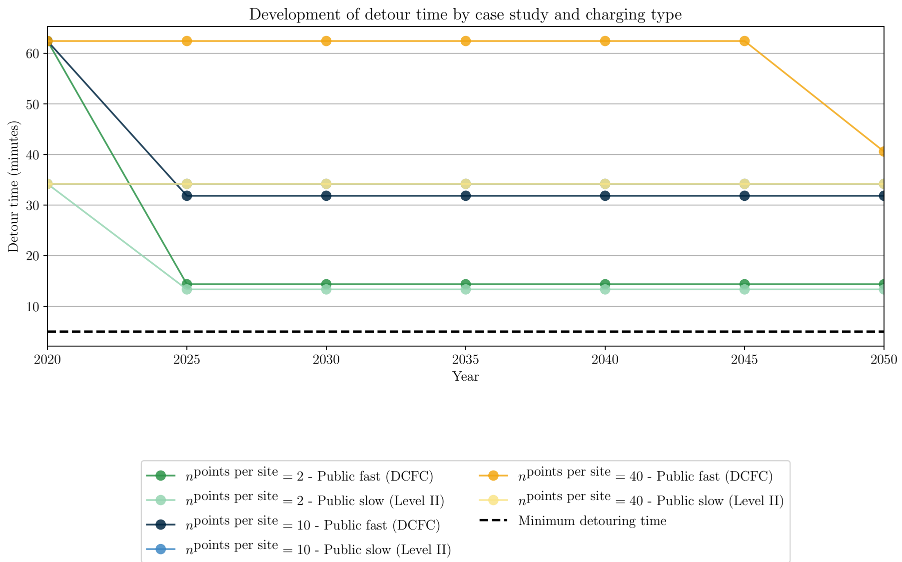

Figure 4.12.: Development of average detouring times for public fast and slow charging in Basque Country for different scenarios of spatial densification.

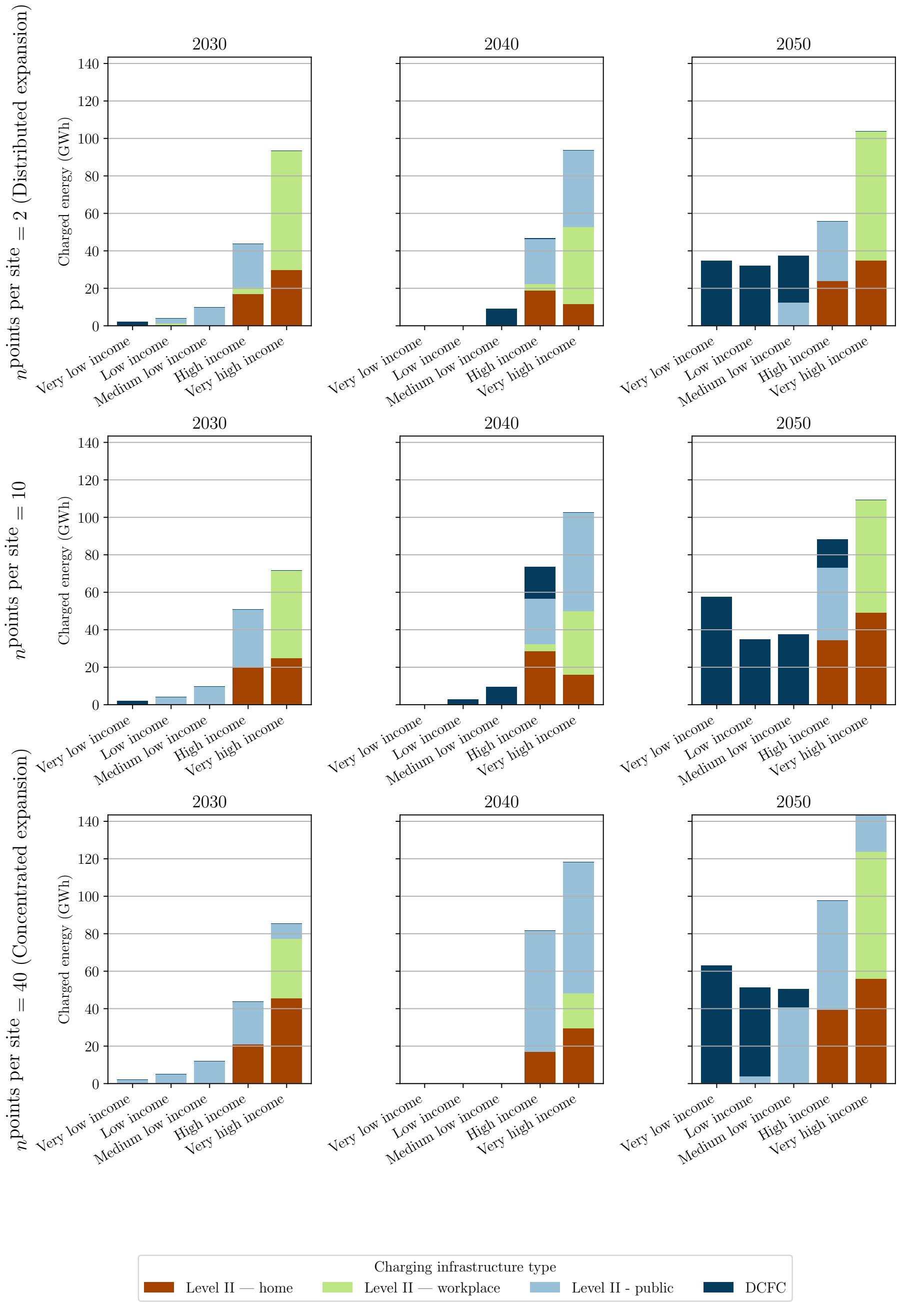

Figure 4.13.: Charged energy by income class and infrastructure type in 2030, 2040, and 2050.

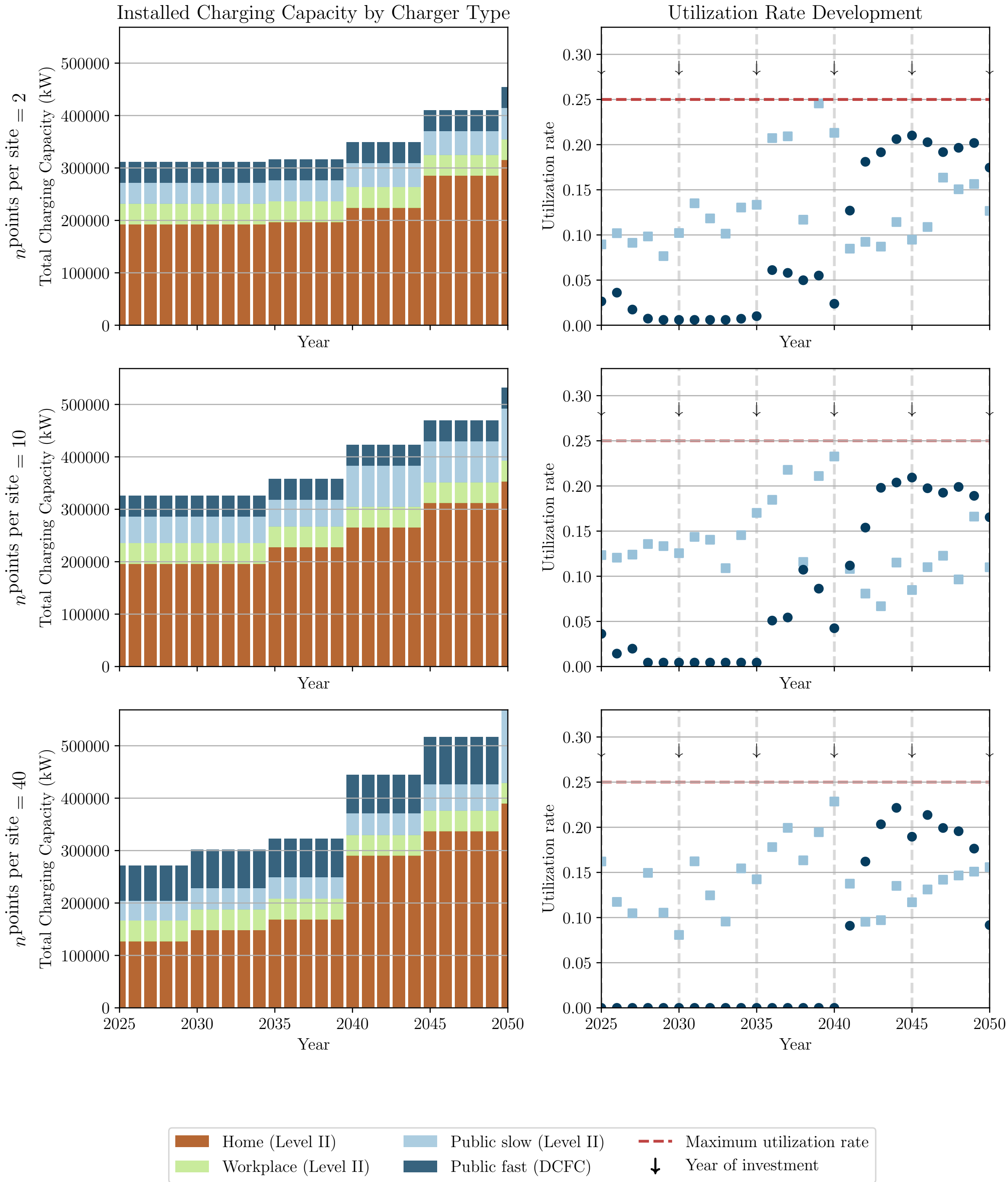

Figure 4.14.: Charging infrastructure expansion for Basque Country region in the scenario of *Concentrated expansion* ($n^{\text{points per site}} = 40$).

4.2.4. Spill-over effect of charging infrastructure planning for destination charging

The results in this section are obtained using the LP extension of Model 2 (Section 3.3.5.2), which introduces consumer group differentiation and intangible costs.

Figure 4.15 displays the development of the adoption of BEVs by years 2030 and 2040. In particular, this is shown for the NUTS-3 regions Bizkaia, Araba, and Gipuzkoa, under the different scenarios in the speed of charging infrastructure expansion. In the *Baseline Roll-out* case, in 2030, the range of the electrification rate varies by region within 2.6-6.0%. By 2040, the region-dependent range becomes wider, 35.5-42.8%, with Araba having the highest rate of electrification and Bizkaia the lowest. With increased charging capacities built in Bizkaia (case *Central Acc. Roll-out*), the electrification share in Bizkaia increases slightly (0.9% by 2030 and 3.3% by 2040), while the electrification rates of Araba and Gipuzkoa are also positively affected (increase by 0.6-1.3%). When the charging infrastructure capacity in Araba and Gipuzkoa (scenarios *Neighbor Slowed Roll-out* and *Central Acc. + Neighbor Slowed Roll-out*) are reduced, the values for the share of electrification in Araba and Gipuzkoa in 2040 are significantly lower than in *Baseline Roll-out*. Figure 4.16 displays the temporal development of the difference in electric vehicle numbers between the *Central Acc. Roll-out* and the *Baseline Roll-out*. A peak in the positive absolute differences occurs in 2045. The maximum relative difference is in 2044 and is 2.1% for Araba and 1.1% for Gipuzkoa. For some time periods, we observe negative spillover. In particular, there is a distinctive decrease in the BEV numbers in 2049.

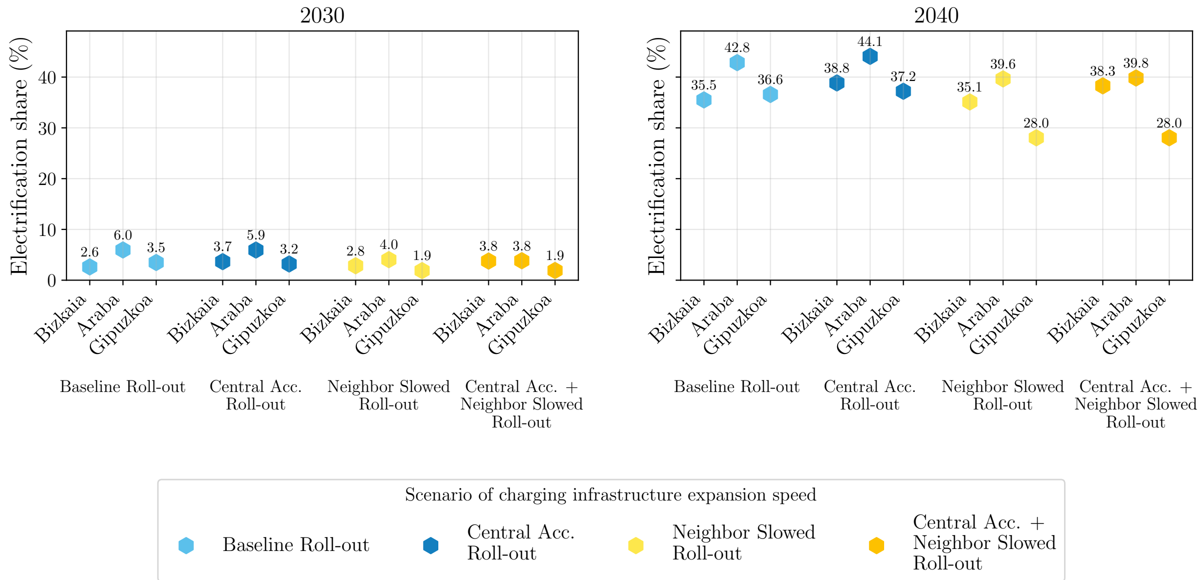

Figure 4.15.: Share of electrification of the passenger car fleet in the regions Bizkaia, Araba, and Gipuzkoa under different scenarios referring to the speed of charging infrastructure roll-out, for years 2030 and 2040.

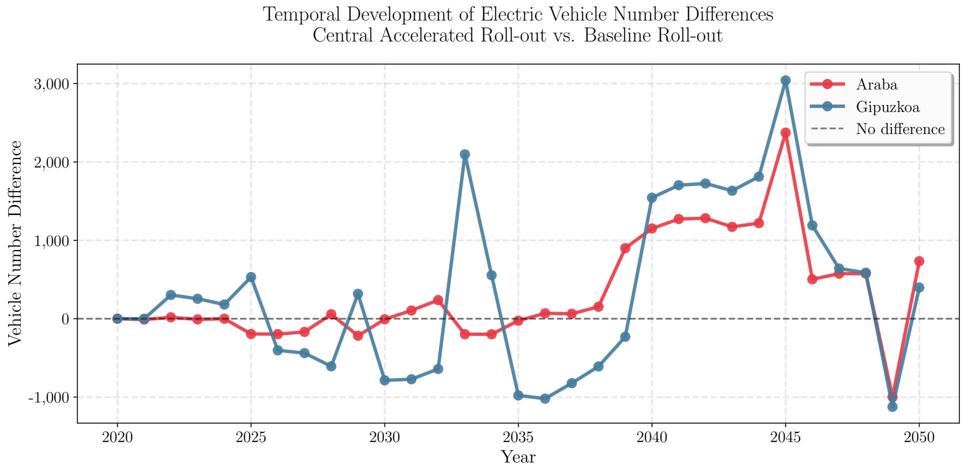

Figure 4.16.: Temporal development of the difference of adopted battery electric vehicles numbers in Araba and Gipuzkoa between the *Baseline Roll-out* and *Central Acc. Roll-out.*

With higher expansion in Bizkaia in *Central Acc. + Neighbor Slowed Roll-out*, the electrification share in Bizkaia is increased by 2.1% by 2040. The BEV adoption shares in Araba and Gipuzkoa are here impacted significantly less, i.e., than in the reference capacity expansion. In 2030, a slight negative impact is observed. A sensitivity analysis testing different levels of capacity expansion (+15%, +30%, +45%) is provided in Appendix C.11.

Figure 4.17 gives an insight into the effect of increased charging capacity in Bizkaia on the electrification of the commuting traffic. The plots display the absolute numbers of BEVs used for commuting trips originating in Araba or Gipuzkoa and traveling to Bizkaia. We zoom in on three years: 2030, 2040, and 2050. The text in blue indicates the relative differences to the *Baseline Roll-out* scenario. The gray text indicates the relative difference to *Neighbor Slowed Roll-out*. The following four key observations are made here:

- The slowed roll-out in Araba and Gipuzkoa has little impact on the BEV adoption in the commuting fleet. Some negative impact is observed during 2040-2050 which is partly compensated with increased capacities in Bizkaia.
- In Araba, the electrification of the commuting fleet is mainly positively accelerated during the years 2030 and 2040 by the increased charging infrastructure deployment in Bizkaia. By 2050, this positive impact is limited to +0.6%.
- In Gipuzkoa, the impact of increased infrastructure expansion in Bizkaia remains not significant.
- In the case of reduction of charging infrastructure in Araba and Gipuzkoa, the increased charging infrastructure in Bizkaia does not fully compensate for the reduced capacities.

In Figure 4.18, we examine the allocation of charging processes for commuting trips between the origin and destination regions. We generally observe a high share of destination charging across all cases, years, and both regions. These shares increase under the accelerated rollout in Bizkaia. One exception is the development in Gipuzkoa between 2030 and 2040, during which there is a 43% increase in charging processes at the origin (between the Central Acc. Roll-out and Baseline Roll-out scenarios).

Figure 4.19 displays utilization rates for the public charging infrastructure in the Baseline Roll-out across all three regions. Here again, utilization rates of public slow charging are consistently higher than those of public fast charging. Overall, utilization rates are significantly higher when spatial density is not considered.

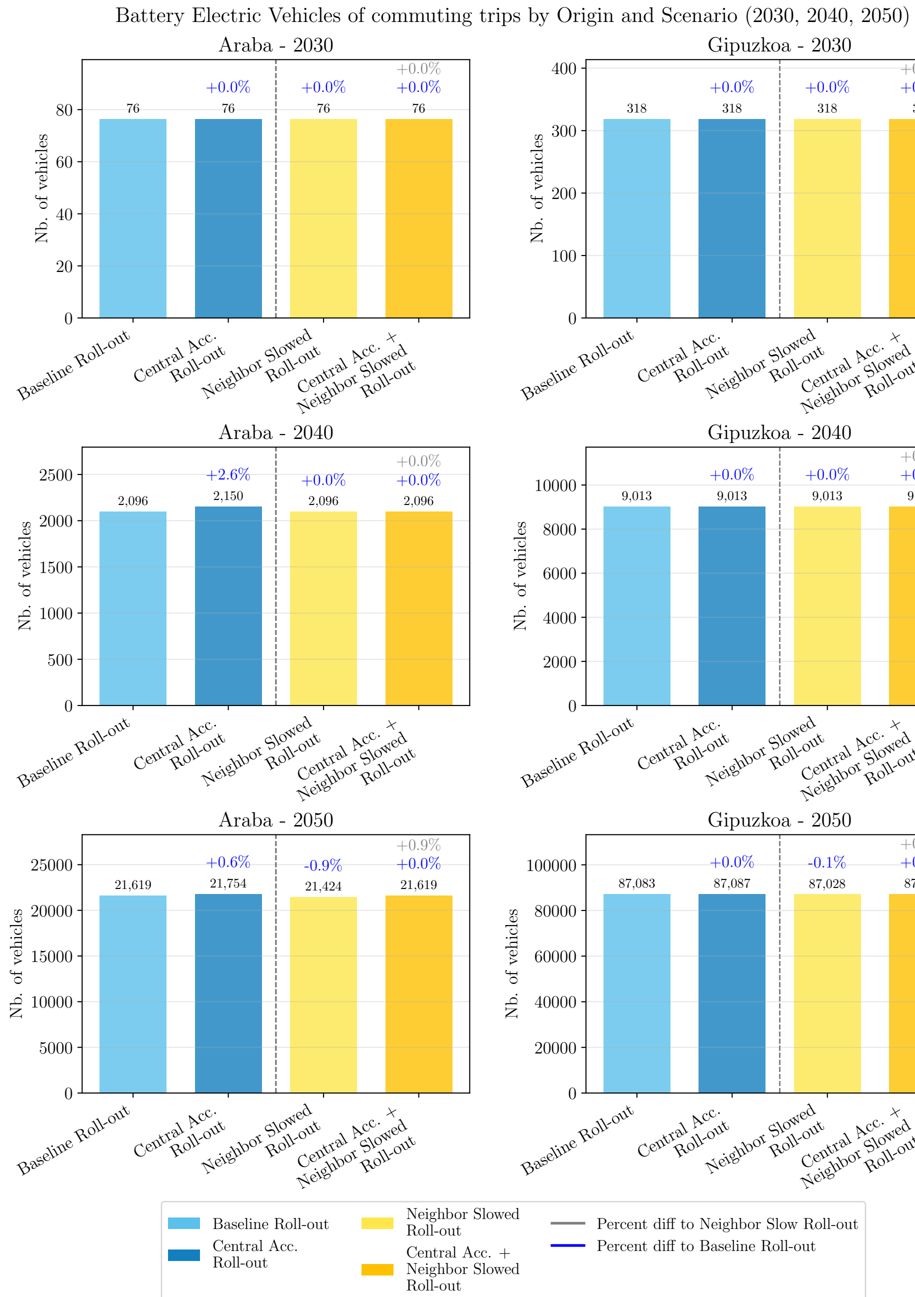

Figure 4.17.: Numbers of BEVs in the commuting trips between Araba or Gipuzkoa and Bizkaia under different scenarios of charging infrastructure roll-out scenarios for years 2030, 2040 and 2050.

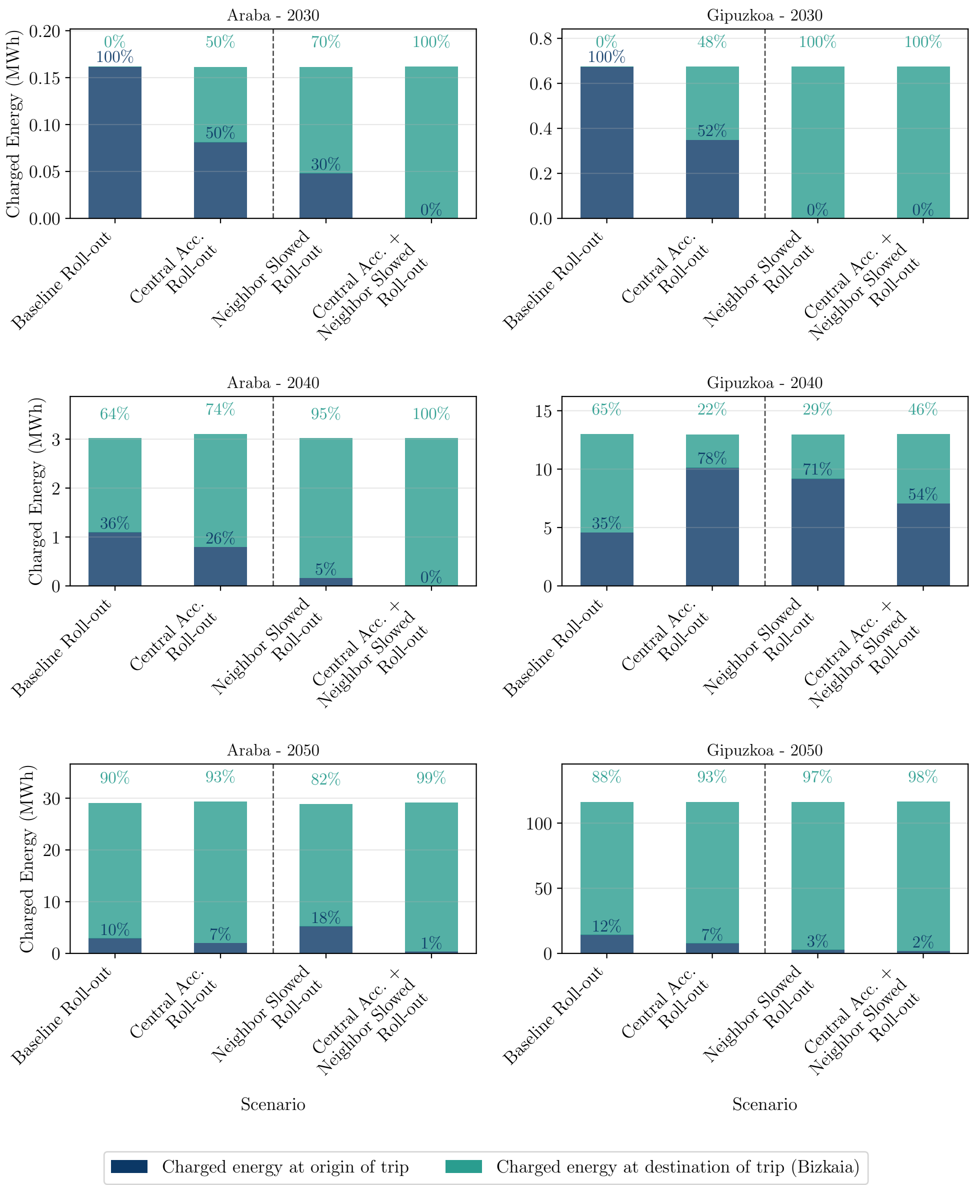

Figure 4.18.: Snapshots for 2030, 2040 and 2050: Allocation of charging processes between origin and destination of commuting trips originating in Araba or Gipuzkoa and going to Bizkaia with text indicating relative percentages of charging processes.

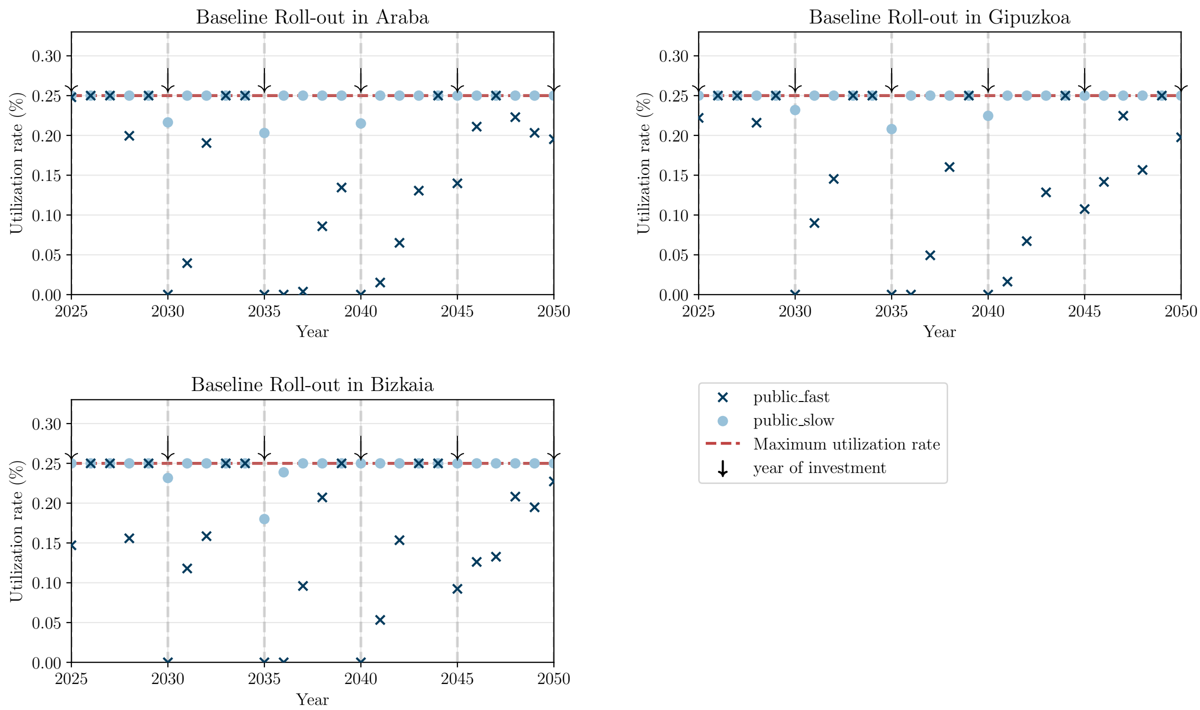

Figure 4.19.: Development of utilization rates for public charging infrastructure network in Bizkaia, Araba and Gipuzkoa.

4.3. Spatial patterns of charging demand allocation in inter-zonal long-haul truck electrification (RQ3)

The following case study and results are taken from the journal publication by Golab et al. (n.d.).

4.3.1. Case study: Scandinavian-Mediterranean corridor

4.3.1.1. The Scandinavian-Mediterranean corridor

The Scandinavian-Mediterranean corridor represents a North-South axis within the Trans-European Transport Network (TEN-T) [170]. It includes parts of the European road and rail network that connect Norway and Finland through Sweden, Denmark, Germany, Austria, and Italy, extending to Sicily. Part of the corridor is also allocated in the South of Finland, which is connected through maritime waterways and airports to the major part of the corridor[5]. Overall, the corridor connects 19 international airports and 25 seaports positioned along the European coastline. The core network encompasses over 6,300 km of road and 9,400 km of rail transport infrastructure [172]. Under the EU Alternative Fuels Infrastructure Regulation (AFIR), TEN-T core corridors must be equipped with publicly accessible high-power charging stations for heavy-

[5]Given the wide extent of this corridor, no image outlining its allocation and its course is included in this manuscript. Please refer to [171] for an overview.

duty vehicles at intervals of no more than 120 km by 2030, making this corridor a priority area for charging infrastructure deployment.

The extent of this corridor is used in this work for the definition of the area of the case study. While the road network of the Scandinavian-Mediterranean corridor connects major demand centers in the given countries, in the present analysis, we also include road transport infrastructure that is not directly part of the corridor infrastructure as the official classification: We also include road infrastructure aside from this that is allocated within the connected regions. Therefore, the extent of the case study is set to all NUTS-2 regions[6] intersected by the Scandinavian-Mediterranean road infrastructure. In total, the case study network comprises 81 NUTS-2 regions (nodes).

4.3.1.2. Transport activity

Transport demand data is derived from the ETISplus European transport statistics database, as processed by Speth et al. [174]. The dataset provides projected road freight flows for 2030, specifying origin-destination nodes at NUTS-3 geographic granularity, annual tonnages between each node pair, and the route course through intermediate regions. Demand for other modeled years is obtained by scaling the 2030 reference values using a GDP-based growth rate derived from historic European road freight activity [175]. This data is aggregated to NUTS-2 level and filtered for exclusively international routes, as cross-border routes are the primary driver of spatial variation in charging costs across electricity market zones. Routes with less than 100 tonnes annual volume or shorter than 360 km are excluded, following Shoman et al. [176], since these routes do not require en-route charging.

The resulting dataset comprises 81 NUTS-2 regions and 2,244 exclusively international OD pairs connecting eight countries (AT, DE, DK, FI, IT, NO, SE). Figure 4.20 shows the freight demand by country pair. The largest share of routes connects Germany and Italy (1,167 OD pairs, 52%), followed by Germany–Scandinavia connections (DE–SE: 459, DE–NO: 118, DK–SE: 76). The optimization horizon is 2020–2060, while the analysis focuses on 2020–2050; the additional decade avoids end-of-horizon artifacts in investment decisions.

Transport demand is extrapolated from the 2030 reference year based on historic growth of European road freight activity [175], growing from 7.63 billion tonne-kilometres (Btkm) in 2020 to 8.93 Btkm in 2060 (compound annual growth rate of 0.4%), applied uniformly across all OD pairs. For each route, mandatory driver rest stops are determined following EU Regulation 561/2006 [176]: starting from the origin, the model traces the route node by node, accumulating driving time at a fixed speed; whenever the cumulative driving time since the last break reaches 4.5 hours, a 45-minute break is placed at the nearest node, and after 9 hours of total daily driving, a 9-hour rest period is assigned. This procedure is applied to every OD pair individually, so each route has its own sequence and number of mandatory stops determined by its length and geometry. These mandatory breaks create natural charging windows for battery-electric trucks, as charging during mandatory breaks can partially or fully overlap with the required rest period, reducing additional time cost. Figure 4.21 shows the resulting spatial distribution

[6] Using the following NUTS definition: [173].

of mandatory break locations. Break locations concentrate in central Germany and the Alpine region, reflecting the convergence of long-haul routes from Scandinavia toward the Brenner and other Alpine crossings.

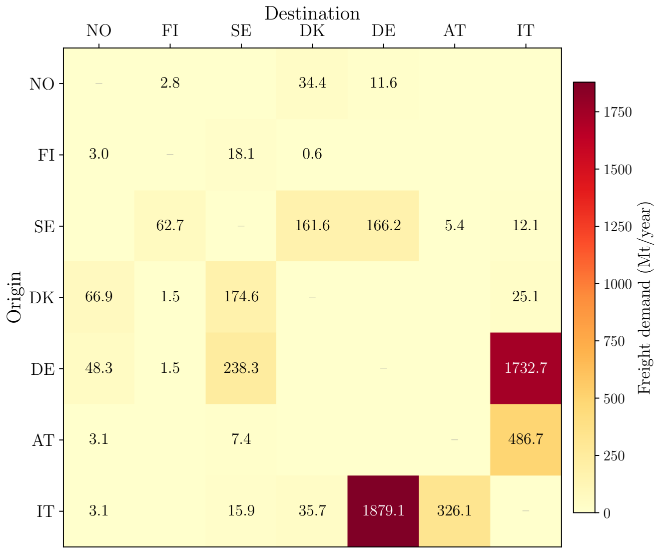

Figure 4.20.: International freight demand (Mt/year) by origin–destination country pair for the base year 2020. The DE–IT corridor dominates with approximately 3.6 Mt bidirectional freight, followed by SE–DE (404 kt) and SE–DK (336 kt). Total international freight along the corridor amounts to approximately 5.5 Mt/year. Routes are filtered for $\geq 100\,\text{t}$ annual volume and $\geq 360\,\text{km}$ length.

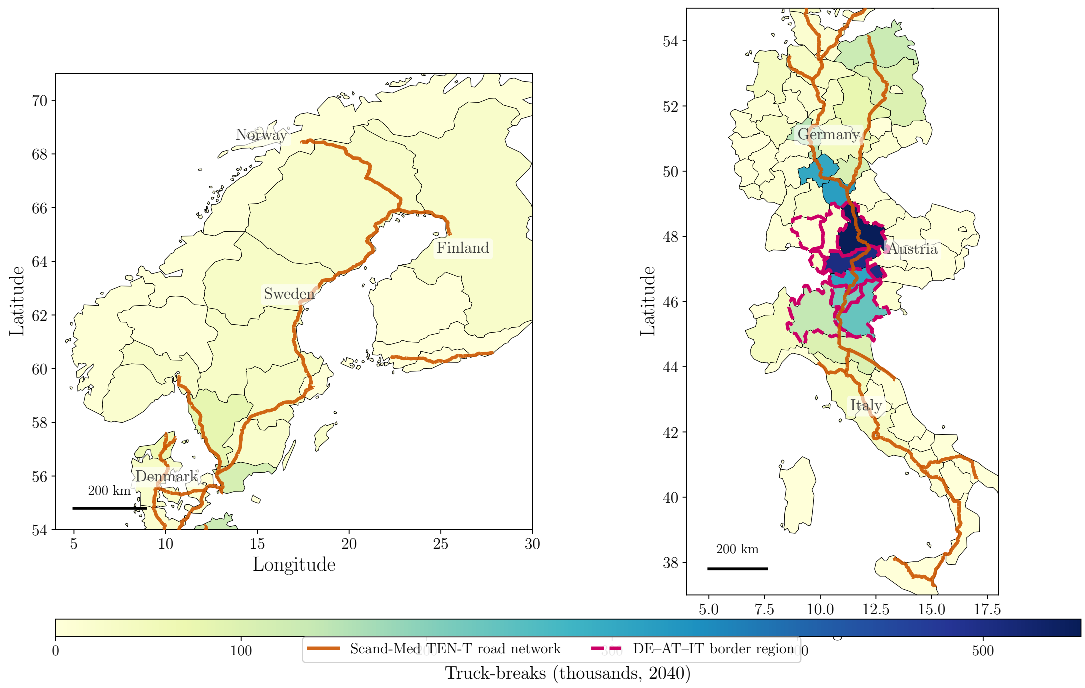

Figure 4.21.: Mandatory driver break locations per NUTS-2 region along the Scandinavian-Mediterranean corridor (2040). Orange lines show the TEN-T Scandinavian-Mediterranean road corridor geometry [177]. Colour indicates the estimated number of annual truck-breaks (thousands) at mandatory rest locations, derived from freight volumes and an average payload of 13.6 tonnes per truck. Break locations follow EU driving time regulations (45-min break after 4.5 h, 9-h rest after 9 h). Dashed magenta borders indicate the DE-AT-IT border region.

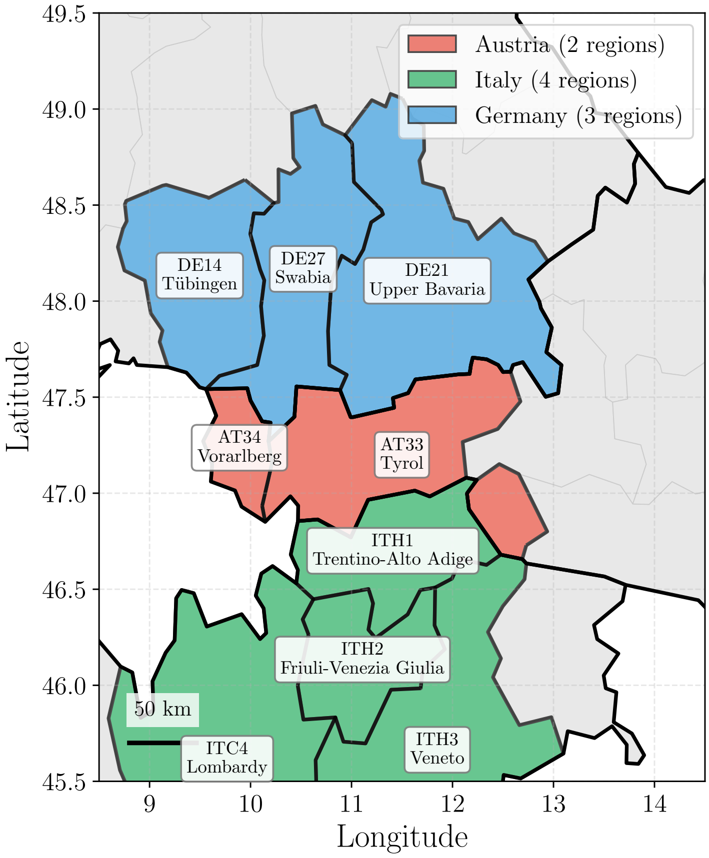

Figure 4.22.: Geographic overview of the DE-AT-IT border region with NUTS-2 boundaries and node locations used in the model. The border region encompasses the Alpine crossing between Southern Germany, Western Austria, and Northern Italy. The dashed magenta border in Figure 4.21 indicates this region.

4.3.1.3. Cost parameters

Drive-train technologies Table 4.15 describes cost assumptions and sources for internal combustion engine trucks and BETs.

Table 4.15.: Assumptions on fleet-related parameters.

Fleet-related costs and parameters	2020	2035	2050	Unit	Reference
Diesel-fueled combustion engine					
Vehicle purchase	105,484	115,252	115,252	€	[178]
Maintenance & Repair (M&R)	20,471	20,608	20,608	€/a	[178]
Specific energy consumption	2.90	2.65	2.40	kWh/km	[84]
Emission factor	86	79	71	gCO_2/Tkm	[179]
Tank size	5600	5600	5600	kWh	[174]
Yearly mileage	136,750	136,750	136,750	km	[164]
Average load per truck	13.6	13.6	13.6	T	[176]
Battery-electric					
Vehicle Purchase	220,000	177,381	146,293	€	[180]
Maintenance & Repair (M&R)	14,000	14,000	14,000	€/a	[178]
Specific energy consumption	1.44	1.15	1.05	kWh/km	[178]
Tank size	560	1120	1400	kWh	[178]
Yearly mileage	136,750	136,750	136,750	km	[178]
Average load per truck	11.56	12.38	13.6	T	[176]

Table 4.16 summarizes additional operational and infrastructure parameters common to all scenarios.

Table 4.16.: Additional operational and infrastructure parameters.

Parameter	**Value**	**Unit**	**Reference**
Discount rate ρ	0.04	—	—
Investment period length Δ	4	years	—
Initial / final state of charge (SOC)	50 / 50	% of battery capacity	—
Average driving speed (road)	80	km/h	—
Charging power (MCS)	1	MW	[181]
Utilization factor γ_t	0.3	—	—
Vehicle lifetime (ICEV & BET)	10	years	—
Base diesel price (pre-tax)	0.051	€/kWh	—
Carbon price	Scenario-specific (Table 4.19)		[182]

BETs depart each trip with a state of charge of 50% of battery capacity and must arrive at the destination with at least 50%, ensuring that the net energy consumed along the route equals the total energy charged en route. The utilization factor $\gamma_t = 0.3$ reflects the ratio of average to peak power demand at charging stations, analogous to a capacity factor in power system terminology. The base diesel price of 0.051 €/kWh represents the pre-tax wholesale price, to which the scenario-specific carbon price component (Table 4.19) is added.

Rail infrastructure Since the focus of this analysis is the spatial allocation of truck charging demand, the rail mode is modeled in reduced form: a levelized transport cost of 0.057 €/tkm covers rail operation, fuel, rolling stock, labor, and infrastructure amortization [119, 183], while a terminal handling cost of 4 €/tonne accounts for on- and off-loading of freight at intermodal terminals (i.e. transfer facilities where freight containers are moved between road and rail) [184]. Both cost components are assumed uniform across the corridor and constant over time. Rail becomes available from 2026 onward, reflecting the lead time for intermodal terminal deployment along the corridor. The maximum corridor-wide rail modal share is capped at 30%, and the transition rate is bounded by approximately 1 percentage point per four-year period, reflecting infrastructure and market constraints on rapid modal shift. The initial fleet is 100% internal combustion engine vehicles (ICEVs, i.e. conventional diesel trucks) in the base year 2020.

Charging infrastructure Regarding charging infrastructure, exclusively electric charging infrastructure is considered, assuming that existing diesel refueling infrastructure is of sufficient capacity and does not need to be expanded. One representative charging infrastructure type is modeled with cost parameters presented in Table 4.17.

Table 4.17.: Assumptions of parameters related to the deployment of charging infrastructure.

Charging-infrastructure-related parameters	2020	2035	2050	Unit	Reference
Charging station (CAPEX)	400	300	250	€/kW	[178]
Charging station (O&M)	16.2	15.7	15.2	€/kW/a	[178]

In addition to infrastructure hardware costs, grid connection charges vary by country and are applied as an annual capacity-based fee (Table 4.18).

Table 4.18.: Grid charges assumed constant over time [25].

Country	**Grid charges (€/kW/a)**
Austria	14.69
Denmark	12.80
Finland	12.33
Germany	16.77
Italy	8.96
Norway	8.58
Sweden	12.19

4.3.1.4. Scenario definition

To account for uncertainties surrounding future electricity and carbon prices, four scenarios are defined based on projections from GENeSYS-MOD (Global Energy System Model), an open-source linear energy system model built on the Open Source Energy Modelling System (OSeMOSYS) that jointly optimizes electricity, heat, and transport sectors at country-level resolution [185]. Each scenario pairs a distinct electricity price trajectory with an internally consistent carbon price trajectory, capturing a range of European energy transition pathways.

GENeSYS-MOD energy system scenarios To capture uncertainty in future energy system development, four scenarios from the GENeSYS-MOD EU-EnVis-2060 framework [182] are used: *Go RES* (ambitious renewable expansion achieving carbon neutrality before 2050), *NECPEssentials* (extension of current national energy and climate plans), *REPowerEU++* (self-sufficient European energy system), and *Trinity* (internal EU divisions affecting climate ambition). Each scenario embeds an internally consistent carbon budget constraint that determines both the electricity generation mix and resulting wholesale prices, as well as a distinct carbon price trajectory (Table 4.19) and diesel price development. Country-specific and year-specific electricity price ratios are applied relative to the Go RES baseline, preserving the temporal trajectory of each scenario. Compared to Go RES, the alternative scenarios yield lower average wholesale electricity prices — approximately 70% of the Go RES level for NECPEssentials, 66% for REPowerEU++, and 62% for Trinity, averaged across all countries and years. Go RES thus represents the highest-electricity-price scenario, reflecting aggressive renewable deployment costs, while Trinity exhibits the lowest electricity prices due to reduced climate ambition.

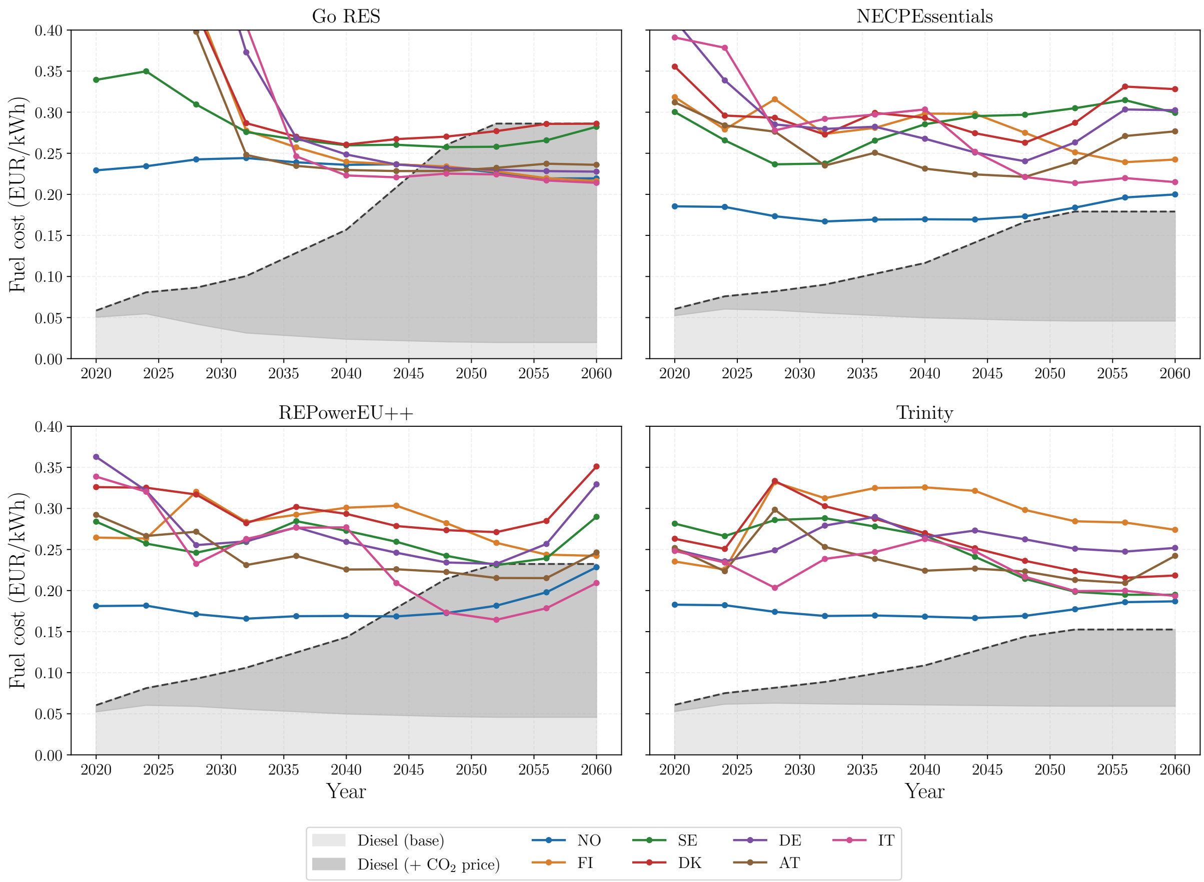

Figure 4.23.: Electricity prices (€/kWh) by country and GENeSYS-MOD scenario over the modelling horizon. Coloured lines show country-specific electricity prices; the light grey area shows the base diesel price and the dark grey area the diesel price including the scenario-specific carbon price component. Go RES exhibits the highest carbon price trajectory, causing diesel costs to exceed electricity prices in most countries by 2045, whereas Trinity's lower carbon prices keep diesel competitive longer.

Carbon price trajectories Each GENeSYS-MOD scenario is associated with a distinct carbon price trajectory reflecting its climate ambition level (Table 4.19). Carbon prices are interpolated annually to provide consistent year-by-year values synchronized with each energy system scenario, maintaining internal consistency between energy system decarbonization and carbon pricing.

Table 4.19.: Carbon price trajectories by GENeSYS-MOD scenario (€/tCO_2).

Scenario	2020	2030	2040	2050
Go RES	30	200	500	1000
REPowerEU++	30	150	350	700
NECPEssentials	30	100	250	500
Trinity	30	80	180	350

Charging and fuel cost parameters The charging infrastructure markup represents the margin charged by Charging Point Operators (CPOs) on top of wholesale electricity costs, covering equipment amortization, maintenance, grid connection fees, and profit margin. Based on the levelized cost of charging framework by Lanz et al. [25], a markup of 30% is applied, representing current public charging conditions consistent with observed pricing at major European truck charging networks. The value of time (VoT) for road freight transport is set to 30 €/h per vehicle.

BET purchase costs follow the reference trajectory from Table 4.15, declining from €220,000 in 2020 to €146,293 by 2050.

4.3.1.5. Spatial reallocation mechanisms

Two mechanisms are evaluated for their effectiveness in redistributing charging demands along the corridor (Table 4.20). These levers operate through different cost channels and affect both the spatial distribution of charging infrastructure and the aggregate level of electrification.

Table 4.20.: Spatial reallocation mechanisms and their effects on charging demand distribution.

Mechanism	Cost channel	Primary effect	Expected outcome
Electricity price differentiation	Variable costs (per-kWh)	Spatial arbitrage	Incentivizes charging in lower-price regions; effect scales with charging volume
Grid charge differentiation	Fixed costs (per-kW)	Infrastructure siting	Affects economic attractiveness of charging infrastructure deployment; stronger effect on high-capacity stations

4.3.1.6. Border region locational cost premium scenarios

To investigate the effectiveness of cost-based policy instruments in relieving charging demand concentration, the analysis focuses on the Germany–Austria–Italy border region (Figure 4.22). Transportation routes between Germany and Italy traverse a small number of connections through Austria, creating a potential bottleneck for charging infrastructure. Austria's electricity price at fast chargers (0.159 €/kWh in 2040 under Go RES, including the 30% CPO markup) lies between Italy (0.155 €/kWh) and Germany (0.173 €/kWh), with a total spread of only 0.018 €/kWh across the border region. Similarly, Austria's grid charge (14.69 €/kW/yr) lies between Italy (8.96 €/kW/yr) and Germany (16.77 €/kW/yr).

Six locational cost premium scenarios are tested, applying graduated premiums to the two Austrian border nodes (Tyrol, Vorarlberg) while keeping all other inputs at REPowerEU++ baseline values. REPowerEU++ is selected as the baseline because it exhibits the highest early-stage charging demand concentration in the border region:

- *Electricity price premiums:*
  - Low (+0.01 €/kWh, +6%): Austria at 0.169 €/kWh — narrows the gap to Germany but remains below it.
  - Medium (+0.02 €/kWh, +13%): Austria at 0.179 €/kWh — flips the ordering so Austria becomes more expensive than Germany.
  - High (+0.04 €/kWh, +25%): Austria at 0.199 €/kWh — makes Austria clearly the most expensive border node.
- *Grid charge premiums:*
  - Low (+2 €/kW/yr, +14%): Austria at 16.69 €/kW/yr — nearly equal to Germany.
  - Medium (+5 €/kW/yr, +34%): Austria at 19.69 €/kW/yr — above Germany.
  - High (+10 €/kW/yr, +68%): Austria at 24.69 €/kW/yr — significantly above both neighbours.

The premium levels are chosen to test whether incremental cost adjustments can meaningfully redistribute charging demand without undermining BET competitiveness on routes that must traverse Austria. The electricity price premiums range from a modest narrowing of the AT–DE price gap to a clear reversal of the price ordering. The grid charge premiums follow the same logic applied to the fixed capacity-based cost component, testing whether annual infrastructure charges — which affect long-term investment decisions rather than per-trip costs — have a comparable steering effect. Each locational cost premium scenario is run both in road-only mode and with modal shift to rail enabled.

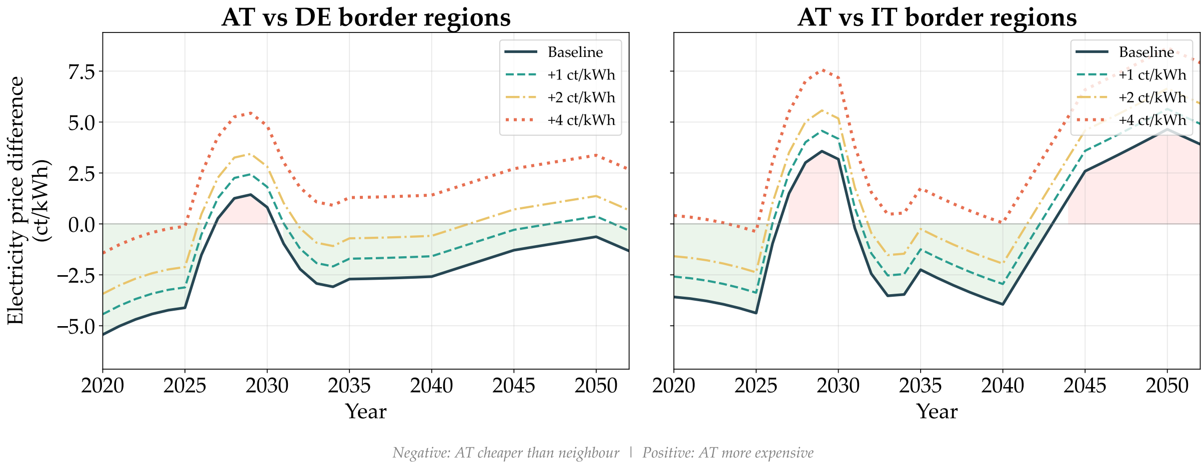

Figure 4.24.: Relative electricity price difference (ct/kWh) between Austrian and neighbouring border regions under locational cost premiums (fast charging, REPowerEU++ scenario). Left: AT vs. DE border regions; right: AT vs. IT border regions. Negative values indicate AT is cheaper; positive values indicate AT is more expensive. Shaded areas highlight the baseline cost advantage or disadvantage.

4.3.1.7. Analytical setup for charging allocation comparison

To investigate the spatially flexible charging demand allocation, two allocation strategies are compared (Table 4.21):

- **System-optimal:** Charging is permitted at all geographic locations along each route, with both charging allocation and fleet adoption jointly optimized.
- **Heuristic:** Charging is restricted to mandatory break locations with energy distributed according to a state-of-charge-aware rule, using the fleet from the System-optimal case. At each break, the rule checks whether the current battery state of charge is sufficient to reach the next break location; if not, exactly enough energy is charged to bridge that gap. This represents a plausible real-world charging pattern where drivers charge reactively based on remaining range rather than optimizing for lowest-cost locations.

Table 4.21.: Overview of analytical setup for spatial charging demand allocation.

	Charging locations	Charging allocation	Fleet adoption	Optimized
System-optimal	All locations	Optimized	Jointly optimized	Yes
Heuristic	Mandatory breaks only	SOC-aware rule	Fixed (from System-optimal)	No

In total, the analysis comprises: (i) four GENeSYS-MOD energy scenarios run in both road-only and road-with-rail configurations (8 system-optimal runs), each paired with a Heuristic counterpart (8 Heuristic runs); and (ii) six border region locational cost premium scenarios, each run in road-only and road-with-rail mode (12 runs). This yields 28 model configurations in total.

4.3.2. Geographic distribution of charging demand across energy system scenarios

BET fleet adoption grows approximately linearly across all four scenarios, reaching 19.5% of the corridor fleet by 2032 and 44.1% by 2040 under Go RES. The scenarios group into two pairs: Go RES and REPowerEU++ exhibit higher adoption levels, while NECPEssentials and Trinity show roughly 10–15 pp lower BET shares throughout the horizon, reflecting different fuel cost advantages. Figure 4.25 shows the BET fleet composition by origin–destination country pair in 2040. The Germany–Italy connection accounts for the largest share of BET adoption, followed by Austria–Italy, consistent with the DE–IT corridor's role as the primary freight transit route. The bottom row highlights relative differences between Go RES and the remaining three scenarios.

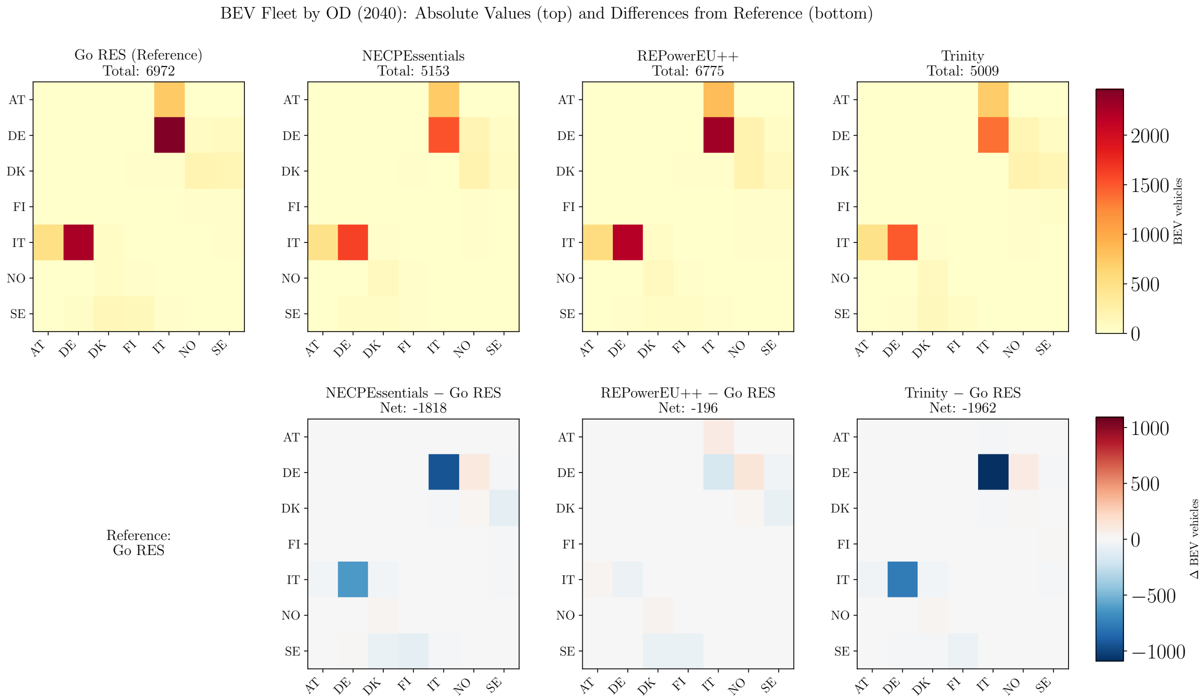

Figure 4.25.: BET fleet by origin-destination country pair in 2040. Top row: absolute values for each scenario. Bottom row: differences relative to the Go RES reference scenario.

Figure 4.26 shows how this fleet adoption translates into the geographic distribution of charging demand along the corridor. A pronounced peak in the southern section (Germany–Austria–Italy) is persistent across years and scenarios. Over time, the relative distribution of charging demand across latitudes converges between scenarios: the total corridor-wide BET charging demand varies by up to 135% between the most extreme scenarios in 2032 (Go RES: 202 GWh vs. REPowerEU++: 474 GWh), narrowing to 24% by 2048 as all scenarios converge toward high electrification.

To examine these cross-scenario variations more closely, Figure 4.27 shows the maximum spread in charging demand by NUTS-2 region along the corridor. The following patterns are notable:

- The timing of peak cross-scenario divergence varies across regions, with no single year or time horizon dominating.

- Throughout Germany, substantial variations are visible, with larger spreads in southern regions than in northern ones.
- The highest absolute spreads concentrate around the DE–AT–IT border region (ITH1: 44 GWh, DE21: 40 GWh, AT33: 37 GWh). In relative terms, however, peripheral regions such as DED4 exhibit spreads exceeding 700%.

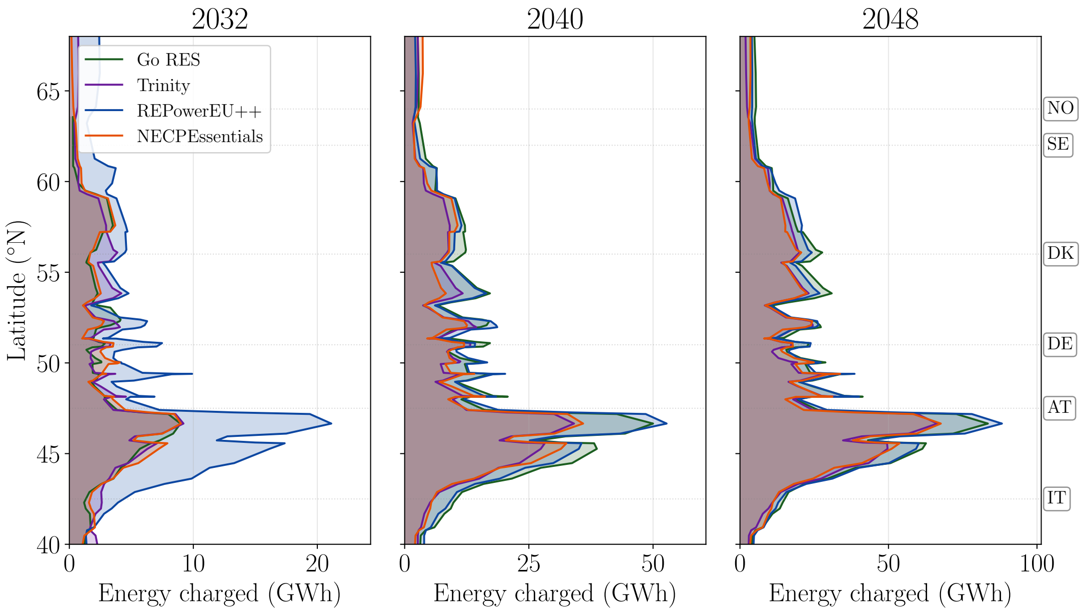

Figure 4.26.: BET energy demand by latitude for the four energy system scenarios across three time horizons (2032, 2040, 2048). Country codes indicate approximate latitude along the Scandinavian-Mediterranean corridor.

Figure 4.27.: Maximum cross-scenario spread in BET charging demand (GWh) by NUTS-2 region, ordered north to south along the corridor. Bar color indicates the year in which the maximum spread across the four road-only energy system scenarios is reached. Regions with spread >15 GWh are labeled.

4.3.3. Spatial patterns of charging demand distribution

The preceding subsection presented the cost-optimized (System-optimal) charging allocation, in which the model freely chooses where trucks charge along their routes to minimize total system costs. This subsection quantifies the degree of spatial flexibility inherent in this solution by comparing it against the Heuristic benchmark defined in Section 4.3.1.7 (Table 4.21), in which charging is restricted to mandatory break locations and energy is distributed by a reactive state-of-charge rule rather than cost optimization. The difference between these two strategies reveals how much the geographic distribution of charging demand can shift when location choice is optimized.

Figure 4.28 compares the geographic distribution of charging demand by latitude for all four energy system scenarios in 2032. The Heuristic shows pronounced demand peaks across all scenarios, while the cost-optimized allocation smooths these peaks by redistributing charging demand to locations with lower costs. The largest differences between the two strategies arise around the DE–AT–IT border and the DK–DE border.

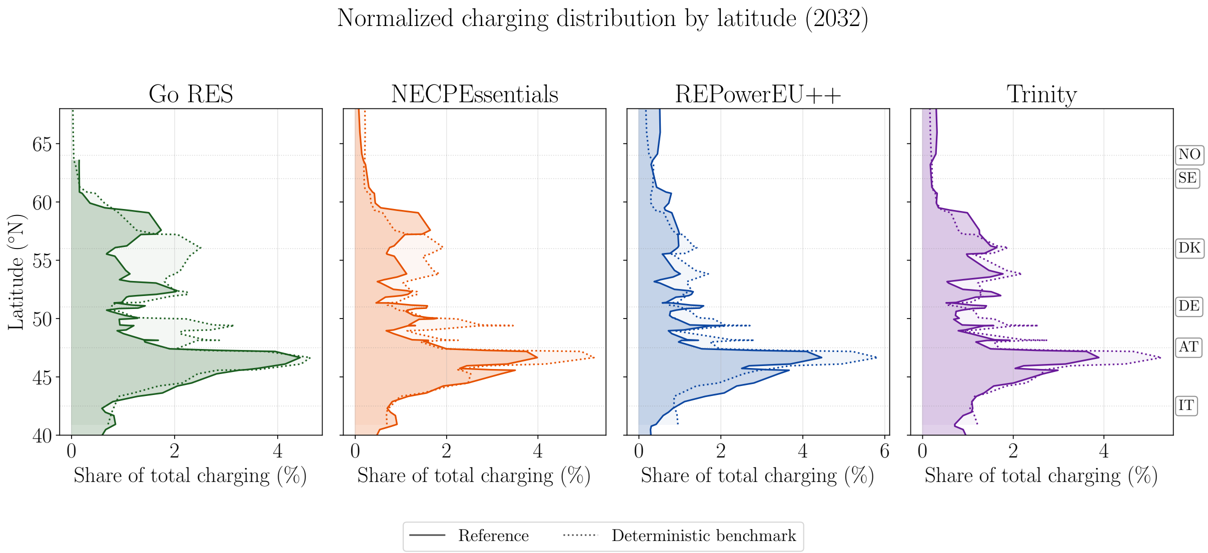

Figure 4.28.: Normalized charging distribution by latitude for the four energy system scenarios in 2032. The System-optimal (cost-optimized, solid) and Heuristic (dash-dotted) strategies are compared. Values show the share of total corridor-wide charging at each latitude.

The Go RES scenario is examined in greater detail to illustrate the charging-cost-optimized reallocation of demand. Country-level differences between the two allocation strategies are summarized in Table 4.22. Compared to the System-optimal, the Heuristic concentrates more charging in Germany (+9.0 pp) at the expense of Italy (−6.4 pp) and Austria (−4.2 pp), where pp denotes percentage points. Overall, the relative difference in charging demand magnitudes across NUTS-2 regions amounts to 39% when comparing the cost-optimized and heuristic allocation strategies, indicating that charging demands along the corridor are substantially spatially flexible.

Table 4.23 compares average charging prices between the two strategies across three time horizons. Total charging costs — comprising location-dependent electricity prices and charging infrastructure costs influenced by local grid charges — are consistently higher under the Heuristic, with a premium of +0.20 ct/kWh in 2032 and +0.17 ct/kWh in 2040.

Table 4.22.: Country-level share of BET charging demand (%) for the Go RES scenario, 2032. Left: System-optimal vs. Heuristic. Right: road-only vs. with modal shift to rail. Δ: difference in percentage points.

	Allocation strategy			**Modal shift to rail**		
Country	**Sys.-opt. (%)**	**Heur. (%)**	**Δ (pp)**	**Road (%)**	**+Rail (%)**	**Δ (pp)**
NO	0.6	0.5	−0.1	1.2	2.9	+1.7
SE	8.8	10.2	+1.4	11.6	5.0	−6.6
DK	3.8	4.1	+0.3	2.8	2.7	−0.1
DE	43.6	52.6	+9.0	43.2	56.3	+13.1
AT	9.9	5.7	−4.2	6.5	7.0	+0.5
IT	33.3	26.9	−6.4	34.7	25.9	−8.8
Relocated			*39.0%*			*15.4%*

Table 4.23.: Charging cost per kWh: System-optimal vs. Heuristic — Go RES scenario, 2032–2040.

Strategy	**Elec. (ct/kWh)**	**Infra. (ct/kWh)**	**Total (ct/kWh)**	**Δ (ct/kWh)**
2032 (BET share 19.5%)				
System-optimal	27.62	5.69	33.30	—
Heuristic	27.76	5.75	33.50	+0.20
2036 (BET share 32.8%)				
System-optimal	19.77	1.97	21.74	—
Heuristic	19.84	2.06	21.90	+0.16
2040 (BET share 44.1%)				
System-optimal	18.35	1.61	19.96	—
Heuristic	18.44	1.70	20.13	+0.17

4.3.4. Role of modal shift to rail

The spatial distribution of rail transport work by latitude is robust across all four energy system scenarios: beyond 2028 the curves largely overlap. Rail is allocated primarily in two latitude bands: Scandinavia (Norway, Sweden, Finland) and Italy, while Germany and Austria see comparatively less rail shift, suggesting that intermediate corridor segments remain more efficiently served by road freight.

Figure 4.29 isolates the Go RES scenario to show the effect of modal shift on BET electricity consumption more clearly. The shaded area between the solid (baseline) and dashed (with rail) curves highlights the reduction in charging demand at each latitude. The largest reductions appear in northern Scandinavia (60–68°N) and at the Austria–Italy border (46–48°N), where long distances and high local charging costs make rail economically attractive. By 2048, the cumulative reduction extends across the full corridor as rail capacity grows.

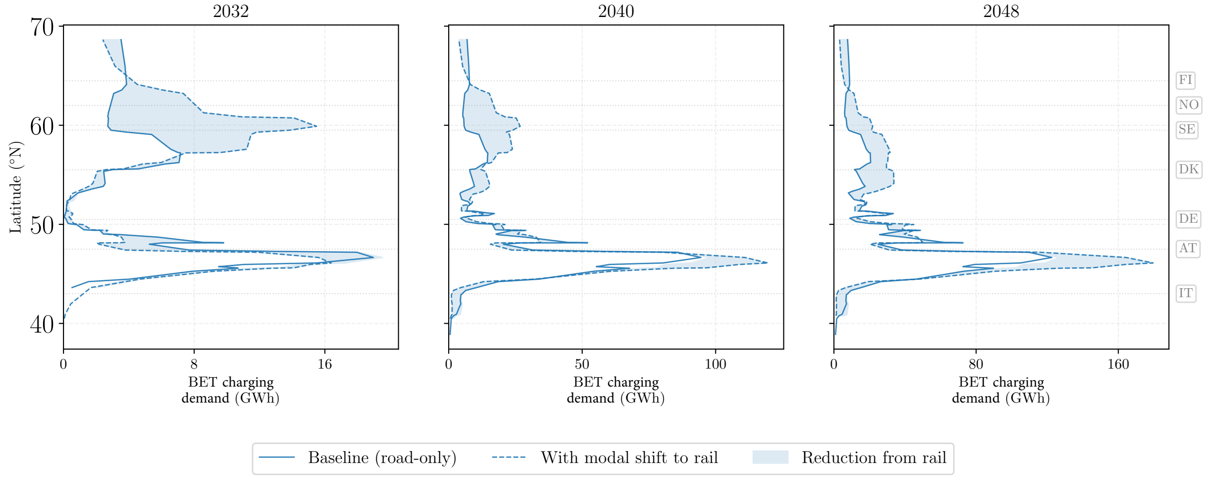

Figure 4.29.: BET electricity consumption by latitude for the Go RES scenario in 2032, 2040, and 2048. Solid line: road-only baseline; dashed line: with modal shift to rail. The shaded area indicates the reduction in charging demand from modal shift. Country centroids are annotated on the right.

The modal shift is strongly distance-dependent: connections between distant countries — such as SE↔FI (88%), IT↔DK (100%), and IT↔SE (100%) — are shifted almost entirely to rail by 2048, while shorter connections (e.g., AT↔IT, DK↔SE) remain on road (Figure 4.30). The DE↔IT corridor contributes the largest rail volume (831+602=1,433 kt) despite a modest rail share (6–9%), due to its high total freight volume.

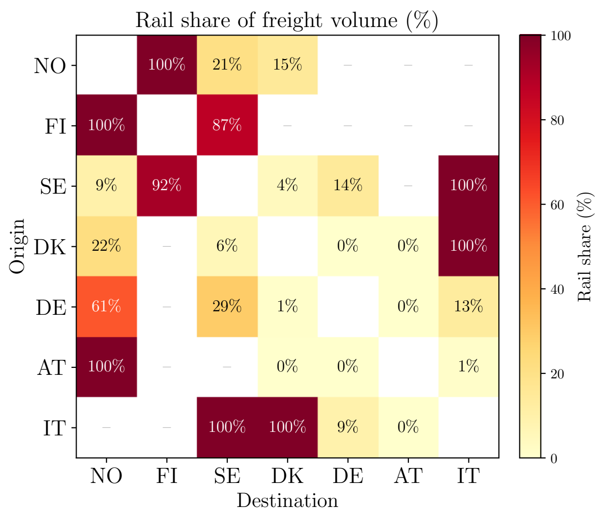

Figure 4.30.: Rail share of freight volume (%) by origin–destination country pair for the Go RES scenario in 2048. Connections between distant countries (e.g., SE↔FI, IT↔DK) show near-complete rail shift, while short corridors (e.g., AT↔IT, DK↔SE) remain road-dominated.

Table 4.22 (right) compares the country-level distribution of BET charging demand between road-only and road-plus-rail for the Go RES scenario. Enabling modal shift to rail concentrates BET charging in Germany (+13.1 pp), as the long-haul routes through Italy — which accounted for a large share of BET demand in the road-only case — are preferentially shifted to rail. Italy's share drops by 8.8 pp, and Sweden's by 6.6 pp, reflecting the disproportionate shift of long-distance Nordic–Italian freight to rail. Austria's share remains nearly unchanged (+0.5 pp), while Norway increases slightly (+1.7 pp) as the remaining BET routes concentrate on shorter northern segments. Overall, the charging demand distribution is changed by 15.4%.

Table 4.24 complements this share-based view with an infrastructure-normalized measure: BET charging demand per highway-kilometer (MWh/hw-km) by country. While the share-based analysis highlights Germany's growing dominance, the intensity metric reveals that *Austria* becomes the most infrastructure-stressed country under modal shift. Austria's charging intensity nearly doubles by 2048 (+82%, from 189 to 343 MWh/hw-km), driven by its short highway network (695 km). Germany's intensity also increases substantially (+44%), but from a much lower base due to its large highway network (13,210 km).

Denmark shows the most dramatic relative increase from 2040 onward (+445% in 2040, +266% in 2048): as long-haul Scandinavian freight shifts to rail, the remaining road traffic through Denmark is dominated by medium-distance DK–DE routes that electrify rapidly, raising Denmark's intensity from 9 to 48 MWh/hw-km in 2040 and to 80 MWh/hw-km by 2048. Conversely, Finland and Sweden see sharp intensity reductions as their long-distance freight shifts to rail. The overall spatial concentration of charging intensity per highway-km decreases with rail, as the extreme Scandinavian hotspots are relieved.

Notably, this intensification in Austria exhibits a temporal reversal. In the early transition (2032), Austrian border region charging *decreases* with rail (−16% to −25% across scenarios), because the first routes shifted to rail are the long DE↔IT connections that previously charged in Tyrol (AT33) while transiting Austria. Removing these long-haul routes from road eliminates their charging stops in Austria. From 2040 onward, however, the effect reverses: the remaining shorter road routes electrify more aggressively — since rail has absorbed the costliest long-distance freight — and Austria's border regions end up with substantially *more* charging than without rail (+8% to +82% by 2048). This pattern is consistent across all four GENeSYS-MOD scenarios, indicating that modal shift to rail initially relieves the Austrian transit bottleneck but ultimately intensifies it as the residual road fleet accelerates its electrification.

Table 4.24.: BET charging demand intensity (MWh/highway-km) by country for the Go RES scenario: road-only baseline vs. with modal shift to rail. Highway-km from Eurostat motorway statistics.

	2032			**2040**			**2048**		
Country	**Road**	**+Rail**	**Δ**	**Road**	**+Rail**	**Δ**	**Road**	**+Rail**	**Δ**
NO	1.8	9.9	+455%	42.5	26.2	−38%	44.5	37.6	−16%
FI	0.0	4.4	—	28.4	0.4	−99%	55.8	1.1	−98%
SE	8.1	5.8	−28%	43.3	33.9	−22%	89.3	61.4	−31%
DK	5.7	4.8	−16%	8.9	48.3	+445%	22.0	80.3	+266%
DE	6.7	9.7	+45%	39.6	57.1	+44%	69.9	100.5	+44%
AT	28.8	24.2	−16%	106.7	172.5	+62%	189.1	343.4	+82%
IT	7.9	7.3	−8%	53.1	38.0	−28%	82.0	66.8	−19%

4.3.5. Border region analysis: Germany–Austria–Italy

The preceding subsections showed that the DE–AT–IT border region consistently exhibits the highest charging demand concentration across all scenarios, and that modal shift to rail ultimately intensifies rather than relieves this concentration in Austria. This subsection analyses whether localized cost instruments can manage this demand. The analysis focuses on the REPowerEU++ scenario because it exhibits the highest early-stage charging demand concentration in the border region among all four scenarios, providing the most demanding test case for localized cost instruments. The two borders are in close geographic proximity, as the Austrian transit corridor is relatively short. Using the locational cost premium scenarios defined in Section 4.3.1.6, the analysis investigates how the two spatially varying cost components — electricity prices and fixed grid charges — affect the local concentration of charging demand.

Figures 4.31 and 4.32 show the effect of different cost premiums applied to AT33 on charging allocation in the border region. The Austrian locational cost premiums have negligible impact on corridor-wide BET fleet shares. In contrast, the spatial distribution of *where* BETs charge shifts substantially (Figure 4.31).

At an electricity price premium of +0.04 €/kWh, charging demand in Austria drops by 97% (from 120 GWh to 3 GWh in 2032), with demand redirected primarily to Italy. AT33 (Tyrol) — the primary transit node for DE–IT freight — sees a 99.9% reduction in charged energy. This reallocation effect saturates at low premium levels: even +0.01 €/kWh, which merely narrows the AT–DE price gap without reversing it, reduces Austrian charging by approximately 10%.

Grid charge premiums produce a markedly weaker effect: even the highest tested level (+10 €/kW/yr, raising Austrian fees 50% above baseline) results in negligible changes in charging allocation.

Figure 4.32 maps this reallocation in terms of charging intensity (GWh per highway-km), which captures the spatial concentration of demand by region. The left panel shows the System-optimal allocation, where demand concentrates in AT33 and ITH1. The middle and right panels demonstrate the effectiveness of electricity price premiums: charging demand is reduced not only in AT33 but also in ITH1. Grid charge premiums show a weaker spatial reallocation effect.

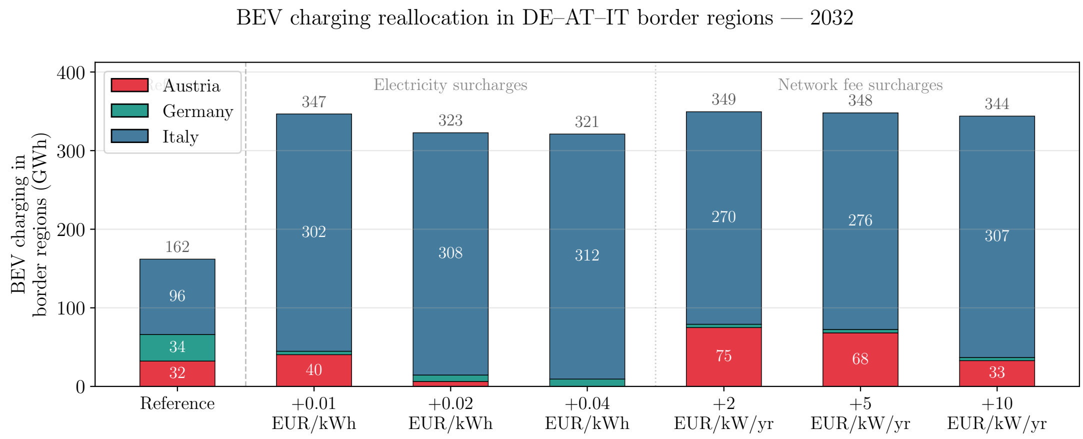

Figure 4.31.: BET charging reallocation in the DE-AT-IT border region under Austrian electricity price and grid charge locational cost premiums (2032, REPowerEU++ scenario, road-only). Stacked bars show charged energy (GWh) by country for the System-optimal (unconstrained) allocation followed by graduated premium levels.

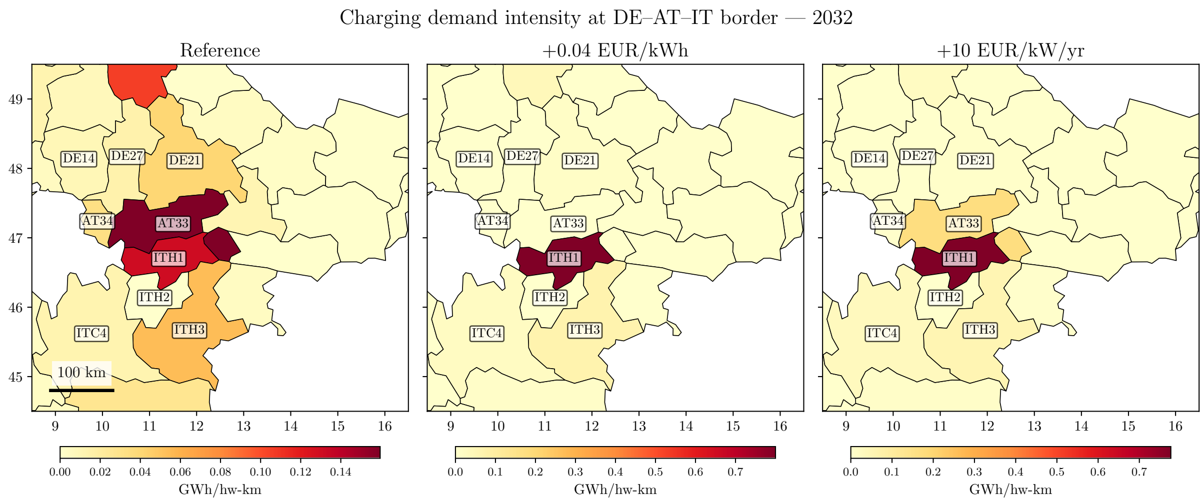

Figure 4.32.: Charging demand intensity (GWh/highway-km) at the DE-AT-IT border region in 2032 (REPowerEU++ scenario). Three panels compare the System-optimal (unconstrained) allocation and the highest electricity price and grid charge premiums. Each panel uses an independent colour scale.

4.4. Hourly charging demand profiles of the commercial fleet in Austria in 2040 (RQ4)

The following case study and results are taken from the journal publication by Golab et al. (2025).

4.4.1. Case study: Austria's commercial fleet at federal state level

The representation of the commercial fleet is focused on three fleet types: light-duty truck (LDT), heavy-duty truck (HDT), and buses. These are chosen based on their substantial fleet size and transport activity. Their electrification is therefore expected to play a significant role in the electricity system. Based on the mobility patterns of the different fleet types and expected charging strategies, future charging profiles are generated together with plug-in times and installed charging power. The charging profiles are generated for an average week in 2040, i.e. representing seven days in hourly resolution. In the geographic aggregation of charging profiles, Austria is represented by nine nodes ($|\mathcal{N}_1| = 9$) mirroring the nine federal states of Austria.

Light-duty trucks The LDT fleet is classified as vehicles that have a maximum total weight of 3.5 tons [186]. The Austrian vehicle fleet of this type amounted to a size of around 510,000 vehicles by the end of 2023 and is expected to be 535,000 by 2040 in decarbonization scenarios by Angelini et al. [187]. According to [188], the average daily range of a light-duty vehicle operated in Germany is 70 km. This value is adopted for Austria. Further assumptions are summarized in Table 4.25. Based on this, routes of vehicles of this fleet type are assumed to remain within one federal state and, therefore, the charging profile of the LDT fleet is modeled based on the operational pattern of *local transport*.

The assumed charging strategy is that vehicles are charged during nighttime and weekends at depots, with the charging process performed at the lowest possible power level and evenly distributed during the plug-in period. Departure times are evenly distributed between 7 am and 9 am, and arrival times between 4 pm and 6 pm. The expected number of LDT is distributed proportionally to the population size of each federal state.

Heavy-duty truck The HDT has a total weight greater than 3.5 tons. Technical assumptions for this fleet are summarized in Table 4.26. Origin-destination flow data on the European level is used to capture charging demand from national as well as transit transport. The ETISplus dataset provides estimations of the annual tons transported by trucks in the year 2030 [174]. The charging profile generation for *interregional transport* is applied under the following considerations:

The assumed charging strategy considers that European regulation dictates a maximum driving shift of 4.5 hours [189], and that in Austria, driving is prohibited between 10 pm and 5 am for multiple transport segments [190] as well as on Saturdays and Sundays. HDTs are charged at depots and highway service areas. For trips longer than 4.5 hours, the mandatory 45-minute break takes place at a highway service area, during which the HDT is connected to a megawatt charging station at full power. The remaining energy demand is covered at origin and destination depots during nighttime. For trips shorter than 4.5 hours, charging is fully compensated at depots. Additional depot charging during daytime occurs if the driving range is insufficient for a single battery charge. During multi-day trips, charging demand is also covered at service areas during nighttime. Assumptions on charging infrastructure sizing and temporal distribution of fast-charging are provided in Appendix E.3.

Buses For the bus fleet, the application case for public transport in both cities and rural regions is considered. Similar to the modeling of light-duty vehicles, the routes of buses do not

Table 4.25.: Parameter settings for light-duty truck charging demand modeling.

Parameter	Value	Reference
Fleet size (Austria 2040)	535,000	[187]
Battery capacity	145 kWh	[144]
Specific energy consumption	0.27 kWh/km	[187]
Daily range	70 km/day	[188]
Charging power at charging station depot	11 kW	[188]

Table 4.26.: Parameter settings for heavy-duty truck charging demand modeling.

Parameter	Value	Reference
Battery capacity	350 kWh	[191]
Specific energy consumption	1.3 kWh/km	[192]
Average load	10.63 tons	[192]
Average driving speed	50 km/h	[193]
Peak charging power at highway/depot charging station (fast charging during daytime)	1000 kW	
Peak charging power at highway/depot charging station (slow charging during nighttime)	100 kW	

extend across multiple regions. Table 4.27 comprises technical assumptions for an average bus in public transport.

The assumed charging strategy is that most charging demand is covered during nighttime at the bus depot; if required, buses enter the depot for fast charging during daytime. The operation is assumed to be similar during each day of the week. City buses are considered for Austrian cities that surpass a population of 100,000, with each city's bus fleet scaled by population size. The bus fleet in rural areas is sized using the total area of the rural region within a federal state.

Table 4.27.: Parameter settings for bus charging demand modeling in urban and rural areas.

Parameter	Value	Reference
Fleet size (Austria 2040)	12,900	extrapolated based on [194], [195]
Battery capacity	350 kWh	[196]
Specific energy consumption	1.3 kWh/km	[192]
Daily range - city	236 km	[194]
Daily range - rural	138 km	[195]
Peak charging power at depot charging station (fast charging during daytime)	1000 kW	
Peak charging power at depot charging station (slow charging during nighttime)	100 kW	

4.4.2. Scenarios on the electrification of the commercial fleet

The variation in the share of electrification of the commercial fleet in 2040 is explored. Table 4.28 displays the shares of electrification for the three considered electrification scenarios, *LOW*, *MEDIUM*, and *HIGH*.

Table 4.28.: Electrification scenarios for the commercial fleet in Austria in 2040.

	Share of electrification in fleet		
Scenario	Light-duty vehicles	Heavy-duty vehicles	Buses
LOW	100%	30%	100%
MEDIUM	100%	50%	100%
HIGH	100%	100%	100%

4.4.3. Spatio-temporal demand distribution

Table 4.29 summarizes the annual demand estimates for the commercial BEV fleet under different electrification scenarios in Austria for 2040. The annual charging demands are 7.4 TWh / 9.6 TWh / 15.2 TWh, accounting for 8.2% / 10.7% / 16.9% of the estimated total electricity load in 2040. The highest charging demand stems from the charging of HDTs which is mostly allocated to depot locations. In the electrification scenarios *MEDIUM* and *HIGH* which indicate a respective electrification rate of 50% and 100% of the HDT fleet, the fleet is the clear dominant demand segment with 5.6 TWh/ 11.2 TWh of annual charging demand.

Figure 4.33 displays peak power levels along the highway for the *HIGH* scenario. Positions of service areas along the Austrian highway network are shown together with the coloring indicating approximated peak power levels. The peak power level at a service area is derived from the estimated accumulated demand within the federal state, which is then equally distributed among all existing service areas. It is important to note that this figure is an approximation of the peak power distribution aimed to illustrate geographical variations and provide an estimate for peak power levels at service areas in 2040. The figure indicates a wide range in required power levels at highway service areas, between 7 and 29 MW. The illustration suggests distinctly higher power levels at the service areas located in the northeast than in the western regions of Austria.

The accumulated load profile for the charging of the commercial BEV fleet is shown in Figure 4.34 for the whole of Austria. The figure displays an average week of charging load. During daytime, most charging at highway service areas by the HDT fleet is conducted around noon. The load peaks at nighttime around midnight due to accumulated night charging of the LDT fleet, city and regional buses, and the HDT fleet. During weekend days, the charging demand is substantially lower than during workdays as vehicle operation is mostly performed during workdays.

Table 4.29.: Annual charging demand of three commercial fleet types under different electrification rate assumptions for Austria in 2040. HDT demand is further disaggregated by physical charging location.

		Total annual demand of fleet types 2040 (TWh)					
	Total	LDT	Bus	HDT			
				Charging location			
Scenario				Depot	Highway (day time)	Highway (night time)	Total
LOW	**7.4**	2.7	1.3	1.8	1.1	0.5	3.4
MEDIUM	**9.6**	2.7	1.3	3.1	1.7	0.8	5.6
HIGH	**15.2**	2.7	1.3	6.2	3.5	1.5	11.2

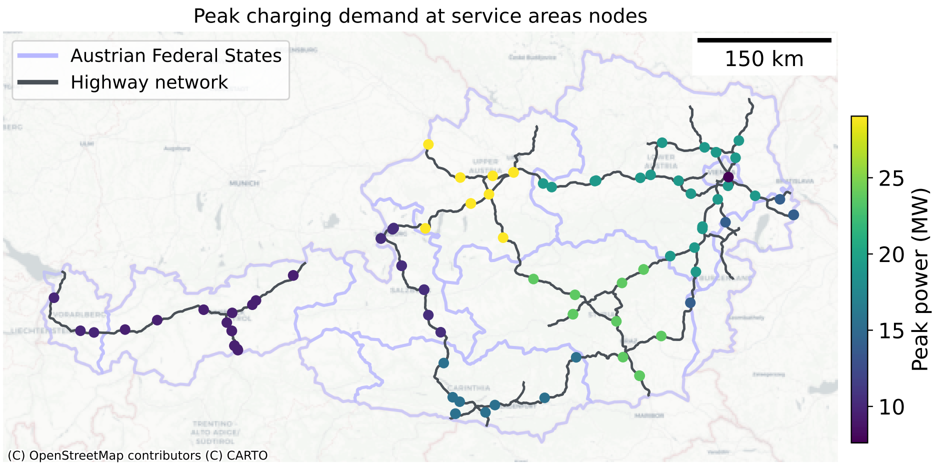

Figure 4.33.: Approximation of charging peaks at Austrian service areas for a 100% electrified commercial vehicle fleet (positions retrieved from [140]).

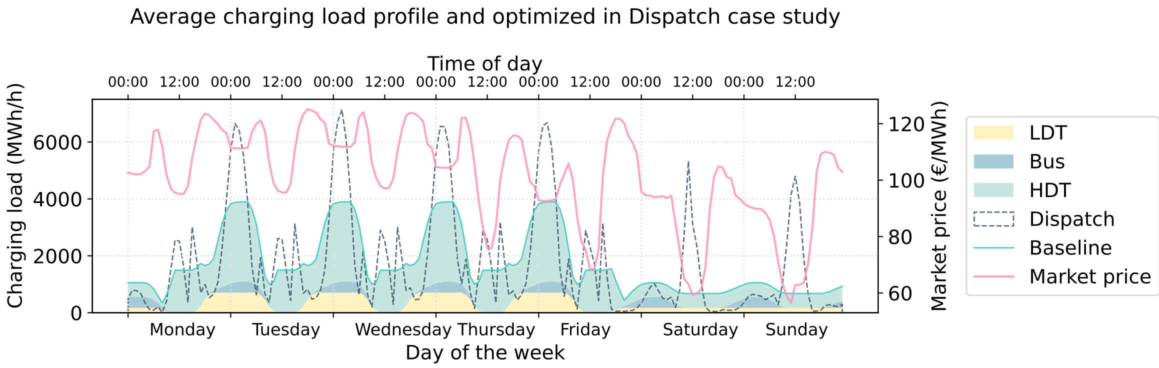

Figure 4.34.: Aggregated weekly charging load profile of the commercial BEV fleet under baseline assumptions and with market-price-coordinated charging, together with the average day-ahead market price.

4.4.3.1. Composition of heavy-duty truck charging demand by traffic type

The traffic-flow-based estimation of heavy-duty truck charging demand enables a decomposition by the type of traffic generating the demand within Austria. Table 4.30 classifies the origin-destination flows from the ETISplus dataset into three categories: inland transport between Austrian regions, transit flows that neither originate nor terminate in Austria, and import/export flows with one end in Austria. Of the 319,447 origin-destination pairs considered, the vast majority (239,209) are transit flows. The total transport volume amounts to 529.5 Mio. tons, of which transit and import/export contribute similar shares (175.4 and 185.0 Mio. tons, respectively). In terms of ton-kilometers traveled within Austria, transit traffic accounts for 32.6 Mrd. tkm (37.4%), import/export for 32.2 Mrd. tkm (36.9%), and inland transport for 22.5 Mrd. tkm (25.7%). The resulting daily charging demand within Austria follows this distribution, with transit traffic generating 15.3 GWh/d and import/export 15.1 GWh/d, together accounting for 74.3% of the total daily demand of 40.9 GWh/d.

Table 4.30.: Classification of heavy-duty truck charging demand in Austria by traffic type, based on ETISplus 2030 transport flow projections. AT tkm refers to ton-kilometers traveled within Austria. Daily demand represents electricity required for charging within Austria.

Traffic Type	**Flows** [-]	**Transport Volume** [Mio. t]	**AT tkm** [Mrd. tkm]	**Daily Demand** [GWh/d]	**Share** [%]
Inland (AT → AT)	1,160	169.0	22.5	10.5	25.7
Transit (non-AT → non-AT)	239,209	175.4	32.6	15.3	37.4
Import/Export (AT ↔ non-AT)	79,078	185.0	32.2	15.1	36.9
Total	**319,447**	**529.5**	**87.3**	**40.9**	**100.0**

Table 4.31 presents the annual charging demand for the *HIGH* scenario (100% HDT electrification), disaggregated by traffic type and charging occasion. The total annual HDT charging demand of 11.24 TWh is predominantly covered by end-of-trip charging after the driving day (10.97 TWh), while break charging during mandatory driving breaks accounts for 0.27 TWh. Break charging is concentrated in transit and import/export traffic (0.12 TWh each), whereas inland transport generates negligible break charging demand (0.02 TWh). This reflects the shorter route segments of inland transport within Austria, for which end-of-trip charging at origin and destination is sufficient without en-route stops.

Table 4.31.: Annual heavy-duty truck charging demand in Austria by traffic type and charging occasion under 100% electrification. Break charging refers to energy recharged during mandatory driving breaks (>4.5 h driving time); end-of-trip charging covers the remaining energy demand recharged after the driving day. Annual values are calculated with 261 workdays and 95% charging efficiency.

Traffic Type	**Break charging** [TWh/a]	**End-of-trip charging** [TWh/a]	**Total** [TWh/a]	**Share** [%]
Inland (AT → AT)	0.02	2.87	2.89	25.7
Transit (non-AT → non-AT)	0.12	4.08	4.20	37.4
Import/Export (AT ↔ non-AT)	0.12	4.03	4.15	36.9
Total	**0.27**	**10.97**	**11.24**	**100.0**

4.4.3.2. Regional charging load profiles

Figure 4.35 displays the weekly charging load profiles disaggregated by Austrian federal state. The profiles reveal substantial regional differences in both the magnitude and temporal shape of charging load. Regions with high shares of transit and long-haul truck traffic exhibit higher daytime peaks due to en-route highway charging, while regions with predominantly local commercial vehicle fleets show load profiles dominated by nighttime depot charging. The weekday-weekend contrast is visible across all regions, with weekend charging load being substantially lower. The relative magnitude of the charging load varies across regions, reflecting the heterogeneous distribution of commercial vehicle fleets and transit traffic volumes across Austria.

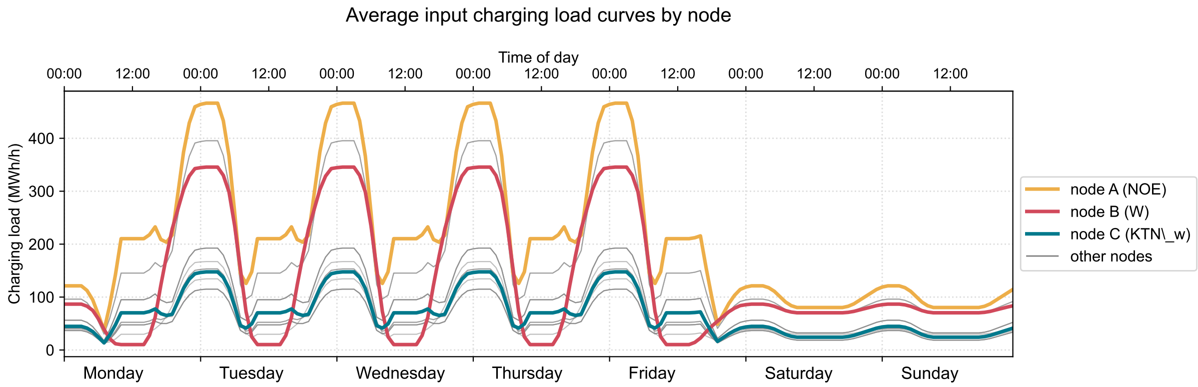

Figure 4.35.: Weekly charging load profiles of the commercial BEV fleet by Austrian federal state.

Figure 4.36 shows the composition of the weekly charging load by fleet type for three selected locations in Austria. The stacked profiles illustrate the relative contributions of light-duty trucks, heavy-duty trucks, and buses to the total charging load at each location. The composition varies across the selected regions: locations along major transit corridors show a higher share of heavy-duty truck charging, whereas regions with large urban centers exhibit a more balanced composition due to the presence of bus fleets and light-duty commercial vehicles. The temporal patterns also differ by fleet type, with bus charging showing a more uniform daily pattern compared to the pronounced weekday peaks of truck charging.

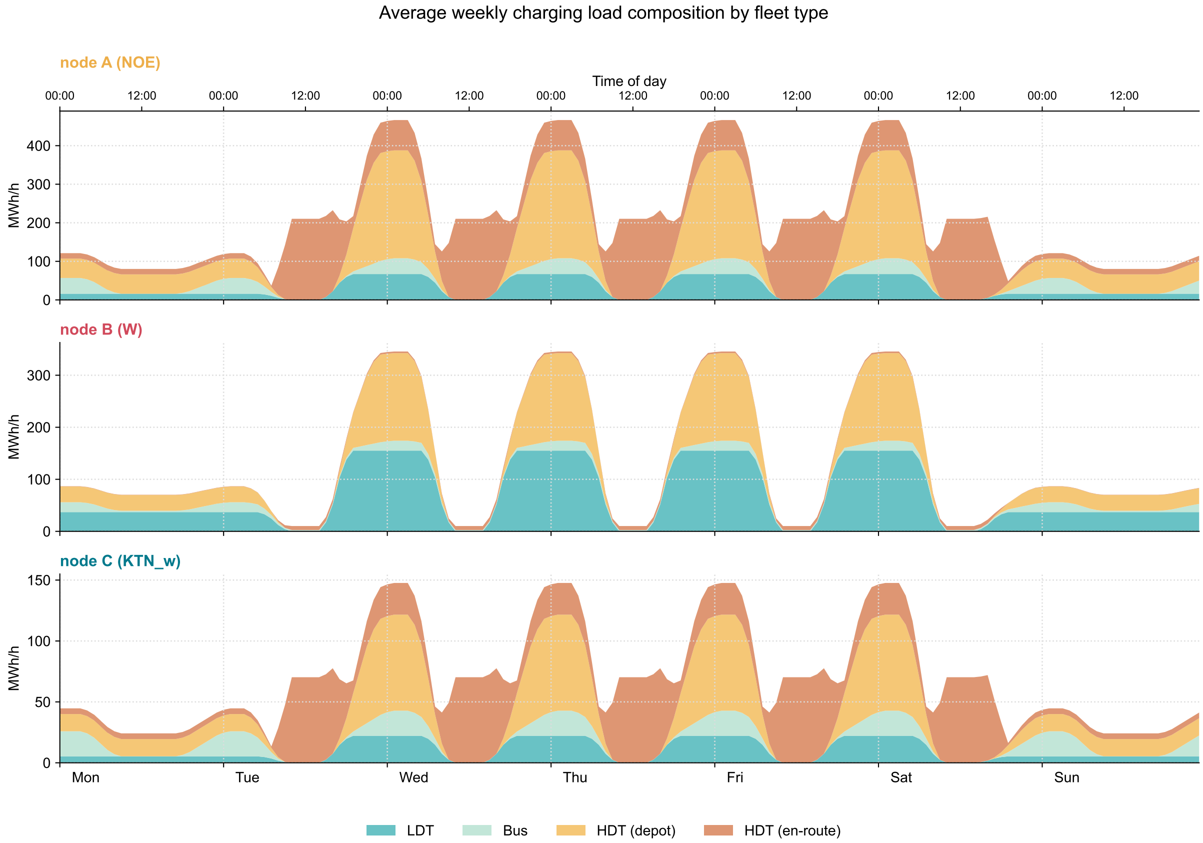

Figure 4.36.: Composition of the weekly charging load by fleet type (light-duty trucks, heavy-duty trucks, buses) for three selected locations in Austria.

The results of integrating the spatio-temporal charging load profiles into the electricity market model for dispatch and redispatch optimization are presented in Appendix E.

5. Discussion & Synthesis of Results

This chapter discusses and synthesizes the findings of the four contributions presented in this thesis. Section 5.1 contextualizes the results of each research question within the existing literature, comparing estimates against policy targets and prior studies, and discussing the limitations of the applied methodologies. Section 5.2 synthesizes the findings across contributions, drawing out cross-cutting insights. Parts of this discussion have been adapted, reformulated, and extended from the four respective journal publications.

5.1. Contextualizing findings in existing literature

5.1.1. Contribution 1 (RQ1)

RQ1: *What are the required fast-charging capacities along Austria's highway network for battery-electric passenger cars in 2030 under different decarbonization scenarios, and where are these allocated?*

Planned capacities Along the Austrian highway network, the results indicate 160–240 MW of fast-charger capacities distributed across 53–56 stations to serve passenger BEVs traveling in 2030.

The required values for the total capacity vary by decarbonization scenario. The following discussion first contextualizes the results within policy-based expansion goals for charging infrastructure, particularly the Alternative Fuels Infrastructure Regulation (AFIR), and the estimates from the study by Jochem et al. [98]. Subsequently, the discussion focuses on influential parameters of the decarbonization scenarios.

Table 5.1 provides a quantitative comparison of the results. Since the studies differ in their assumptions regarding passenger car fleet development, the adjustments applied to ensure comparability are summarized in the column "Key Assumptions".

Comparison with AFIR In the AFIR, there are two charger-capacity related specifications that are relevant for the present discussion. The first specifies required capacities along highways until 2027; the second defines national targets for public charging infrastructure.

Regarding the first specification, the AFIR mandates 600 kW of charging capacity every 60 km for light-duty vehicles along the European TEN-T core network by 2027 [11]. Given a 2,258 km highway network in Austria, this translates to a minimum requirement of 22.6 MW. This is substantially lower than the estimated range of 160–240 MW.

Regarding the national targets, the AFIR mandates that public charging infrastructure is continuously expanded to at least 1.3 kW of power output per BEV [11]. Since the AFIR establishes minimum deployment requirements, the resulting capacity targets represent a policy-mandated lower bound. Assuming 30% electrification of Austria's current passenger car fleet of 5.2 million vehicles, this translates to approximately 2 GW of required public charging infrastructure. However, only a small share of public charging is highway-based fast charging. Drawing on Norwegian deployment data[1], highway charging capacity is estimated to account for roughly 210 MW. That the optimization-based estimates of 160–240 MW are in the same order of magnitude as, and partially exceed, this policy-derived lower bound is consistent with the expectation that cost-optimal infrastructure sizing meets or surpasses minimum regulatory requirements.

Comparison with previous research To the best of the author's knowledge, the study by Jochem et al. [98] is, alongside Golab et al. [28], the only prior work that estimates concrete locations and capacities for fast chargers along Austria's highway network. It is important to note here that the focus of the study by Jochem et al. [98] is not specifically on Austrian highway charging but rather on European highway charging. Under 15% BEV electrification, Jochem et al. [98] identify 15 charging sites with approximately eight 150 kW points each (18 MW total, scaling to 36 MW at 30% electrification)—roughly one-tenth of the capacity estimated by the present approach. Several factors explain this divergence.

First, the two studies differ fundamentally in their modeling approach. Jochem et al. [98] employ a Flow Refueling Location Model, which minimizes the number of stations required to cover origin-destination flows, with station capacity determined ex post through queuing models. The proposed approach, by contrast, jointly optimizes both location and capacity within a single cost-minimization framework. This integrated formulation allows infrastructure costs to enter the objective directly, yielding capacity estimates that are driven by economic trade-offs rather than minimum-coverage requirements. However, this also means that the resulting capacity sizing is highly sensitive to the assumed peak demand, as station dimensions are directly determined by the traffic volumes at each node rather than based on an operational modeling of the station utilization.

Second, the studies differ in how charging demand is derived. Jochem et al. [98] estimate demand from Europe-wide origin-destination flows, applying two filtering thresholds that systematically exclude demand: only O/D pairs with a trip length of at least 100 km are considered, and flows below 5,000 vehicles per year are removed to reduce computational complexity. This excludes shorter highway trips that may still require en-route charging as well as low-volume routes that collectively contribute to aggregate demand. The traffic count-based approach applied in this work, by contrast, observes aggregate volumes at each highway link and therefore captures a broader share of charging demand, including shorter-distance and domestic trips that fall below these thresholds. The assumption of a constant share of long-distance traffic along the highway network neglects the heterogeneously distributed long-distance traffic that is particularly captured by origin-destination data. Furthermore, the international scope of Jochem et al. [98]

[1] Norway is globally an exception for the relative share of fast chargers. According to IEA [197], fast chargers account for 5–6% of public charging points globally. In Norway, highway chargers represent 20% of total charging points but 35% of installed capacity, reflecting a 1.75× capacity-to-points ratio due to higher power ratings. Applying this ratio yields an estimated highway charging capacity of approximately 210 MW (6% × 1.75 × 2,000 MW).

allows demand to be served at charging stations outside Austria's national borders, further reducing the capacity allocated domestically.

The authors themselves, Jochem et al. [98], acknowledge the tendency of their approach to underestimate, by noting that their results for Germany of 128 stations appear "absurdly small" compared to actual deployment of 315 stations from a single operator alone [98, p. 126].

A key advantage of the proposed approach is that it relies on traffic count data that are directly observed at actual service area locations along the highway network, rather than on origin-destination flows for which up-to-date data are difficult to obtain[2]. However, this comes at the cost of information on individual vehicle trajectories: aggregate link volumes do not reveal where a vehicle entered the network, its battery state of charge, or whether it has already charged upstream. Charging demand is therefore inferred from traffic volumes rather than from vehicle-level charging needs, which may lead to spatial misallocation of capacity across the network rather than a systematic over- or underestimation of total demand. The absence of explicit route information remains a crucial limitation relative to origin-destination-based formulations.

Regarding the overall magnitude, the assumed 30% BEV electrification by 2030 is optimistic relative to Austria's current share of 3.8%, suggesting that the total capacity estimates likely represent an upper bound. Under more conservative projections of approximately 15% electrification, required highway charging capacity would be 80–120 MW. The resulting range of 80–240 MW thus brackets realistic capacity requirements between conservative and ambitious electrification scenarios.

Geographic allocation of charging sites The optimized allocation distributes capacity across 54–55 charging sites. This aligns with observed deployment trends: 31 sites are operational at the time of model calibration, expanding to 47 sites at service areas by early 2026 [140], suggesting continued convergence toward the modeled 55-site network. The highest modeled capacity for a single passenger BEV charging site is 4.9 MW, equivalent to approximately 14 charging points at 350 kW each. This capacity level has precedent in Austrian highway infrastructure: one existing service area features 4.8 MW of installed capacity across twelve 400 kW charging points, primarily targeting battery-electric trucks but demonstrating the technical and operational feasibility of multi-megawatt charging sites.

The modeled geographic allocation is primarily driven by existing charging site locations, highway network topology, and a uniform maximum capacity constraint applied across all service areas. This uniform constraint does not account for site-specific limitations, such as available parking spaces or grid connection capacity, which may limit deployment at individual locations. A notable spatial pattern emerges: charging capacity concentrates near Austria's borders, particularly at network entry and exit points. This edge effect likely constitutes a modeling artifact arising from the constraint that all charging demand must be satisfied within the Austrian network, thereby neglecting the option for cross-border charging in neighboring countries. The allocation ensures at least one charging site per highway segment.

[2] This statement exclusively applies to the Austrian case study for which ASFINAG provides traffic count data for free [198].

Drivers in decarbonization scenarios Different scenarios are used to account for uncertainties in the possible developments in the transport sector. Further, sensitivity analyses are conducted on specific parameters. The focus is on uncertainties in the charging demand and technological parameters, reflecting diverging developments in vehicle battery technology, driving range, and charging power.

Investment costs scale nearly proportionally with charging demand. A 31% demand reduction decreases total investment from 100 million €to 68 million €(32% cost reduction), while a 17% demand reduction yields 18 million €savings (18% cost reduction). Higher charging power produces comparable cost benefits: increasing charging power beyond the baseline 350 kW per charging point reduces required capacity similarly to demand reduction or vehicle efficiency improvements, as fewer charging points are needed to serve the same traffic volume. By contrast, increasing driving range has a limited impact on required capacities. This is in line with the observations by Wang et al. [99] and Nugroho et al. [104]. Overall, the two defining parameters for the estimation are, on the one hand, the share of long-distance-traveling passenger BEVs and, on the other hand, charging efficiency.

Table 5.1.: Comparison of fast-charging infrastructure capacity estimates for Austrian highways.

Approach	BEV share	Capacity (MW)	Key assumptions
Golab et al. [28] (2022)	30%	160–240	53–56 stations, 350 kW points, traffic count-based, demand reduction 21%, cost optimization (2030)
Golab et al. [28] (adj.)	15%[a]	80–120	Adjusted for realistic fleet growth trajectory, includes demand reduction (2030)
AFIR distance-based	—	22.6	600 kW per 60 km on TEN-T core, 2,258 km network, minimum connectivity (2027)
AFIR fleet-based	30%	∼210	1.3 kW per BEV (1.56M BEVs), highway share 35%[b], total public: 2.03 GW (2030)
Jochem et al. [98]	15%	18	15 stations, 8 pts/station, 150 kW, FRLM approach, excludes flows <5,000 cars/yr (2030)
Jochem et al. [98] (scaled)	30%	36	Linear scaling from 15%, likely underestimate per authors[c] (2030)

[a] Accounts for actual electrification trajectory (current: 3.8% in Austria).
[b] Based on IEA (2025) global average of 5–6% highway share of charging points and Norwegian capacity ratio (1.75×).
[c] Authors note: “our results [...] seem absurdly small. Today only a single platform provides already 315 FCS [for Germany vs. modeled 128]” [98, p. 126].

5.1.2. Contribution 2 (RQ2)

RQ2: *How does public charging infrastructure expansion influence the adoption of passenger BEVs across income groups and neighboring regions?*

RQ2.a: *To what extent does the spatial density of public charging infrastructure influence battery-electric vehicle adoption across consumers with different income levels?*

Impact of spatial density on BEV adoption across income levels Scenarios that differ in the spatial densification strategy of charging infrastructure rollout lead to distinct solutions for cost-optimal infrastructure expansion, BEV adoption across consumer groups, and the allocation of charging processes to different charger types. The following relation is evident: Higher spatial density of public charging infrastructure leads to higher adoption among lower-income groups.

This aligns with the objectives of existing charging planning tools that prioritize equitable charger distribution at the neighborhood level [116, 117]. The joint optimization of BEV adoption and charging capacities further emphasizes that, beyond enabling uniform charging access across neighborhoods, this equity consideration is also crucial for supporting the adoption of passenger cars across all consumer groups.

Charger utilization Despite facing both longer detour times to reach fast chargers and longer charging durations, cost-optimal solutions indicate BEV adoption in lower-income groups. This occurs because their lower valuation of time makes the time spent on fast charging less costly in monetary terms than for higher-income groups. From the system perspective of meeting passenger car travel demand at minimum cost, BEV adoption by lower-income consumers using fast charging remains economically efficient, even accounting for these time costs. Overall, in the proposed optimization model, the expansion of public fast charging infrastructure is mainly driven by the constraint that ensures the simultaneous expansion of both slow and fast charging capacities, and by the achievable reduction in average detour time, as modeled by the fueling detour time reduction function.

Throughout all scenarios, fast charger utilization only grows significantly after 2040. Realistically, this would lead to very high charging prices to cover installations, maintenance, and operation costs. The pricing and consumer-specific constraints to charging costs are not reflected in the modeling.

It should be noted that the cost-optimal allocation of fast-charger utilization to low-income classes is a system-level outcome that does not account for current market pricing structures. In practice, fast charging prices are significantly higher than slow charging prices per kilowatt-hour. Realizing the modeled allocation would therefore require a shift in pricing to be better aligned with system costs — specifically, lowering fast charging prices and raising slow charging prices — which would also increase the utilization of fast charger capacities.

Concerning the interpretation of the differences in scenarios, the following limitations should be noted. The approach to linking infrastructure capacity with access time operates at the NUTS-2 level, which introduces two important simplifications. First, the modeled regions are treated as spatially homogeneous: charger locations are implicitly distributed uniformly, and road network characteristics and residential patterns are held constant. This abstraction overlooks variation in urban density and ignores the strategic placement of charging infrastructure at trip attractors (shopping centers, workplaces, etc.), which in practice substantially reduces access times without requiring additional detours. Second, the formulation does not capture congestion dynamics at individual charging sites. The model assumes drivers possess perfect information about charger availability and can always access their nearest station without queuing or searching for an available plug. In reality, sites with limited capacity may force drivers to visit multiple locations before finding an open charging point—a phenomenon analyzed by [110]. This effectively assumes zero probability of encountering occupied chargers, implying that drivers never incur additional search time beyond the initial detour.

A further assumption with significant implications for the results is the limited availability of home charging for lower-income groups. In the model, lower-income households are assumed to have restricted access to home chargers, reflecting the empirical observation that these households are more likely to reside in multi-family housing without private parking or dedicated electrical infrastructure [18]. This assumption is deliberately imposed to isolate the role of public charging infrastructure in driving BEV adoption across income classes. However, it simultaneously suppresses adoption in lower-income groups, as it removes the most convenient and cost-effective charging option from their consideration set. In practice, the availability of home charging varies considerably within income groups depending on housing type, tenure, and local infrastructure investments. Relaxing this constraint — for instance, through subsidized installation of shared charging facilities in multi-family buildings or employer-provided workplace charging — likely yields higher BEV adoption numbers among lower-income consumers, as the dependence on public infrastructure for daily charging needs diminishes. The main finding that higher spatial density of public charging supports adoption in lower-income groups is expected to hold even with relaxed home charging constraints, as public infrastructure remains critical for consumers without home charging access and for trips exceeding daily commuting distances.

RQ2.b.: *How does a local rapid charging infrastructure expansion affect the adoption of battery-electric vehicles in neighboring regions with slower charging infrastructure expansion?*

Cross-border effects Enhanced charging deployment in one region, Bizkaia, generates spillover effects in neighbouring regions, Araba and Gipuzkoa, though the magnitude remains modest (electrification share increases by at most 2.1%) and varies substantially between the two neighboring regions. The asymmetric response across regions likely stems from differences in mobility patterns captured in the input data. When examining commuting-specific adoption versus fleet-wide electrification trends, a mechanism emerges: expanded destination-region capacity enables commuters to charge at their workplace or destination, thereby releasing origin-region infrastructure to serve local, non-commuting vehicle use. The elevated utilization rates observed for public charging infrastructure support this reallocation dynamic. However, fully characterizing these cross-regional dynamics and explaining the observed asymmetries would require addressing several methodological and data limitations. First, resolving the inconsistent spillover patterns demands methodological extensions in two directions: expanding the geographic scope to include many more regions would enable statistical analysis of spillover determinants, while isolating commuter flows from other trip purposes would clarify their specific contribution. The current framework optimizes system-wide infrastructure deployment and adoption trajectories subject to endogenous capacity decisions and fleet electrification rate constraints—assumptions that presume strong coordination between infrastructure availability and individual vehicle choices, potentially masking heterogeneous regional dynamics. Second, a critical data limitation concerns commuter trip volumes. The baseline draws on 2010 calibrations from the ETISplus project [151] and extrapolates all trip categories proportionally to GDP growth, as temporal trends distinguishing commuting from other purposes remain unavailable. Spillover magnitudes are highly sensitive to these assumed commuter flows. Additionally, regional trip distance distributions lack validation from Spanish-specific mobility surveys, introducing further uncertainty. Despite these limitations, the findings provide a complementary perspective to previous studies that report significant spillover effects in the early adoption phase based on empirical evidence [21], [22]. The purely techno-economic mechanisms captured by the model likely account for only a small share of the spillover effects observed empirically, where social and behavioral factors are expected to dominate.

5.1.3. Contribution 3 (RQ3)

> **RQ3:** *To what extent do geographically varying charging costs — arising from electricity price differentials and grid charges across market zones — affect the spatial patterns of charging demand distribution for international long-haul battery-electric trucks?*

Geographically varying electricity prices affect the distribution of charging demand at multiple levels. For the Scandinavian-Mediterranean corridor, the following is observed. First, energy system scenarios with different relative electricity price rankings across corridor countries produce substantially different spatial charging patterns over time, with allocated charging loads varying by up to 135% between the most divergent scenarios. Second, comparing the cost-optimized allocation against a deterministic benchmark with uniform charging distribution reveals that 39% of total demand shifts location under cost optimization. Third, localized surcharges on charging costs effectively redirect demand to surrounding regions, demonstrating that even modest price differentials can reshape the spatial concentration of charging load across the corridor.

Two effects drive these spatial differences in charging demand distribution: the spatially and temporally varying cost advantage of BETs over conventional trucks, and the geographic variation in charging costs.

Comparison with Pagnier et al. [120] These cross-scenario differences are consistent with findings by Pagnier et al. [120], who draw electricity-zone-dependent prices from a Norwegian energy system model and quantify their impact on regional charging infrastructure investment. They observe that, across different electricity pricing levels, investments in charging infrastructure differ by up to a factor of three at a given location. In the present analysis, the maximum cross-scenario spread in charging demand at the NUTS-2 level in a selected region reaches a factor of seven in 2032, though this reduces substantially over the optimization horizon to approximately two.

A key methodological difference is that Pagnier et al. [120] iterate between a freight transport model and an energy system model, thereby capturing the feedback effect of regionally developing charging demand on electricity prices. The present analysis does not account for this feedback, which may lead to some overestimation of the magnitude of demand reallocation and local concentration. However, the total charging loads in the case study remain small relative to national electricity demand (less than 0.35% across all countries), suggesting that price feedback effects would be limited in practice. A further simplification in the analysis is the use of country-level electricity prices, which is reasonable given that most countries along the corridor correspond to a single electricity market zone. Exceptions are Norway and Italy, which comprise multiple price zones — a distinction the energy system model input does not capture. Incorporating more granular price zones, including locational signals from balancing markets, could refine the spatial allocation results and represents an avenue for future work.

Modal shift to rail When modal shift to rail is enabled, approximately 15.4% of total charging demand along the corridor is spatially redistributed across regions. This shift predominantly reduces charging demand in the corridor's peripheral regions — Scandinavia and Italy — where long-distance routes are most cost-effectively shifted to rail. As a result, charging demand concentrates in the centrally located countries, particularly Germany and Austria, while overall electrification levels remain comparable. On a per-highway-km basis, Austrian highways face the highest demand concentration, reflecting the country's short transit corridor and high freight volumes.

These findings have direct implications for infrastructure planning. Cost-optimal decarbonization of the corridor requires not only consistent expansion of rail freight capacity along its full length, but also accelerated deployment of charging infrastructure in Denmark, Austria, and Germany to accommodate the concentrated road freight electrification.

These results should be interpreted in light of implementation constraints. The optimization model includes constraints based on the assumption that the pace of vehicle electrification is independent of the pace of modal shift — effectively treating rail as an accelerator for road freight decarbonization. In practice, both transitions face capacity and regulatory constraints that may slow their parallel deployment. Consequently, while the spatial concentration of charging demand in central regions is a robust finding, the required pace of charging infrastructure expansion may be slower than indicated here.

5.1.4. Contribution 4 (RQ4)

> **RQ4:** *What is the expected spatio-temporal distribution of charging demand from the commercial battery electric vehicle fleet in Austria in 2040?*

In 2040, the estimated annual charging demand of the commercial BEV fleet in Austria ranges from 7.4 TWh (*LOW* scenario, 30% HDT electrification) to 15.2 TWh (*HIGH* scenario, 100% HDT electrification), corresponding to 8–17% of the projected total electricity demand in Austria. Heavy-duty trucks account for the bulk of commercial charging demand. In the *HIGH* scenario, HDT charging accounts for 11.2 TWh of the total 15.2 TWh (74%). The decomposition by traffic type reveals that 74.3% of the HDT charging demand within Austria stems from transit and import/export traffic (Table 4.30). This underscores the importance of the traffic-flow-based methodology.

Regarding temporal distribution, the findings indicate substantial daytime charging activity across most Austrian regions, though peak demand predominantly occurs during nighttime hours, with lower intensity on weekends. Daytime charging demand may be overestimated, as the model assumes that highway charging infrastructure is the preferred option for heavy-duty trucks during mandatory rest periods, potentially introducing an upward bias in projected daytime loads at motorway service areas. Conversely, the model excludes operators without access to depot charging, whose vehicles would rely entirely on public infrastructure during daytime hours. These opposing effects may partially offset each other.

In the absence of empirical data on the spatio-temporal distribution of commercial charging demand across Austrian regions, the estimates are contextualized by comparing peak loads and charging demand against policy targets and scientific literature. Table 5.2 summarizes these comparisons, which are discussed in detail below.

Comparison of charging load estimations The estimated increase in electricity demand from the commercial fleet is close to estimates by Li et al. [121]. Li et al. [121] estimate that EV charging in China would increase national peak demand by 5.2–12.5% under unmanaged charging scenarios. While a direct comparison of demand between China and Austria is limited by the vastly different fleet sizes and electricity systems, the share of total electricity demand attributable to commercial vehicle charging is comparable. In the present analysis, the commercial fleet alone—excluding private passenger BEVs—accounts for 8–17% of total demand, highlighting the significance of commercial vehicle electrification for electricity system planning in a European transit country.

For heavy-duty vehicles specifically, the Transition Mobility 2040 scenario implies an energy demand of approximately 25,000 TJ ($\approx$6.9 TWh) for SNF in 2040 (Figure 16 in [187]). The HIGH scenario estimates 11.2 TWh for HDT charging — approximately 1.6 times higher. This difference can be attributed to several factors: the traffic-flow-based methodology captures transit and international freight, which constitutes 74.3% of HDT charging demand on Austrian highways; the Transition Mobility 2040 scenario assumes a modal shift in freight transport from 70% to 62% road share (by tonne-kilometres), with rail increasing from 27% to 34%; and it includes a

15% FCEV share for heavy-duty vehicles, whose energy demand would not appear as stationary charging.

Comparison with European truck charging infrastructure studies Shoman et al. [176] estimate daily charged energy for battery-electric long-haul trucks across European countries at 15% fleet electrification. For Austria, they report a daily charged energy of 3.5 GWh and identify 71 charging areas, with approximately 1,169 CCS and 240 MCS charging points required. Linearly scaling the *HIGH* scenario (40.9 GWh/d at 100% electrification) to 15% yields approximately 6.1 GWh/d — noting that linear scaling is an approximation, as charging patterns and infrastructure utilization may not scale proportionally with fleet size. This is higher than the estimate by Shoman et al. [176] by a factor of approximately 1.7. The difference is primarily attributable to the broader scope of the estimation: the methodology includes all HDT traffic types—inland, transit, and import/export—and encompasses the full range of heavy-duty vehicle weights, whereas Shoman et al. [176] focus specifically on long-haul truck operations. Additionally, differences in assumptions on specific energy consumption, average loads, and the allocation between en-route and depot charging contribute to the divergence.

Comparison with planned infrastructure in Austria The modeled peak power levels at highway service areas range between 7 and 29 MW for the *HIGH* scenario. As a reference, the Austrian highway infrastructure operator ASFINAG has communicated plans to provide 5 MW of grid connection capacity at each service area by 2035/2040 [199]. This planned capacity corresponds to a commercial fleet electrification rate of approximately 30% in the modeling framework. At higher electrification rates, the modeled peak power levels substantially exceed the planned grid connection capacities, particularly in the northeastern regions of Austria where transit traffic is concentrated. This indicates that grid reinforcement beyond current plans will be required to support full electrification of the commercial fleet.

The AFIR defines progressively increasing HDV recharging infrastructure targets along the TEN-T network [11]. By 2030, recharging pools must be deployed every 60 km on the TEN-T core network, each providing a minimum of 3,600 kW. For Austria, this translates to approximately 135 MW across the network (assuming roughly 38 pools over 2,258 km). The modeled peak power levels of 7–29 MW per service area substantially exceed these per-pool minimums, reflecting the high concentration of freight traffic on Austrian transit corridors. Even in aggregate, the AFIR minimum is an order of magnitude below the modeled total peak demand.

Table 5.2.: Quantitative comparison of commercial fleet charging demand and infrastructure estimates for Austria.

Source	**Electrification**	**Metric**	**Key assumptions & notes**
Charging demand estimates			
This work (*LOW*)	30% HDT	7.4 TWh/a	All commercial fleet segments; traffic-flow-based; Austria 2040
This work (*HIGH*)	100% HDT	15.2 TWh/a (40.9 GWh/d)	74% from HDT; 74.3% of HDT demand from transit + import/export
This work (scaled)	15% HDT	∼6.1 GWh/d	Linear scaling of *HIGH* for comparability with Shoman et al. [176]
Shoman et al. [176]	15% BET	3.5 GWh/d	Long-haul trucks only; 71 areas; 1,169 CCS + 240 MCS points
Transition Mobility 2040[a]	100%[b]	127 PJ/a (≈35 TWh/a)	Total transport energy; 85% BEV/ERS + 15% FCEV
Infrastructure capacity per service area			
This work (*HIGH*)	100% HDT	7–29 MW	Peak power at highway service areas; highest in NE Austria
ASFINAG (planned)[c]	—	5 MW	Grid connection per service area by 2035/2040
AFIR HDV (2030)[d]	—	3.6 MW/pool	Min. per pool every 60 km on TEN-T core; ≈135 MW for Austria

[a] Angelini et al. [187]. Projects truck freight modal share declining from 70% to 62%, rail increasing from 27% to 34%.
[b] 100% zero-emission fleet, but only 85% require stationary charging (BEV/ERS); 15% are FCEV.
[c] [199]. Corresponds to ≈30% electrification in the modeling framework.
[d] [11]. Minimum connectivity requirement; 2025 interim target: 1,400 kW/pool along 15% of TEN-T.

5.2. Synthesis of results

The four contributions of this thesis address different vehicle segments, geographic scopes, and planning dimensions. Yet they share a common analytical perspective: charging infrastructure planning at scales that extend beyond individual cities or regions — spanning national highway networks, international transport corridors, and cross-regional adoption dynamics. This section draws out three cross-cutting insights that emerge from considering the contributions jointly rather than in isolation.

Driving range and spatial density As framed by the review paper by Metais et al. [95], the overcoming of *range anxiety* is a primary design goal for most charging infrastructure capacity planning algorithms. Throughout the proposed model, the range of battery-electric vehicles is a core parameter determining the maximum distance between charging opportunities. The objective in most graph-based optimization algorithms applied for planning charging infrastructure at large scale focuses on establishing a minimum required set of charging sites (as in Contribution 1), while range dictates the maximum distance between them. Although battery capacities have grown to cover the vast majority of daily travel needs — largely rendering this anxiety no longer technically justified — and substantial charging infrastructure is deployed across European countries, the modeling tools applied in large-scale analyses remain largely unchanged, for both passenger and commercial vehicle fleets. In Contribution 1, range has no significant influence on siting decisions. In the regional consideration of both passenger and commercial BEVs addressed in Contributions 2 and 4, the impact of range on the relative siting of charging stations is inherently negligible by design. Results of Contribution 3 demonstrate that the expected range of long-haul trucks allows for substantial flexibility in the allocation of charging demand, suggesting that local electricity prices are more important determinants for infrastructure planning than range. Contribution 2 analytically demonstrates the benefit of increased spatial density of charging sites.

With charging infrastructure for passenger cars along European highway networks now largely established, the central problem in geographic charging capacity allocation is no longer the siting of initial charging sites, but rather the densification of the charger network and the expansion of on-site capacity, accounting for the spatial configuration of chargers. For trucks, while more foundational infrastructure efforts are still required, future tools should increasingly target the densification of charging sites, expansion of on-site capacity, and integration of local pricing signals.

Spatial flexibility Spatial flexibility in the allocation of charging demand is a key feature across the methodologies of all four contributions. Contribution 4 demonstrates that substantial charging loads arise from interregional transport flows, underscoring the cross-border dimension of spatial demand allocation. Contribution 3 quantifies significant spatial flexibility in charging demand allocation along transport corridors. Contribution 1, by contrast, finds that the siting of charging stations is not significantly affected by extending the driving range of vehicles — that is, by increasing the spatial extent of this flexibility. While the study area in Contribution 1 does not extend across borders, nor does it draw on origin-destination flows rather than traffic-count-derived demand, the findings of Contribution 3 — albeit for a different vehicle segment and geographic scope — point in the same direction, suggesting that the limited impact of range on spatial allocation may generalize beyond the specific setting of Contribution 1. Contribution 2 shows that spatial flexibility does not significantly affect BEV adoption from a techno-economic perspective, but reveals flexibility in the utilization of different charging infrastructure types within a region, depending on the spatial density of the network.

Taken together, these findings challenge the prevailing exogenous treatment of charging demand in large-scale analyses, which neglects flexibility in vehicle operation, the relative configuration of infrastructure, and the favorable spatial reallocation of demand that this flexibility enables. While endogenous modeling of charging demand and vehicle operation flexibility is explored at smaller scopes in the literature, i.e., for smaller geographic regions and for individual routes, its application to large-scale contexts remains limited and warrants further development.

Value of time in consumer segmentation In the commercial freight sector, the monetization of time is relatively straightforward, expressed directly through labor costs, as in Contribution 3, and delivery schedules. This is reflected in the vehicle-type segmentation applied in the charging load estimation of Contribution 4, where demand is characterized by energy consumption, installed battery capacity, and mobility patterns. For passenger vehicles, however, the trade-off between time and money is individual and closely tied to income level. The results of Contribution 2 illustrate this concretely: income-specific behavioral differences produce divergent preferences in charging infrastructure utilization, suggesting that a homogeneous representation of passenger BEV users is an oversimplification with practical planning consequences. Contribution 1 further demonstrates the high sensitivity of highway charging infrastructure efficiency to installed charging power levels.

Incorporating income-class-dependent value of time into charging infrastructure planning could therefore translate into two distinct implications for highway network operators. First, higher-income drivers may systematically prefer faster, higher-cost charging, shifting the cost-optimal configuration of charging power levels at highway sites. Second, lower-income drivers may prefer cheaper off-highway charging alternatives, which could reduce utilization of highway charging infrastructure. Neither behavioral pattern is captured by current aggregate, homogeneous modeling approaches.

6. Conclusions

The findings of the four contributions presented in the previous chapters address the three interconnected dimensions of charging infrastructure planning outlined in the Introduction (Section 1), with implications for transport policy makers, transport and charging infrastructure planners, as well as electricity grid and system operators:

(i) **From coverage-based siting to operations-aware capacity planning:** The optimization of charging infrastructure siting should account endogenously for the interactions between capacity sizing, station utilization, and vehicle operation. Doing so shifts the basis for infrastructure siting from capital expenditures alone toward a fuller consideration of operational expenditures. Current large-scale methodologies predominantly optimize for coverage, meaning that sites are selected to serve maximum demand, with utilization implicitly assumed rather than explicitly modeled. While smaller-scope studies of individual vehicle movements do capture these operational dynamics, they are rarely incorporated at the geographic scales relevant for international corridor planning.

(ii) **Grounding charging infrastructure expansion targets in spatial accessibility**: Spatially dense charging infrastructure supports BEV adoption across all income groups, making geographic reach the decisive factor for large-scale uptake. This finding implies that expansion targets framed solely in terms of charging points per BEV or installed capacity are insufficient — such metrics do not capture whether infrastructure is actually accessible to drivers where and when they need it. Instead, definitions should be grounded in minimum spatial density requirements within the road transport network. However, the AFIR applies this logic only to highway charging through distance-based deployment mandates, while country-level definitions remain count-based — a gap that risks underserving the very regions and consumer groups for which spatial density is most critical. Extending spatial-density-based requirements beyond highways would better support equitable and effective fleet electrification.

(iii) **Leveraging spatially flexible load as a demand-side resource for the electricity system**: Charging infrastructure is the physical interface between electrified vehicle fleets and the electricity grid, meaning that its spatial and temporal operation directly shapes the load imposed on the electricity system. The flexibility perspective for the transport sector should extend beyond timing-based demand-response to encompass the spatial dimension of charging demand allocation. Grid operators and regulators should leverage this potential, and the findings of this work demonstrate that it is substantial along international corridors.

Overall, cost-optimal BEV adoption for long-haul and international transport requires coordinated action from transport policy makers, transport and charging infrastructure planners, as well as electricity grid and system operators beyond the national level. International transport flows along the TEN-T corridors encompass substantial freight and passenger movements, but the resulting charging demand does not follow transport activity directly — it is shaped by the relative configuration of charging infrastructure and costs across national boundaries.

7. Future Work

In addition to the limitations identified in Section 5, the spatial allocation of charging infrastructure for battery-electric vehicle fleets — the central focus of this work — opens several directions for further research that extend its scope. Many of the aspects discussed below have been addressed in the literature at smaller scopes — for individual vehicles, local networks, or selected routes — but increasingly need to be incorporated into analyses of aggregate vehicle fleets at larger geographic scales. The following discussion first outlines research directions that address broader questions at the intersection of transport electrification and energy system planning, and then identifies open methodological challenges for advancing the analytical tools.

Logistical flexibility In the aggregated representation of vehicle movements, individual decision-making is strongly simplified and system-optimal charging behavior is assumed. However, such behavior can only be initiated through significant pricing signals or incentives. For example, cost-optimal charging allocation along a TEN-T corridor only emerges when charging costs differ substantially across locations. Moreover, reaching the system optimum requires flexibility in mobility and logistics patterns, which in practice are constrained by commercial and operational processes. Further research is needed to understand what incentives are required to drive cost-optimal charging behavior, particularly in the context of rigid logistics schedules and mobility patterns, and to what extent individual behavioral responses translate to the aggregate representation adopted in this work. A closely related question concerns the design of charging price structures that can steer spatial and temporal demand allocation toward system-optimal outcomes — bridging the gap between the modeled cost-optimal solutions and implementable market mechanisms.

Charging infrastructure as a lever for second-hand vehicle markets This work extends the role of charging infrastructure beyond covering existing demand, towards serving as a lever for the adoption of battery-electric vehicles. A particularly relevant further extension concerns the second-hand vehicle market. In Europe, substantial volumes of used trucks are traded across borders, making second-hand markets a key pathway for fleet renewal. As battery-electric trucks enter fleets in early-adopting countries, they will eventually feed into these cross-border trade flows. The question arises whether the deployment of charging infrastructure in the receiving countries can enable the adoption of second-hand battery-electric trucks — by ensuring charging availability and contributing to reasonable total costs of ownership. Further research should examine this relationship, including challenges associated with battery degradation and state of health, which directly affect the economic viability of used battery-electric vehicles.

Cross-modal perspective Contribution 3 demonstrates that modal shift to rail can substantially redistribute charging demand along transport corridors. This finding points toward a broader cross-modal perspective on charging and fueling infrastructure planning. At intermodal nodes — such as rail terminals, airports, ports, logistics hubs, and park-and-ride facilities — energy demand from road-based freight and passenger transport converges with that of other transport modes, creating potential synergies in infrastructure investment and grid connection capacity. Analyzing the geographic allocation of shared energy infrastructure at these nodes, and how it interacts with road-based charging demand, is a natural extension of the corridor-level analysis conducted in this work. This is further motivated by policy efforts to shift both freight and passenger transport towards rail and other collective modes, which increase the relevance of intermodal transfer points as shared infrastructure investment sites.

Along with the research directions outlined above, several *modeling challenges* need to be addressed to advance the analytical tools. The joint optimization framework for charging infrastructure allocation, as extended in scope towards broader system interactions, requires methodological improvements in the following areas:

- **Sub-annual temporal resolution in transport demand data:** Origin-destination data is predominantly available for aggregate annual flows, and both up-to-date data and forward-looking projections are scarce. Sub-annual fluctuations in transport demand — such as seasonal variations in freight and passenger volumes — are typically neglected, yet become increasingly important when considering operational interactions with the electricity grid and system.

- **Uncertainty in origin-destination transport flow projections:** Current approaches place high trust in origin-destination data, with little consideration for the inherent uncertainty in these projections. However, future transport flows are shaped by spatial planning, infrastructure development, socio-economic dynamics, and labor market evolution — involving multiple layers of decision-making that are difficult to predict. Rather than extending the modeling scope towards macroeconomic representations, which would result in prohibitively large systems, a more tractable approach is to incorporate demand uncertainty directly into the optimization, enabling more robust infrastructure investment decisions.

- **Spatio-temporal complexity in optimization:** Jointly resolving spatial and temporal dimensions introduces substantial computational challenges, well known from energy system modeling. These are further amplified by vehicle mobility, where the discretization of time steps must account for moving consumers whose behaviors may interact, for example through queuing at charging sites. While such dynamics are typically addressed through agent-based approaches, their integration into system-optimization frameworks requires careful identification of which spatio-temporal elements are essential to represent at the aggregate level and which can be simplified. Two complementary strategies emerge: adopting aggregation techniques from other optimization domains, and identifying the minimal set of dynamic representations required — for instance, to capture intra-day demand response of long-haul trucks to variable charging prices.

Bibliography

[1] European Commission, *The European Green Deal*, COM, Communication from the Commission to the European Parliament, the European Council, the Council, the European Economic and Social Committee and the Committee of the Regions, Brussels, 2019. [Online]. Available: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52019DC0640 (cit. on p. 1).

[2] European Environment Agency, *Greenhouse gas emissions from transport in Europe*, Indicator published 06 November 2025, 2025. Accessed: Jan. 5, 2026. [Online]. Available: https://www.eea.europa.eu/en/analysis/indicators/greenhouse-gas-emissions-from-transport (cit. on p. 1).

[3] European Commission, *Sustainable and smart mobility strategy*, 2024. [Online]. Available: https://transport.ec.europa.eu/transport-themes/mobility-strategy_en (cit. on p. 1).

[4] B. van der Zwaan, I. Keppo, and F. Johnsson, "How to decarbonize the transport sector?", *Energy Policy*, vol. 61, pp. 562–573, 2013, ISSN: 0301-4215. DOI: 10.1016/j.enpol.2013.05.118 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0301421513004734 (cit. on p. 1).

[5] O. Y. Edelenbosch, D. L. McCollum, D. P. van Vuuren, C. Bertram, S. Carrara, H. Daly, S. Fujimori, A. Kitous, P. Kyle, E. Ó Broin, P. Karkatsoulis, and F. Sano, "Decomposing passenger transport futures: Comparing results of global integrated assessment models", *Transportation Research Part D: Transport and Environment*, vol. 55, pp. 281–293, 2017. DOI: 10.1016/j.trd.2016.07.003 (cit. on pp. 1, 10).

[6] IEA, "Global EV outlook 2025", International Energy Agency, Paris, Tech. Rep., 2025, Licence: CC BY 4.0. [Online]. Available: https://www.iea.org/reports/global-ev-outlook-2025 (cit. on pp. 1, 11, 62).

[7] M. Monteforte, P. Mock, and U. Tietge, "European vehicle market statistics 2024/25", International Council on Clean Transportation, Tech. Rep., Dec. 2024, Accessed: 2026-01-09. [Online]. Available: https://theicct.org/publication/european-vehicle-market-statistics-2024-25/ (cit. on p. 1).

[8] European Automobile Manufacturers' Association, "Vehicles on European roads", ACEA, Tech. Rep., Feb. 2025. [Online]. Available: https://www.acea.auto/files/ACEA-Report-Vehicles-on-European-roads-.pdf (cit. on p. 1).

[9] European Commission, "Sustainable and smart mobility strategy – putting European transport on track for the future", European Commission, Brussels, COM(2020) 789 final, Dec. 2020, Accessed: 2026-01-09. [Online]. Available: https://transport.ec.europa.eu/transport-themes/mobility-strategy_en (cit. on p. 1).

[10] European Commission, "Fit for 55: Delivering the EU's 2030 climate target on the way to climate neutrality", European Commission, COM(2021) 550 final, Jul. 2021, Accessed: 2026-01-09. [Online]. Available: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52021DC0550 (cit. on p. 1).

[11] European Parliament and Council of the European Union, *Regulation (EU) 2023/1804 of the European Parliament and of the Council of 13 September 2023 on the deployment of alternative fuels infrastructure, and repealing Directive 2014/94/EU*, Official Journal of the European Union, L 234, pp. 1–47, Alternative Fuels Infrastructure Regulation (AFIR), 22 September 2023, Sep. 2023. [Online]. Available: `http://data.europa.eu/eli/reg/2023/1804/oj` (cit. on pp. 1, 102, 103, 112, 113).

[12] European Parliament and Council of the European Union, *Regulation (EU) 2024/1679 of the European Parliament and of the Council of 13 June 2024 on Union guidelines for the development of the trans-European transport network, amending Regulations (EU) 2021/1153 and (EU) No 913/2010 and repealing Regulation (EU) No 1315/2013*, Official Journal of the European Union, L 2024/1679, Revised TEN-T Regulation, 28 June 2024, Jun. 2024. [Online]. Available: `http://data.europa.eu/eli/reg/2024/1679/oj` (cit. on p. 2).

[13] G. Rainieri, C. Buizza, and A. Ghilardi, "The psychological, human factors and socio-technical contribution: A systematic review towards range anxiety of battery electric vehicles' drivers", *Transportation Research Part F: Traffic Psychology and Behaviour*, vol. 99, pp. 52–70, 2023, ISSN: 1369-8478. DOI: `10.1016/j.trf.2023.10.001` [Online]. Available: `https://www.sciencedirect.com/science/article/pii/S1369847823002012` (cit. on pp. 2, 11, 12).

[14] E. Hopkins, D. Potoglou, S. Orford, and L. Cipcigan, "Can the equitable roll out of electric vehicle charging infrastructure be achieved?", *Renewable and Sustainable Energy Reviews*, vol. 182, p. 113 398, 2023, ISSN: 1364-0321. DOI: `10.1016/j.rser.2023.113398` [Online]. Available: `https://www.sciencedirect.com/science/article/pii/S1364032123002551` (cit. on pp. 2, 7, 12).

[15] Z. Du, L. Zheng, and B. Lin, "Influence of charging stations accessibility on charging stations utilization", *Energy*, vol. 298, p. 131 374, 2024, ISSN: 0360-5442. DOI: `10.1016/j.energy.2024.131374` [Online]. Available: `https://www.sciencedirect.com/science/article/pii/S0360544224011472` (cit. on p. 2).

[16] Q. Ye, P. Bansal, and B. Adey, "Multi-period charging infrastructure planning under uncertainty: Challenges and opportunities", *Sustainable Cities and Society*, vol. 116, p. 105 908, 2024, ISSN: 2210-6707. DOI: `10.1016/j.scs.2024.105908` [Online]. Available: `https://www.sciencedirect.com/science/article/pii/S2210670724007327` (cit. on pp. 2, 6).

[17] BMK, "Mobilitätsmaster- plan 2030 für Österreich", Tech. Rep., 2021 (cit. on p. 6).

[18] G. Brückmann, F. Willibald, and V. Blanco, "Battery electric vehicle adoption in regions without strong policies", *Transportation Research Part D: Transport and Environment*, vol. 90, p. 102 615, 2021, ISSN: 1361-9209. DOI: `10.1016/j.trd.2020.102615` [Online]. Available: `https://www.sciencedirect.com/science/article/pii/S1361920920308002` (cit. on pp. 7, 11, 107).

[19] J. Tattini, K. Ramea, M. Gargiulo, C. Yang, E. Mulholland, S. Yeh, and K. Karlsson, "Improving the representation of modal choice into bottom-up optimization energy system models - the mocho-times model", *Applied Energy*, vol. 212, pp. 265–282, 2018, ISSN: 0306-2619. DOI: `10.1016/j.apenergy.2017.12.050` [Online]. Available: `https://www.sciencedirect.com/science/article/pii/S030626191731766X` (cit. on pp. 7, 16, 28, 60, 166).

[20] F. Spaven, Y. Liu, and M. Baghdadi, “Going further with smaller evs: System-level battery range, emissions and charging infrastructure analysis”, *Journal of Cleaner Production*, vol. 369, p. 133 349, 2022, ISSN: 0959-6526. DOI: 10.1016/j.jclepro.2022.133349 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0959652622029341 (cit. on p. 7).

[21] L. Qian, C. Zhang, Y. Chen, and Z. Xiao, “Spatial cross-side network effect of charging stations on electric vehicle adoption”, *Transportation Research Part D: Transport and Environment*, vol. 136, p. 104 400, 2024, ISSN: 1361-9209. DOI: 10.1016/j.trd.2024.104400 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1361920924003572 (cit. on pp. 7, 12, 108).

[22] S. M. Sheng, L. Wen, B. Sharp, B. Du, P. Ranjitkar, and D. Wilson, “A spatio-temporal approach to electric vehicle uptake: Evidence from new zealand”, *Transportation Research Part D: Transport and Environment*, vol. 105, p. 103 256, 2022, ISSN: 1361-9209. DOI: 10.1016/j.trd.2022.103256 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1361920922000864 (cit. on pp. 7, 12, 108).

[23] Eurostat, *NUTS - Nomenclature of Territorial Units for Statistics*, Accessed: 2025-05-20, 2025. [Online]. Available: https://ec.europa.eu/eurostat/web/nuts (cit. on pp. 7, 58).

[24] European Commission. “Weekly oil bulletin”. Accessed: 2025-05-20. [Online]. Available: https://energy.ec.europa.eu/data-and-analysis/weekly-oil-bulletin_en (cit. on p. 8).

[25] L. Lanz, B. Noll, T. S. Schmidt, and B. Steffen, “Comparing the levelized cost of electric vehicle charging options in europe”, *Nature Communications*, vol. 13, no. 1, p. 5277, 2022, ISSN: 2041-1723. DOI: 10.1038/s41467-022-32835-7 [Online]. Available: https://doi.org/10.1038/s41467-022-32835-7 (cit. on pp. 8, 12, 61, 81, 83).

[26] J. Hildermeier and A. Jahn, *The power of moving loads: Cost analysis of megawatt charging in europe*, 2024 (cit. on p. 8).

[27] C. Crozier, T. Morstyn, and M. McCulloch, “The opportunity for smart charging to mitigate the impact of electric vehicles on transmission and distribution systems”, *Applied Energy*, vol. 268, p. 114 973, 2020, ISSN: 0306-2619. DOI: 10.1016/j.apenergy.2020.114973 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0306261920304852 (cit. on p. 8).

[28] A. Golab, S. Zwickl-Bernhard, and H. Auer, “Minimum-cost fast-charging infrastructure planning for electric vehicles along the austrian high-level road network”, *Energies*, vol. 15, no. 6, 2022, ISSN: 1996-1073. DOI: 10.3390/en15062147 [Online]. Available: https://www.mdpi.com/1996-1073/15/6/2147 (cit. on pp. 10, 22, 45, 103, 105, 141).

[29] A. Golab, S. Zwickl-Bernhard, and H. Auer, “Public charger expansion: Impacts across income classes and beyond regional borders”, *Transportation Research Part D: Transport and Environment*, 2026, In press (cit. on pp. 10, 22, 45, 58, 147).

[30] A. Golab, S. J. Bakker, S. Zwickl-Bernhard, and H. Auer, “Spatial patterns of charging demand distribution in inter-zonal long-haul truck electrification”, *Energy*, n.d. Submitted (cit. on pp. 10, 22, 45, 76, 172).

[31] A. Golab, C. Loschan, S. Zwickl-Bernhard, and H. Auer, "The value of flexibility of commercial electric vehicle fleets in the redispatch of congested transmission grids", *Energy*, vol. 316, p. 134 385, 2025, ISSN: 0360-5442. DOI: 10.1016/j.energy.2025.134385 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0360544225000271 (cit. on pp. 10, 22, 45, 94, 186).

[32] R. Sims, S. R., X. Cruz-Núñez, M. D'Agosto, Dimitriu, M. J. Figueroa Meza, L. Fulton, S. Kobayashi, O. Lah, A. McKinnon, P. Newman, M. Ouyang, J. J. Schauer, D. Sperling, and T. G., "Transport", in O. Edenhofer et al., Eds., 2014 (cit. on p. 10).

[33] A. Milovanoff, R. Balasubramanian, L. Minet, L. Cheah, I. D. Posen, and H. L. MacLean, "Greenhouse gas emission mitigation pathways for urban passenger land transport under ambitious climate targets", *Environmental Science and Technology*, vol. 55, no. 12, pp. 8236–8246, 2021, ISSN: 15205851. DOI: 10.1021/acs.est.0c06671 (cit. on p. 10).

[34] L. Winkler, D. Pearce, J. Nelson, and O. Babacan, "The effect of sustainable mobility transition policies on cumulative urban transport emissions and energy demand", *Nature Communications*, vol. 14, no. 1, p. 2357, 2023. DOI: 10.1038/s41467-023-37728-x [Online]. Available: https://doi.org/10.1038/s41467-023-37728-x (cit. on p. 10).

[35] J. Bauer, P. Letmathe, and R. Woeste, "Total cost of ownership for battery electric vehicles: The role of energy prices", *Applied Energy*, vol. 389, p. 125 764, 2025, ISSN: 0306-2619. DOI: 10.1016/j.apenergy.2025.125764 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0306261925004945 (cit. on p. 10).

[36] D. C. Rosenfeld, J. Lindorfer, and K. Fazeni-Fraisl, "Comparison of advanced fuels—which technology can win from the life cycle perspective?", *Journal of Cleaner Production*, vol. 238, p. 117 879, 2019, ISSN: 0959-6526. DOI: 10.1016/j.jclepro.2019.117879 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0959652619327490 (cit. on p. 10).

[37] P. Korkmaz, D. Schmid, and U. Fahl, "Incorporating uncertainties towards a sustainable european energy system: A stochastic approach for decarbonization paths focusing on the transport sector", *Energy Strategy Reviews*, vol. 38, p. 100 707, 2021, ISSN: 2211-467X. DOI: 10.1016/j.esr.2021.100707 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2211467X21000936 (cit. on p. 10).

[38] IEA, "Global ev outlook 2022", International Energy Agency, Paris, Tech. Rep., 2022, Licence: CC BY 4.0. [Online]. Available: https://www.iea.org/reports/global-ev-outlook-2022 (cit. on p. 10).

[39] N. Berkeley, D. Bailey, A. Jones, and D. Jarvis, "Assessing the transition towards battery electric vehicles: A multi-level perspective on drivers of, and barriers to, take up", *Transportation Research Part A: Policy and Practice*, vol. 106, pp. 320–332, 2017, ISSN: 0965-8564. DOI: 10.1016/j.tra.2017.10.004 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0965856416304785 (cit. on p. 11).

[40] S. Wappelhorst, D. Hall, M. Nicholas, and N. Lutsey, "Analyzing policies to grow the electric vehicle market in European cities", International Council on Clean Transportation, Washington, DC, White Paper, 2020. [Online]. Available: https://theicct.org/wp-content/uploads/2021/06/EV_city_policies_white_paper_fv_20200224.pdf (cit. on p. 11).

[41] International Energy Agency, “Global EV outlook 2024”, International Energy Agency, Paris, Tech. Rep., 2024, Licence: CC BY 4.0. [Online]. Available: https://www.iea.org/reports/global-ev-outlook-2024 (cit. on p. 11).

[42] N. J. Klein, R. Basu, and M. J. Smart, “In the driver’s seat: Pathways to automobile ownership for lower-income households in the united states”, *Transportation Research Interdisciplinary Perspectives*, vol. 18, p. 100 787, 2023, ISSN: 2590-1982. DOI: 10.1016/j.trip.2023.100787 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2590198223000349 (cit. on p. 11).

[43] G. Pedrosa and H. Nobre, “Second-hand electrical vehicles: A first look at the secondary market of modern evs”, *International Journal of Electric and Hybrid Vehicles*, vol. 10, no. 3, pp. 236–252, 2018. DOI: 10.1504/IJEHV.2018.097378 eprint: https://www.inderscienceonline.com/doi/pdf/10.1504/IJEHV.2018.097378. [Online]. Available: https://www.inderscienceonline.com/doi/abs/10.1504/IJEHV.2018.097378 (cit. on p. 11).

[44] V. Schiaroli and L. Fraccascia, “Driving the change: How do personal factors and socio-economic context influence electric vehicles adoption across europe?”, *Environmental Innovation and Societal Transitions*, vol. 58, p. 101 044, 2026, ISSN: 2210-4224. DOI: 10.1016/j.eist.2025.101044 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2210422425000838 (cit. on p. 11).

[45] N. Rauh, T. Franke, and J. F. Krems, “Understanding the impact of electric vehicle driving experience on range anxiety”, *Human Factors*, vol. 57, no. 1, pp. 177–187, 2015, PMID: 25790577. DOI: 10.1177/0018720814546372 eprint: https://doi.org/10.1177/0018720814546372. [Online]. Available: https://doi.org/10.1177/0018720814546372 (cit. on p. 11).

[46] M. E. Biresselioglu, M. Demirbag Kaplan, and B. K. Yilmaz, “Electric mobility in europe: A comprehensive review of motivators and barriers in decision making processes”, *Transportation Research Part A: Policy and Practice*, vol. 109, pp. 1–13, 2018, ISSN: 0965-8564. DOI: 10.1016/j.tra.2018.01.017 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0965856417311771 (cit. on p. 11).

[47] IEA, *Number of available electric car models and driving range by type, 2015-2021*, International Energy Agency, Paris, Licence: CC BY 4.0, 2022. [Online]. Available: https://www.iea.org/data-and-statistics/charts/number-of-available-electric-car-models-and-driving-range-by-type-2015-2021 (cit. on p. 11).

[48] N. S. Pearre, W. Kempton, R. L. Guensler, and V. V. Elango, “Electric vehicles: How much range is required for a day’s driving?”, *Transportation Research Part C: Emerging Technologies*, vol. 19, no. 6, pp. 1171–1184, 2011, ISSN: 0968-090X. DOI: 10.1016/j.trc.2010.12.010 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0968090X1100012X (cit. on p. 11).

[49] K. B. Baden, “More than just miles: Range anxiety unplugged”, *Transportation Research Part A: Policy and Practice*, vol. 200, p. 104 661, 2025, ISSN: 0965-8564. DOI: 10.1016/j.tra.2025.104661 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0965856425002897 (cit. on p. 11).

[50] S. LaMonaca and L. Ryan, “The state of play in electric vehicle charging services - a review of infrastructure provision, players, and policies”, *Renewable and Sustainable Energy Reviews*, vol. 154, p. 111 733, 2022, ISSN: 1364-0321. DOI: 10.1016/j.rser.2021.111733 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1364032121010066 (cit. on pp. 11, 61).

[51] M. Baresch and S. Moser, “Allocation of e-car charging: Assessing the utilization of charging infrastructures by location”, *Transportation Research Part A: Policy and Practice*, vol. 124, no. April, pp. 388–395, 2019, ISSN: 09658564. DOI: 10.1016/j.tra.2019.04.009 [Online]. Available: https://doi.org/10.1016/j.tra.2019.04.009 (cit. on pp. 11, 46, 62).

[52] H. Karimianfard, “A robust optimization framework for smart home energy management: Integrating photovoltaic storage, electric vehicle charging, and demand response”, *Journal of Energy Storage*, vol. 110, p. 115 259, 2025, ISSN: 2352-152X. DOI: 10.1016/j.est.2024.115259 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2352152X2404845X (cit. on pp. 11, 12).

[53] M. van der Kam, A. Pena-Bello, and D. Parra, “Pv for what? a pv self-sufficiency analysis of electric vehicles, heat pumps, and home storage”, *Energy*, vol. 334, p. 137 774, 2025, ISSN: 0360-5442. DOI: 10.1016/j.energy.2025.137774 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0360544225034164 (cit. on p. 11).

[54] G. M. Fetene, G. Hirte, S. Kaplan, C. G. Prato, and S. Tscharaktschiew, “The economics of workplace charging”, *Transportation Research Part B: Methodological*, vol. 88, pp. 93–118, 2016, ISSN: 0191-2615. DOI: 10.1016/j.trb.2016.03.004 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0191261515302009 (cit. on p. 11).

[55] R. Kataoka, K. Hidaka, and T. Nishi, “Bridging the gap in workplace charging requirements between demand and supply sides”, *Transportation Research Part D: Transport and Environment*, vol. 151, p. 105 136, 2026, ISSN: 1361-9209. DOI: 10.1016/j.trd.2025.105136 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1361920925005462 (cit. on p. 11).

[56] J. Kester, L. Noel, G. Zarazua de Rubens, and B. K. Sovacool, “Policy mechanisms to accelerate electric vehicle adoption: A qualitative review from the nordic region”, *Renewable and Sustainable Energy Reviews*, vol. 94, pp. 719–731, 2018. DOI: 10.1016/j.rser.2018.05.067 [Online]. Available: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85049480962&doi=10.1016%2fj.rser.2018.05.067&partnerID=40&md5=98ba8b7c61cda6abcdc448c86fc700fd (cit. on p. 11).

[57] S. Ghanbari Motlagh and L. Li, “A review on electric vehicle charging station planning: Infrastructure placement, sizing, upgrades, and uncertainties”, *Journal of Energy Storage*, vol. 141, p. 119 325, 2026, ISSN: 2352-152X. DOI: 10.1016/j.est.2025.119325 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2352152X25040381 (cit. on p. 11).

[58] H. Bian, C. Zhou, Z. Guo, X. Wang, Y. He, and S. Peng, “Planning of electric vehicle fast-charging station based on poi interest point division, functional area, and multiple temporal and spatial characteristics”, *Energy Reports*, vol. 8, pp. 831–840, 2022, 2022 The 5th International Conference on Renewable Energy and Environment Engineering,

ISSN: 2352-4847. DOI: 10.1016/j.egyr.2022.10.161 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2352484722020960 (cit. on p. 11).

[59] G. Yu, X. Ye, D. Gong, and X. Xia, “Stochastic planning for transition from shopping mall parking lots to electric vehicle charging stations”, *Applied Energy*, vol. 379, p. 124 894, 2025, ISSN: 0306-2619. DOI: 10.1016/j.apenergy.2024.124894 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0306261924022773 (cit. on p. 11).

[60] R. Philipsen, T. Schmidt, J. V. Heek, and M. Ziefle, “Fast-charging station here , please ! User criteria for electric vehicle fast-charging locations”, *Transportation Research Part F: Psychology and Behaviour*, vol. 40, no. 2016, pp. 119–129, 2020, ISSN: 1369-8478. DOI: 10.1016/j.trf.2016.04.013 [Online]. Available: http://dx.doi.org/10.1016/j.trf.2016.04.013 (cit. on p. 12).

[61] J. Serradilla, J. Wardle, P. Blythe, and J. Gibbon, “An evidence-based approach for investment in rapid-charging infrastructure”, *Energy Policy*, vol. 106, no. April, pp. 514–524, 2017, ISSN: 03014215. DOI: 10.1016/j.enpol.2017.04.007 [Online]. Available: http://dx.doi.org/10.1016/j.enpol.2017.04.007 (cit. on pp. 12, 46).

[62] A. Ermagun and J. Tian, “Charging into inequality: A national study of social, economic, and environment correlates of electric vehicle charging stations”, *Energy Research & Social Science*, vol. 115, p. 103 622, 2024, ISSN: 2214-6296. DOI: 10.1016/j.erss.2024.103622 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2214629624002135 (cit. on p. 12).

[63] A. Stefaniec, R. Egan, K. Hosseini, and B. Caulfield, “The challenge of making evs just affordable enough: Assessing the impact of subsidies on equity and emission reduction in ireland”, *Research in Transportation Economics*, vol. 109, p. 101 495, 2025, ISSN: 0739-8859. DOI: 10.1016/j.retrec.2024.101495 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0739885924000908 (cit. on p. 12).

[64] H. Gazmeh, Y. Guo, and X. Qian, “Understanding the opportunity-centric accessibility for public charging infrastructure”, *Transportation Research Part D: Transport and Environment*, vol. 131, p. 104 222, 2024, ISSN: 1361-9209. DOI: 10.1016/j.trd.2024.104222 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1361920924001792 (cit. on p. 12).

[65] L. M. Austmann, “Drivers of the electric vehicle market: A systematic literature review of empirical studies”, *Finance Research Letters*, vol. 41, p. 101 846, 2021, ISSN: 1544-6123. DOI: 10.1016/j.frl.2020.101846 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1544612320316603 (cit. on p. 12).

[66] R. Liu, Z. Ding, X. Jiang, J. Sun, Y. Jiang, and W. Qiang, “How does experience impact the adoption willingness of battery electric vehicles? the role of psychological factors”, *Environmental Science and Pollution Research*, vol. 27, no. 20, pp. 25 230–25 247, 2020, ISSN: 1614-7499. DOI: 10.1007/s11356-020-08834-w [Online]. Available: https://doi.org/10.1007/s11356-020-08834-w (cit. on p. 12).

[67] U. Illmann and J. Kluge, “Public charging infrastructure and the market diffusion of electric vehicles”, *Transportation Research Part D: Transport and Environment*, vol. 86, no. November, 2020, ISSN: 13619209. DOI: 10.1016/j.trd.2020.102413 (cit. on p. 12).

[68] L. V. White, A. L. Carrel, W. Shi, and N. D. Sintov, “Why are charging stations associated with electric vehicle adoption? untangling effects in three united states metropolitan areas”, *Energy Research & Social Science*, vol. 89, p. 102 663, 2022, ISSN: 2214-6296. DOI: 10.1016/j.erss.2022.102663 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2214629622001670 (cit. on p. 12).

[69] J. Globisch, P. Plötz, E. Dütschke, and M. Wietschel, “Consumer preferences for public charging infrastructure for electric vehicles”, *Transport Policy*, vol. 81, no. June, pp. 54–63, 2019, ISSN: 0967-070X. DOI: 10.1016/j.tranpol.2019.05.017 [Online]. Available: https://doi.org/10.1016/j.tranpol.2019.05.017 (cit. on p. 12).

[70] F. Gebauer, R. Vilimek, A. Keinath, and C.-c. Carbon, “Technological Forecasting & Social Change Changing attitudes towards e-mobility by actively elaborating fast-charging technology”, *Technological Forecasting & Social Change*, vol. 106, pp. 31–36, 2016, ISSN: 0040-1625. DOI: 10.1016/j.techfore.2016.02.006 [Online]. Available: http://dx.doi.org/10.1016/j.techfore.2016.02.006 (cit. on p. 12).

[71] Y. Shevtsova, J. Díaz-Lanchas, D. Persyn, and G. Mandras, “Trade spillover effects of transport infrastructure investments: A structural gravity analysis for eu regions”, *Regional Studies*, vol. 59, no. 1, p. 2 441 231, 2025. DOI: 10.1080/00343404.2024.2441231 eprint: https://doi.org/10.1080/00343404.2024.2441231. [Online]. Available: https://doi.org/10.1080/00343404.2024.2441231 (cit. on p. 12).

[72] J. Sporkmann, Y. Liu, and S. Spinler, “Carbon emissions from european land transportation: A comprehensive analysis”, *Transportation Research Part D: Transport and Environment*, vol. 121, p. 103 851, 2023, ISSN: 1361-9209. DOI: 10.1016/j.trd.2023.103851 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1361920923002481 (cit. on p. 12).

[73] D. Montolio and A. Solé-Ollé, “Road investment and regional productivity growth: The effects of vehicle intensity and congestion*”, *Papers in Regional Science*, vol. 88, no. 1, pp. 99–119, 2009, ISSN: 1056-8190. DOI: 10.1111/j.1435-5957.2008.00167.x [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1056819023014537 (cit. on p. 12).

[74] ICCT, “European vehicle market statistics 2025/26”, International Council on Clean Transportation, Berlin, Germany, Tech. Rep., Dec. 2025, Published December 14, 2025. [Online]. Available: https://theicct.org/publication/european-vehicle-market-statistics-2025-dec25/ (cit. on p. 13).

[75] B. Anderhofstadt and S. Spinler, “Factors affecting the purchasing decision and operation of alternative fuel-powered heavy-duty trucks in germany – a delphi study”, *Transportation Research Part D: Transport and Environment*, vol. 73, pp. 87–107, 2019, ISSN: 1361-9209. DOI: 10.1016/j.trd.2019.06.003 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1361920918310599 (cit. on p. 13).

[76] Federal Ministry for Climate Action, Environment, Energy, Mobility, Innovation and Technology, *Austria's 2030 mobility master plan*, Accessed: 2025-08-11, 2021. [Online]. Available: https://www.bmimi.gv.at/en/topics/mobility/mobilitymasterplan2030.html (cit. on p. 13).

[77] A. Karam, A. J. K. Jensen, and M. Hussein, “Analysis of the barriers to multimodal freight transport and their mitigation strategies”, *European Transport Research Review*, vol. 15, no. 1, p. 43, 2023. DOI: 10.1186/s12544-023-00614-0 (cit. on pp. 13, 17).

[78] D. M. Z. Islam, S. Ricci, and B.-L. Nelldal, “How to make modal shift from road to rail possible in the European transport market, as aspired to in the EU transport white paper 2011”, *European Transport Research Review*, vol. 8, no. 3, p. 18, 2016. DOI: 10.1007/s12544-016-0204-x (cit. on p. 13).

[79] H. Auer, P. C. del Granado, P.-Y. Oei, K. Hainsch, K. Löffler, T. Burandt, D. Huppmann, and I. Grabaak, “Development and modelling of different decarbonization scenarios of the european energy system until 2050”, *e & i Elektrotechnik und Informationstechnik*, vol. 137, no. 7, pp. 346–358, 2020. DOI: 10.1007/s00502-020-00832-7 (cit. on p. 13).

[80] P. Plötz, “Hydrogen technology is unlikely to play a major role in sustainable road transport”, *Nature Electronics*, vol. 5, no. 1, pp. 8–10, 2022. DOI: 10.1038/s41928-021-00706-6 (cit. on p. 13).

[81] K. H. Baek, X. Wu, Y. Zhou, R. Vijayagopal, N. Kim, and A. Elgowainy, “Total cost of ownership of vehicle electrification and fuel switching options for light-duty and heavy-duty vehicles”, *eTransportation*, vol. 27, p. 100 512, 2026, ISSN: 2590-1168. DOI: 10.1016/j.etran.2025.100512 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2590116825001195 (cit. on p. 13).

[82] B. Noll, S. del Val, T. S. Schmidt, and B. Steffen, “Analyzing the competitiveness of low-carbon drive-technologies in road-freight: A total cost of ownership analysis in europe”, *Applied Energy*, vol. 306, p. 118 079, 2022. DOI: 10.1016/j.apenergy.2021.118079 (cit. on p. 13).

[83] IEA, “Global ev outlook 2024”, International Energy Agency (IEA), Paris, Tech. Rep., 2024, Licence: CC BY 4.0. [Online]. Available: https://www.iea.org/reports/global-ev-outlook-2024 (cit. on p. 13).

[84] J. Martin, E. Dimanchev, and A. Neumann, “Carbon abatement costs for renewable fuels in hard-to-abate transport sectors”, *Advances in Applied Energy*, vol. 12, p. 100 156, 2023, ISSN: 2666-7924. DOI: 10.1016/j.adapen.2023.100156 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2666792423000355 (cit. on pp. 13, 80).

[85] A. Scherrer, M. Helferich, D. Speth, and S. Link, “Requirements for charging infrastructure for battery-electric heavy-duty vehicles”, Tech. Rep., 2024. DOI: 10.24406/publica-2615 (cit. on p. 13).

[86] L. Xiao, L. Chen, P. Sun, G. Laporte, and R. Baldacci, “The container drayage problem for electric trucks with charging resource constraints”, *Transportation Research Part C: Emerging Technologies*, vol. 174, p. 105 100, 2025, ISSN: 0968-090X. DOI: 10.1016/j.trc.2025.105100 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0968090X25001044 (cit. on p. 14).

[87] M. R. Alam and Z. Guo, “Charging station planning and fleet operation for electric freight vehicles: A state-of-the-art review”, *Journal of Transportation Engineering, Part A: Systems*, vol. 151, no. 11, p. 03 125 005, 2025. DOI: 10.1061/JTEPBS.TEENG-8856 eprint: https://ascelibrary.org/doi/pdf/10.1061/JTEPBS.TEENG-8856. [Online]. Available: https://ascelibrary.org/doi/abs/10.1061/JTEPBS.TEENG-8856 (cit. on p. 14).

[88] H. Gillström, "Barriers and enablers: How logistics companies could tackle the transition to electrified road freight transport", *Cleaner Logistics and Supply Chain*, vol. 13, p. 100 172, 2024, ISSN: 2772-3909. DOI: 10.1016/j.clscn.2024.100172 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2772390924000349 (cit. on p. 14).

[89] B. Borlaug, M. Muratori, M. Gilleran, D. Woody, W. Muston, T. Canada, A. Ingram, H. Gresham, and C. McQueen, "Heavy-duty truck electrification and the impacts of depot charging on electricity distribution systems", *Nature Energy*, vol. 6, no. 6, pp. 673–682, 2021. DOI: 10.1038/s41560-021-00855-0 (cit. on pp. 14, 189).

[90] G. Wang, M. R. Miller, and L. M. Fulton, "The infrastructure cost for depot charging of battery electric trucks", *The Electricity Journal*, vol. 38, no. 3, p. 107 490, 2025, ISSN: 1040-6190. DOI: 10.1016/j.tej.2025.107490 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1040619025000351 (cit. on p. 14).

[91] B. Al-Hanahi, I. Ahmad, D. Habibi, and M. A. Masoum, "Smart charging strategies for heavy electric vehicles", *eTransportation*, vol. 13, p. 100 182, 2022, ISSN: 2590-1168. DOI: 10.1016/j.etran.2022.100182 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2590116822000285 (cit. on p. 14).

[92] R. Miyabe, Y. Fujimoto, and Y. Hayashi, "Low-carbon routing and charging planning for electric freight trucks utilizing local surplus solar power", *Journal of Energy Storage*, vol. 132, p. 117 626, 2025, ISSN: 2352-152X. DOI: 10.1016/j.est.2025.117626 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2352152X25023394 (cit. on p. 14).

[93] B. Al-Hanahi, I. Ahmad, D. Habibi, and M. A. S. Masoum, "Charging infrastructure for commercial electric vehicles: Challenges and future works", *IEEE Access*, vol. 9, pp. 121 476–121 492, 2021. DOI: 10.1109/ACCESS.2021.3108817 (cit. on p. 14).

[94] Y. Wan, Z. He, Y. Gao, and Y. Xue, "Long-haul truck charging planning problem considering time flexibility and energy flexibility", *Energy*, vol. 306, p. 132 361, 2024, ISSN: 0360-5442. DOI: 10.1016/j.energy.2024.132361 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0360544224021352 (cit. on p. 14).

[95] M. O. Metais, O. Jouini, Y. Perez, J. Berrada, and E. Suomalainen, "Too much or not enough ? Planning electric vehicle charging infrastructure : A review of modeling options", *Renewable and Sustainable Energy Reviews*, vol. 153, no. November 2020, p. 111 719, 2022, ISSN: 1364-0321. DOI: 10.1016/j.rser.2021.111719 [Online]. Available: https://doi.org/10.1016/j.rser.2021.111719 (cit. on pp. 14, 114).

[96] J. Li, A. Chew, and H. Wang, "Investigating state-of-the-art planning strategies for electric vehicle charging infrastructures in coupled transport and power networks: A comprehensive review", *Progress in Energy*, vol. 6, no. 4, p. 042 004, 2024. DOI: 10.1088/2516-1083/ad6be1 (cit. on pp. 14, 15).

[97] M. Kuby and S. Lim, "The flow-refueling location problem for alternative-fuel vehicles", *Socio-Economic Planning Sciences*, vol. 39, no. 2, pp. 125–145, 2005, ISSN: 0038-0121. DOI: 10.1016/j.seps.2004.03.001 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0038012104000175 (cit. on p. 15).

[98] P. Jochem, E. Szimba, and M. Reuter-Oppermann, "How many fast-charging stations do we need along European highways?", *Transportation Research Part D: Transport and Environment*, vol. 73, no. January, pp. 120–129, 2019, ISSN: 13619209. DOI: 10.1016/j.trd.2019.06.005 [Online]. Available: https://doi.org/10.1016/j.trd.2019.06.005 (cit. on pp. 15, 102–105).

[99] Y. Wang, J. Shi, R. Wang, Z. Liu, and L. Wang, "Siting and sizing of fast charging stations in highway network with budget constraint", *Applied Energy*, vol. 228, no. June, pp. 1255–1271, 2018, ISSN: 03062619. DOI: 10.1016/j.apenergy.2018.07.025 [Online]. Available: https://doi.org/10.1016/j.apenergy.2018.07.025 (cit. on pp. 15, 16, 105).

[100] L. Yan, "Optimal layout and scale of charging stations for electric vehicles", in *2016 China International Conference on Electricity Distribution (CICED)*, 2016, pp. 1–5. DOI: 10.1109/CICED.2016.7575934 (cit. on p. 15).

[101] G. Napoli, A. Polimeni, S. Micari, L. Andaloro, and V. Antonucci, "Optimal allocation of electric vehicle charging stations in a highway network: Part 1. Methodology and test application", *Journal of Energy Storage*, vol. 27, no. April 2019, p. 101 102, 2020, ISSN: 2352152X. DOI: 10.1016/j.est.2019.101102 [Online]. Available: https://doi.org/10.1016/j.est.2019.101102 (cit. on pp. 15, 46).

[102] C. Csiszár, B. Csonka, D. Földes, E. Wirth, and T. Lovas, "Location optimisation method for fast-charging stations along national roads", *Journal of Transport Geography*, vol. 88, no. May 2019, p. 102 833, 2020, ISSN: 0966-6923. DOI: 10.1016/j.jtrangeo.2020.102833 [Online]. Available: https://doi.org/10.1016/j.jtrangeo.2020.102833 (cit. on p. 15).

[103] P. K. Rose, R. Nugroho, T. Gnann, P. Plötz, M. Wietschel, and M. Reuter-Oppermann, "Optimal development of alternative fuel station networks considering node capacity restrictions", *Transportation Research Part D: Transport and Environment*, vol. 78, p. 102 189, 2020, ISSN: 1361-9209. DOI: 10.1016/j.trd.2019.11.018 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1361920919306479 (cit. on p. 15).

[104] R. I. Nugroho, T. Gnann, D. Speth, W. W. Purwanto, J. Hanafi, and S. Soehodho, "Designing optimal fast-charging infrastructure for various electric vehicle ranges in emerging-market city", *Transportation Research Interdisciplinary Perspectives*, vol. 31, p. 101 470, 2025, ISSN: 2590-1982. DOI: 10.1016/j.trip.2025.101470 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2590198225001496 (cit. on pp. 15, 16, 105).

[105] W. u. Rehman, S. Wang, L. Lv, J. W. Kimball, and R. Bo, "Multi-period coordinated planning of xfcs in coupled tn-pdn networks: Integrating demand charge reduction and pre-existing infrastructure", *eTransportation*, vol. 27, 2026, Cited by: 0. DOI: 10.1016/j.etran.2025.100521 [Online]. Available: https://www.scopus.com/inward/record.uri?eid=2-s2.0-105024212175&doi=10.1016%2fj.etran.2025.100521&partnerID=40&md5=e8ff8e6f2e5008a113de593d59ff6895 (cit. on pp. 15, 16).

[106] M. Kavianipour, F. Fakhrmoosavi, M. H. Shojaei, A. Zockaie, M. Ghamami, J. Wang, and R. Jackson, "Impacts of technology advancements on electric vehicle charging infrastructure configuration: a Michigan case study", *International Journal of Sustainable Transportation*, vol. 0, no. 0, pp. 1–14, 2021, ISSN: 15568334. DOI: 10.1080/15568318.2021.

1914789 [Online]. Available: https://doi.org/10.1080/15568318.2021.1914789 (cit. on p. 15).

[107] H. M. Abdullah, A. Gastli, L. Ben-Brahim, and S. O. Mohammed, "Planning and optimizing electric-vehicle charging infrastructure through system dynamics", *IEEE Access*, vol. 10, pp. 17 495–17 514, 2022. DOI: 10.1109/ACCESS.2022.3149944 (cit. on p. 16).

[108] J. Gao, X. Xu, and T. Zhang, "Forecasting the development of clean energy vehicles in large cities: A system dynamics perspective", *Transportation Research Part A: Policy and Practice*, vol. 181, p. 103 969, 2024, ISSN: 0965-8564. DOI: 10.1016/j.tra.2024.103969 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S096585642400017X (cit. on p. 16).

[109] Y. Liao, Ç. Tozluoğlu, F. Sprei, S. Yeh, and S. Dhamal, "Impacts of charging behavior on bev charging infrastructure needs and energy use", *Transportation Research Part D: Transport and Environment*, vol. 116, p. 103 645, 2023, ISSN: 1361-9209. DOI: 10.1016/j.trd.2023.103645 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1361920923000421 (cit. on p. 16).

[110] R. Wolbertus, R. van den Hoed, M. Kroesen, and C. Chorus, "Charging infrastructure roll-out strategies for large scale introduction of electric vehicles in urban areas: An agent-based simulation study", *Transportation Research Part A: Policy and Practice*, vol. 148, pp. 262–285, 2021, ISSN: 0965-8564. DOI: 10.1016/j.tra.2021.04.010 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0965856421001087 (cit. on pp. 16, 107).

[111] P. Patil, K. Kazemzadeh, and P. Bansal, "Integration of charging behavior into infrastructure planning and management of electric vehicles: A systematic review and framework", *Sustainable Cities and Society*, vol. 88, p. 104 265, 2023, ISSN: 2210-6707. DOI: 10.1016/j.scs.2022.104265 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2210670722005704 (cit. on p. 16).

[112] S. Luh, R. Kannan, T. J. Schmidt, and T. Kober, "Behavior matters: A systematic review of representing consumer mobility choices in energy models", *Energy Research & Social Science*, vol. 90, p. 102 596, 2022 (cit. on pp. 16, 28).

[113] G. Venturini, J. Tattini, E. Mulholland, and B. Ó. Gallachóir, "Improvements in the representation of behavior in integrated energy and transport models", *International Journal of Sustainable Transportation*, vol. 13, no. 4, pp. 294–313, 2019. DOI: 10.1080/15568318.2018.1466220 eprint: https://doi.org/10.1080/15568318.2018.1466220. [Online]. Available: https://doi.org/10.1080/15568318.2018.1466220 (cit. on p. 16).

[114] S. Luh, R. Kannan, R. McKenna, T. J. Schmidt, and T. Kober, "Quantifying the impact of travel time duration and valuation on modal shift in swiss passenger transportation", *Applied Energy*, vol. 356, p. 122 412, 2024, ISSN: 0306-2619. DOI: 10.1016/j.apenergy.2023.122412 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0306261923017762 (cit. on pp. 17, 28, 62).

[115] A. Loni and S. Asadi, "Data-driven equitable placement for electric vehicle charging stations: Case study san francisco", *Energy*, vol. 282, p. 128 796, 2023, ISSN: 0360-5442. DOI: 10.1016/j.energy.2023.128796 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0360544223021904 (cit. on p. 17).

[116] D. Batic, V. Stankovic, and L. Stankovic, “Geodemographic aware electric vehicle charging location planning for equitable placement using graph neural networks: Case study of scotland metropolitan areas”, *Energy*, vol. 324, p. 135 834, 2025, ISSN: 0360-5442. DOI: 10.1016/j.energy.2025.135834 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0360544225014768 (cit. on pp. 17, 106).

[117] G.-D. Voinea, F. Gîrbacia, M. Duguleană, and C.-C. Postelnicu, “Planning future ev charging infrastructure by forecasting spatio-temporal adoption trends across heterogeneous user segments”, *Information*, vol. 16, no. 11, 2025, ISSN: 2078-2489. DOI: 10.3390/info16110933 [Online]. Available: https://www.mdpi.com/2078-2489/16/11/933 (cit. on pp. 17, 106).

[118] R. Wang, H.-K. Kwon, K. H. Jordan, S. J. Moura, M. Boloor, and M. L. Machala, “Leveraging commuting patterns and workplace charging to advance equitable ev charger access”, *Transportation Research Part D: Transport and Environment*, vol. 150, p. 105 089, 2026, ISSN: 1361-9209. DOI: 10.1016/j.trd.2025.105089 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1361920925004997 (cit. on p. 17).

[119] S. J. Bakker, J. Martin, E. R. van Beesten, I. S. Brynildsen, A. Sandvig, M. Siqveland, and A. Golab, “Stram: A strategic network design model for national freight transport decarbonization”, *Transportation Research Part E: Logistics and Transportation Review*, vol. 197, p. 104 076, 2025, ISSN: 1366-5545. DOI: 10.1016/j.tre.2025.104076 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1366554525001176 (cit. on pp. 17, 18, 28, 80).

[120] C. Pagnier, S. J. Bakker, and P. Seljom, “Linking freight transport and energy system models for strategic investments decisions”, *Transportation Research Part D: Transport and Environment*, vol. 150, p. 105 114, 2026, ISSN: 1361-9209. DOI: 10.1016/j.trd.2025.105114 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1361920925005243 (cit. on pp. 18, 109).

[121] B. Li, Z. Ma, P. Hidalgo-Gonzalez, A. Lathem, N. Fedorova, G. He, H. Zhong, M. Chen, and D. M. Kammen, “Modeling the impact of evs in the chinese power system: Pathways for implementing emissions reduction commitments in the power and transportation sectors”, *Energy Policy*, vol. 149, p. 111 962, 2021, ISSN: 0301-4215. DOI: 10.1016/j.enpol.2020.111962 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S030142152030673X (cit. on pp. 18, 111).

[122] A. Suski, T. Remy, D. Chattopadhyay, C. S. Song, I. Jaques, T. Keskes, and Y. Li, “Analyzing electric vehicle load impact on power systems: Modeling analysis and a case study for maldives”, *IEEE Access*, vol. 9, pp. 125 640–125 657, 2021. DOI: 10.1109/ACCESS.2021.3111001 (cit. on p. 18).

[123] F. Safdarian, D. Wallison, J. L. Wert, O. O. Oshinkoya, J. Kyo Jung, T. J. Overbye, and Y. Xu, “Technical impacts of light-duty and heavy-duty transportation electrification on a coordinated transmission and distribution system”, *IEEE Transactions on Transportation Electrification*, vol. 11, no. 1, pp. 3404–3417, 2025. DOI: 10.1109/TTE.2024.3440334 (cit. on p. 18).

[124] Z. Wang, J.-P. Sasse, and E. Trutnevyte, “Home or workplace charging? spatio-temporal flexibility of electric vehicles within swiss electricity system”, *Energy*, vol. 320, p. 135 452, 2025, ISSN: 0360-5442. DOI: 10.1016/j.energy.2025.135452 [Online]. Available: https:

//www.sciencedirect.com/science/article/pii/S0360544225010941 (cit. on pp. 18, 28).

[125] H. Yang and M. G. H. Bell, "Models and algorithms for road network design: A review and some new developments", *Transport Reviews*, vol. 18, no. 3, pp. 257–278, 1998. DOI: 10.1080/01441649808717016 eprint: https://doi.org/10.1080/01441649808717016. [Online]. Available: https://doi.org/10.1080/01441649808717016 (cit. on pp. 26, 28).

[126] L. Fridstrøm, V. Østli, and K. W. Johansen, "A stock-flow cohort model of the national car fleet", *European Transport Research Review*, vol. 8, no. 3, p. 22, 2016, ISSN: 1866-8887. DOI: 10.1007/s12544-016-0210-z [Online]. Available: https://doi.org/10.1007/s12544-016-0210-z (cit. on p. 28).

[127] J. Martin, A. Golab, G. Durakovic, S. Zwickl-Bernhard, H. Auer, and A. Neumann, "Modeling cost-optimal fuel choices for truck, ship, and airplane fleets: The impact of sustainability commitments", *Energy*, vol. 308, p. 132 882, 2024, ISSN: 0360-5442. DOI: 10.1016/j.energy.2024.132882 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0360544224026562 (cit. on p. 28).

[128] United Nations Framework Convention on Climate Change, *Paris declaration on electromobility and climate change and call to action*, 2015 (cit. on p. 45).

[129] BEO, *E-Mobilität in Zahlen*, https://www.beoe.at/statistik/, (accessed on 27 January 2022), 2022 (cit. on p. 45).

[130] BMK, *Österreich unterwegs 2013/2014*, https://www.bmk.gv.at/themen/verkehrsplanung/statistik/oesterreich_unterwegs.html, (accessed on 27 January 2022), 2014 (cit. on p. 45).

[131] P. Jochem, C. Brendel, M. Reuter-Oppermann, W. Fichtner, and S. Nickel, "Optimizing the allocation of fast charging infrastructure along the German autobahn", *Journal of Business Economics*, vol. 86, no. 5, pp. 513–535, 2016, ISSN: 18618928. DOI: 10.1007/s11573-015-0781-5 (cit. on pp. 45, 46).

[132] A. Frei, T. G. Kuhnimhof, and K. W. Axhausen, "Long distance travel in europe today: Experiences with a new survey", *Arbeitsberichte Verkehrs-und Raumplanung*, vol. 611, 2010 (cit. on p. 45).

[133] ASFINAG, *Verkehrsentwicklung*, https://www.asfinag.at/verkehr-sicherheit/verkehrszahlung/, (accessed on 27 January 2022), 2022 (cit. on p. 45).

[134] BMK, *Mobilitätsmasterplan 2030 Neuausrichtung des Mobilitätssektors*, https://www.bmk.gv.at/themen/mobilitaet/mobilitaetsmasterplan/mmp2030.html, (accessed on 14 February 2022), 2018 (cit. on pp. 46, 48, 143, 144).

[135] M. Gholamrezaei and K. Ghorbanian, "Rotated general regression neural network", 6, vol. 2, 2007, pp. 1959–1964, ISBN: 142441380X. DOI: 10.1109/IJCNN.2007.4371258 (cit. on pp. 46, 50).

[136] J. Neubauer and E. Wood, "The impact of range anxiety and home, workplace, and public charging infrastructure on simulated battery electric vehicle lifetime utility", *Journal of Power Sources*, vol. 257, pp. 12–20, 2014, ISSN: 0378-7753. DOI: 10.1016/j.jpowsour.2014.01.075 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0378775314000998 (cit. on p. 46).

[137] Österreichischer Verein für Elektrotechnik, "Ladeinfrastruktur für Elektrofahrzeuge : Bedarf , Kosten und Auswirkungen auf die Energieversorgung in Österreich bis 2030", Tech. Rep., 2019 (cit. on p. 46).

[138] C. Suarez and W. Martinez, "Fast and Ultra-Fast Charging for Battery Electric Vehicles - A Review", IEEE, 2019, pp. 569–575, ISBN: 9781728103952. DOI: 10.1109/ECCE.2019.8912594 (cit. on p. 46).

[139] OpenStreetMap contributors, *Planet dump retrieved from https://planet.osm.org*, https://www.openstreetmap.org, 2017 (cit. on pp. 46, 144).

[140] ASFINAG, *Rastanlagensuche*, https://www.asfinag.at/parken-rasten/rastanlagensuche/, Accessed: 2026-01-29, 2026 (cit. on pp. 46, 98, 104, 144).

[141] openEntrance, *openEntrance*, https://openentrance.eu/, (accessed on 18 January 2022), 2019 (cit. on p. 46).

[142] H. Auer, P. C. del Granado, S. Backe, P. Pisciella, and K. Hainsch, "Storylines for Low Carbon Futures of the European Energy System. Deliverbale 7.1.", Tech. Rep. 7, 2019 (cit. on p. 47).

[143] K. Hainsch, K. Löffler, T. Burandt, H. Auer, P. Crespo del Granado, P. Pisciella, and S. Zwickl-Bernhard, "Energy transition scenarios: What policies, societal attitudes, and technology developments will realize the eu green deal?", *Energy*, vol. 239, p. 122 067, 2022, ISSN: 0360-5442. DOI: 10.1016/j.energy.2021.122067 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S036054422102315X (cit. on p. 47).

[144] P. Andersen et al., "Innovation outlook: Smart charging for electric vehicles", *Int. Renewable Energy Agency*, 2019 (cit. on pp. 48, 96).

[145] A. Thielmann, M. Wietschel, S. Funke, A. Grimm, T. Hettesheimer, S. Langkau, A. Loibl, C. Moll, C. Neef, P. Plötz, L. Sievers, L. T. Espinoza, and J. Edler, "Batterien für Elektroautos: Faktencheck und Handlungsbedarf", Tech. Rep., 2020, p. 36. [Online]. Available: //hdl.handle.net/10419/233461 (cit. on p. 48).

[146] E. Suomalainen and F. Colet, "A corridor-based approach to estimating the costs of electric vehicle charging infrastructure on highways", *World Electric Vehicle Journal*, vol. 10, no. 4, 2019, ISSN: 20326653. DOI: 10.3390/wevj10040068 (cit. on p. 48).

[147] F. Baumgarte, M. Kaiser, and R. Keller, "Policy support measures for widespread expansion of fast charging infrastructure for electric vehicles", *Energy Policy*, vol. 156, no. May, p. 112 372, 2021, ISSN: 03014215. DOI: 10.1016/j.enpol.2021.112372 [Online]. Available: https://doi.org/10.1016/j.enpol.2021.112372 (cit. on p. 48).

[148] A. Genovese, F. Ortenzi, and C. Villante, "On the energy efficiency of quick DC vehicle battery charging", vol. 7, pp. 570–576, 2015 (cit. on p. 49).

[149] Eurostat, *Road transport by type of vehicle and region [tran_r_vehst]*, https://ec.europa.eu/eurostat/databrowser/view/tran_r_vehst__custom_16758639/default/table?lang=en, Accessed: 2025-05-19, 2024 (cit. on p. 58).

[150] Eurostat. "Stock of vehicles by category, fuel type and nuts 2 regions (road_eqs_zev)", Accessed: May 19, 2025. [Online]. Available: https://ec.europa.eu/eurostat/databrowser/view/road_eqs_zev/default/table?lang=en&category=road.road_eqs (cit. on p. 58).

[151] ETISplus, *ETISplus - the european transport policy information system*, https://cordis.europa.eu/project/id/233596, Funded by the European Union under FP7-TRANSPORT, Grant agreement ID: 233596, 2012 (cit. on pp. 58, 108).

[152] Arval, *Annual fleet of vehicles in use in spain from 2012 to 2022, with a forecast through 2026 (in million units) [graph]*, https://www.statista.com/statistics/454353/fleet-of-vehicles-in-use-in-spain-projection/, In Statista. Retrieved May 19, 2025, Jul. 2023 (cit. on p. 58).

[153] Statista, *Gross domestic product (gdp) growth in spain from 2004 to 2024*, https://www.statista.com/statistics/263610/gross-domestic-product-gdp-growth-in-spain/, Accessed: 2025-04-28, 2024 (cit. on p. 58).

[154] I. Tikoudis, A. Papu Carrone, R. Mba Mebiame, N. Lamhauge, K. Hassett, and O. Bystrom, "Household transport choices: New empirical evidence and policy implications for sustainable behaviour", OECD Publishing, OECD Environment Working Papers 246, 2024. DOI: 10.1787/0e8469ed-en [Online]. Available: https://dx.doi.org/10.1787/0e8469ed-en (cit. on p. 59).

[155] T. Chatterton, J. Anable, S. Cairns, and R. Wilson, "Financial implications of car ownership and use: A distributional analysis based on observed spatial variance considering income and domestic energy costs", *Transport Policy*, vol. 65, pp. 30–39, 2018, Household transport costs, economic stress and energy vulnerability, ISSN: 0967-070X. DOI: 10.1016/j.tranpol.2016.12.007 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0967070X16308435 (cit. on p. 59).

[156] Eurostat, *Income, Consumption and Wealth - Experimental Statistics (ICW)*, Accessed: 2025-05-20, 2024. [Online]. Available: https://ec.europa.eu/eurostat/cache/metadata/en/icw_esms.htm (cit. on p. 59).

[157] G. Mattioli, M. Dugato, and I. Philips, "Vulnerability to motor fuel price increases: Socio-spatial patterns in italy", in *Vulnerable Households in the Energy Transition: Energy Poverty, Demographics and Policies*, R. Bardazzi and M. G. Pazienza, Eds. Cham: Springer International Publishing, 2023, pp. 89–115, ISBN: 978-3-031-35684-1. DOI: 10.1007/978-3-031-35684-1_5 [Online]. Available: https://doi.org/10.1007/978-3-031-35684-1_5 (cit. on p. 59).

[158] Eurostat, *Distribution of income by different income groups (eu-silc and echp surveys)*, Last update: 16 June 2025; coverage years: 1995-2024, 2025. DOI: 10.2908/ILC_DI02 [Online]. Available: https://ec.europa.eu/eurostat/databrowser/view/ilc_di02/default/table (cit. on p. 60).

[159] Statista Research Department, *Average weekly household expenditure on new car and van purchases in the united kingdom (uk) in 2023, by gross income decile group*, https://www.statista.com/statistics/285881/new-car-and-van-purchases-weekly-uk-household-expenditure-by-gross-income/, Accessed: 2025-04-22, 2023 (cit. on p. 60).

[160] Office for National Statistics (UK), *Average weekly household expenditure on vehicle purchases in the united kingdom (uk) in 2023, by gross income decile group (in gbp) [graph]*, Retrieved April 28, 2025, from Statista, Aug. 2024. [Online]. Available: https://www.statista.com/statistics/285877/car-and-vehicle-purchases-weekly-uk-household-expenditure-by-gross-income (cit. on p. 60).

[161] Statista, *Spain: Share of homeowners 2022*, Accessed: 2025-05-20, 2022. [Online]. Available: https://www.statista.com/statistics/543390/house-owners-among-population-spain/ (cit. on p. 60).

[162] T. Grube, S. Kraus, J. Reul, and D. Stolten, "Passenger car cost development through 2050", *Transportation Research Part D: Transport and Environment*, vol. 101, p. 103 110, 2021, ISSN: 1361-9209. DOI: 10.1016/j.trd.2021.103110 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1361920921004065 (cit. on pp. 60, 61).

[163] K. Löffler, N. Moskalenko, P. Herpich, J. Hanto, K. Hainsch, J. Bornemann, A. Diesing, R. Dupke, and M. Barani, *The european energy vision 2060 (eu envis-2060): Scenario parametrization (3.1) [data set]*, Zenodo, 2025. DOI: 10.5281/zenodo.17079324 [Online]. Available: https://doi.org/10.5281/zenodo.17079324 (cit. on pp. 60, 61).

[164] J. Martin, A. Neumann, and A. Ødegård, "Renewable hydrogen and synthetic fuels versus fossil fuels for trucking, shipping and aviation: A holistic cost model", *Renewable and Sustainable Energy Reviews*, vol. 186, p. 113 637, 2023, ISSN: 1364-0321. DOI: 10.1016/j.rser.2023.113637 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S136403212300494X (cit. on pp. 60, 80).

[165] W. Jia, Z. Jiang, Q. Wang, B. Xu, and M. Xiao, "Preferences for zero-emission vehicle attributes: Comparing early adopters with mainstream consumers in california", *Transport Policy*, vol. 135, pp. 21–32, 2023 (cit. on p. 60).

[166] Dirección General de Tráfico, *Anuario Estadístico General 2023*, Gobierno de España, 2024. [Online]. Available: https://www.dgt.es/menusecundario/dgt-en-cifras/dgt-en-cifras-resultados/dgt-en-cifras-detalle/Anuario-Estadistico-General-2023/ (cit. on p. 61).

[167] K. Löffler, K. Hainsch, T. Burandt, P.-Y. Oei, C. Kemfert, and C. Von Hirschhausen, "Designing a model for the global energy system-genesys-mod: An application of the open-source energy modeling system (osemosys)", *Energies*, vol. 10, no. 10, p. 1468, 2017 (cit. on p. 61).

[168] EAFO, *Quarterly public electric vehicle charging point volume in spain between q1 2020 and q4 2023, by current type [graph]*, https://www.statista.com/statistics/1385003/spain-quarterly-public-ev-charging-point-volume-by-current-type/, In Statista. Retrieved May 21, 2025, Jan. 2024 (cit. on p. 62).

[169] Eurostat, *Number of zero-emission vehicles (zevs)*, https://ec.europa.eu/eurostat/databrowser/view/road_eqs_zev/default/table?lang=en&category=road.road_eqs, Retrieved May 21, 2025, 2025 (cit. on p. 62).

[170] European Commission. "Trans-European Transport Network (TEN-T)". Accessed: 2025. [Online]. Available: https://transport.ec.europa.eu/transport-themes/infrastructure-and-investment/trans-european-transport-network-ten-t_en (cit. on p. 76).

[171] European Commission. "TEN-T Maps". Accessed: 2025. [Online]. Available: https://ec.europa.eu/transport/infrastructure/tentec/tentec-portal/map/maps.html (cit. on p. 76).

[172] European Commission. “Strategic Corridor Management - Final Report on Elements of Work Plan”. Accessed: 2025. [Online]. Available: https://transport.ec.europa.eu/system/files/2018-07/scm_corridor_study_final_report_on_elements_of_work_plan.pdf (cit. on p. 76).

[173] Eurostat. “Nomenclature of Territorial Units for Statistics (NUTS) - Overview”. Accessed: 2025. [Online]. Available: https://ec.europa.eu/eurostat/web/nuts/overview (cit. on p. 77).

[174] D. Speth, V. Sauter, P. Plötz, and T. Signer, *Synthetic european road freight transport flow data based on etisplus*, Mendeley Data, version V1, 2021. DOI: 10.17632/py2zkrb65h.1 (cit. on pp. 77, 80, 95, 180).

[175] Eurostat, *Road freight transport by journey characteristics*, Statistics Explained, Data extracted August 2024. Accessed: 16 December 2025, 2024. [Online]. Available: https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Road_freight_transport_by_journey_characteristics (cit. on pp. 77, 180).

[176] W. Shoman, D. Speth, and P. Plötz, “Public charging infrastructure for battery electric trucks”, Tech. Rep., 2023 (cit. on pp. 77, 80, 112, 113, 180).

[177] European Commission, *TENtec interactive map viewer*, https://webgate.ec.europa.eu/tentec-maps/web/public/screen/home, Accessed: 2025-02-28, 2024 (cit. on p. 78).

[178] Transport & Environment, “How to decarbonise long-haul trucking in germany. an analysis of available vehicle technologies and their associated costs”, Tech. Rep., 2021 (cit. on pp. 80, 81).

[179] Umweltbundesamt, *Environmental impact of different modes of transportation*, PDF Document, 2023. [Online]. Available: https://www.umweltbundesamt.at/fileadmin/site/themen/mobilitaet/daten/ekz_pkm_tkm_verkehrsmittel.pdf (cit. on p. 80).

[180] H. Basma and F. Rodríguez, “A total cost of ownership comparison of truck decarbonization pathways in Europe”, International Council on Clean Transportation, Working Paper 2023-28, Nov. 2023. [Online]. Available: https://theicct.org/wp-content/uploads/2023/11/ID-54-EU-HDV-TCO-paper-working-paper-28-A4-50145-v2.pdf (cit. on p. 80).

[181] J. Karlsson and A. Grauers, “Agent-based investigation of charger queues and utilization of public chargers for electric long-haul trucks”, *Energies*, vol. 16, no. 12, 2023, ISSN: 1996-1073. DOI: 10.3390/en16124704 [Online]. Available: https://www.mdpi.com/1996-1073/16/12/4704 (cit. on p. 80).

[182] M. Barani, K. Löffler, P. Crespo del Granado, N. Moskalenko, E. Panos, F. M. Hoffart, C. von Hirschhausen, M. Kannavou, H. Auer, and K. Hainsch, “European energy vision 2050 and beyond: Designing scenarios for europe’s energy transition”, *Renewable and Sustainable Energy Reviews*, vol. 225, p. 115 758, 2025. DOI: 10.1016/j.rser.2025.115758 (cit. on pp. 80, 82).

[183] M. Janić, “Modelling the full costs of an intermodal and road freight transport network”, *Transportation Research Part D: Transport and Environment*, vol. 12, no. 1, pp. 33–44, 2007. DOI: 10.1016/j.trd.2006.10.004 (cit. on p. 80).

[184] B. Wiegmans and B. Behdani, “A review and analysis of the investment in, and cost structure of, intermodal rail terminals”, *Transport Reviews*, vol. 38, no. 1, pp. 33–51, 2018. DOI: 10.1080/01441647.2017.1297867 (cit. on p. 81).

[185] K. Löffler, K. Hainsch, T. Burandt, P.-Y. Oei, C. Kemfert, and C. von Hirschhausen, "Designing a model for the global energy system—GENeSYS-MOD: An application of the open-source energy modeling system (OSeMOSYS)", *Energies*, vol. 10, no. 10, p. 1468, 2017. DOI: 10.3390/en10101468 (cit. on p. 81).

[186] Bundesministerium für Finanzen, *Leichte nutzfahrzeuge - normverbrauchsabgabe (nova)*, Accessed: 2024-08-08, 2023. [Online]. Available: https://www.usp.gv.at/steuern-finanzen/weitere-steuern-und-abgaben/normverbrauchsabgabe-nova/sonstige-faelle/leichte-nutzfahrzeuge.html (cit. on p. 95).

[187] A. Angelini, H. Heinfellner, P. Pfaffenbichler, and M. Schwingshackl, "Transition mobility 2040", Umweltbundesamt, Tech. Rep., 2022, Accessed: 2024-08-08. [Online]. Available: https://www.umweltbundesamt.at/fileadmin/site/publikationen/rep0808.pdf (cit. on pp. 95, 96, 111, 113).

[188] M. Scorrano, R. Danielis, and M. Giansoldati, "Electric light commercial vehicles for a cleaner urban goods distribution. are they cost competitive?", *Research in Transportation Economics*, vol. 85, p. 101 022, 2021, Policies prompting sustainable transport in urban areas, ISSN: 0739-8859. DOI: 10.1016/j.retrec.2020.101022 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0739885920302201 (cit. on pp. 95, 96).

[189] European Commission, *Driving time and rest periods*, Accessed: 2024-08-08, 2023. [Online]. Available: https://transport.ec.europa.eu/transport-modes/road/social-provisions/driving-time-and-rest-periods_en (cit. on p. 95).

[190] Bundesministerium für Finanzen, *Lkw-fahrverbote*, Accessed: 2024-08-08, 2023. [Online]. Available: https://www.usp.gv.at/betrieb-und-umwelt/transport-und-verkehr/lkw-fahrverbote.html (cit. on p. 95).

[191] R. Teoh, K. Craig, D. Parkes, J. Pagenkopf, and R. Hoefnagels, "Decarbonising road freight transport", *Energy Policy*, 2022 (cit. on p. 96).

[192] Umweltbundesamt, "Emissionskennzahlen datenbasis 2022", Umweltbundesamt, Tech. Rep., 2024, Accessed: 2024-08-08. [Online]. Available: https://www.umweltbundesamt.at/fileadmin/site/themen/mobilitaet/daten/ekz_pkm_tkm_verkehrsmittel.pdf (cit. on p. 96).

[193] K. K. Fjær, V. Lakshmanan, B. N. Torsæter, and M. Korpås, "Heavy-duty electric vehicle charging profile generation method for grid impact analysis", in *2021 International Conference on Smart Energy Systems and Technologies (SEST)*, 2021, pp. 1–6. DOI: 10.1109/SEST50973.2021.9543135 (cit. on p. 96).

[194] Wiener Linien, *Die wiener öffis in zahlen*, Accessed: 2024-08-08, 2024. [Online]. Available: https://www.wienerlinien.at/die-wiener-oeffis-in-zahlen (cit. on p. 96).

[195] VMobil, "Gf-bericht 2022", VMobil, Tech. Rep., 2023, Accessed: 2024-08-08. [Online]. Available: https://www.vmobil.at/sites/default/files/2023-07/Gf-Bericht%202022_klein.pdf (cit. on p. 96).

[196] Daimler Truck AG and C. Will, "Flexibility marketing options for charging processes of electric medium-duty and heavy-duty commercial vehicles", Tech. Rep., 2022 (cit. on p. 96).

[197] IEA, *Type and location of public charging in selected regions and countries and the share of population living within 1 kilometre of a charger, 2024*, International Energy Agency, Paris, Licence: CC BY 4.0, 2025. [Online]. Available: https://www.iea.org/data-and-statistics/charts/type-and-location-of-public-charging-in-selected-regions-and-countries-and-the-share-of-population-living-within-1-kilometre-of-a-charger-2024 (cit. on p. 103).

[198] ASFINAG, *Verkehrszählung*, Accessed: 2026-02-19, 2025. [Online]. Available: https://www.asfinag.at/verkehr-sicherheit/verkehrszaehlung/ (cit. on p. 104).

[199] IÖB-Innovationsplattform, *Energiespeicher für rastanlagen*, Accessed: 2024-08-08, 2023. [Online]. Available: https://www.ioeb-innovationsplattform.at/challenges/detail/energiespeicher-fuer-rastanlagen/ (cit. on pp. 112, 113, 188).

[200] *Meistverkaufte elektroautos in österreich nach modell 2020*, de, https://de.statista.com/statistik/daten/studie/575997/umfrage/meistverkaufte-elektrofahrzeuge-in-oesterreich-nach-modell/, (accessed on 14 February 2022) (cit. on p. 143).

[201] P. Hezel, *Anzahl der monatlichen Neuzulassungen von Elektroautos in Österreich nach Modell von Januar bis November 2021*, https://de.statista.com/statistik/daten/studie/993281/umfrage/pkw-neuzulassungen-mit-elektroantrieb-in-oesterreich-nach-modell/, [Online; accessed 18-January-2022], 2021 (cit. on p. 143).

[202] E. Database, *EV Database - v4.3*, https://ev-database.org/, (accessed on 18 January 2022), 2021 (cit. on p. 143).

[203] Environmental Systems Research Institute, *Arcgis desktop: Release 10*, 2011 (cit. on p. 144).

[204] R. Tomschy, M. Herry, G. Sammer, R. Klementschitz, S. Riegler, R. Follmer, D. Gruschwitz, F. Josef, S. Gensasz, R. Kirnbauer, et al., "Österreich unterwegs 2013/2014: Ergebnisbericht zur österreichweiten mobilitätserhebung „österreich unterwegs 2013/2014 "", Tech. Rep., 2016 (cit. on p. 157).

[205] M. Lubin, O. Dowson, J. Dias Garcia, J. Huchette, B. Legat, and J. P. Vielma, "JuMP 1.0: Recent improvements to a modeling language for mathematical optimization", *Mathematical Programming Computation*, 2023. DOI: 10.1007/s12532-023-00239-3 (cit. on pp. 158, 184).

[206] Gurobi Optimization, LLC, *Gurobi Optimizer Reference Manual*, 2021. [Online]. Available: https://www.gurobi.com (cit. on pp. 158, 184).

[207] M. Costa, G. Valença, C. Adorean, and F. Moura, "The relation between circuity and mobility cultures: A study of 41 european cities", *Cities*, vol. 166, p. 106 087, 2025, ISSN: 0264-2751. DOI: 10.1016/j.cities.2025.106087 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0264275125003877 (cit. on p. 159).

[208] H. Kweon, "Comparisons of estimated circuity factor of forest roads with different vertical heights in mountainous areas, republic of korea", *Forests*, vol. 10, no. 12, 2019, ISSN: 1999-4907. DOI: 10.3390/f10121147 [Online]. Available: https://www.mdpi.com/1999-4907/10/12/1147 (cit. on p. 159).

[209] Open Charge Map Contributors, *Open charge map: The global public registry of electric vehicle charging locations*, https://openchargemap.org, Accessed: 2026-01-25, 2011 (cit. on pp. 164, 165).

[210] Eurostat, *Gisco: Geographical information and maps – territorial units statistics*, https://ec.europa.eu/eurostat/web/gisco/geodata/statistical-units/territorial-units-statistics, Accessed: 2026-01-25, European Commission, 2024 (cit. on p. 164).

[211] T. Litman, "Travel time costs", Victoria Transport Policy Institute, Victoria, BC, Transportation Cost and Benefit Analysis II 5.2, 2023, Accessed: 2026-01-25. [Online]. Available: https://www.vtpi.org/tca/tca0502.pdf (cit. on p. 166).

[212] M. Wardman, K. Neki, and R. M. Humphreys, "Meta-analysis of the value of travel time savings in low- and middle-income countries", World Bank, Washington, DC, Mobility and Transport Connectivity Series, Jun. 2023, Accessed: 2026-01-25. [Online]. Available: https://documents1.worldbank.org/curated/en/099032124211022462/pdf/P1775471cf971e0ee1becd1a1faab55d1d7.pdf (cit. on p. 166).

[213] Eurostat, *Income quintile share ratio (s80/s20) for disposable income [ilc_di11]*, Eurostat Database, Accessed: 2026-01-25, 2024. [Online]. Available: https://ec.europa.eu/eurostat/databrowser/view/ilc_di11/default/table (cit. on p. 166).

[214] G. de Rus, J. Campos, D. Graham, M. P. Socorro, and J. Valido, "Evaluación económica de proyectos y políticas de transporte: Metodología y aplicaciones", FEDEA, Madrid, Working Paper 2020-11, 2020. [Online]. Available: https://www.airef.es/wp-content/uploads/2020/07/INFRAESTRUCTURAS/Anexo-2.-Metodolog%5C%C3%5C%ADa-para-el-an%5C%C3%5C%A1lisis-coste-beneficio-de-proyectos-y-pol%5C%C3%5C%ADticas-de-transporte.pdf (cit. on p. 167).

[215] M. Wardman, V. P. K. Chintakayala, and G. de Jong, "Values of travel time in europe: Review and meta-analysis", *Transportation Research Part A: Policy and Practice*, vol. 94, pp. 93–111, 2016, ISSN: 0965-8564. DOI: 10.1016/j.tra.2016.08.019 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0965856415302810 (cit. on p. 167).

[216] C. Loschan, D. Schwabeneder, G. Lettner, and H. Auer, "Flexibility potential of aggregated electric vehicle fleets to reduce transmission congestions and redispatch needs: A case study in austria", *International Journal of Electrical Power & Energy Systems*, vol. 146, p. 108 802, 2023. DOI: 10.1016/j.ijepes.2022.108802 (cit. on pp. 186, 187).

[217] C. Loschan, H. Auer, and G. Lettner, "Synergies and competition: Examining flexibility options in the european electricity market", *International Journal of Electrical Power & Energy Systems*, vol. 159, p. 109 992, 2024. DOI: 10.1016/j.ijepes.2024.109992 (cit. on p. 186).

Appendix A.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the author used Claude Code (Anthropic) in order to format and spell-check the written text, ensure terminological consistency across chapters, assist with grammar and language editing, and support LaTeX typesetting. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published work.

Appendix B.

Supplementary Material for Contribution 1 (RQ1)

The content of this appendix is based on Golab et al. (2022).

B.1. Nomenclature

Table B.1 lists the complete nomenclature for the fast-charging infrastructure allocation model described in Section 3.2.

Table B.1.: Nomenclature for the fast-charging infrastructure allocation model (RQ1).

Sets and indices	Description		
$n, l \in \{0, \dots, N\}$	node index		
$k \in \{0, 1\}$	driving direction		
$s \in \{0, \dots, M\}$	highway segment index		
A_{n_k}	set of nodes accessible within $dist_{max}$ from node n in direction k		

Decision variables	Description	Unit	Type / Domain
X_l	whether a charging station is installed at node l	–	$\{0, 1\}$ (binary)
Y_{l_k}	number of charging points at node l_k	–	$\mathbb{Z}_{\geq 0}$ (integer)
$\hat{E}_{l_k}^{charged,n_k}$	charged energy at node l_k of demand from node n_k during peak hour	kWh/h	$[0, \infty[$
$\hat{D}_{l_k}^{in,n_k}$	energy demand from node n_k, incoming to node l_k during peak hour	kWh/h	$[0, \infty[$
$\hat{D}_{l_k}^{out,n_k}$	energy demand from node n_k, not covered and outgoing from l_k during peak hour	kWh/h	$[0, \infty[$
$\hat{D}_{l_k}^{demand,n_k}$	energy demand at node l_k originating from demand source n_k during peak hour	kWh/h	$[0, \infty[$

Parameters – car fleet	Description	Unit	Domain
ϵ	share of BEVs in total passenger car fleet	%	$[0, 1]$
μ	share of cars traveling long-distance	%	$[0, 1]$
γ_h	share of daily car count traveling during peak hour	%	$[0, 1]$
$\hat{f}_{n_k}$	maximum daily amount of cars passing at node n in direction k during a year	vehicles/day	$[0, \infty[$
α	scaling factor of overall road traffic load relative to year of survey of traffic counts	–	$[0, \infty[$

Parameters – BEV technology	Description	Unit	Domain
$\overline{d}_{spec,BEV}$	average specific energy demand per km of BEVs	kWh/km	$[0, \infty[$
$\overline{dist}_{range,BEV}$	average driving range of BEVs	km	$[0, \infty[$
$\overline{P}_{charge,BEV}$	average charging capacity of BEVs	kW	$[0, \infty[$

Parameters – infrastructure	Description	Unit	Domain
$\hat{P}_{CP}$	peak power level of a charging point	kW	$[0, \infty[$

P_{max}	maximum capacity installed at a charging station	kW	$[0, \infty[$
c_X	investment costs for installation of a charging station	€	$[0, \infty[$
c_Y	investment costs for installation of one charging point	€	$[0, \infty[$

Derived parameters	Description	Unit	Domain
$dist_{ns_k}$	distance of node n along segment s in direction k	km	$[0, \infty[$
$GRNN_{s_k}(dist)$	GRNN function of peak daily traffic load along segment s in direction k	vehicles/day	$[0, \infty[$
$\hat{D}_{n_k}$	average energy demand at node n in direction k during peak hour	kWh/h	$[0, \infty[$
$dist_{max}$	maximum distance between charging stations	km	$[0, \infty[$
s_{l_k}	highway segment to which node l in direction k belongs	–	$\{0, \ldots, M\}$
r_{l_k, l'_k}	rationing parameter for demand flow split at junction nodes	–	$[0, 1]$

B.2. Computational details

The optimization model is implemented in Python 3.8 using Pyomo 6.2 as the algebraic modeling language and Gurobi 9.1.2 as the solver. The Austrian highway network is represented by 249 nodes (139 service areas and 110 junctions) along 55 highway segments. Depending on the assumed BEV driving range, the resulting mixed-integer linear program comprises approximately 1.3 million constraints. Solve times range from 650 seconds (at a driving range of 400 km) to 1,360 seconds (at 800 km), with a maximum optimality gap (MIPgap) of 0.2%.

The source code and input data are publicly available at: `https://github.com/antoniagolab/HighCharge`.

B.3. Determination of average technological parameters for BEVs

To evaluate the technological parameters of an average battery-electric vehicle (BEV) in the Austrian car fleet, the technological attributes of the most sold car models from the last three years were used. These are displayed in Table B.2 with the respective number of sales. Using sale numbers as weights, the weighted averages for the driving range and charging capacity were evaluated.

Table B.2.: Top 10 sold car models in Austria in 2019–2021 and technical attributes [200, 201, 202] (average driving range $\overline{dist}_{range,BEV}$, average charging power $\overline{P}_{charge,BEV}$, average specific energy demand $\overline{d}_{spec,BEV}$).

Car model	Sales 2019	Sales 2020	Sales 2021	$\overline{dist}_{range,BEV}$ (km)	$\overline{P}_{charge,BEV}$ at $\hat{P}_{CP} = 150kW$ (kW)	$\overline{d}_{spec,BEV}$ highway - cold weather (kWh/km)
Tesla Model 3	2342	2892	3304	380	110	0.23
Renault Zoe	944	2071	1778	310	41	0.22
Kia Niro	1125	421	924	370	77	0.24
Hyundai Kona	897	861	-	395	64	0.28
Audi e-Tron	364	782	1192	285	114	0.29
BMW i3	1191	697	-	235	47	0.23
VW e-Golf	805	401	-	190	39	0.24
VW Up!	-	376	-	205	30	0.22
Mazda MX-30	-	370	-	170	34	0.25
Peugeot 208	-	355	-	285	78	0.23
Hyundai Ioniq	361	-	-	250	36	0.22
Nissan Leaf	557	-	-	225	40	0.23
Tesla Model S	389	-	-	560	110	0.23
VW ID.3	-	-	2361	350	85	0.23
VW ID.4	-	-	2361	400	103	0.27
Skoda Enyaq	-	-	2208	420	103	0.26
Fiat 500	-	-	1356	235	67	0.23
Seat Mii	-	-	936	205	30	0.22
Tesla Model Y	-	-	921	435	108	0.24
Weighted average values				**340**	**81**	**0.24**

B.4. Details on projected developments for 2030

For the quantification of the scenarios used in the analysis, projections on the development of BEV share had to be made. For the *Societal Commitment* and *Techno-Friendly* scenarios an *optimistic* projection was made based on the decarbonization goals for the transport sector in the governmental document “Austria’s Mobility Master Plan 2030” [134]. A *pessimistic* projection is made for the *Directed Transition* and *Gradual Development* scenarios based on the trend of new registrations of BEVs in the years between 2018 and 2020.

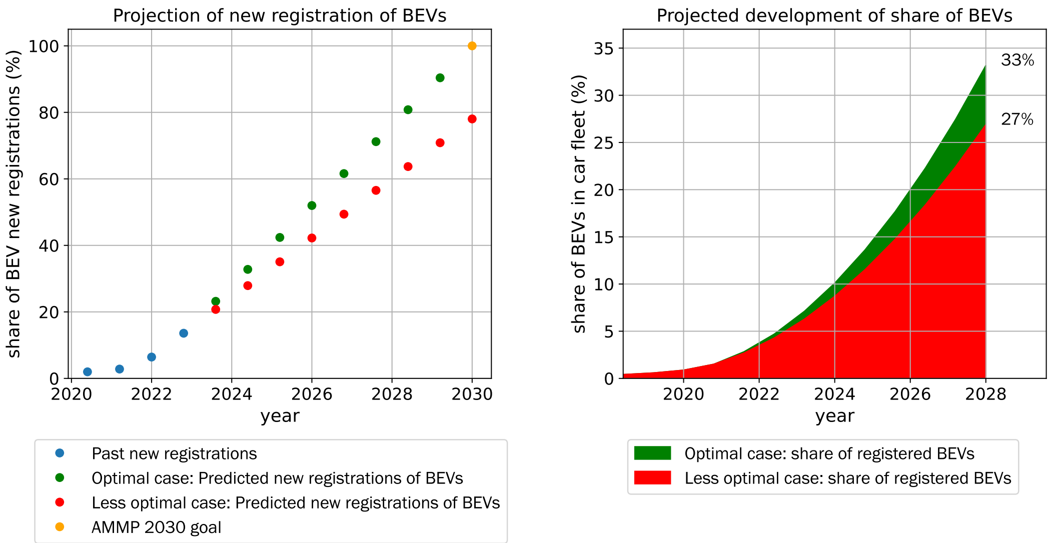

Figure B.1.: Projections of linearly increasing new registrations of BEVs (**left**, *AMMP=Austria Mobility Master Plan* [134]) and based on this, developments of the share of BEVs (**right**).

B.5. Details on data preparation for the Austrian case study

The underlying data pre-processing procedures of this work encompassed the preparation of the graph representation of the Austrian high-level road network and the potential sites for the installment of charging stations. Geographic data representing the Austrian highways and motorways was obtained based on OpenStreetMap data [139]. For this, all *way* instances with tags *highway='motorway'* and *highway='trunk'* were retrieved and further processed using ArcGIS Pro [203] in order to obtain a clean shape representation of the road network, including the representation of both driving directions through parallel lines. Geographic data implying positions of service areas were retrieved using tags *highway='services'* and *highway='rest_area'*. This geographic information was paired with data on service areas obtained through ASFINAG [140] which provide information on the driving direction a service area is accessible from and on the services offered on-site.

B.6. Model validation

The model is validated by modeling present-day infrastructure requirements and comparing the results with the setup of the existing infrastructure. The current infrastructure requirements are modeled using input parameters displayed in Table 4.1. In Figure B.2, the capacities of the currently installed fast-charging infrastructure along Austria's highways (as of December 2021) are illustrated along with a comparison to the model output. Table B.3 presents the numbers of fast-charging points of different peak power levels and total capacity.

While the overall number of charging stations is in a similar range, a gap exists between the total installed capacity estimated by the model and the existing capacity. There are several possible reasons for this. First, the proposed modeling framework aims to design a minimum

Table B.3.: Comparison of existing fast-charging infrastructure along the Austrian highway network (as of December 2021) and model output based on the number of charging stations and summed peak power levels ($\hat{P}_{CP}$).

	Existing infrastructure	Model output
Nb. charging stations	31	43
Nb. charging points with $\hat{P}_{CP} = 44$ kW (AC)	8	-
Nb. charging points with $\hat{P}_{CP} = 50$ kW (DC)	72	-
Nb. charging points with $\hat{P}_{CP} = 75$ kW (DC)	4	-
Nb. charging points with $\hat{P}_{CP} = 150$ kW (DC)	22	98
Nb. charging points with $\hat{P}_{CP} = 350$ kW (DC)	40	-
Total capacity (MW)	20.1	14.7

required charging infrastructure for a given point in time; no pre-planned charging capacities with regard to future developments are modeled. In existing infrastructure, this is not the case, as, for example, IONITY has installed most of its charging points with $\hat{P}_{CP} = 350$ kW while currently only few cars on the market are able to charge at a power level above 150 kW. Therefore, in this context, the modeled infrastructure reflects the minimum requirements for a fast-charging infrastructure. Moreover, the difference could stem from the fact that the model assumes charging points of a singular peak power level and all BEVs of the Austrian car fleet having the same technical parameters, which makes it difficult to directly compare existing and modeled capacity because the charging points of different peak power levels have different efficiencies and the average charging capacity varies significantly among car models.

The difference in the location distribution of capacities can be attributed to the model assumption that the proportion of long-distance travelers among highway traffic is equal throughout the whole highway network. Further, local differences of the cost parameter c_X are neglected, whereas in reality this cost component significantly varies depending on the location. Another reason for the difference in site allocation is that within the model, all energy demand is required to be covered within the geographic boundary of Austria; charging stations lying right outside the country's borders are not considered. Therefore, the model tends to install charging stations at service areas that are closest to the end points of the highway network. Another striking difference between existing and modeled fast-charging infrastructure is the accumulation of many small charging stations in the vicinity of Vienna in the modeled infrastructure. Some of them may stem from the tendency of installing charging infrastructure near borders, whereas others are most likely placed there due to the dense topology of the highway network around Vienna.

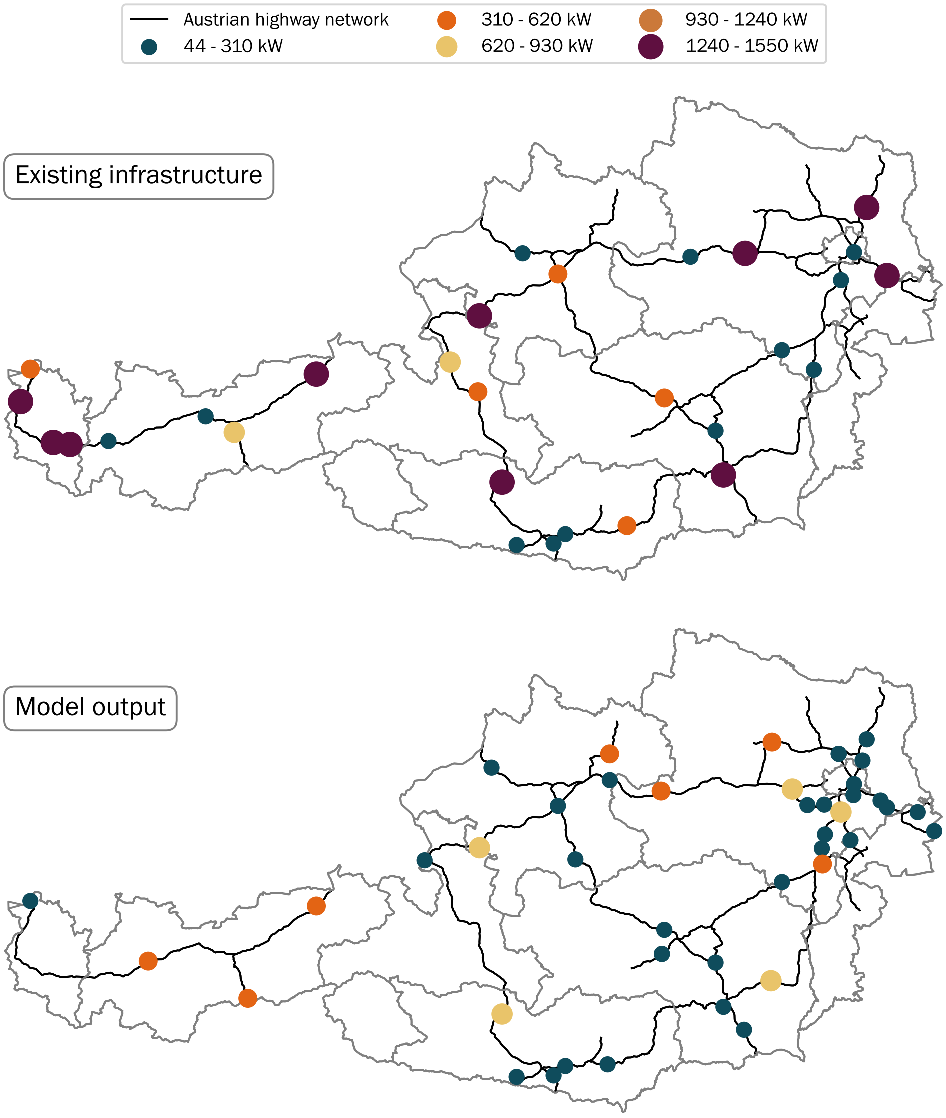

Figure B.2.: Model validation for Model 1 (RQ1). Visual comparison of the capacities, i.e., summed peak power of all onsite charging points, in existing fast-charging infrastructure (**top**) and the modeled infrastructure (**bottom**) using the values displayed in Table 4.1.

Appendix C.

Supplementary Material for Contribution 2 (RQ2)

The content of this appendix is based on Golab et al. (2026).

C.1. Extended nomenclature

Table C.1 lists the extended nomenclature for the transport sector decarbonization model described in Section 3.3.5.

Table C.1.: Extended nomenclature for the transport sector decarbonization model (RQ2).

Sets and indices	Description
$y \in \mathcal{Y}$	year
$u \in \mathcal{U}$	consumer group
$p \in \mathcal{P}$	trip purpose
$r \in \mathcal{R}_p$	trip between origin-destination (OD) pair for a specified purpose
$e \in \mathcal{E}$	geographic location
$v \in \mathcal{V}$	type of vehicle
$t \in \mathcal{T}_v$	drivetrain technology fueled by a specific fuel
$g \in \mathcal{G}$	year that vehicle was introduced to market
$l \in \mathcal{L}$	types of fueling infrastructure (by capacity)
$i \in \mathcal{I}_l$	index for detour time reduction potential for type of fueling infrastructure l
$\mathcal{Y}^{exp}$	set of years of investment decisions
$\mathcal{R}_e$	set of routes that go through location e
$\mathcal{L}^{nothome}$	fueling infrastructure that is not allocated at home of vehicle owner
$\mathcal{L}^{home}$	fueling infrastructure that is allocated at home of vehicle owner
$\mathcal{L}_t$	types of infrastructure by drivetrain technology
$\mathcal{E}_{rl}$	geographic location along route and available fueling type
S_j^u	subsets of time periods for which a monetary budget is defined by consumer group

Decision variables	Description	Unit	Domain
f_{yurvtg}	number of person-trips	–	$[0, \infty[$
h_{yurvtg}	number of vehicles	–	$[0, \infty[$
h^{+}_{yurvtg}	newly purchased vehicles	–	$[0, \infty[$
h^{-}_{yurvtg}	number of vehicles leaving the vehicle stock	–	$[0, \infty[$
$s_{yurvtgle}$	energy fueled	kWh	$[0, \infty[$
n_{yurlge}	number of charging vehicles (aggregated over v, t)	–	$[0, \infty[$
b_{yiel}	VoT-weighted detour cost at location e for charging type l in year y	€	$[0, \infty[$

w_{yiel}	binary decision variable indicating whether detour reduction i is active	–	$\{0,1\}$
$q^{+,\text{charg\_infr,home}}_{yrte}$	added fueling infrastructure at home	kW	$[0,\infty[$
$q^{+,\text{charg\_infr}}_{ytle}$	added fueling infrastructure	kW	$[0,\infty[$
$q^{\Delta+,\text{charg\_infr}}_{ytle}$	relative difference of fueling capacity expansion to fixed capacity size values	kW	$[0,\infty[$
budget^{+}_{yur}	budget overrun	€	$[0,\infty[$
budget^{-}_{yur}	budget underrun	€	$[0,\infty[$

Parameters	Description	Unit	Domain
LoS	level of service	h	$[0,\infty[$
VoT	value of time	€/h	$[0,\infty[$
γ_l	factor between annual peak hour and total annual demand	–	$[0,1]$
$Q^{\text{charg\_infr,nothome}}_{urte}$	initial fueling infrastructure (not home)	kW	$[0,\infty[$
$Q^{\text{charg\_infr,home}}_{urtle}$	initial fueling infrastructure at home	kW	$[0,\infty[$
$\hat{Q}^{\text{charg\_infr,home}}_{utle}$	upper limit for home charging infrastructure capacity by consumer group	kW	$[0,\infty[$
$\hat{Q}^{\text{charg\_infr}}_{tle}$	upper limit for charging infrastructure capacity	kW	$[0,\infty[$
α_{iel}	reduction i of detour time	%	$[0,100]$
cap^{lb}_{iel}	lower bound of capacity level i for detour time reduction α_i	kW	$[0,\infty[$
cap^{ub}_{iel}	upper bound of capacity level i for detour time reduction α_i	kW	$[0,\infty[$
τ^{u}	vehicle replacement period by consumer group	a	$[0,\infty[$
β	relative threshold for budget under- or overrun	%	$[0,100]$
B^{init}_{el}	initial detour time	h	$[0,\infty[$
D^{spec}_{vtg}	specific energy consumption	kWh/km	$[0,\infty[$
W_{vtg}	load factor	persons/vehicle	$[0,\infty[$
$\text{Cap}^{\text{tank}}_{vtg}$	tank capacity	kWh	$[0,\infty[$
L^{annual}_{vtg}	annual range	km	$[0,\infty[$
L_r	length of route	km	$[0,\infty[$
C^{CAPEX}_{yvtg}	expenditure costs for vehicles	€	$[0,\infty[$
Budget_u	monetary budget of consumer group	€/trip	$[0,\infty[$
H^{init}_{urvtg}	initial vehicle stock	vehicles	$[0,\infty[$
Λ_{vtg}	vehicle lifetime	a	$[0,\infty[$
μ_{yvt}	maximum market share of technology t	–	$[0,1]$
$\sigma^{\text{commuting}}_{e}$	commuting share at location e	–	$[0,1]$
$\sigma^{\text{private}}_{e}$	private trip share at location e	–	$[0,1]$
M	sufficiently large constant (big-M)	–	$[0,\infty[$

C.2. Extended mathematical formulation

This section provides the complete set of constraints of the transport sector decarbonization model. Constraints that are also presented with detailed explanations in Section 3.3.5 are included here for reference.

Objective function:

The objective function comprises costs for the vehicle stock ($C^{\text{vehicle\_stock}}$), required fueling (C^{fueling}), costs for fueling infrastructure ($C^{\text{infrastructure}}$), intangible costs ($C^{\text{intangiblecosts}}$), penalty costs ($C^{\text{penaltycosts}}$), and the VoT-weighted detour cost (b). The explicit definitions of each cost component are provided in the following subsections of this appendix. All cost components are discounted to net present value using the discount rate δ:

$$
\underset{f,h,h^+,h^-,s,n,b,w,q^+,\text{budget}^+,\text{budget}^-}{minimize} \quad z = \sum_y \frac{1}{(1+\delta)^{y-y_0}} \Big(C_y^{\text{vehicle\_stock}} + C_y^{\text{fueling}} \\ + C_y^{\text{infrastructure}} + C_y^{\text{intangiblecosts}} + C_y^{\text{penaltycosts}} + \sum_{i,e,l} b_{yiel} \Big) \tag{C.1}
$$

Fueling infrastructure sizing:

Fueling infrastructure is expanded for each investment period $y' \in \mathcal{Y}^{\text{exp}}$ and sized based on a peak-to-annual demand ratio γ_l. We differentiate between fueling infrastructure independent from the particular trip ($\mathcal{L}^{\text{not home}}$) and home-based fueling infrastructure ($\mathcal{L}^{\text{home}}$):

$$
\begin{aligned}
Q_{tle}^{\text{charg\_infr}} + \sum_{y' \in \mathcal{Y}_y^{\text{exp}}} q_{y'tle}^{+,\text{charg\_infr}} &\geq \gamma_l \sum_{r \in \mathcal{R}_e} \sum_{u \in \mathcal{U}} \sum_{v \in \mathcal{V}} \sum_{g \in \mathcal{G}} s_{yurvtgle} \quad &: \forall y, t, e, l \in \mathcal{L}^{\text{not home}} \\
Q_{urtle}^{\text{charg\_infr,home}} + \sum_{y' \in \mathcal{Y}_y^{\text{exp}}} q_{y'urtle}^{+,\text{charg\_infr,home}} &\geq \gamma_l \sum_{v \in \mathcal{V}} \sum_{g \in \mathcal{G}} s_{yurvtgle} \quad &: \forall y, u, t, r, e, l \in \mathcal{L}^{\text{home}}
\end{aligned} \tag{C.2}
$$

Demand coverage:

Decision variable f expresses the number of trips covered by a specific vehicle type and technology-fuel pair of a specific generation. The sum over all vehicle types, technology-fuel pairs and vehicle generations equals the exogenously given travel demand:

$$
\sum_{v \in \mathcal{V}} \sum_{t \in \mathcal{T}_v} \sum_{g \in \mathcal{G}} f_{yurvtg} = F_{yur} \quad : \forall y, u, r \tag{C.3}
$$

The right side of the equation expresses the coverage of the demand by different vehicle types and drivetrain technologies of different generations.

Fleet sizing:

Based on the trips covered by a vehicle type of a technology, the vehicle stock is sized as follows:

$$
h_{yurvtg} \geq \frac{L_r}{W_{vtg} L_{vtg}^{\text{annual}}} f_{yurvtg} \quad : \forall y, u, r, v, t, g \tag{C.4}
$$

Vehicle stock balance with vintage tracking:

The vehicle stock of each generation g is tracked using the carried-over stock variable h^{exist}. The stock balance for each vintage is:

$$h_{yurvtg} = h^{\text{exist}}_{yurvtg} + h^{+}_{yurvtg} - h^{-}_{yurvtg} \quad : \forall y, u, r, v, t,\ g \leq y \tag{C.5}$$

The carried-over stock h^{exist} represents the vehicles of generation g that are inherited from the previous year. Its definition depends on the relationship between the model year y, the vehicle generation g, the initial year y_0, and the vehicle lifetime Λ_{vtg}:

$$h^{\text{exist}}_{yurvtg} = \begin{cases} H^{\text{init}}_{urvtg} & \text{if } y = y_0,\ g < y_0,\ y_0 - g \leq \Lambda_{vtg} \\ h_{(y-1)urvtg} & \text{if } y > y_0,\ g < y,\ y - g \leq \Lambda_{vtg} \\ 0 & \text{if } g = y \text{ (new generation)} \\ 0 & \text{if } y - g > \Lambda_{vtg} \text{ (lifetime exceeded)} \end{cases} \tag{C.6}$$

where H^{init}_{urvtg} is the exogenously given initial vehicle stock for pre-existing generations. For the initial year, pre-existing generations are initialized from data; for subsequent years, the stock is carried over from the previous year. Vehicles of generation $g = y$ (newly produced) have no carry-over. Vehicles that have exceeded their maximum lifetime Λ_{vtg} are set to zero.

New vehicle purchases are restricted to the current generation:

$$h^{+}_{yurvtg} = 0 \quad : \forall y, u, r, v, t,\ g \neq y \tag{C.7}$$

Equation C.4 expresses the sizing of the vehicle stock via a vehicle's annual range L^{annual} and load factor W. The stock balance (Equation C.5) with the vintage-tracked carry-over (Equation C.6) governs the aging of the vehicle fleet. Vehicles exit the stock when their maximum lifetime is reached, and new purchases (Equation C.7) are only possible for the current generation.

Fueling demand:

Fueling demand is derived from the number of trips f and via the specific demand D^{spec} of the vehicle:

$$\sum_{l \in \mathcal{L}_t} \sum_{e \in \mathcal{E}_{rl}} s_{yurvtgle} \geq \frac{D^{\text{spec}}_{vtg} \cdot L_r}{W_{vtg}} f_{yurvtg} \quad : \forall y, u, r, v, t, g \tag{C.8}$$

Number of charging vehicles:

n indicates the number of charging vehicles and is derived from the charged energy, accounting for the usable state of charge $\text{SOC}^{+,\text{charging}}$:

$$n_{yurlge} \geq \sum_{v,t} \frac{1}{\text{SOC}^{+,\text{charging}} \cdot \text{Cap}_{vtg}^{\text{tank}}} s_{yurvtgle} \quad : \forall y,u,r,g,e,l \in \mathcal{L}^{vt} \tag{C.9}$$

Maximum capacity constraints:

The geographic allocation of the fueling demand is given by the summation over all geographic elements that are traveled by the vehicle. The summation over all geographic locations along the travel path and different types of fueling infrastructures l ensures that the allocation of fueled energy along the path is an endogenous decision. The expansion of fueling capacities is also limited by maximum values:

$$\begin{aligned} Q_{tle}^{\text{charg\_infr}} + \sum_{y' \in \mathcal{Y}_y^{\text{exp}}} q_{y'tle}^{+,\text{charg\_infr}} &\leq \hat{Q}_{tle}^{\text{charg\_infr}} &&: \forall y,t,l,e \\ \sum_{r \in \mathcal{R}_u} \left(Q_{urtle}^{\text{charg\_infr,home}} + \sum_{y' \in \mathcal{Y}_y^{\text{exp}}} q_{y'urtle}^{+,\text{charg\_infr,home}} \right) &\leq \hat{Q}_{utle}^{\text{charg\_infr,home}} &&: \forall y,u,t,l,e \end{aligned} \tag{C.10}$$

Intangible costs and level of service:

The intangible costs represent non-monetary costs monetized via the consumer-group-dependent value of time (VoT). In the MILP extension, the detour cost component is captured separately by the variable b_{yiel} (see below), so the intangible costs reduce to the VoT-weighted travel time:

$$C^{\text{intangiblecosts,total}} = \sum_{y,u,r,v,t,g} \text{VoT}_u \cdot \text{LoS}_{yrvt}^{f} \cdot f_{yurvtg} \tag{C.11}$$

The level of service (LoS) is defined as the total travel time:

$$\text{LoS}_{yrvt}^{f} = \frac{L_r}{\text{speed}_{vt}} + \text{fueling\_time}_{yrvt} \quad : \forall y,r,v,t \tag{C.12}$$

Monetary budget constraints:

Monetary budgets constrain expenditures for vehicles and home charging infrastructure per consumer group and purchase cycle $\mathcal{S}_j^u$:

$$\sum_{y \in \mathcal{S}_j^u} \left(\sum_{vtg} C_{yvtg}^{\text{CAPEX}} \cdot h_{yurvtg}^{+} + \sum_{tl} \sum_{y' \in \mathcal{Y}_y^{\text{exp}}} q_{yurtle}^{+,\text{charg\_infr,home}} \right) \\ \leq \beta \cdot \tau_u \cdot \text{Budget}_u \cdot \sum_{y \in \mathcal{S}_j^u} \sum_{vtg} f_{yurvtg} + \sum_{y \in \mathcal{S}_j^u} \text{budget}_{yur}^{+} \quad : \forall u, r, j \tag{C.13}$$

$$\sum_{y \in \mathcal{S}_j^u} \left(\sum_{vtg} C_{yvtg}^{\text{CAPEX}} \cdot h_{yurvtg}^{+} + \sum_{tl} \sum_{y' \in \mathcal{Y}_y^{\text{exp}}} q_{yurtle}^{+,\text{charg\_infr,home}} \right) \\ \geq \beta \cdot \tau_u \cdot \text{Budget}_u \cdot \sum_{y \in \mathcal{S}_j^u} \sum_{vtg} f_{yurvtg} - \sum_{y \in \mathcal{S}_j^u} \text{budget}_{yur}^{-} \quad : \forall u, r, j \tag{C.14}$$

where budget_{yur}^{+} and budget_{yur}^{-} are decision variables for budget overrun and underrun, penalized in the objective function. The penalty cost term in the objective function is defined as:

$$C_y^{\text{penaltycosts}} = M^{\text{penalty}} \cdot \sum_{u,r} \left(\text{budget}_{yur}^{+} + \text{budget}_{yur}^{-} \right) \tag{C.15}$$

where M^{penalty} is a large penalty coefficient that ensures budget constraints are nearly binding while preserving model feasibility.

MILP detour time extension:

The total installed charging capacity is decomposed into a discrete component tied to detour-reduction levels and a continuous residual. Let the total installed capacity at location e for charging type l up to year y be defined as the sum of the initial capacity $Q_{tle}^{\text{charg\_infr}}$ and the cumulative expansion $\sum_{y'} q_{y'tle}^{+,\text{charg\_infr}}$. This total capacity is linked to a discrete capacity level selected by the binary variable w_{yiel}:

$$Q_{tle}^{\text{charg\_infr}} + \sum_{y' \in \mathcal{Y}_y^{\text{exp}}} q_{y'tle}^{+,\text{charg\_infr}} = \sum_{i} w_{yiel} \cdot cap_{iel}^{\text{lb}} + \sum_{y' \in \mathcal{Y}_y^{\text{exp}}} q_{y'tle}^{\Delta+,\text{charg\_infr}} \quad : \forall y, t, e, l \in \mathcal{L}^{\text{not home}} \tag{C.16}$$

The left-hand side is the total installed capacity (initial plus cumulative expansion). The right-hand side decomposes it into a discrete component cap_{iel}^{lb}, selected by the binary variable w_{yiel}, and a continuous residual $q^{\Delta+,\text{charg\_infr}}$ that captures the capacity above the discrete threshold. The indicator constraint ensures that when level i is active, the total capacity lies within the associated bounds:

$$w_{yiel} = 1 \implies cap^{\text{lb}}_{iel} \leq Q^{\text{charg\_infr}}_{tle} + \sum_{y' \in \mathcal{Y}^{\text{exp}}_y} q^{+,\text{charg\_infr}}_{y'tle} \leq cap^{\text{ub}}_{iel} \quad : \forall y, i, e, l \in \mathcal{L}^{\text{not home}} \tag{C.17}$$

Exactly one detour-reduction level is selected per location, year, and charging type:

$$\sum_i w_{yiel} = 1 \quad : \forall y, e, l \in \mathcal{L}^{\text{not home}} \tag{C.18}$$

VoT-weighted detour cost:

The detour cost captures the additional travel time needed to reach charging infrastructure, monetized via the consumer-specific VoT. Based on an initial detour time B^{init}_{el} and its reduction α_{iel} at the active capacity level $[cap^{\text{lb}}_{iel},\ cap^{\text{ub}}_{iel}]$, the decision variable b_{yiel} is defined via an indicator constraint:

$$w_{yiel} = 1 \implies b_{yiel} \geq B^{\text{init}}_{el} \cdot (1 - \alpha_{iel}) \cdot \sum_{u,r,g} \text{VoT}_u \cdot n_{yurlge} \quad : \forall y, i, e, l \in \mathcal{L}^{\text{not home}} \tag{C.19}$$

The variable b_{yiel} enters the objective function directly and represents the total monetized detour cost at location e for charging type l in year y. The implementation uses indicator constraints natively supported by the solver. For solvers without indicator support, the equivalent big-M reformulation is:

$$b_{yiel} \geq B^{\text{init}}_{el} \cdot (1 - \alpha_{iel}) \cdot \sum_{u,r,g} \text{VoT}_u \cdot n_{yurlge} - M \cdot (1 - w_{yiel}) \quad : \forall y, i, e, l \in \mathcal{L}^{\text{not home}} \tag{C.20}$$

$$b_{yiel} \leq M \cdot w_{yiel} \quad : \forall y, i, e, l \in \mathcal{L}^{\text{not home}} \tag{C.21}$$

where M is a sufficiently large constant. Equation C.20 activates the lower bound on b_{yiel} when $w_{yiel} = 1$ and relaxes it otherwise. Equation C.21 forces $b_{yiel} = 0$ when $w_{yiel} = 0$.

The following additional constraints complete the formulation.

Technology stock transition rate:

The year-over-year change in the vehicle stock of a drivetrain technology t is bounded to prevent unrealistically rapid technology transitions:

$$\left|\sum_v \sum_g h_{yurvtg} - \sum_v \sum_g h_{(y-1)urvtg}\right| \leq \alpha_t^h \sum_{v,t',g} h_{yurvt'g} + \beta_t^h \sum_v \sum_g h_{(y-1)urvtg} \qquad : \forall y \in \mathcal{Y} \setminus \{y_0\}, u, r, t \tag{C.22}$$

where α_t^h and β_t^h are the maximum rate-of-change parameters relative to the total vehicle stock and the previous stock of the respective technology.

Vehicle-type stock transition rate:

The year-over-year change in the stock of an individual vehicle type v and technology t combination is similarly bounded:

$$\left|\sum_g h_{yurvtg} - \sum_g h_{(y-1)urvtg}\right| \leq \alpha^h \sum_{v',t',g} h_{yurv't'g} + \beta^h \sum_g h_{(y-1)urvtg} \qquad : \forall y \in \mathcal{Y} \setminus \{y_0\}, u, r, v, t \tag{C.23}$$

Market share constraints:

For vehicle technologies with exogenously defined market shares μ_{yvt}, the share of newly purchased vehicles is fixed:

$$\sum_{u,r,g} h^+_{yurvtg} = \mu_{yvt} \cdot \sum_{u,r,v',t',g} h^+_{yurv't'g} \qquad : \forall (y, v, t) \in \mathcal{MS} \tag{C.24}$$

where $\mathcal{MS}$ denotes the set of year–technology–vehicle type combinations with defined market shares.

Trip purpose ratio:

The ratio between commuting and private trip vehicle stocks is fixed per geographic location to maintain the observed trip-purpose distribution:

$$\sigma_e^{\text{commuting}} \cdot \sum_{\substack{r \in \mathcal{R}_e^{\text{private}} \\ v,t}} h_{yurvtg} = \sigma_e^{\text{private}} \cdot \sum_{\substack{r \in \mathcal{R}_e^{\text{commuting}} \\ v,t}} h_{yurvtg} \qquad : \forall y, e, g \tag{C.25}$$

where $\sigma_e^{\text{commuting}}$ and $\sigma_e^{\text{private}}$ are the exogenously given trip shares for commuting and private purposes at location e, and $\mathcal{R}_e^{\text{commuting}}$ and $\mathcal{R}_e^{\text{private}}$ denote the respective subsets of OD pairs originating at e.

BEV fleet growth:

The total BEV fleet within each consumer group must not decrease over time:

$$\sum_{r\in\mathcal{R}_u}\sum_{v}\sum_{t\in\mathcal{T}^{\text{BEV}}}\sum_{g} h_{yurvtg} \geq \sum_{r\in\mathcal{R}_u}\sum_{v}\sum_{t\in\mathcal{T}^{\text{BEV}}}\sum_{g} h_{(y-1)urvtg} \quad : \forall y \in \mathcal{Y}\setminus\{y_0\}, u \tag{C.26}$$

Minimum home charging share:

For consumer groups with a defined minimum home charging share $\sigma^{\text{home}}_{yu}$, the energy charged at home must be at least a specified fraction of the total electricity consumed by that consumer group:

$$\sum_{\substack{r\in\mathcal{R}_u,\, v,\, t,\, g\\ e=\text{origin}(r),\, l\in\mathcal{L}^{\text{home}}}} s_{yurvtgle} \geq \sigma^{\text{home}}_{yu}\cdot \sum_{\substack{r\in\mathcal{R}_u,\, v,\, t,\, g\\ e,\, l}} s_{yurvtgle} \quad : \forall y, u \tag{C.27}$$

Forced fast charging for long-distance trips:

Trips whose path length exceeds the usable driving range of the vehicle are restricted to fast charging infrastructure only:

$$n_{yurlge} = 0 \quad : \forall y, u, r, l \notin \mathcal{L}^{\text{fast}}, g, e \quad \text{if } L_r > \frac{\text{SOC}^{+,\text{driving}}\cdot \text{Cap}^{\text{tank}}_{vtg}}{D^{\text{spec}}_{vtg}} \tag{C.28}$$

Slow and fast charging expansion ratio:

The cumulative expansion of public slow charging infrastructure is bounded relative to the expansion of public fast charging at each location:

$$\sum_{y'\in\mathcal{Y}^{\text{exp}}_y} q^{+,\text{charg\_infr}}_{y'(l=\text{slow})e} \leq 2\cdot \sum_{y'\in\mathcal{Y}^{\text{exp}}_y} q^{+,\text{charg\_infr}}_{y'(l=\text{fast})e} \quad : \forall y \in \mathcal{Y}^{\text{exp}}, e \tag{C.29}$$

Infrastructure expansion rate limits:

The expansion of public charging infrastructure is bounded by initial-period capacity limits and subsequent growth rate constraints:

$$
\begin{aligned}
&q^{+,\text{charg\_infr}}_{y_0 le} \leq \overline{q}^{\text{init}} \qquad : \forall e, l \in \mathcal{L}^{\text{pub}} \\
&q^{+,\text{charg\_infr}}_{yle} - q^{+,\text{charg\_infr}}_{(y-\Delta y)le} \leq \rho \cdot \Delta y \cdot \left(Q^{\text{charg\_infr,init}}_{le} + \sum_{y'<y} q^{+,\text{charg\_infr}}_{y'le} \right) \\
&\qquad : \forall y \in \mathcal{Y}^{\text{exp}} \setminus \{y_0\}, e, l \in \mathcal{L}^{\text{pub}}
\end{aligned}
\tag{C.30}
$$

where $\overline{q}^{\text{init}}$ is the maximum new infrastructure investment in the initial period and ρ is the maximum growth rate per investment period.

C.3. Model parametrization

Table C.2 summarizes the key constants and parametric assumptions used in the model formulation. Values marked as "input data" are read from the scenario-specific input files and may vary across sensitivity analyses.

Table C.2.: Model constants and parametric assumptions.

Symbol	Description	Value	Unit
Vehicle and battery			
$\text{SOC}^{+,\text{charging}}$	Usable battery capacity for charging	0.8	–
$\text{SOC}^{+,\text{driving}}$	Usable battery capacity for driving range	0.8	–
W_{vtg}	Load factor (persons per vehicle)	1.4	pers./veh.
L^{annual}_{vtg}	Annual driving range	11 600	km/a
Fleet dynamics			
α^h, β^h	Maximum stock transition rate (vehicle type)	0.1	–
α^h_t, β^h_t	Maximum stock transition rate (drivetrain technology)	0.1	–
$\sigma^{\text{home}}_{yu}$	Minimum home charging share (Q1–Q3 / Q4–Q5)	0.0 / 0.30	–
τ_u	Vehicle purchase cycle (Q1 / Q2 / Q3 / Q4 / Q5)	10 / 8 / 6 / 4 / 2	a
Infrastructure constraints			
	Slow/fast charging expansion ratio	2	–
ρ	Maximum public infrastructure growth rate per period	0.3	–
γ_l	Peak-to-annual demand ratio (home / work / public fast / public slow / fossil)	0.05 / 0.20 / 0.25 / 0.25 / 0.10	–
Δy	Investment period length	5	a
Economic			
δ	Discount rate	0.05	–
β	Budget tolerance factor	input data	–
M^{penalty}	Budget penalty coefficient	10^7	€/€
Temporal scope			
y_0	Initial year	2020	–
Y	Model horizon	41	a

C.4. Pre-processing of travel data

C.4.1. Trip lengths by purpose

Two groups for the environmental setting of a trip are introduced: *urban* and *rural*. The segregation is based on the population distribution between cities with population size greater than 100,000 people of the Basque Country, together with the respective mode share between urban and rural areas (20%/50% of trips by private car). This, together with the information on the trip purpose, serves as information to approximate the path length.

These assumptions are drawn from a survey on mobility for Austria [204], as this survey contains detailed information on average trip lengths by purpose for all NUTS-2 regions in Austria. Specifically, for urban areas, trip lengths recorded for Vienna are used, and for rural areas, the trip lengths recorded for suburban areas are used due to the high population density in the Basque Country.

C.4.2. Clustering of trips to vehicles

The clustering of trips to vehicles introduces a constraint linking commuting and private vehicle stocks. To consider the technology decision vehicle-based rather than trip-based, the share of technology must follow a ratio related to the application of the vehicles. Trips are grouped by income class and origin to M route clusters, resulting in sets of routes $\mathcal{R}_m$. Based on empirical values on the relative frequency of trips per purpose, the vehicle stocks are related to each other with the following constraint:

$$\zeta_1 h_{yur(p=\text{commuting})vtg} \leq \zeta_2 h_{yur(p=\text{private})vtg} \quad : \forall y, u, v, t, g, r \in \mathcal{R}^m, m = 1, 2, \ldots, M \tag{C.31}$$

Table C.3 displays the parameter values by region.

Table C.3.: Parameter setting for the relationship between number of vehicles used for commuting and private purposes by region.

Parameter	Bizkaia	Araba	Gipuzkoa
ζ_1	1	1	1
ζ_2	0.86	0.85	1.05

C.5. Discretization of the detour-fueling function

To introduce the relationship between installed charging infrastructure capacity and the detouring time to the closest available charging station into the mixed-integer linear program, the continuous detour-fueling function is discretized into piecewise steps. This discretization balances the trade-off between the loss of information due to discretization and the increase in model complexity (number of binary variables) with finer resolution. Figure C.1 illustrates the sampling of the fueling detour time functions used in the MILP formulation.

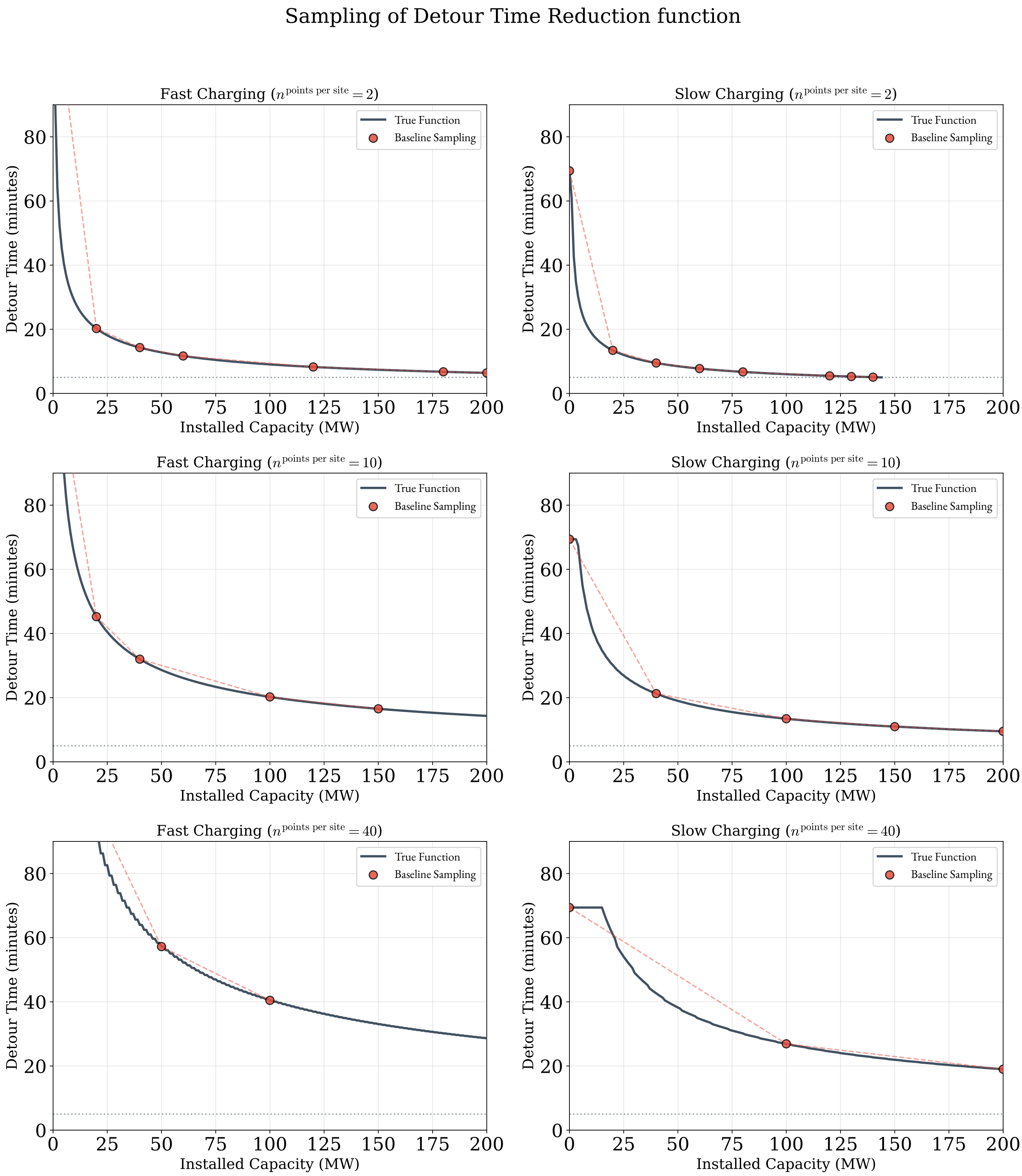

Figure C.1.: Sampling of the fueling detour time functions to introduce the relation between charging infrastructure capacity expansion and fueling detour time to the MILP.

C.6. Computational details

The model is implemented in Julia using JuMP 1.9 [205] and solved with Gurobi 12.0 [206]. Simulations are performed on an AMD Ryzen Threadripper 7980X (64 cores, 3.20 GHz base clock, 5.2 GHz boost) with 512 GB RAM. The model comprises approximately 32 million continuous variables and 189 binary variables. After presolve, the problem is reduced to approximately 2 million variables. A single scenario solves in approximately 6.5 hours, terminating at an MIP gap below 0.005%.

The source code and input data are publicly available at: `https://github.com/antoniamgolab/iDesignRES_transcompmodel`.

C.7. Sensitivity analysis: circuity factor

A sensitivity analysis is conducted on the circuity parameter μ, which translates the aerial distance to the driving distance in the road network. The tested range is $\mu \in \{1.0, 1.2, 1.4, 1.6, 1.8, 2.0, 2.2, 2.4\}$. The base case uses $\mu = 1.4$, which aligns with circuity values reported for European cities [207] and non-urban road networks [208]. Figure C.2 displays the circuity values used as reference.

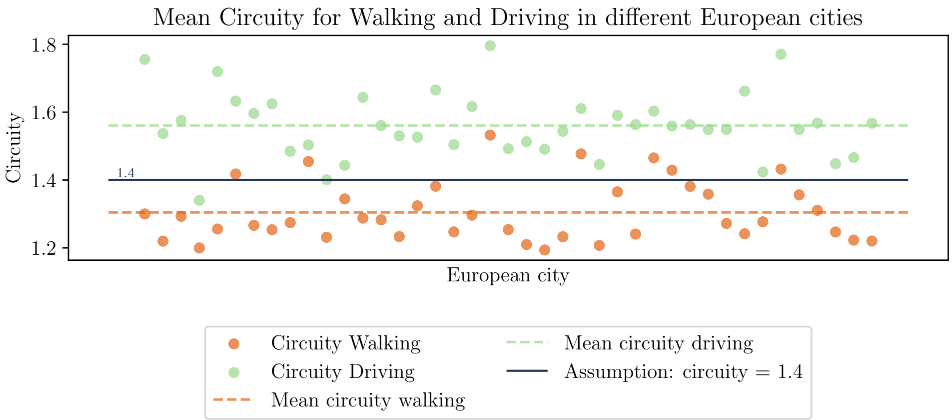

Figure C.2.: Circuity values for European cities published by Costa et al. [207].

The resulting alternative fueling detour time curves for the tested circuity values are displayed in Figure C.3. The sensitivity analysis is conducted based on the densification scenario $n^{\text{points per site}} = 10$.

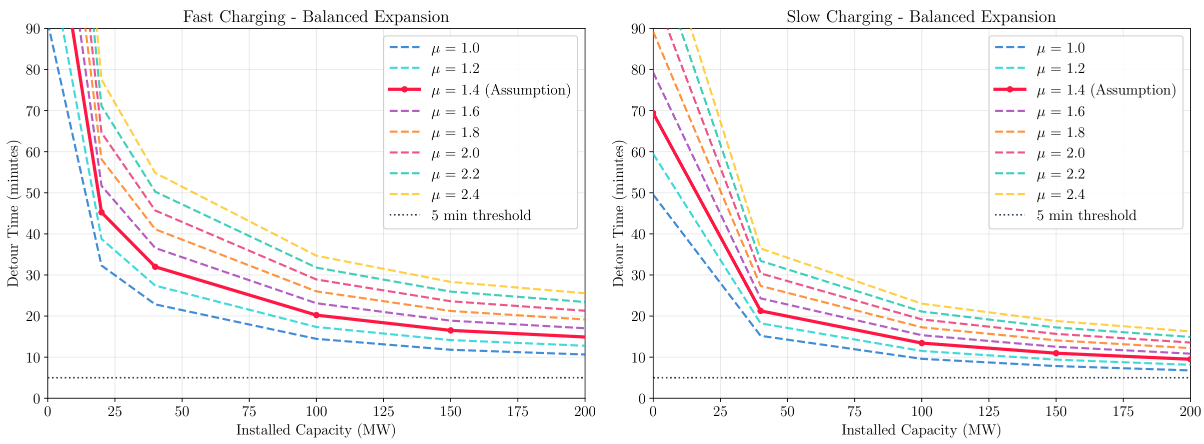

Figure C.3.: Tested fueling detour time curves for different values of circuity μ.

Figure C.4 displays the distribution of electrification shares across the tested circuity values. No direct correlation is found between the increase of μ and either utilization or adoption by income class level.

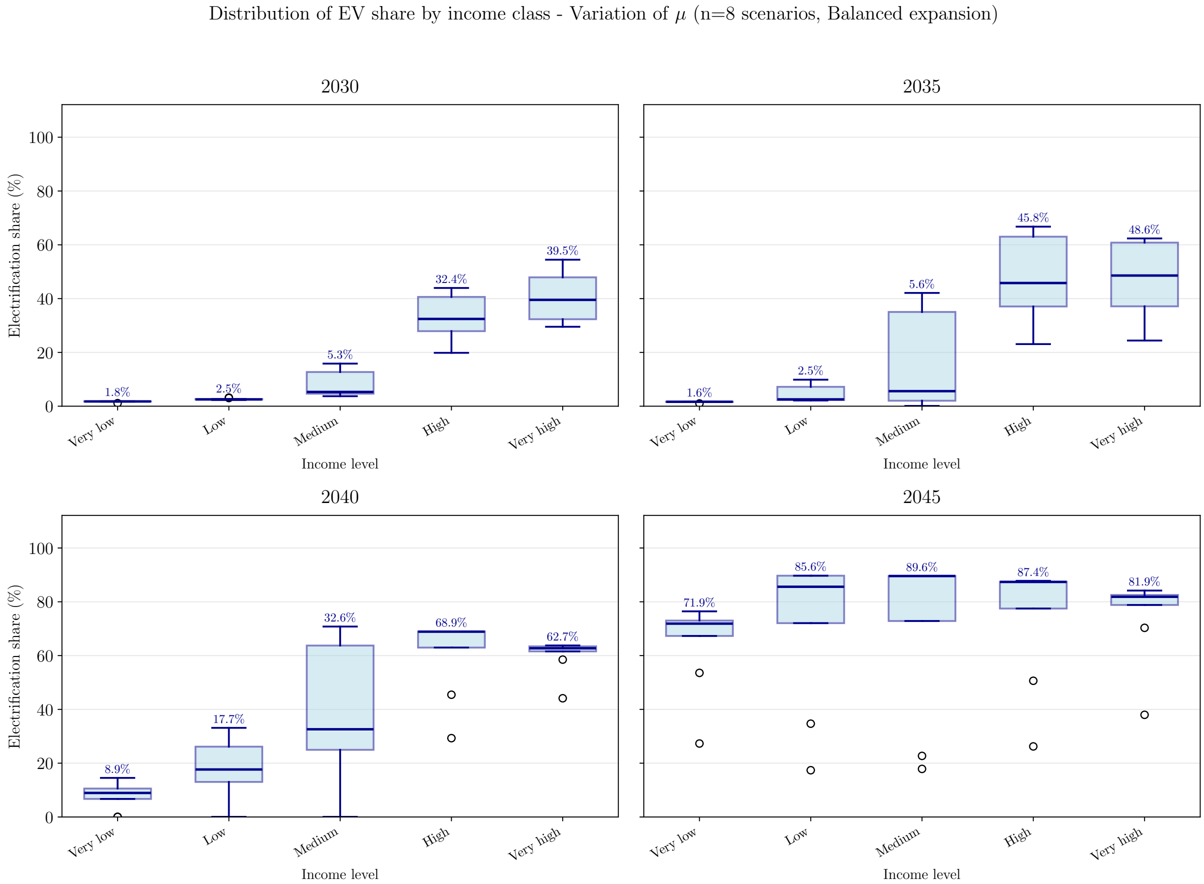

Figure C.4.: Distribution of the development of electrification of the passenger car fleet by income levels for different circuity values μ.

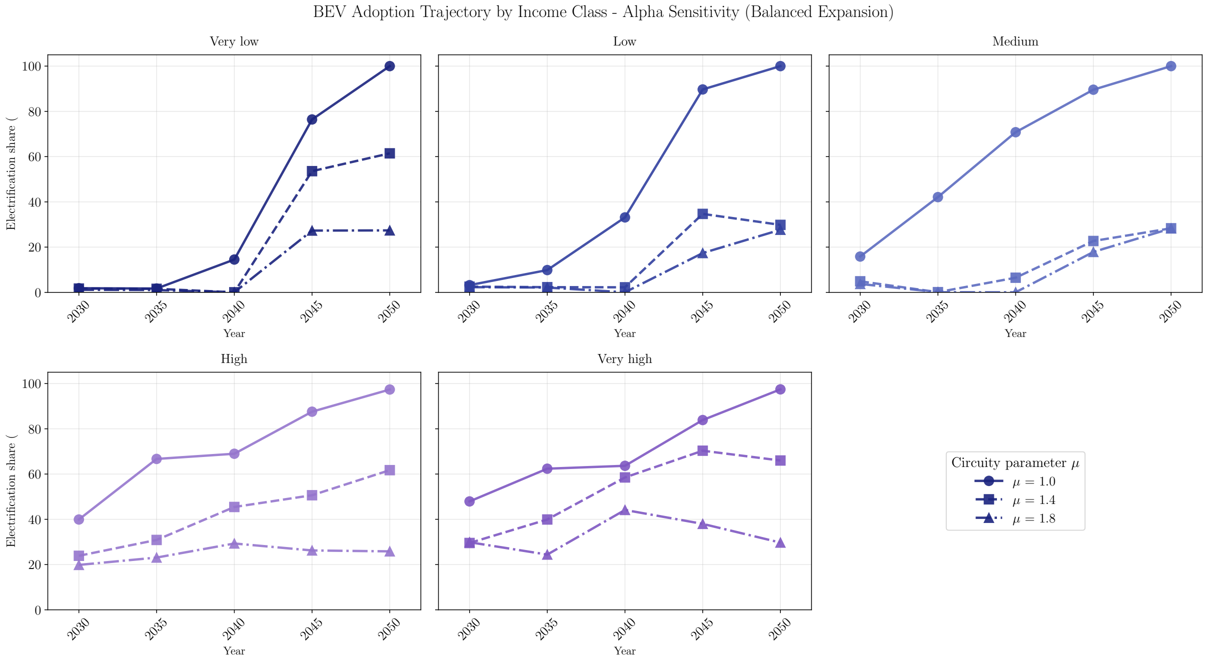

Figure C.5.: Development of electrification shares of the passenger car fleet by income classes for different parameters of μ.

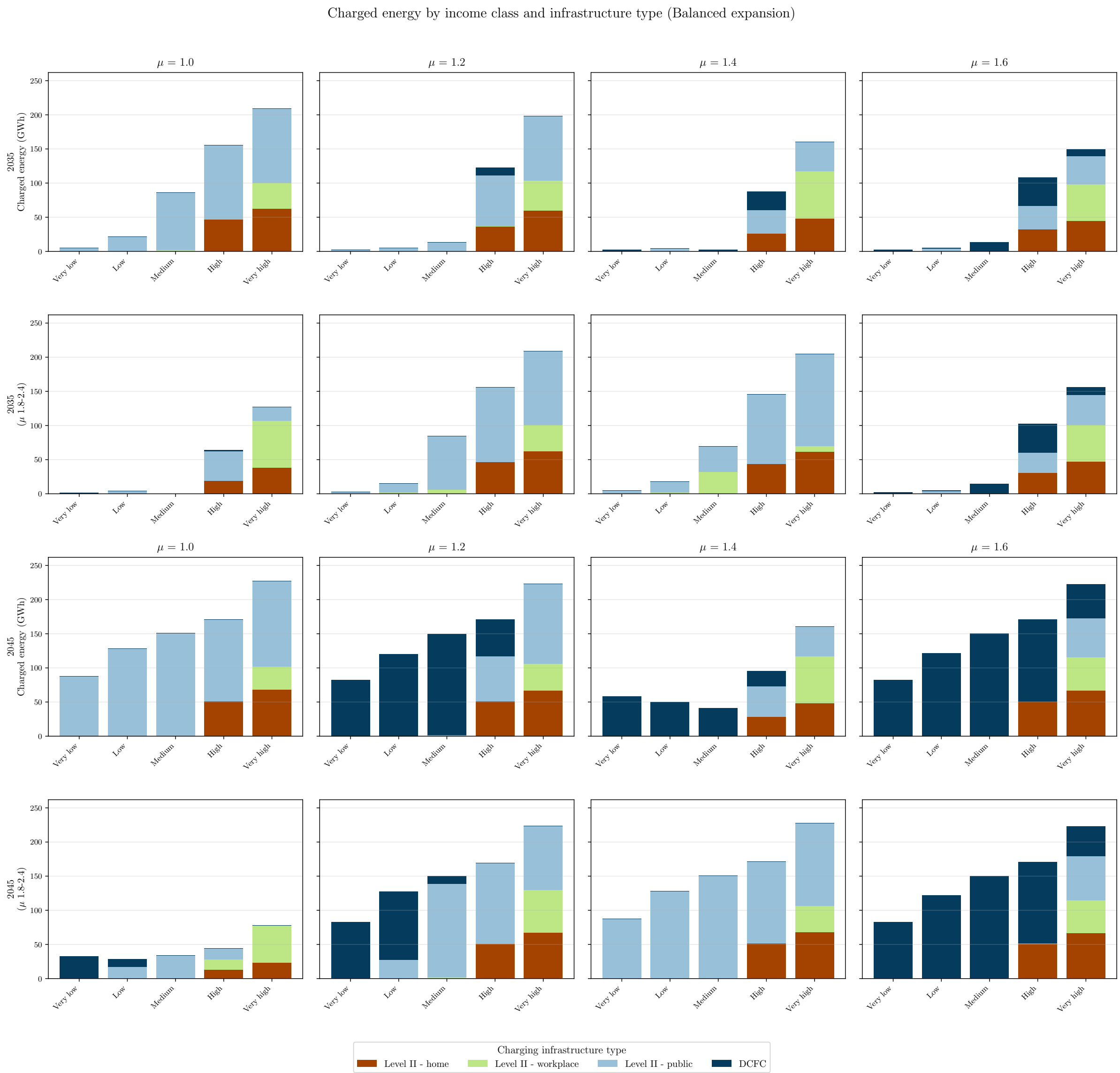

Figure C.6.: Charger utilization in 2050 by income levels for different circuity values μ.

Figure C.7 shows a negative correlation between detour times and expanded charging capacities, which is the intended behavior of the model. Higher availability of public slow charging increases adoption of the *Medium low income* consumer group (Figure C.8).

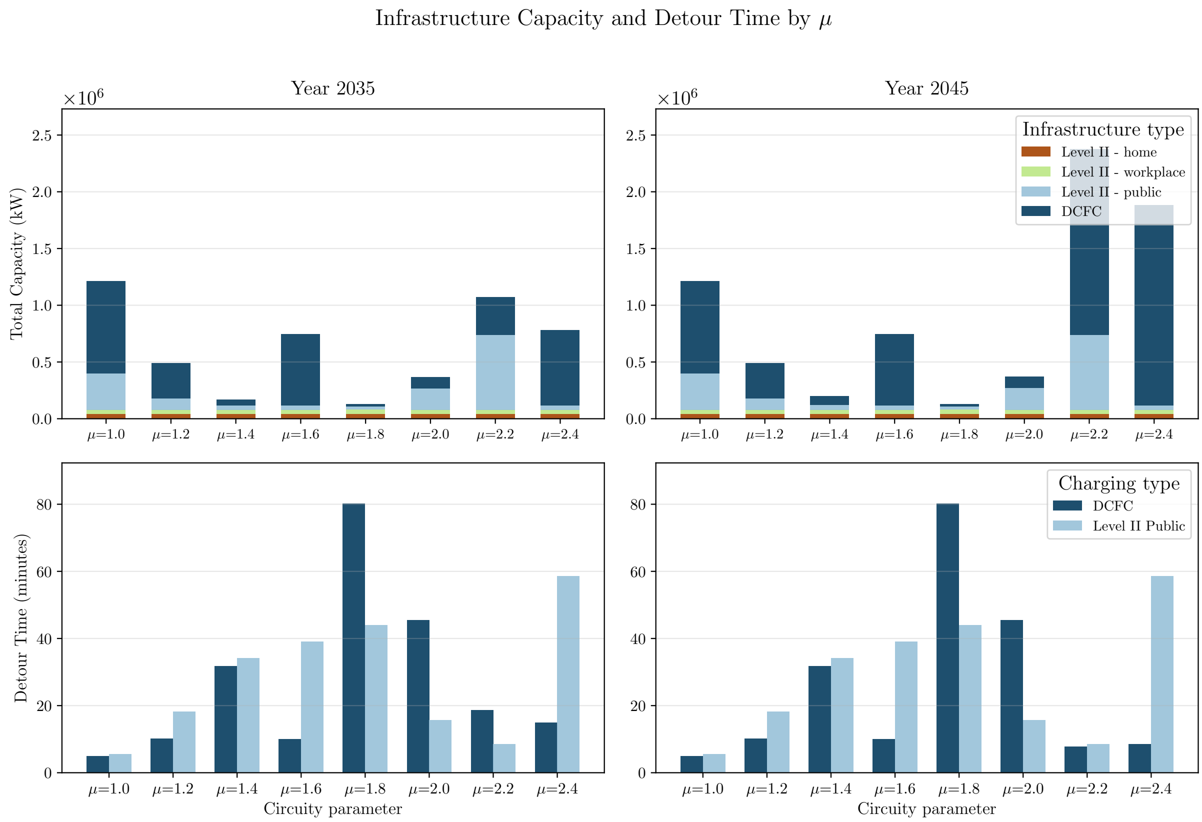

Figure C.7.: Comparison of total installed charging infrastructure and average detour times for different circuity values μ.

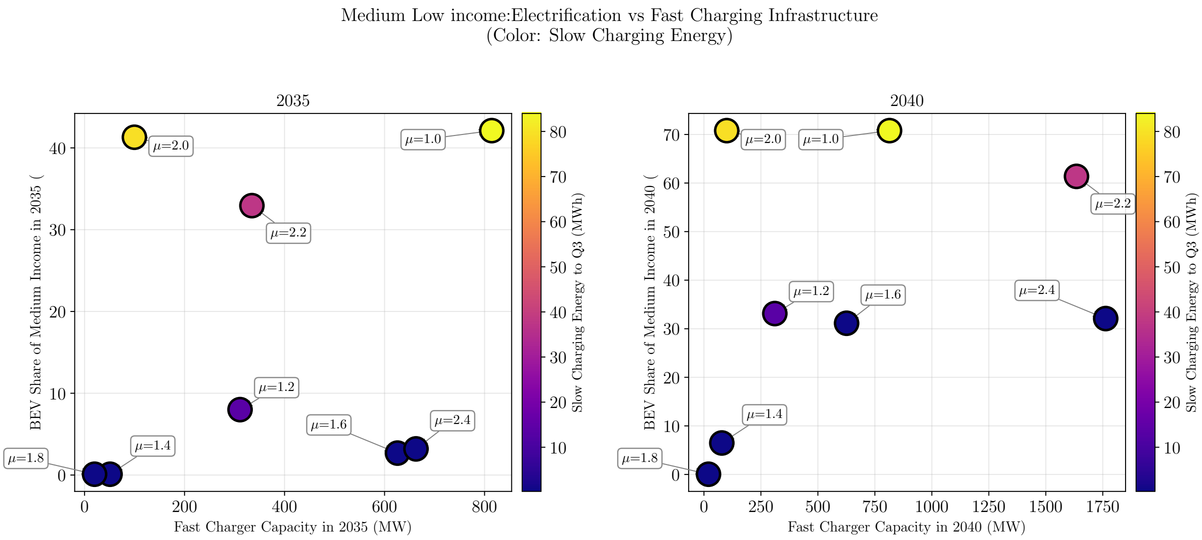

Figure C.8.: Fast charging capacity versus medium-income BEV adoption share for 2035 and 2040 across circuity factor sensitivity scenarios.

C.8. Validation of fueling detour time function

The assumed shape of the fueling detour time function is validated using a Monte Carlo-based approach with real charger data from OpenChargeMap [209] and official Basque Country boundaries [210]. The procedure samples 1,000 random driver locations within the region, identifies the nearest charging station for each location, and calculates the detour time using the same circuity factor and driving speed assumptions as in the model. The validation is performed separately for slow and fast charging networks.

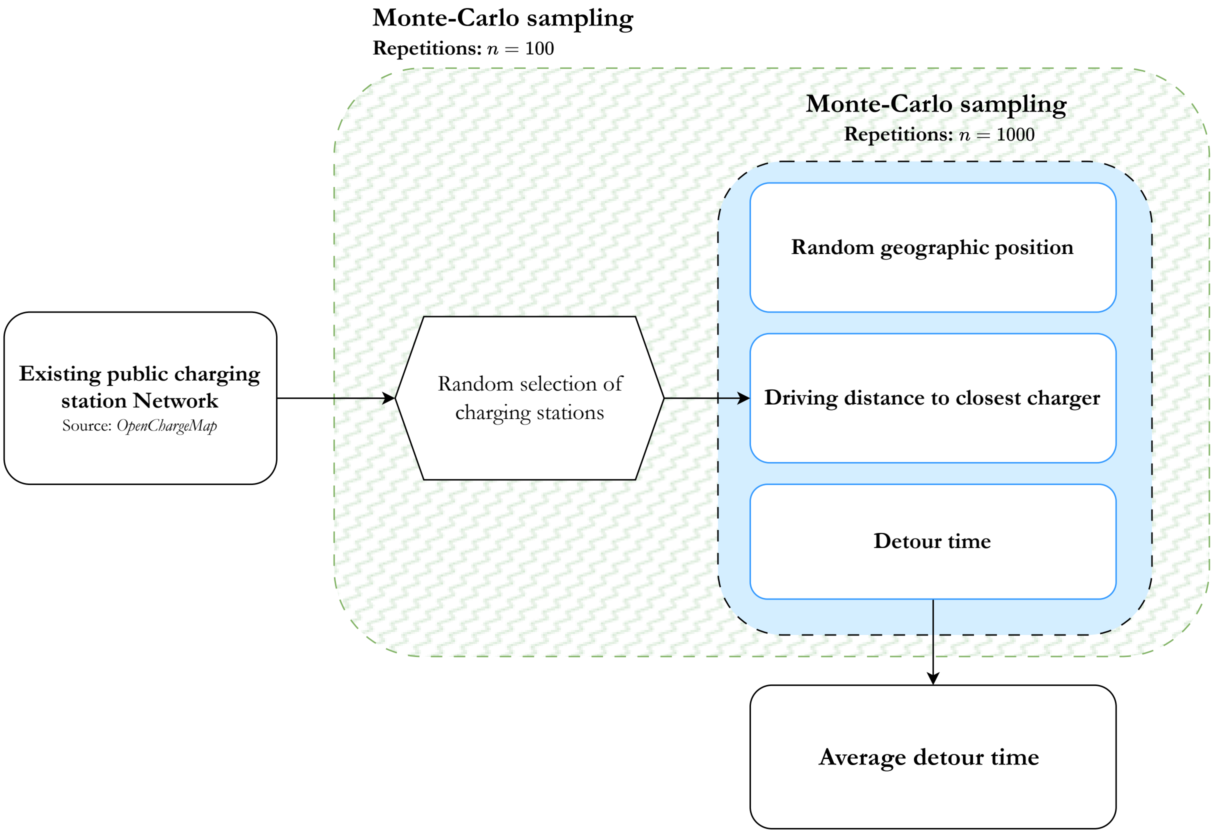

Figure C.9.: Overview of the sampling-based verification of fueling detour time assumptions for the Basque Country.

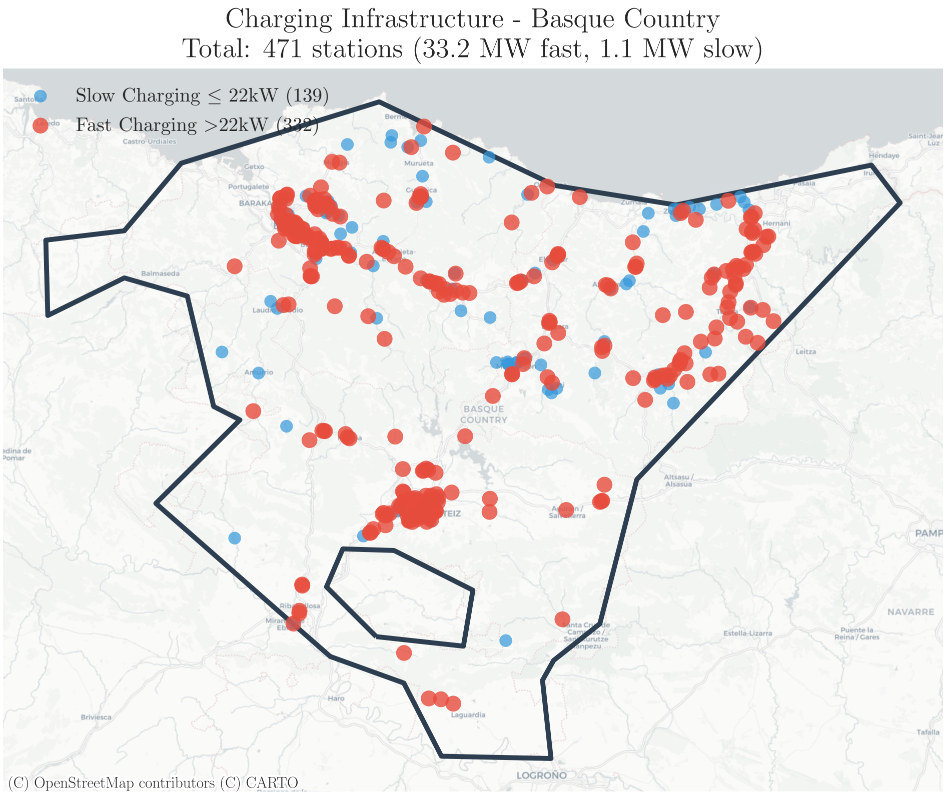

Figure C.10.: Public charger network in the Basque Country, retrieved from OpenChargeMap [209].

Figures C.11 and C.12 display the validation results for slow and fast chargers, respectively. The results show strong correspondence between the analytical assumptions and the empirically-derived curves from the real charging infrastructure.

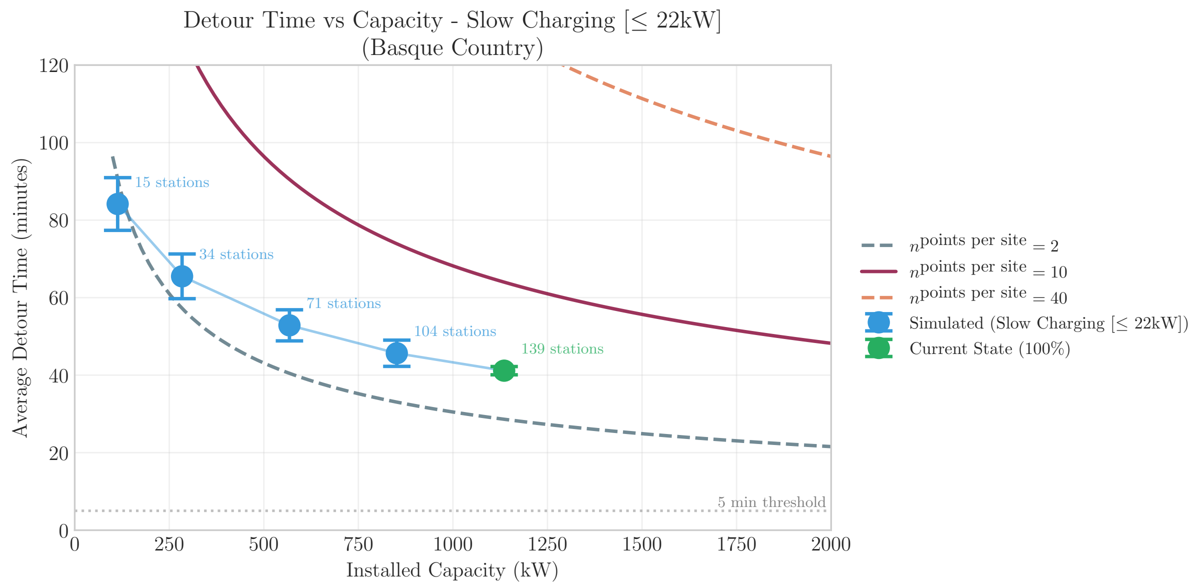

Figure C.11.: Validation of the fueling detour time function for slow chargers in the Basque Country: comparison between Monte Carlo simulation results and assumed functional form.

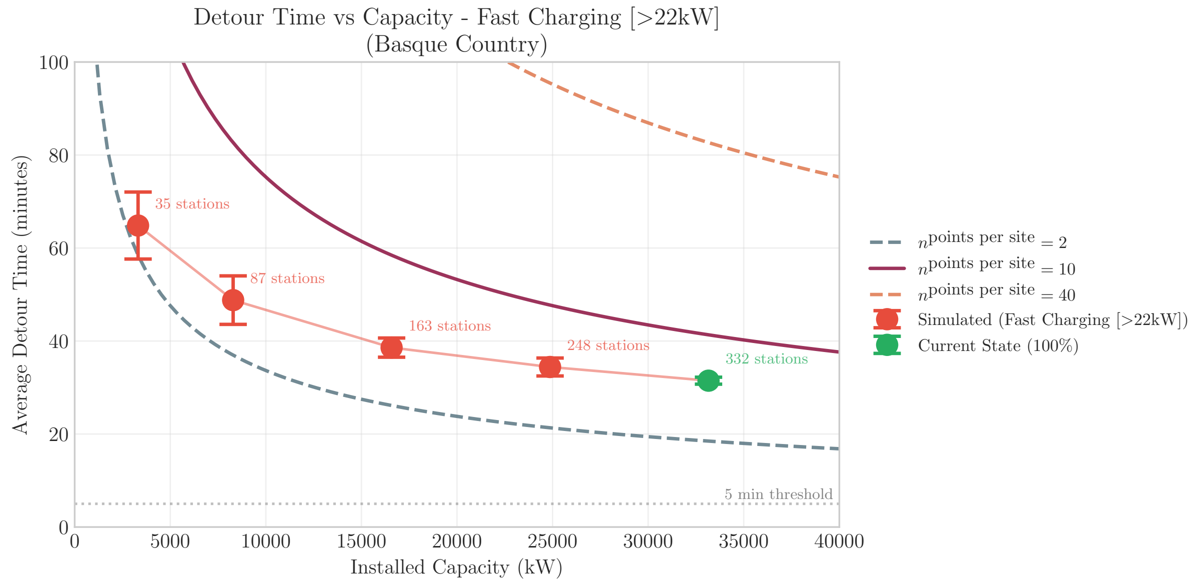

Figure C.12.: Validation of the fueling detour time function for fast chargers in the Basque Country: comparison between Monte Carlo simulation results and assumed functional form.

C.9. Sensitivity analysis: value of time

C.9.1. Quantification of value of time

The value of time (VoT) represents the monetary value that consumers assign to the time spent detouring to reach a charging station. The Victoria Transport Policy Institute recommends valuing personal travel, including commuting, at approximately 50% of the hourly wage rate [211]. For transit under comfortable conditions, recommended values range from 25–35% of wage, while crowded conditions can justify values up to 100%. Similarly, a recent World Bank meta-analysis of VoT studies in low- and middle-income countries found that VoT corresponds to approximately 50% of after-tax wages [212].

Table C.4 shows that the Danish VoT values used in the original source [19] fall within these empirically supported ranges, with VoT/wage ratios spanning from 13% (Q1) to 63% (Q5).

Table C.4.: Danish value of time by income quintile.

Quintile	VoT (€/h)	VoT/Avg Wage (%)
Q1	6.60	13
Q2	12.77	26
Q3	18.93	38
Q4	25.10	50
Q5	31.27	63

Income quintile share ratios (S80/S20) reported in Eurostat [213] show different inequality levels between Denmark and Spain. To better reflect Spanish income distribution while maintaining the empirically grounded VoT/wage relationship from the Danish study, the VoT values are adjusted according to Spain's income ratio. Table C.5 presents the adjusted values.

Table C.5.: Comparison of value of time estimates using Danish and Spain-adjusted income ratios.

Quintile	VoT Before		VoT After (Spain ratio)	
	VoT (€/h)	VoT/Avg Wage (%)	VoT (€/h)	VoT/Avg Wage (%)
Q1	4.37	17	3.86	15
Q2	8.45	33	8.19	32
Q3	12.53	49	12.53	49
Q4	16.61	65	16.87	66
Q5	20.69	81	21.20	83

As an additional validation, the Spanish Independent Authority for Fiscal Responsibility (AIRef) uses VoT values of 12.1 €/h for short-distance and 15.0 €/h for long-distance travel [214], as shown in Table C.6. These values fall within the third quintile range, supporting the reasonableness of the income-differentiated estimates.

Table C.6.: Value of time by trip type [214].

Trip Type	VoT (€/h)
Urban commuting	12.1
Intercity commuting	15.0

C.9.2. Sensitivity analysis of value of time

The sensitivity analysis tests absolute shifts of ±3 €/h, which correspond approximately to the absolute differences between VoT values for inter-city versus urban car trips in Spain [215], and relative compression/expansion of VoT differences between income classes (factors of 0.5, 0.75, 1.25, and 1.5). Figure C.13 illustrates the tested VoT scenarios.

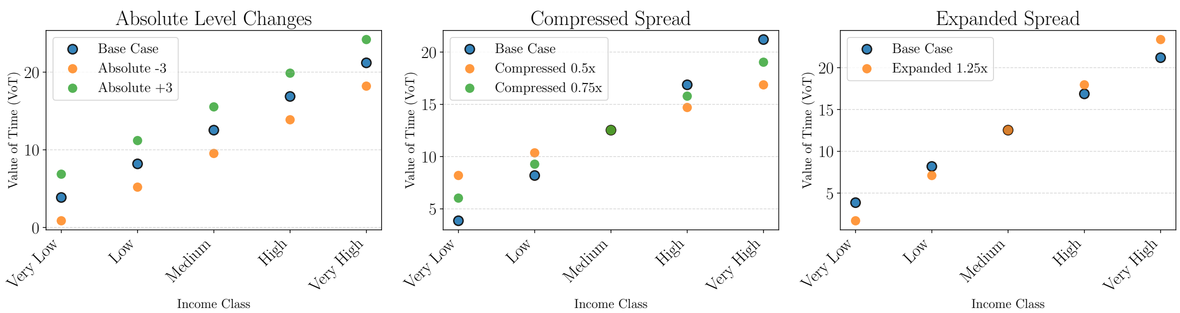

Figure C.13.: Base case VoT values for the Basque Country with tested absolute and relative changes.

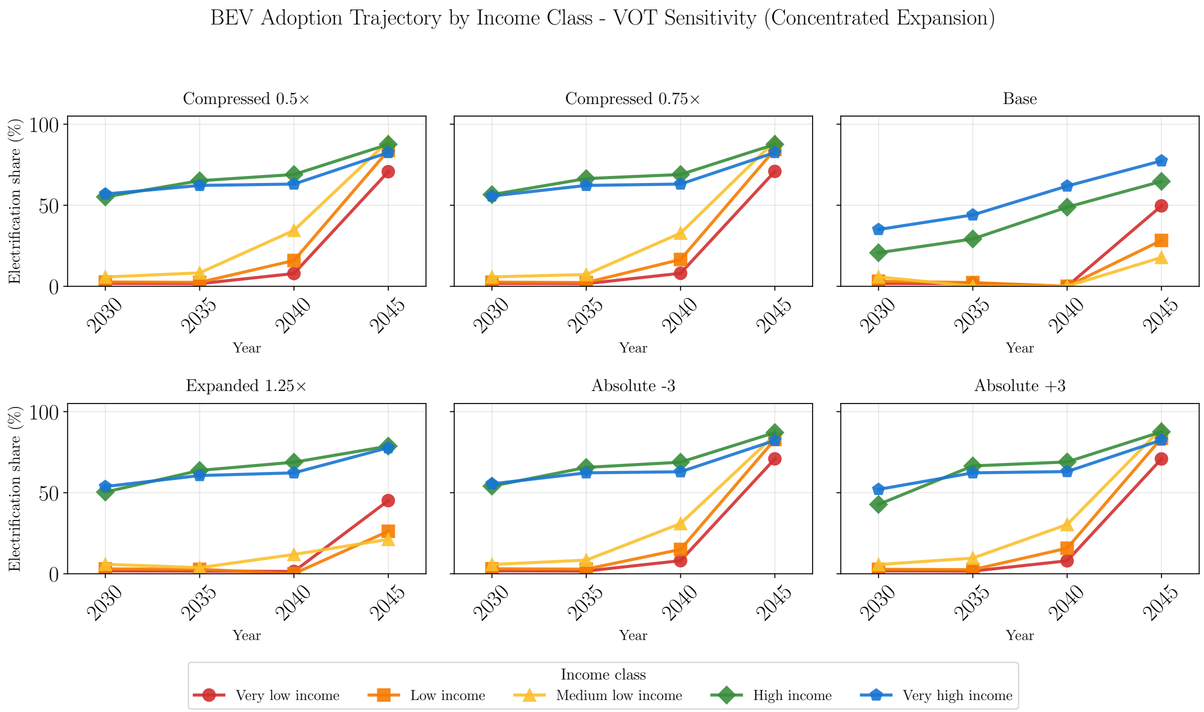

Figure C.14.: Development of electrification of the passenger car fleet by income levels during VoT sensitivity analysis.

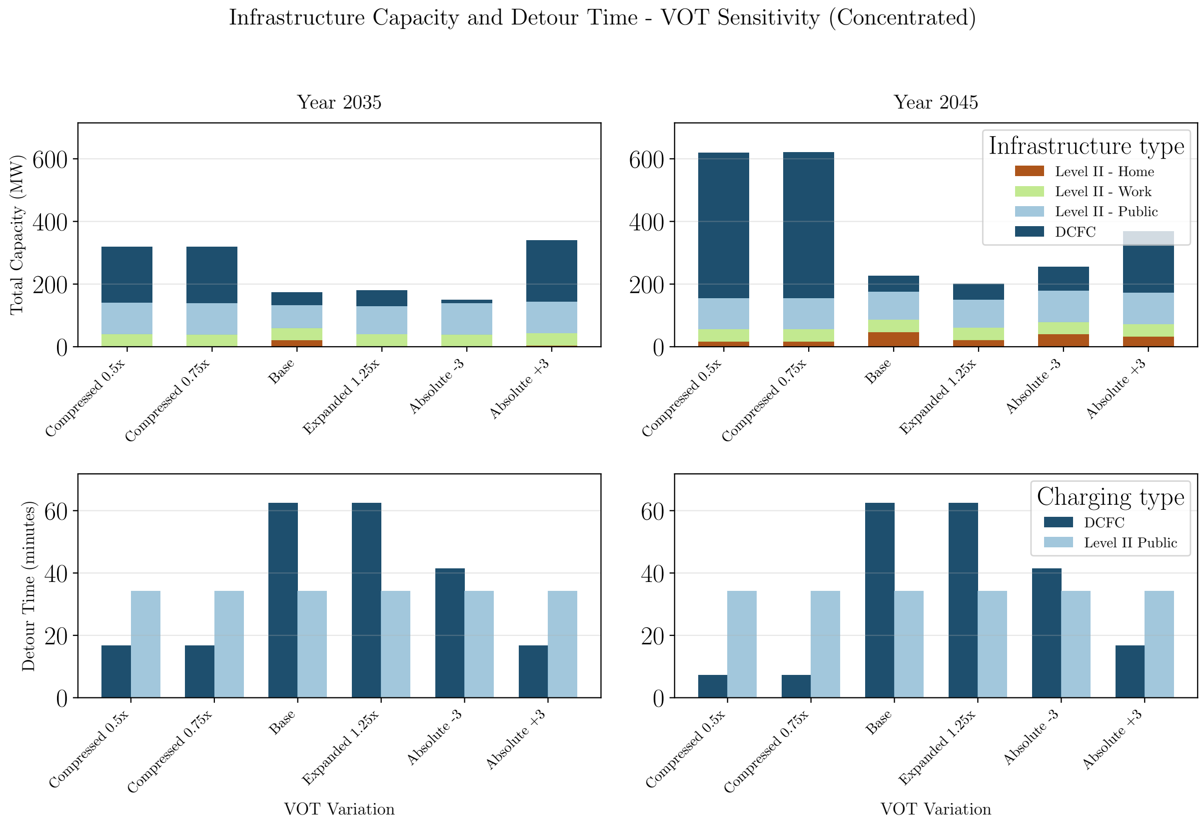

Figure C.15.: Comparison of total installed charging infrastructure and average detour times for different VoT scenarios.

The highest sensitivity is observed in the *Medium low income* consumer group. Fast charging utilization by the *Very low income* and *Low income* groups varies significantly with VoT settings.

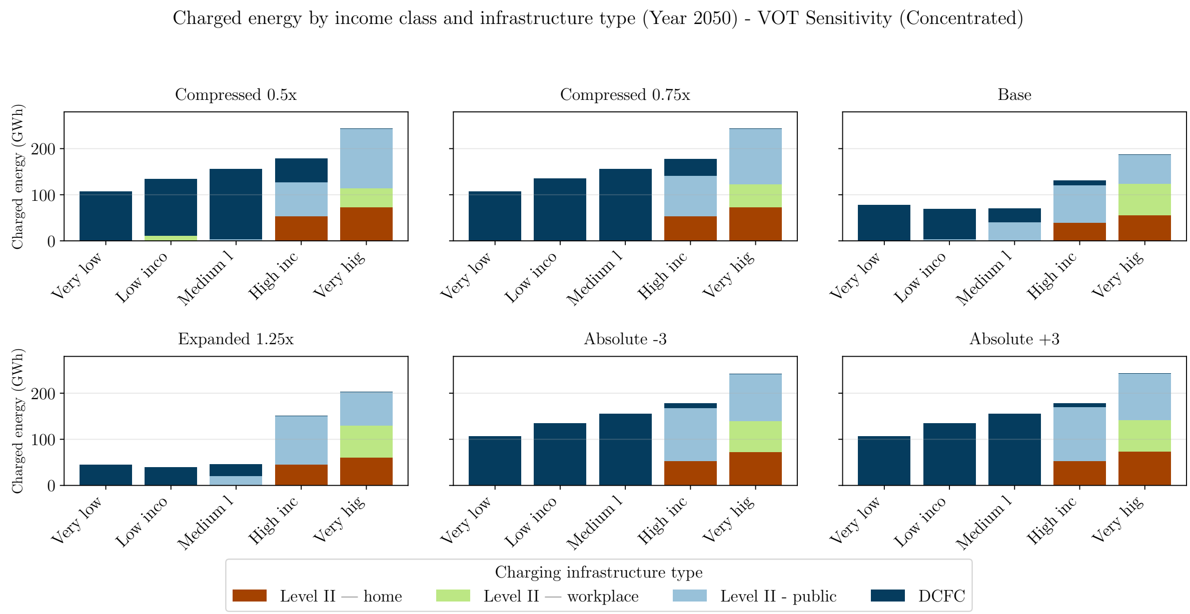

Figure C.16.: Charged energy by charger type in 2050 for different VoT scenarios.

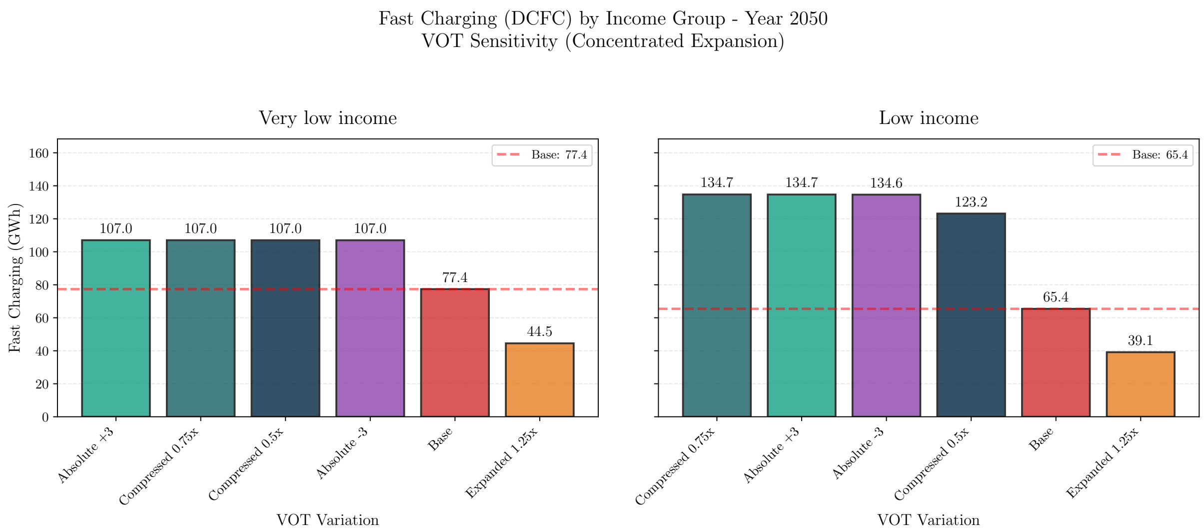

Figure C.17.: Charged energy at fast chargers by the *Very low income* and *Low income* groups in 2050 for different VoT scenarios.

C.10. Sensitivity analysis: home charger capacities

The expansion of home charging is treated as an exogenous parameter reflecting the share of maximum home charging capacity that is reached by 2050. Five scenarios are defined: 20%, 40%, 60%, 80%, and 100% of maximum capacity. Higher home charging availability shifts utilization away from public charging.

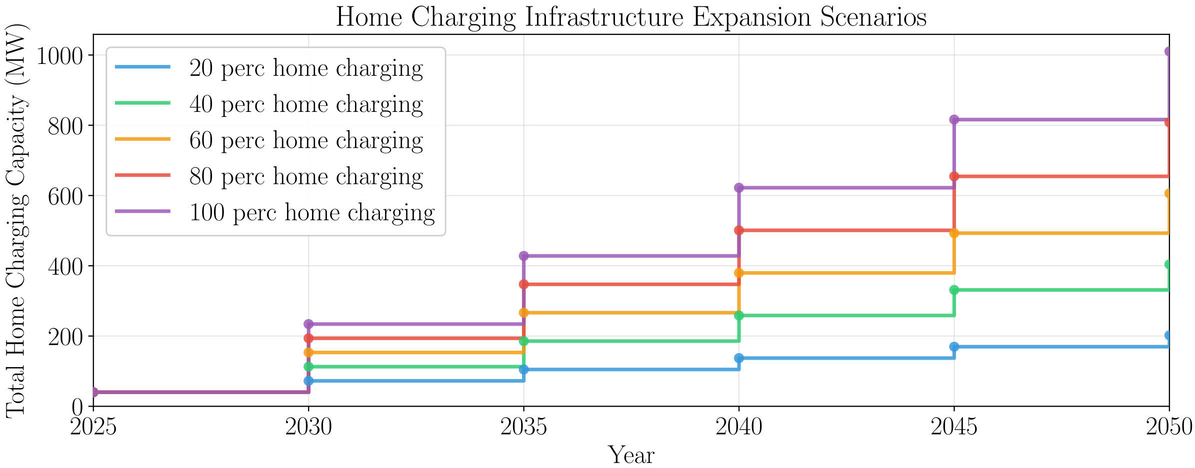

Figure C.18.: Exogenous home charging capacity expansion scenarios.

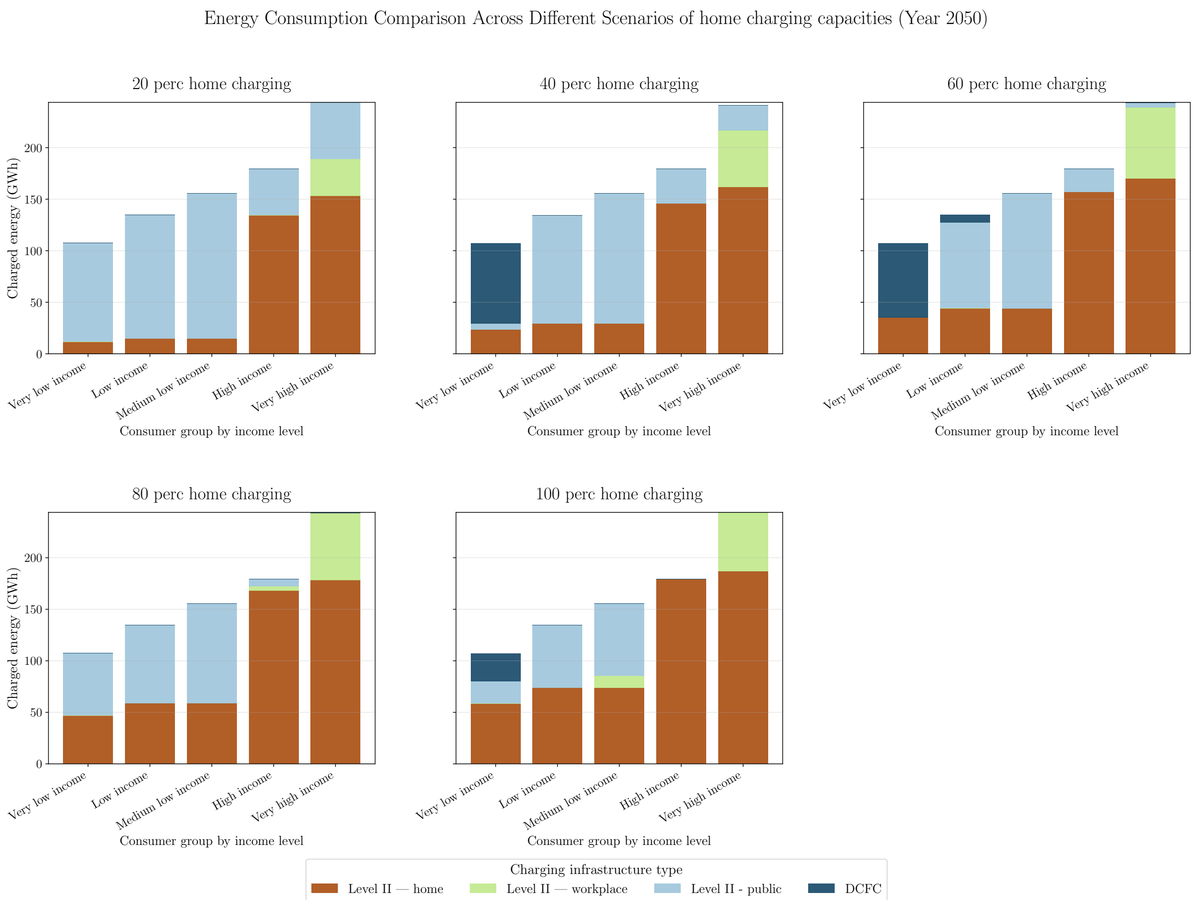

Figure C.19.: Energy consumption patterns across income classes for different exogenous home charging capacity scenarios.

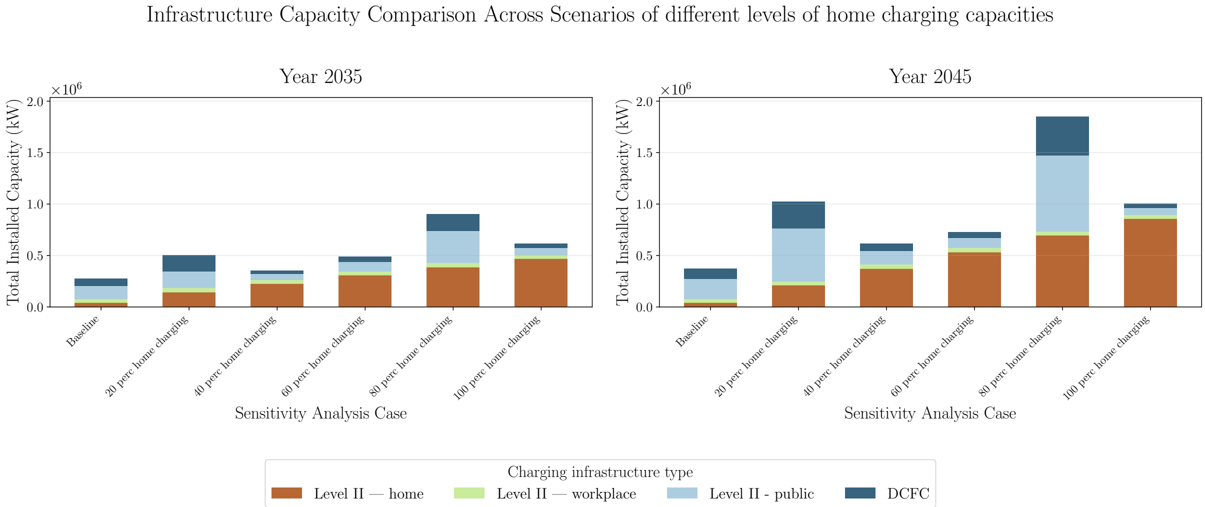

Figure C.20.: Charging infrastructure capacities by charger type for 2035 and 2045 across home charging scenarios.

C.11. Sensitivity analysis: central accelerated roll-out

To test the robustness of the spillover effects observed in the main analysis, additional scenarios with +15%, +30%, and +45% capacity expansion in Bizkaia are tested. The spillover effect increases with larger capacities but remains modest overall.

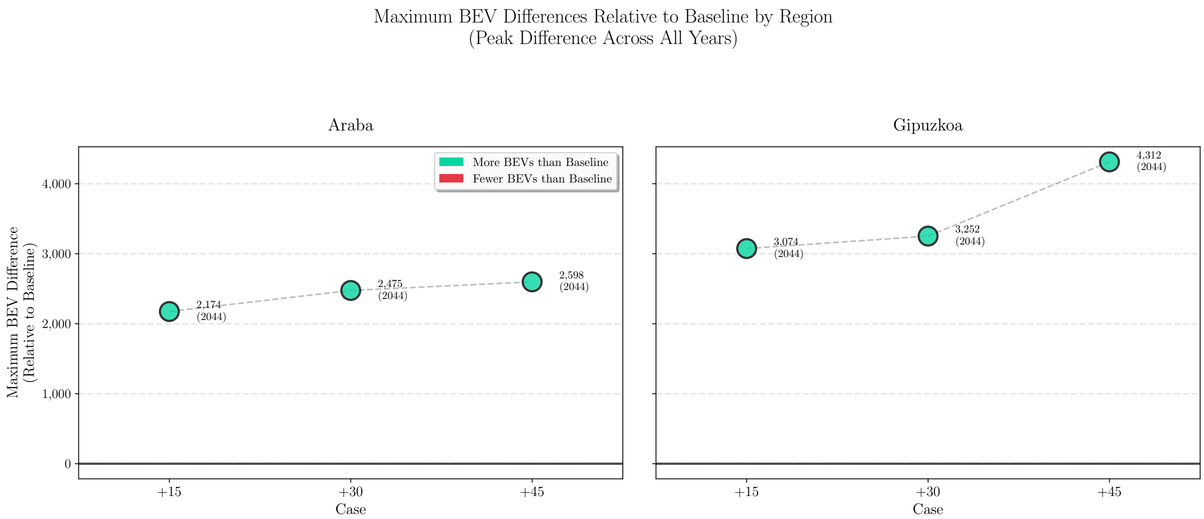

Figure C.21.: Peak absolute differences in BEV adoption between the *Baseline Roll-out* and *Central Acc. Roll-out* scenarios across capacity expansion levels (+15%, +30%, +45%).

Appendix D.

Supplementary Material for Contribution 3 (RQ3)

The content of this appendix is based on Golab et al. (n.d.).

D.1. Extended nomenclature

Table D.1 lists the extended nomenclature for the long-distance freight transport decarbonization model described in Section 3.3.6. Compared to the core model nomenclature (Table 3.4), consumer groups (u) and vehicle types (v) are dropped, as a single vehicle type (trucks) is considered. The technology index t directly distinguishes ICEV and BET for road transport and rail. A mode index $m \in \mathcal{M}$ is introduced for road and rail transport. The flow variable f_{yrmtg} represents annual freight tonnage rather than person-trips, and the payload W_{tg} represents tonnes per vehicle rather than persons per vehicle.

Table D.1.: Extended nomenclature for the long-distance freight transport decarbonization model (RQ3).

Sets and indices	Description
$y \in \mathcal{Y}$	year
$r \in \mathcal{R}$	route between origin-destination (OD) pair
$t \in \mathcal{T}$	drivetrain technology (ICEV, BET for road; rail)
$g \in \mathcal{G}$	year that vehicle was introduced to market
$m \in \mathcal{M}$	transport mode (road, rail)
$e \in \mathcal{E}$	geographic location (NUTS-2 region)
$\mathbf{E}_r$	ordered set of geographic locations along route r
$e_{r,i}$	i-th geographic location along route r
$l \in \mathcal{L}$	types of fueling infrastructure (by capacity)
$\mathcal{Y}^{\text{exp}}$	set of years of investment decisions
$\mathcal{Y}_y^{\text{exp}}$	subset of investment years up to year y
$\mathcal{T}_m$	technologies available for mode m
$\mathcal{T}^{\text{road}}$	road technologies (ICEV, BET)
$\mathcal{T}^{\text{rail}}$	rail technology
$\mathcal{T}^{\text{BET}}$	battery-electric truck technology
$\mathcal{T}^{\text{ICEV}}$	internal combustion engine technology
$\mathcal{R}_e$	set of routes that go through location e
$\mathcal{L}_t$	types of fueling infrastructure by drivetrain technology
$\mathcal{E}_r$	set of geographic locations along route r

Decision variables	Description	Unit	Domain
f_{yrmtg}	freight flow by mode	t/a	$[0, \infty[$
h_{yrtg}	number of vehicles	–	$[0, \infty[$
h^{+}_{yrtg}	newly purchased vehicles	–	$[0, \infty[$
h^{-}_{yrtg}	number of vehicles leaving the vehicle stock	–	$[0, \infty[$
s_{yrtgle}	energy fueled	kWh	$[0, \infty[$

soc_{yrtge}	fleet-aggregate state of charge at location e	kWh	$[0, \infty[$
$travel\_time_{yrtge}$	fleet-aggregate accumulated travel time at location e	h	$[0, \infty[$
$break\_time^{+}_{yrtge}$	time on mandatory break not used for charging	h	$[0, \infty[$
$q^{+,\text{charg\_infr}}_{ytle}$	added charging infrastructure	kW	$[0, \infty[$
h^{exist}_{yrtg}	vehicles carried over from previous period	–	$[0, \infty[$

Parameters	Description	Unit	Domain
y_0	initial (base) year	–	–
F_{yr}	exogenous freight demand	t/a	$[0, \infty[$
F^{rail}_{y}	total rail freight flow in year y	t·km	$[0, \infty[$
F^{total}_{y}	total freight flow in year y	t·km	$[0, \infty[$
D^{spec}_{tg}	specific energy consumption	kWh/km	$[0, \infty[$
W_{tg}	payload capacity (tonnes per vehicle)	t	$[0, \infty[$
L_r	length of route r	km	$[0, \infty[$
$L_{e_{(i-1,i)}}$	distance between consecutive geographic locations	km	$[0, \infty[$
L^{annual}_{tg}	annual mileage	km	$[0, \infty[$
$\text{Cap}^{\text{tank}}_{tg}$	tank capacity	kWh	$[0, \infty[$
Q^{batt}_{tg}	battery capacity	kWh	$[0, \infty[$
$\hat{P}_{tg}$	charging power	kW	$[0, \infty[$
SOC^{init}	initial state of charge (fraction of battery capacity)	–	$[0, 1]$
SOC^{final}	minimum final state of charge (fraction of battery capacity)	–	$[0, 1]$
$\text{SOC}^{+,\text{charging}}$	usable battery capacity for charging	–	$[0, 1]$
$speed$	average driving speed (road)	km/h	$[0, \infty[$
$TravelTime^{\min}_{e_{r,j}}$	minimum cumulative travel time at node j along route r	h	$[0, \infty[$
$\bar{\alpha}^{\text{mode}}$	maximum corridor-wide rail share	–	$[0, 1]$
$\alpha_f,\ \beta_f$	rate-of-change parameters for modal shift	–	$[0, \infty[$
C^{CAPEX}_{ytg}	expenditure costs for vehicles	€	$[0, \infty[$
$C^{\text{O\&M,veh}}_{ytg}$	vehicle maintenance and repair costs	€/a	$[0, \infty[$
$C^{\text{CAPEX,infr}}_{ytl}$	capital expenditure for charging infrastructure	€/kW	$[0, \infty[$
$C^{\text{O\&M,infr}}_{ytl}$	operation and maintenance cost for charging infrastructure	€/kW/a	$[0, \infty[$
c^{fuel}_{ytle}	fuel or electricity cost at location e	€/kWh	$[0, \infty[$
c^{grid}_{e}	annual grid charge at location e	€/kW/a	$[0, \infty[$
VoT	value of time (freight)	€/h	$[0, \infty[$
γ_t	utilization factor (ratio of average to peak power demand)	–	$[0, 1]$
δ	discount rate	–	$[0, 1]$
Δy	time step between consecutive modeled periods	a	$[0, \infty[$
Λ_{tg}	maximum vehicle lifetime	a	$[0, \infty[$
H^{init}_{rtg}	initial vehicle stock	–	$[0, \infty[$
$Q^{\text{charg\_infr}}_{tle}$	initial charging infrastructure	kW	$[0, \infty[$
$\hat{Q}^{\text{charg\_infr}}_{tle}$	upper limit for charging infrastructure capacity	kW	$[0, \infty[$
$\alpha^h,\ \beta^h$	maximum stock transition rate parameters	–	$[0, \infty[$
ρ	maximum infrastructure growth rate per period	–	$[0, \infty[$
$\tau_t(y)$	technology-specific toll rate	€/km	$[0, \infty[$
$\overline{q}^{\text{init}}$	maximum new infrastructure investment in the initial period	kW	$[0, \infty[$
c^{rail}	levelized rail transport cost	€/tkm	$[0, \infty[$

c^{terminal}	rail terminal handling cost	€/t	$[0, \infty[$

D.2. Extended mathematical formulation

This section provides the complete set of constraints of the long-distance freight transport decarbonization model. Constraints that are also presented with detailed explanations in Section 3.3.6 are included here for reference. Compared to the RQ2 formulation (Appendix C.2), the key structural differences are: (i) consumer groups (u) and vehicle types (v) are dropped; (ii) a mode index m distinguishes road and rail transport; (iii) flow f_{yrmtg} represents annual freight tonnage; (iv) new state variables track the battery state of charge, travel time, and break time along routes. For road-specific constraints, f_{yrtg} denotes the road-mode flow, i.e., $f_{yr(m=\text{road})tg}$.

Objective function:

The objective function minimizes the net present value of total system costs comprising vehicle stock costs ($C^{\text{vehicle\_stock}}$), fueling costs (C^{fueling}), toll costs (C^{toll}), charging infrastructure costs ($C^{\text{infrastructure}}$), time-related costs ($C^{\text{time}}$), and rail transport costs (C^{rail}). All cost components are discounted using the discount rate δ:

$$\begin{aligned}\underset{f,h,h^+,h^-,s,soc,travel\_time,break\_time^+,q^+}{minimize} \quad z = \sum_{y} \frac{1}{(1+\delta)^{y-y_0}} \Big(& C_y^{\text{vehicle\_stock}} + C_y^{\text{fueling}} \\ & + C_y^{\text{toll}} + C_y^{\text{infrastructure}} + C_y^{\text{time}} + C_y^{\text{rail}} \Big)\end{aligned} \tag{D.1}$$

The vehicle stock cost sums capital expenditures for newly purchased vehicles and maintenance costs across all routes, technologies, and generations:

$$C_y^{\text{vehicle\_stock}} = \sum_{r,t,g} C_{ytg}^{\text{CAPEX}} \cdot h_{yrtg}^{+} + \sum_{r,t,g} C_{ytg}^{\text{O\&M,veh}} \cdot h_{yrtg} \tag{D.2}$$

The fueling cost sums the location-dependent fuel or electricity charges:

$$C_y^{\text{fueling}} = \sum_{r,t,g,l,e} c_{ytle}^{\text{fuel}} \cdot s_{yrtgle} \tag{D.3}$$

Toll costs represent distance-based road charges differentiated by drivetrain technology:

$$C_y^{\text{toll}} = \sum_{r,t,g} f_{yrtg} \cdot \frac{L_r}{W_{tg}} \cdot \tau_t(y) \tag{D.4}$$

where $\tau_t(y)$ is the technology-specific toll rate (€/km), applied uniformly across all locations.

The infrastructure cost covers capital investment, operation and maintenance, and country-specific annual grid charges:

$$\begin{aligned} C_y^{\text{infrastructure}} &= \sum_{t,l,e} C_{ytl}^{\text{CAPEX,infr}} \cdot q_{ytle}^{+,\text{charg\_infr}} \\ &+ \sum_{t,l,e} \left(C_{ytl}^{\text{O\&M,infr}} + c_e^{\text{grid}} \right) \cdot \left(Q_{tle}^{\text{charg\_infr}} + \sum_{y' \in \mathcal{Y}_y^{\text{exp}}} q_{y'tle}^{+,\text{charg\_infr}} \right) \end{aligned} \tag{D.5}$$

The time cost monetizes the fleet-aggregate travel time via the value of time:

$$C_y^{\text{time}} = \text{VoT} \cdot \sum_{\substack{r \in \mathcal{R},\, t \in \mathcal{T}^{\text{road}} \\ g \in \mathcal{G}}} travel\_time_{yrtge_{r,|\mathbf{E}_r|}} \tag{D.6}$$

The rail transport cost covers the levelized cost of rail freight and terminal handling at origin and destination:

$$C_y^{\text{rail}} = \sum_{\substack{r \in \mathcal{R},\, t \in \mathcal{T}^{\text{rail}} \\ g \in \mathcal{G}}} \left(c^{\text{rail}} \cdot L_r + 2 \cdot c^{\text{terminal}} \right) \cdot f_{yr(m=\text{rail})tg} \tag{D.7}$$

where c^{rail} is the levelized rail transport cost (€/tkm) and c^{terminal} is the terminal handling cost (€/t) incurred at both origin and destination.

Demand coverage:

The total freight flow across all modes, technologies, and vehicle generations must satisfy the exogenous demand for each route:

$$\sum_{m \in \mathcal{M}} \sum_{t \in \mathcal{T}_m} \sum_{g \in \mathcal{G}} f_{yrmtg} = F_{yr} \quad : \forall y, r \tag{D.8}$$

Fueling infrastructure sizing:

Fueling infrastructure is expanded for each investment period $y' \in \mathcal{Y}^{\text{exp}}$ and sized based on the utilization factor γ_t, representing the ratio of average to peak power demand:

$$Q_{tle}^{\text{charg\_infr}} + \sum_{y' \in \mathcal{Y}_y^{\text{exp}}} q_{y'tle}^{+,\text{charg\_infr}} \geq \gamma_t \sum_{r \in \mathcal{R}} \sum_{g \in \mathcal{G}} s_{yrtgle} \quad : \forall y, t, e, l \tag{D.9}$$

Vehicle sizing:

The vehicle stock is sized based on the freight flow covered by each technology, the route length, payload capacity, and annual mileage:

$$h_{yrtg} \geq \frac{L_r}{W_{tg} \cdot L_{tg}^{\text{annual}}} f_{yrtg} \quad : \forall y, r, t \in \mathcal{T}^{\text{road}}, g \tag{D.10}$$

Vehicle stock balance with vintage tracking:

The vehicle stock of each generation g is tracked using the carried-over stock variable h^{exist}:

$$h_{yrtg} = h_{yrtg}^{\text{exist}} + h_{yrtg}^{+} - h_{yrtg}^{-} \quad : \forall y, r, t \in \mathcal{T}^{\text{road}},\ g \leq y \tag{D.11}$$

The carried-over stock h^{exist} is defined analogously to Equation C.6, with consumer groups and vehicle types removed:

$$h_{yrtg}^{\text{exist}} = \begin{cases} H_{rtg}^{\text{init}} & \text{if } y = y_0,\ g < y_0,\ y_0 - g \leq \Lambda_{tg} \\ h_{(y-1)rtg} & \text{if } y > y_0,\ g < y,\ y - g \leq \Lambda_{tg} \\ 0 & \text{if } g = y \text{ (new generation)} \\ 0 & \text{if } y - g > \Lambda_{tg} \text{ (lifetime exceeded)} \end{cases} \tag{D.12}$$

New vehicle purchases are restricted to the current generation:

$$h_{yrtg}^{+} = 0 \quad : \forall y, r, t \in \mathcal{T}^{\text{road}},\ g \neq y \tag{D.13}$$

Fueling demand:

The total energy fueled along a route must cover the energy consumption derived from the specific demand and route length:

$$\sum_{l \in \mathcal{L}_t} \sum_{e \in \mathcal{E}_r} s_{yrtgle} \geq \frac{D_{tg}^{\text{spec}} \cdot L_r}{W_{tg}} f_{yrtg} \quad : \forall y, r, t \in \mathcal{T}^{\text{road}}, g \tag{D.14}$$

Maximum capacity constraints:

The expansion of charging infrastructure is limited by upper bounds:

$$Q_{tle}^{\text{charg\_infr}} + \sum_{y' \in \mathcal{Y}_y^{\text{exp}}} q_{y'tle}^{+,\text{charg\_infr}} \leq \hat{Q}_{tle}^{\text{charg\_infr}} \qquad : \forall y, t, l, e \tag{D.15}$$

State of charge tracking (cf. Equation 3.29):

The battery state of charge is tracked along the ordered sequence of geographic locations visited on each route. All terms are fleet-aggregate quantities: *soc* and *s* represent the total energy stored and charged across all vehicle trips in a given year.

$$\begin{aligned} soc_{yrtge_{r,1}} &= SOC^{\text{init}} \cdot Q_{tg}^{\text{batt}} \cdot \frac{f_{yrtg}}{W_{tg}} \qquad \forall\, y, r, t \in \mathcal{T}^{\text{BET}}, g \\ soc_{yrtge_{r,i}} &= soc_{yrtge_{r,i-1}} + \sum_{l \in \mathcal{L}_t} s_{yrtgle_{r,i}} - D_{tg}^{\text{spec}} \cdot L_{e_{(i-1,i)}} \cdot \frac{f_{yrtg}}{W_{tg}} \qquad \forall\, y, r, t \in \mathcal{T}^{\text{BET}}, g,\ e_{r,i} \in \mathbf{E}_r \setminus \{e_{r,1}\} \end{aligned} \tag{D.16}$$

SOC bounds:

The state of charge must satisfy a minimum final level and remain non-negative at every intermediate node:

$$\begin{aligned} soc_{yrtge_{r,|\mathbf{E}_r|}} &\geq SOC^{\text{final}} \cdot Q_{tg}^{\text{batt}} \cdot \frac{f_{yrtg}}{W_{tg}} \qquad \forall\, y, r, t \in \mathcal{T}^{\text{BET}}, g \\ soc_{yrtge_{r,i-1}} - D_{tg}^{\text{spec}} \cdot L_{e_{(i-1,i)}} \cdot \frac{f_{yrtg}}{W_{tg}} &\geq 0 \qquad \forall\, y, r, t \in \mathcal{T}^{\text{BET}}, g,\ e_{r,i} \in \mathbf{E}_r \setminus \{e_{r,1}\} \\ soc_{yrtge_{r,i}} &\in \left[0,\ Q_{tg}^{\text{batt}} \cdot \frac{f_{yrtg}}{W_{tg}}\right] \qquad \forall\, y, r, t \in \mathcal{T}^{\text{BET}}, g,\ e_{r,i} \in \mathbf{E}_r \end{aligned} \tag{D.17}$$

Travel time tracking (cf. Equation 3.30):

The fleet-aggregate travel time is accumulated along each route, comprising driving time, charging time, and unused break time:

$$\begin{aligned} &travel\_time_{yrtge_{r,1}} = 0 \qquad \forall\, y, r, t \in \mathcal{T}^{\text{road}}, g \\ &travel\_time_{yrtge_{r,i}} = travel\_time_{yrtge_{r,i-1}} + \frac{L_{e_{(i-1,i)}}}{speed} \cdot \frac{f_{yrtg}}{W_{tg}} + \frac{\sum_{l \in \mathcal{L}_t} s_{yrtgle_{r,i}}}{\hat{P}_{tg}} + break\_time^{+}_{yrtge_{r,i}} \\ &\qquad \forall\, y, r, t \in \mathcal{T}^{\text{road}}, g,\ e_{r,i} \in \mathbf{E}_r \setminus \{e_{r,1}\} \end{aligned} \tag{D.18}$$

The charging time term $s_{yrtge_{r,i}}/\hat{P}_{tg}$ divides the total energy charged across all vehicle trips by the per-vehicle charging power, yielding fleet-aggregate charging time (vehicle-hours).

Mandatory break enforcement (cf. Equation 3.31):

At locations where EU driving time regulations require a mandatory break, the accumulated travel time must meet a minimum threshold:

$$travel\_time_{yrtge_{r,j}} \geq TravelTime^{\min}_{e_{r,j}} \cdot \frac{f_{yrtg}}{W_{tg}} \quad \forall\, y, r, t \in \mathcal{T}^{\text{road}}, g,\ e_{r,j} \in \mathbf{E}_r \tag{D.19}$$

where $TravelTime^{\min}_{e_{r,j}}$ is the minimum cumulative travel time at node j, comprising the driving time and the mandatory break time that must have been taken by that point. Since *travel_time* is a fleet-aggregate quantity, the right-hand side is scaled by the number of vehicle trips f_{yrtg}/W_{tg}. The variable $break\_time^{+}_{yrtge}$ in Equation D.18 ensures that the total stop duration at a mandatory break location equals the regulated break duration. Charging during mandatory breaks can partially or fully overlap with the required rest period, reducing additional time cost and creating an economic incentive to co-locate charging at break locations.

Maximum mode share (cf. Equation 3.32):

The share of rail freight is bounded corridor-wide at each modeled year:

$$\sum_{r\in\mathcal{R}} \sum_{t\in\mathcal{T}^{\text{rail}}} \sum_{g\in\mathcal{G}} f_{yr(m=\text{rail})tg} \cdot L_r \ \leq\ \bar{\alpha}^{\text{mode}} \sum_{r\in\mathcal{R}} \sum_{m\in\mathcal{M}} \sum_{t\in\mathcal{T}_m} \sum_{g\in\mathcal{G}} f_{yrmtg} \cdot L_r \quad \forall\, y \tag{D.20}$$

where $\bar{\alpha}^{\text{mode}}$ represents the maximum corridor-wide rail share in tonne-kilometres.

Mode shift rate (cf. Equation 3.33):

The rate of modal shift between consecutive investment periods is bounded:

$$F^{\text{rail}}_{y} - F^{\text{rail}}_{y-\Delta y} \ \leq\ \alpha_f \cdot F^{\text{total}}_{y} + \beta_f \cdot F^{\text{rail}}_{y-\Delta y} \quad \forall\, y \in \mathcal{Y} \setminus \{y_0\} \tag{D.21}$$

where $F^{\text{rail}}_{y} = \sum_{r,t\in\mathcal{T}^{\text{rail}},g} f_{yr(m=\text{rail})tg} \cdot L_r$ and $F^{\text{total}}_{y} = \sum_{r,m,t,g} f_{yrmtg} \cdot L_r$. Parameters α_f and β_f limit the period-over-period growth of rail, reflecting the pace at which intermodal terminal capacity and rail network upgrades can realistically be deployed.

Technology stock transition rate:

The year-over-year change in the vehicle stock of a drivetrain technology t is bounded to prevent unrealistically rapid technology transitions:

$$\left|\sum_{g} h_{yrtg} - \sum_{g} h_{(y-1)rtg}\right| \leq \alpha^h \sum_{t',g} h_{yrt'g} + \beta^h \sum_{g} h_{(y-1)rtg} \quad : \forall y \in \mathcal{Y} \setminus \{y_0\}, r, t \in \mathcal{T}^{\text{road}} \quad \text{(D.22)}$$

where α^h and β^h are the maximum rate-of-change parameters relative to the total vehicle stock and the previous stock of the respective technology.

Infrastructure expansion rate limits:

The expansion of public charging infrastructure is bounded by initial-period capacity limits and subsequent growth rate constraints:

$$\begin{aligned}
&q^{+,\text{charg_infr}}_{y_0tle} \leq \overline{q}^{\text{init}} \quad : \forall t, e, l \\
&q^{+,\text{charg_infr}}_{ytle} - q^{+,\text{charg_infr}}_{(y-\Delta y)tle} \leq \rho \cdot \Delta y \cdot \left(Q^{\text{charg_infr}}_{tle} + \sum_{y'<y} q^{+,\text{charg_infr}}_{y'tle} \right) \\
&\qquad : \forall y \in \mathcal{Y}^{\text{exp}} \setminus \{y_0\}, t, e, l
\end{aligned} \quad \text{(D.23)}$$

where $\overline{q}^{\text{init}}$ is the maximum new infrastructure investment in the initial period and ρ is the maximum growth rate per investment period.

D.3. Model parametrization

Table D.2 summarizes the key constants and parametric assumptions used in the RQ3 model formulation. Detailed cost parameters for vehicle technologies and charging infrastructure are provided in Tables 4.15 and 4.17, respectively. Country-specific grid charges are listed in Table 4.18.

Table D.2.: Model constants and parametric assumptions (RQ3).

Symbol	Description	Value	Unit
Vehicle and battery			
SOC^{init}	Initial state of charge	0.50	–
SOC^{final}	Minimum final state of charge	0.50	–
$\text{SOC}^{+,\text{charging}}$	Usable battery capacity for charging	0.80	–
$\hat{P}_{tg}$	Charging power (MCS)	1 000	kW
$speed$	Average driving speed (road)	80	km/h
W_{tg}	Payload capacity (ICEV / BET 2020 / BET 2050)	13.6 / 11.56 / 13.6	t
L^{annual}_{tg}	Annual mileage	136 750	km/a
Λ_{tg}	Vehicle lifetime (ICEV & BET)	10	a
Fleet dynamics			

Continued from previous page

Symbol	Description	Value	Unit
α^h, β^h	Maximum stock transition rate (drivetrain technology)	0.1	–
Modal shift			
$\bar{\alpha}^{\text{mode}}$	Maximum corridor-wide rail share	0.30	–
α_f, β_f	Modal shift rate-of-change parameters	0.1	–
c^{rail}	Levelized rail transport cost (incl. infrastructure amortization)	0.057	€/tkm
c^{terminal}	Rail terminal handling cost	4	€/t
	Rail availability (first year)	2026	–
Infrastructure constraints			
γ_t	Utilization factor (ratio of average to peak power demand)	0.30	–
ρ	Maximum infrastructure growth rate per period	0.50	–
	CPO charging markup	0.30	–
Economic			
δ	Discount rate	0.04	–
VoT	Value of time (freight)	30	€/h
τ_t	Toll rate (ICEV Euro VI / BET)	0.10 / 0.036	€/km
Temporal scope			
y_0	Initial year	2020	–
Δy	Investment period length	4	a
	Model horizon	41	a

D.4. Pre-processing of transport data

D.4.1. Network construction and OD pair selection

The case study network, transport demand data processing, and OD pair filtering are described in detail in Section 4.3. In summary, the network comprises 81 NUTS-2 regions across eight countries along the Scandinavian-Mediterranean TEN-T corridor. Transport demand is derived from the ETISplus database [174] and scaled using a GDP-based growth rate (CAGR 0.4%) [175]. After filtering for international routes with at least 100 tonnes annual volume and 360 km length [176], the dataset comprises 2,244 OD pairs. The optimization horizon is 2020–2060, with analysis focusing on 2020–2050.

D.4.2. Mandatory break location determination

The determination of mandatory break locations following EU Regulation 561/2006 is described in Section 4.3. Break locations are determined individually per route at the fixed average speed of 80 km/h, with 45-minute breaks after 4.5 hours of driving and 9-hour rest periods after 9 hours of daily driving.

D.5. SOC-aware charging heuristic benchmark

The SOC-aware benchmark estimates where along a route a battery-electric truck would need to charge, given mandatory driving break locations and finite battery capacity. Unlike a uniform distribution of charging across all break nodes, the SOC-aware rule accounts for the truck's state of charge and only charges when necessary.

The benchmark assumes the truck departs with a full battery, whereas the optimization model initializes the fleet-aggregate SOC at 50% of battery capacity ($SOC^{\text{init}} = 0.5$). This conservative starting condition in the model accounts for return trips and depot utilization, whereas the benchmark focuses on a single outbound trip. Consequently, the heuristic may skip slightly more early-route charging stops than the optimization model, marginally overstating the spatial shift toward mid- and late-route charging.

The algorithm proceeds as follows for each origin-destination pair:

1. The truck departs the origin with a full battery ($\text{SOC} = C_{\text{bat}}$), where $C_{\text{bat}} \equiv Q_{tg}^{\text{batt}}$.

2. Energy is consumed proportionally to distance at the technology-specific consumption rate $e_{\text{spec}} \equiv D_{tg}^{\text{spec}}$ (kWh/km), which depends on the vehicle generation.

3. At each mandatory break location b_i, the algorithm computes the energy required to reach the next break b_{i+1} (or the destination if no further break exists):

$$E_{\text{next}} = e_{\text{spec}} \cdot d(b_i,\, b_{i+1}) \tag{D.24}$$

   where $d(b_i,\, b_{i+1})$ is the road distance between consecutive breaks.

4. If $\text{SOC} < E_{\text{next}}$, the truck charges to full at b_i:

$$\Delta E_{b_i} = C_{\text{bat}} - \text{SOC}, \quad \text{SOC} \leftarrow C_{\text{bat}} \tag{D.25}$$

   Otherwise, the break is skipped (no charging).

5. After simulating the entire route, the charging shares at each node are scaled to match the model's actual total energy demand for that OD pair, preserving the spatial distribution pattern while ensuring energy balance.

This lazy charging strategy naturally delays charging toward mid- and late-route nodes, reflecting that trucks departing with a full battery can skip early breaks. For the case study corridor (generation 2032, $C_{\text{bat}} = 1{,}187$ kWh, $e_{\text{spec}} = 1.6$ kWh/km), approximately 43% of mandatory break stops are skipped. The resulting distribution serves as a physically motivated reference against which cost-optimized and breaks-only allocations are compared.

D.6. Additional results

This section presents supplementary results that complement the analysis in the main text. The figures illustrate the fleet transition dynamics, modal split evolution, and the effects of cross-border charging cost surcharges on the geographic allocation of charging infrastructure.

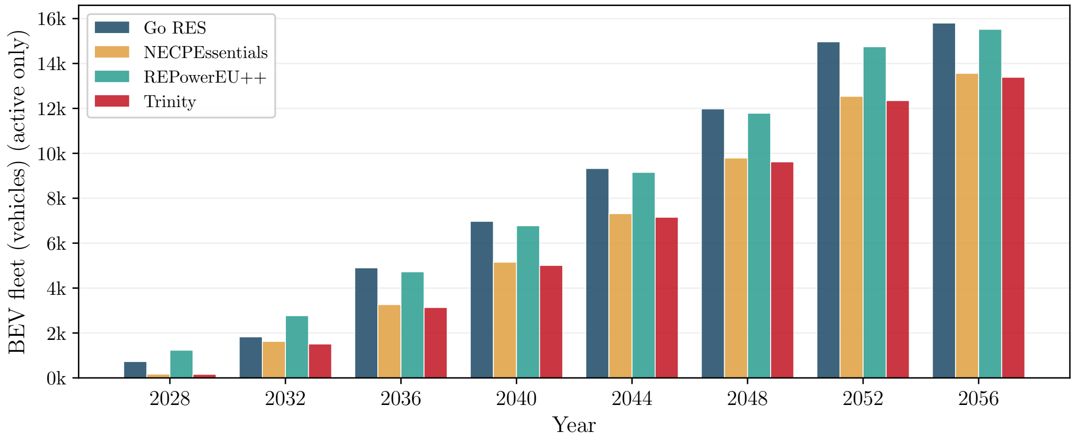

Figure D.1.: Battery-electric truck fleet growth across the four GENeSYS-MOD energy system scenarios along the Scandinavian-Mediterranean corridor. The scenarios group into two pairs: Go RES and REPowerEU++ (higher adoption) and NECPEssentials and Trinity (lower adoption).

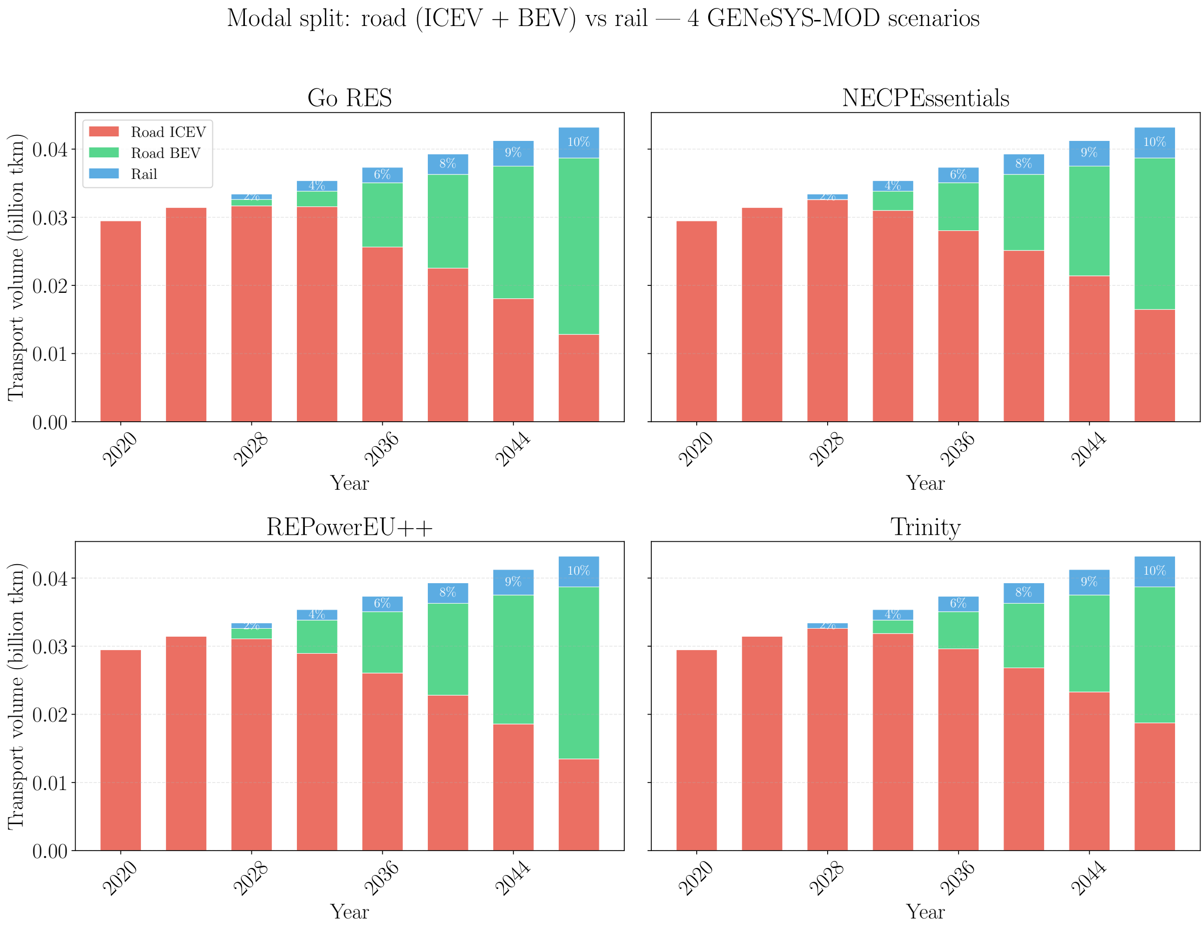

Figure D.2.: Modal split evolution for all four GENeSYS-MOD scenarios: transport volume (Mtkm) by mode and technology (road ICEV, road BET, and rail). The modal shift pattern is consistent across scenarios, with rail reaching approximately 10% by 2048 in all cases.

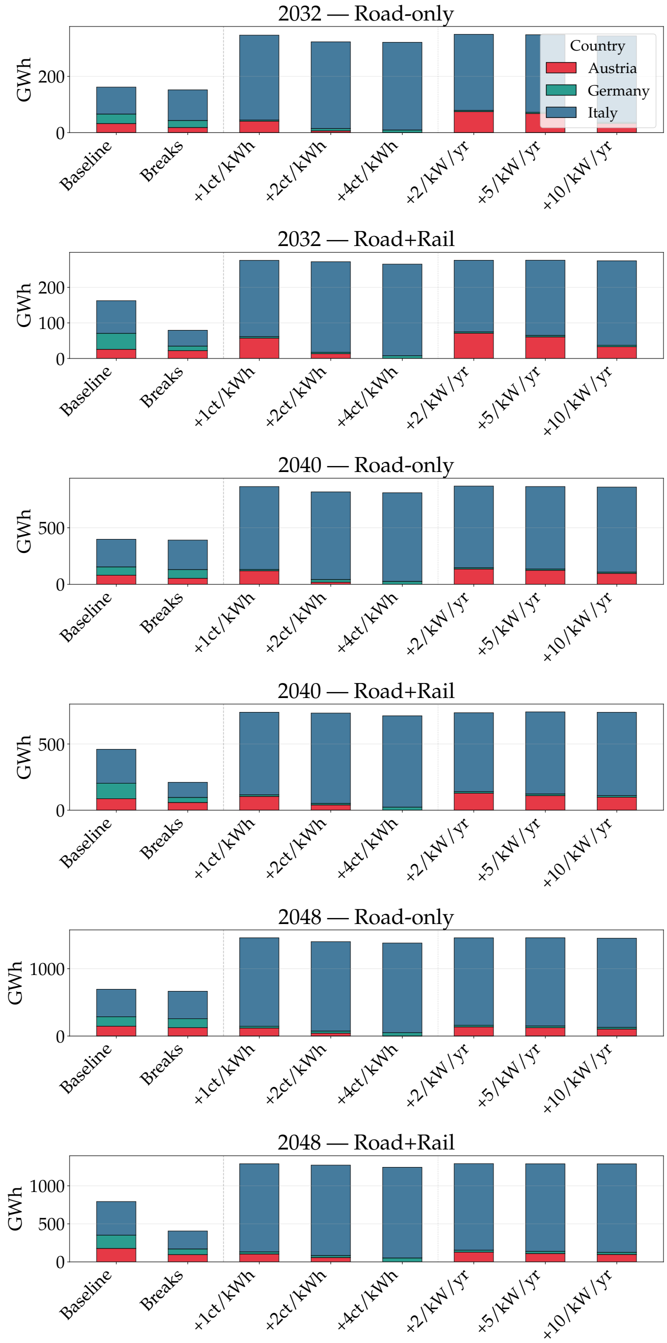

Figure D.3.: BET charging reallocation in the DE-AT-IT border region under Austrian electricity price and grid charge locational cost premiums for three time horizons (2032, 2040, 2048). Each panel shows stacked bars of charged energy (MWh) by country for reference scenarios (baseline, breaks constraint, fixed-fleet unconstrained) and graduated premium levels. Top rows: road-only; bottom rows: road with modal shift to rail.

D.7. Computational details

The model is implemented in Julia using JuMP 1.9 [205] and solved with Gurobi 12.0 [206]. Simulations are performed on an AMD Ryzen Threadripper 7980X (64 cores, 3.20 GHz base clock, 5.2 GHz boost) with 512 GB RAM.

The source code and input data are publicly available at: `https://github.com/antoniamgolab/iDesignRES_transcompmodel`.

Appendix E.

Supplementary Material for Contribution 4 (RQ4)

The content of this appendix is based on Golab et al. (2025). It supplements the results presented in Section 4.4.

This appendix presents results from the integration of the spatio-temporal commercial BEV fleet charging demand profiles—developed as part of this dissertation—into an electricity market model for dispatch and redispatch optimization. The dispatch and redispatch models were developed and implemented by Christoph Loschan; the results shown here illustrate how the bottom-up charging demand profiles interact with market clearing and congestion management mechanisms. Three electrification scenarios are considered for 2040: *LOW* (30% HDT electrification), *MEDIUM* (50%), and *HIGH* (100%), with light-duty vehicles and buses at 100% electrification in all scenarios (see Table 4.28). For each scenario, three case studies are compared: *Baseline* (inflexible charging at constant power), *Dispatch* (charging flexibility coordinated in the day-ahead market to minimize dispatch costs), and *Redispatch* (charging flexibility used for congestion management in the transmission grid).

E.1. Electricity market model

The European electricity market model *EDisOn* [216, 217] is a linear programming unit commitment model that encompasses the European electricity market and national hydrogen markets. The model operates in two sequential stages: dispatch and redispatch.

Dispatch The dispatch optimization minimizes overall costs for electricity generation over a reduced set of nodes with one node per bidding zone ($n \in \mathcal{N}_2$). These costs encompass the short-run marginal costs and start-up costs of thermal power plants, operation and maintenance costs for pumped hydro-power storage and renewable energy sources, penalties for non-supplied energy, and revenues from hydrogen production. The nodal demand balance constraint requires that the sum of exogenous inelastic demand, flexible commercial BEV fleet demand, and hydrogen electrolyzer demand equals the sum of thermal generation, fuel cell generation, storage dispatch, renewable generation (net of curtailment), net exchange, and non-supplied energy at each node and timestep. The dual variable of this constraint provides the day-ahead market price per bidding zone. The flexibility of the commercial BEV fleet enters as positive and negative demand regulation bounded by the maximum and minimum charging power, subject to the constraint that total charged energy within each plug-in period is preserved. The dispatch optimization is performed as a rolling horizon optimization with 26 steps of 336 hours each.

Redispatch Following the dispatch, a detailed model for Austria with 17 nodes ($n \in \mathcal{N}_3$) representing the transmission grid is used to identify congestion and calculate cost-optimal redispatch measures. The previously dispatched generation schedules are fixed and assigned to their respective nodes. Power flows are calculated using a direct current power flow approximation. The redispatch objective minimizes the costs of adjusting thermal generation, storage operation, and renewable curtailment to resolve grid congestion. The cost of increasing thermal generation as a redispatch measure equals the day-ahead market price or the short-run marginal cost, whichever is higher; decreased generation is refunded at short-run marginal cost. The commercial BEV fleet's charging flexibility is also available for redispatch. The redispatch optimization is performed for blocks of 24 hours. Further details on the dispatch and redispatch model constraints are provided by Loschan et al. [216].

E.2. Definition of the flexibility potential

Figure E.1 describes the definition of the flexibility potential in the charging of vehicles graphically as a function of time and the state of charge: A vehicle is connected to a charging station for a defined period during which a certain amount of electricity must be charged. In the present example, the battery needs to be recharged until its maximum capacity, $\mathrm{SOC}^{\mathrm{BEVmax}}$. An initial charging profile is given based on an assumption of the charging strategy of the vehicle. In the illustrated case, this charging strategy is charging at the lowest possible continuous power level ($\mathrm{CAP}^{\mathrm{BEVconst}}$), distributing the charged energy evenly during the plug-in period (indicated with the dark red line). A maximum possible charging power, $\mathrm{CAP}^{\mathrm{BEVmax}}$, defines the inclination of the green lines and, therefore, the time span of the fastest possible charging process and the latest possible point in time at which the charging process must start in order to fully recharge. The orange line illustrates an example of a load curve with a variable power level, with the power level at the beginning and at the end being significantly higher than during the mid-part of the plug-in period. The area bounded by the green oblique lines indicates the extent of possible charging curves to reach $\mathrm{SOC}^{\mathrm{BEVmax}}$ during the plug-in time period, which translates to the flexibility potential of the charging process. For example, in fast-charging processes where the charging happens at $\mathrm{CAP}^{\mathrm{BEVmax}}$ and the plug-in period is closer to the point of "fastest achievable charging completion", flexibility is more limited.

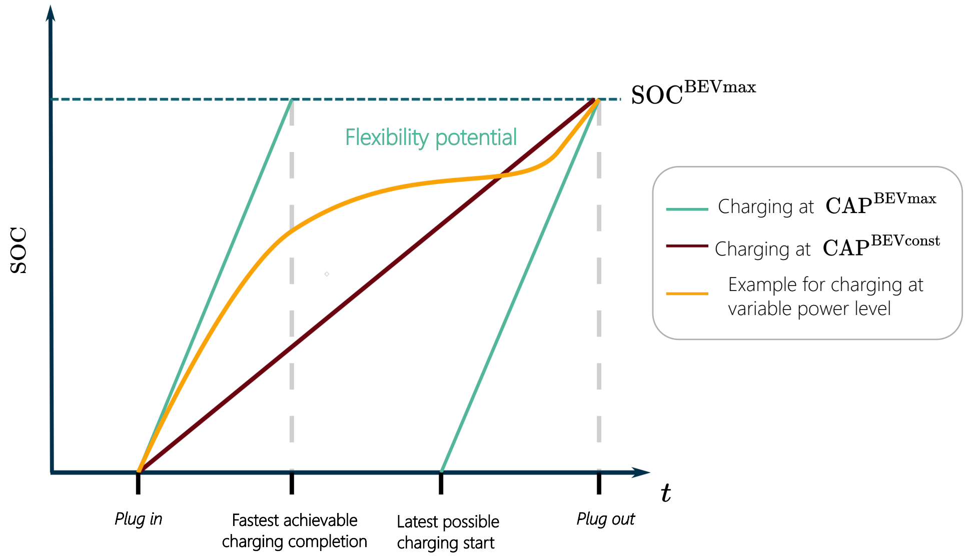

Figure E.1.: Graphic illustration of the flexibility of a charging process.

E.3. Charging infrastructure sizing and temporal distribution assumptions

The following assumptions on charging infrastructure sizing and temporal distribution of charging processes complement the charging strategy descriptions in Section 4.4.

Light-duty trucks The aggregate charging infrastructure for all depots within a federal state region is sized based on assuming that a charging power of 11 kW and charging the vehicle every sixth night is sufficient.

Heavy-duty trucks Charging in depots is assumed to be conducted at 100 kW of peak power and megawatt charging is available at charging stations. As a reference point for the sizing of charging infrastructure, the goals for the installment of 5 MW of grid connection at each charging station until 2035/2040 along the Austrian highway network by the highway infrastructure operator ASFINAG [199][1]. This is assumed to be available for the electrification rate of a heavy-duty fleet of 30%. A relative factor to the charging demand is deducted for each region based on the amount of existing service areas and applied to all charging location demands within the respective region.

Due to the short time frame of the break of 45 minutes and the used temporal resolution of the modeling being one hour, highway charging processes are confined to narrow time windows. To avoid artificially sharp demand spikes caused by the hourly resolution, charging is distributed across three distinct plug-in time frames of two hours each, allocated around noon.

[1]The exact number of 5 MW was stated by a representative of ASFINAG in the course of a stakeholder interview.

Buses During nighttime, each bus is connected to a 100 kW charging station. The power connection of the depot is limited to two-thirds of the summed peak power of the charging stations, which is sufficient to cover the charging demand during downtimes at night and is in line with the peak power achieved with coordinated and, therefore, cost-efficient charging [89].

E.4. Impact of market participation on dispatch

When the commercial BEV fleet's charging flexibility is utilized in the day-ahead market clearing, charging demand shifts toward hours of low electricity prices. During workdays, two price lows occur around noon and around midnight; these coincide with increased charging peaks. The nighttime peak during workdays increases by 80% compared to the *Baseline* charging profile. During weekends, significant price lows around noon create peak load values roughly eight times greater than the *Baseline* charging demand at that hour.

Figure E.2 shows the annual duration curve of total electricity demand in Austria for 2040 under the different case studies and electrification scenarios. The price-oriented dispatch coordination increases peak load values by up to 24% compared to the *Baseline* and decreases the load during hours of higher prices. This effect intensifies with rising electrification rates.

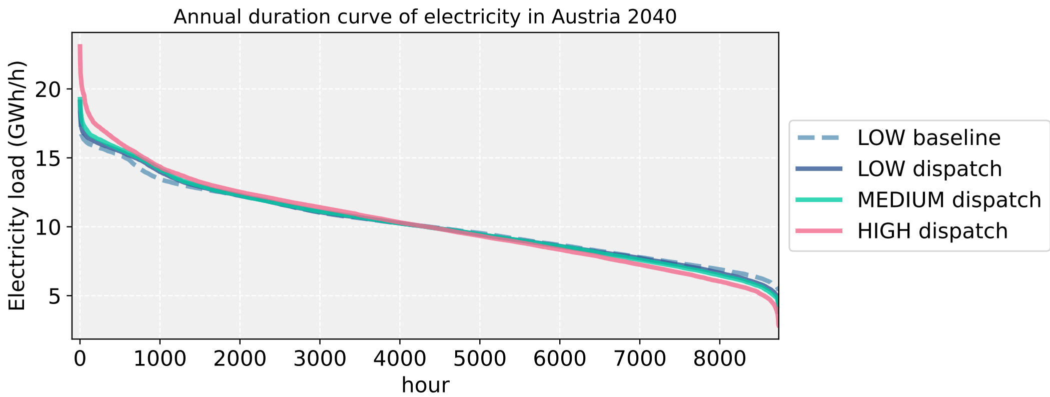

Figure E.2.: Annual duration curve of electricity demand in Austria in 2040 for different case studies and electrification scenarios.

E.5. Participation in redispatch measures

Figure E.3 displays the aggregate weekly charging demand profile resulting from the participation of the commercial BEV fleet in congestion management. The resulting load peaks are significantly lower than in the *Dispatch* case study and, on average, do not significantly exceed the levels of the *Baseline* load profile. On weekend days, the charging load is shifted to around noon, increasing the *Baseline* load by 68%—substantially less than the roughly eightfold increase observed in the *Dispatch* case.

In contrast to the dispatch optimization, where charging processes are aggregated to the bidding zone level, the geographic variations are particularly relevant to the redispatch modeling, where the topology of the transmission grid and geographically varying dispatch and load centers are important factors.

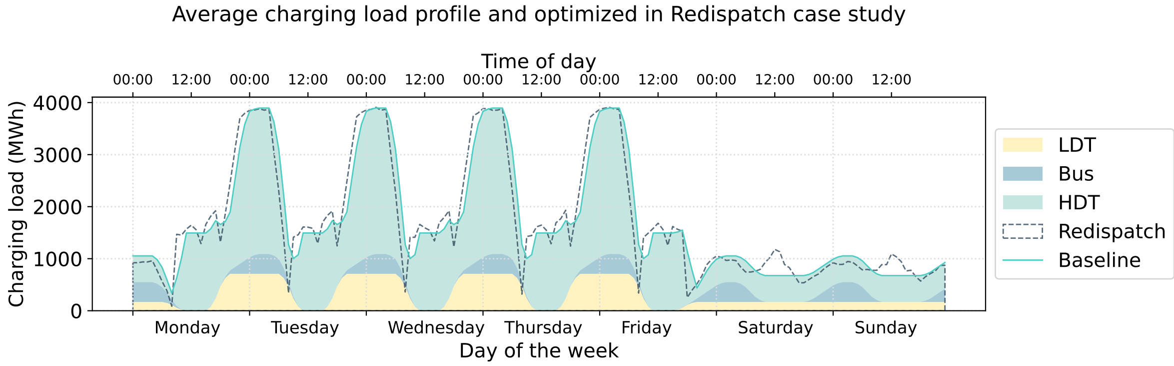

Figure E.3.: Aggregated charging demand profile of the commercial BEV fleet in the *Baseline* and *Redispatch* case studies.

E.6. Spatial variation in redispatch charging profiles

Figure E.4 displays the average weekly charging profiles at all nodes representing the Austrian transmission grid. Compared to the aggregate load profile in Figure E.3, significant variations are visible in both the absolute magnitude of the load profiles and in the shape of the curves, reflecting location-specific grid constraints and generation patterns.

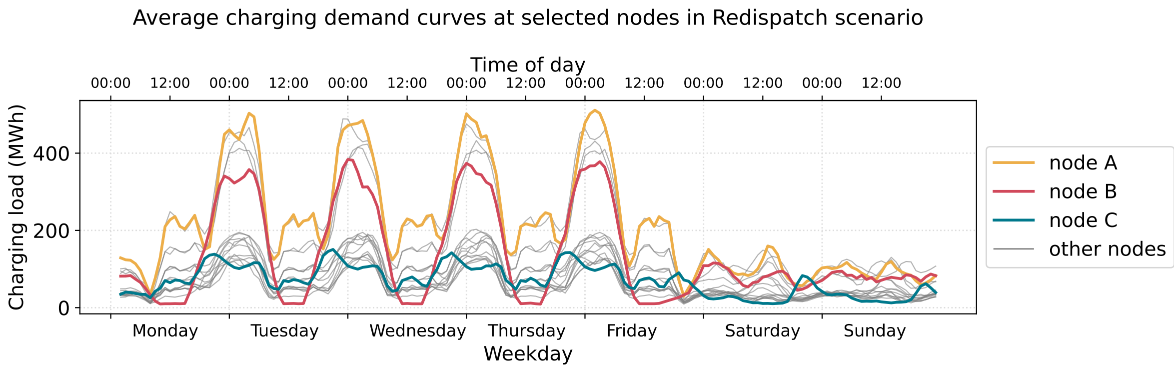

Figure E.4.: Charging load curves resulting from participation of the commercial BEV fleet in redispatch at different nodes in the Austrian transmission grid.

Three nodes are selected to illustrate the geographic differences in detail (Figure E.5):

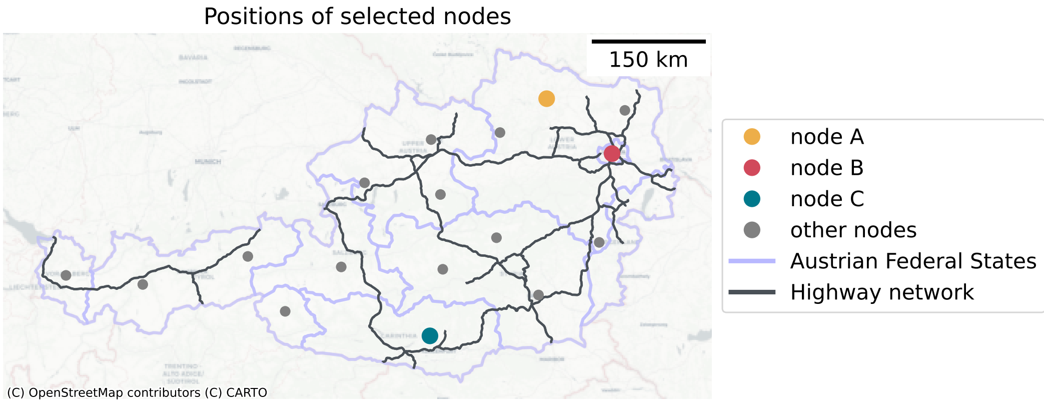

Figure E.5.: Geographical allocation of selected nodes representing the Austrian transmission grid in the redispatch model.

The corresponding charging profiles are shown in Figure E.6.

Node A (Lower Austria): Lower Austria is the federal state with the second-highest population and the largest area, as well as the highest total kilometers of highway and motorway network. Relatively high daytime peak loads are visible. When the commercial BEV fleet participates in redispatch, the charging profile does not significantly deviate from the *Baseline*; however, the absolute peaks during noon (highway charging) and at night around midnight increase. On Saturdays, peaks emerge similarly to the aggregate profile.

Node B (Vienna): Vienna has the highest population and population density, consisting almost exclusively of urban area. This is reflected in the high share of light-duty truck charging demand and the lowest daytime demand for charging at highway service areas. Similarly to Node A, no significant deviations from the *Baseline* profile are observed under redispatch.

Node C (Southern Austria, near Italy): The shape of the charging profile is significantly altered when the commercial BEV fleet participates in redispatch measures. During workdays, the nighttime charging load shifts substantially to earlier times. During weekends, load peaks appear around 6 pm—a pattern distinct from other regions—reflecting the influence of local grid topology and cross-border power flows.

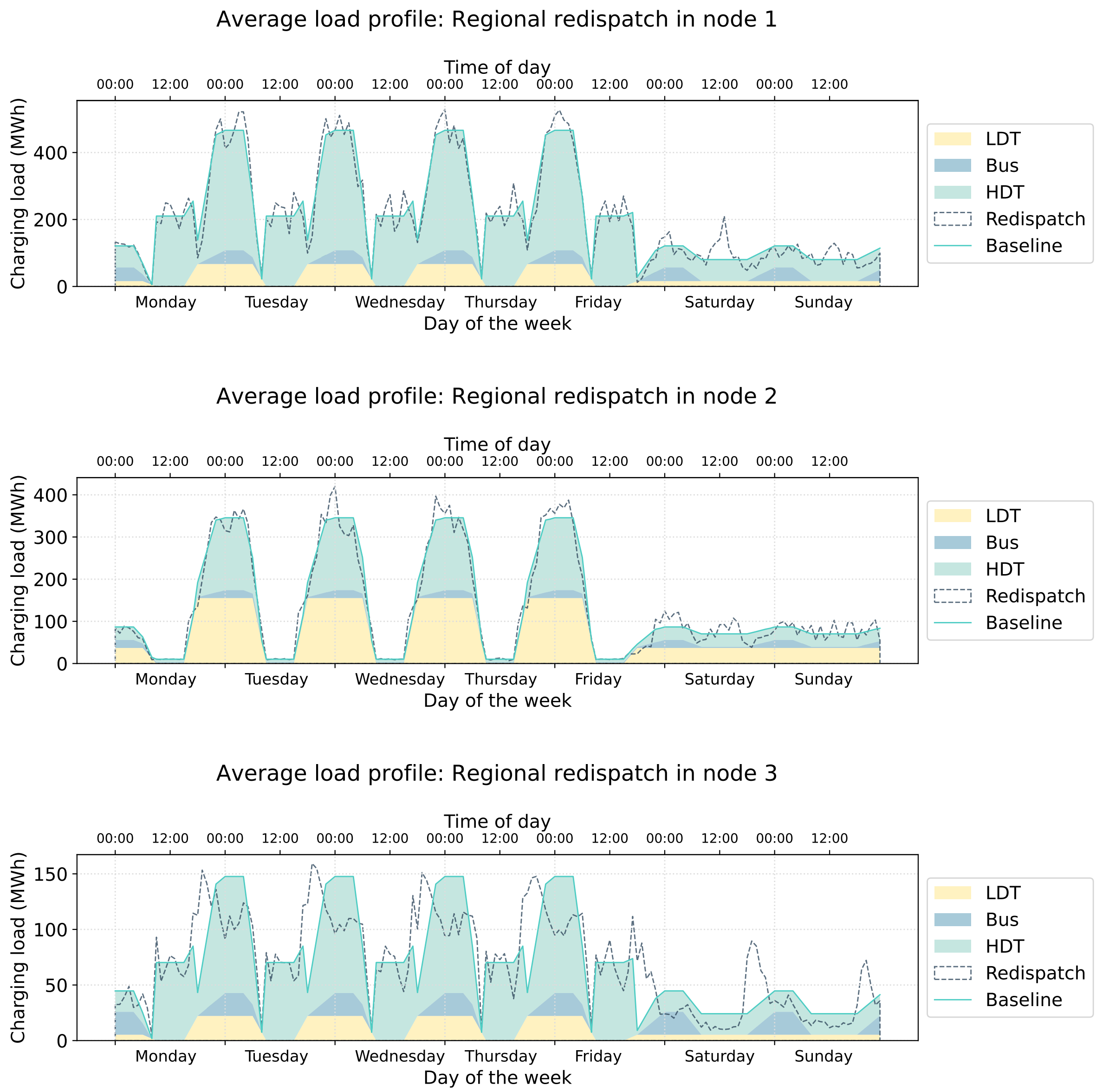

Figure E.6.: Average weekly charging profiles at three selected nodes for the *Baseline* and *Redispatch* case studies: Node A (Lower Austria, top), Node B (Vienna, middle), and Node C (Southern Austria, bottom).

E.7. Comparison of redispatch costs

Figure E.7 compares the redispatch costs across the three case studies and electrification scenarios. Three key observations emerge:

First, in the *Baseline* case, the increasing electrification of the commercial fleet has a limited impact on redispatch costs. The costs decrease slightly by 4 million €between the *LOW* and *HIGH* scenarios.

Second, the increasing electrification of the truck fleet substantially affects the redispatch costs when charging flexibility is utilized in the dispatch. At an electrification rate of 30% of the HDT fleet, costs increase by approximately 40% between the initial (*Baseline*) charging strategy and dispatch-coordinated charging. This demonstrates that market-price-based flexibility utilization, which ignores transmission grid topology, increases grid congestion and the associated redispatch costs.

Third, the coordination of charging profiles to minimize redispatch costs effectively decreases the costs by approximately 30–35% relative to the *Baseline* case, highlighting the value of grid-aware charging coordination.

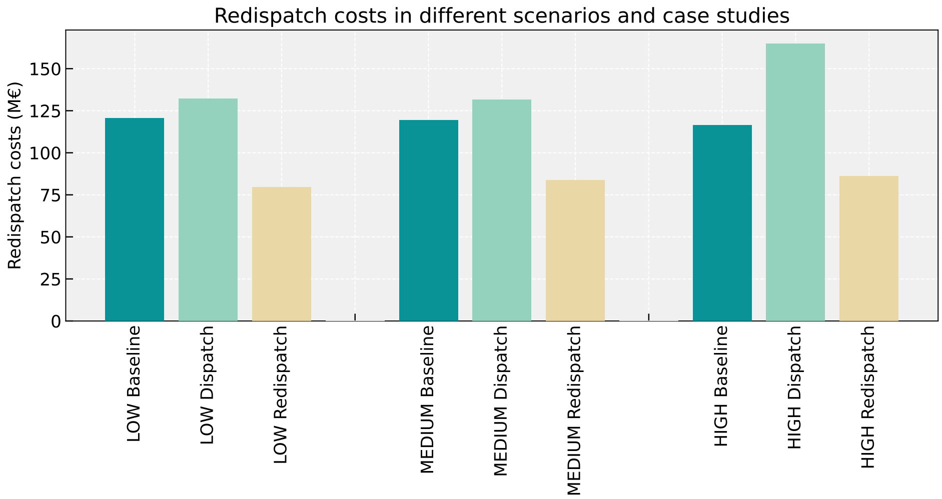

Figure E.7.: Comparison of redispatch costs for different electrification scenarios and case studies for Austria in 2040.

These results emphasize that with the increasing electrification of the transport sector, the cross-sectoral consideration of electricity and transport becomes increasingly important. The activation of demand-side flexibility of the charging processes is not weighted by any additional cost parameter in the dispatch objective and is exclusively limited by plug-in times and charging capacities. This implies that if fleet operators coordinate fleet charging based on market prices alone, substantial peak loads occur and redispatch costs increase. Targeted regulatory incentives—such as peak or dynamic pricing—are needed, and these would have to be dependent on the geographical allocation to be effective.